Inteligencia Artificial

Oscar Bedoya

oscar.bedoya@correounivalle.edu.co

* Algoritmos de búsqueda informada
* Heurísticas
* Búsqueda Avara
* Algoritmo A*



Un lamento de pájaro,

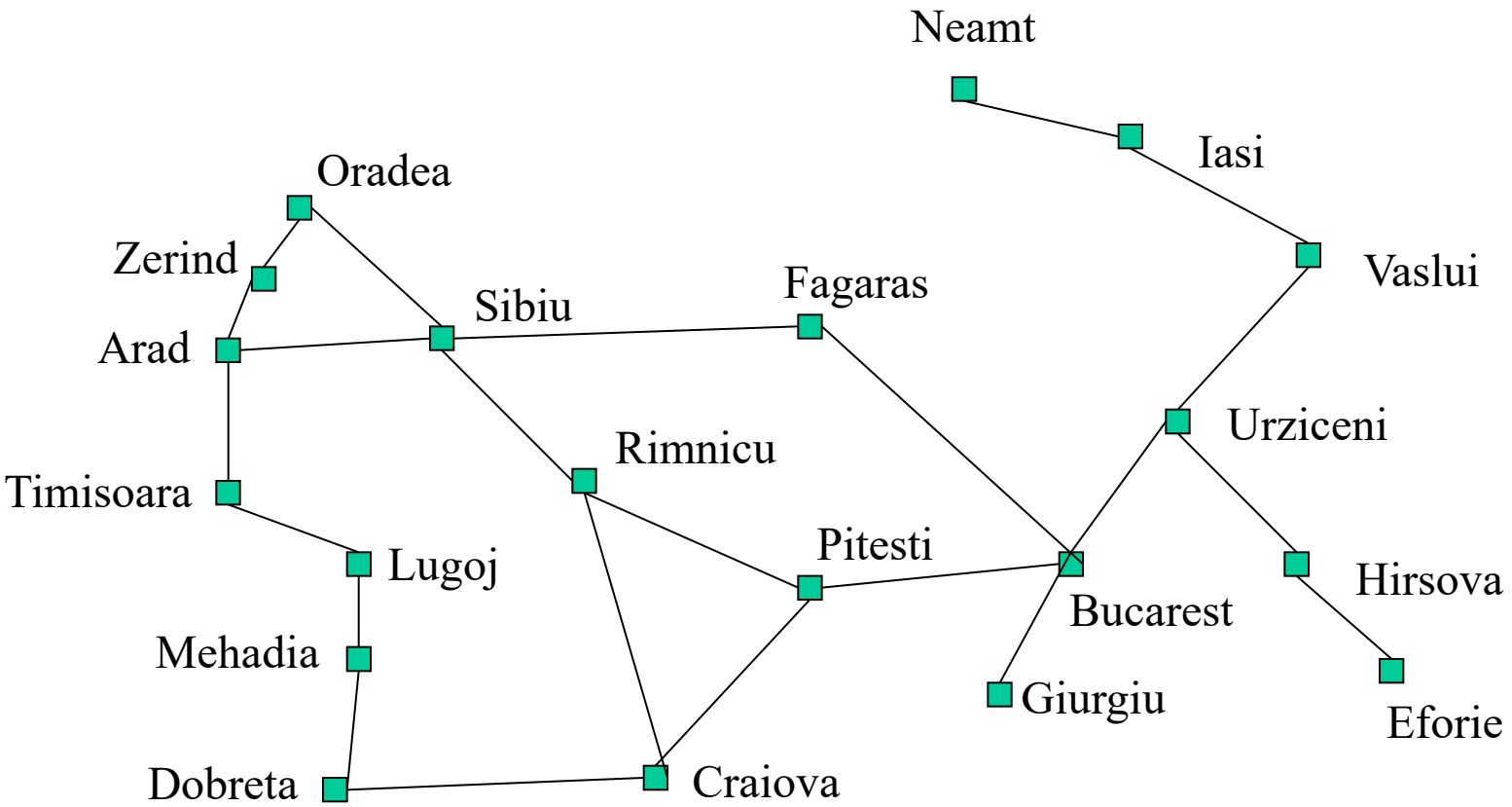
Esta vez, muy de mañana,

En el fondo del barranco.

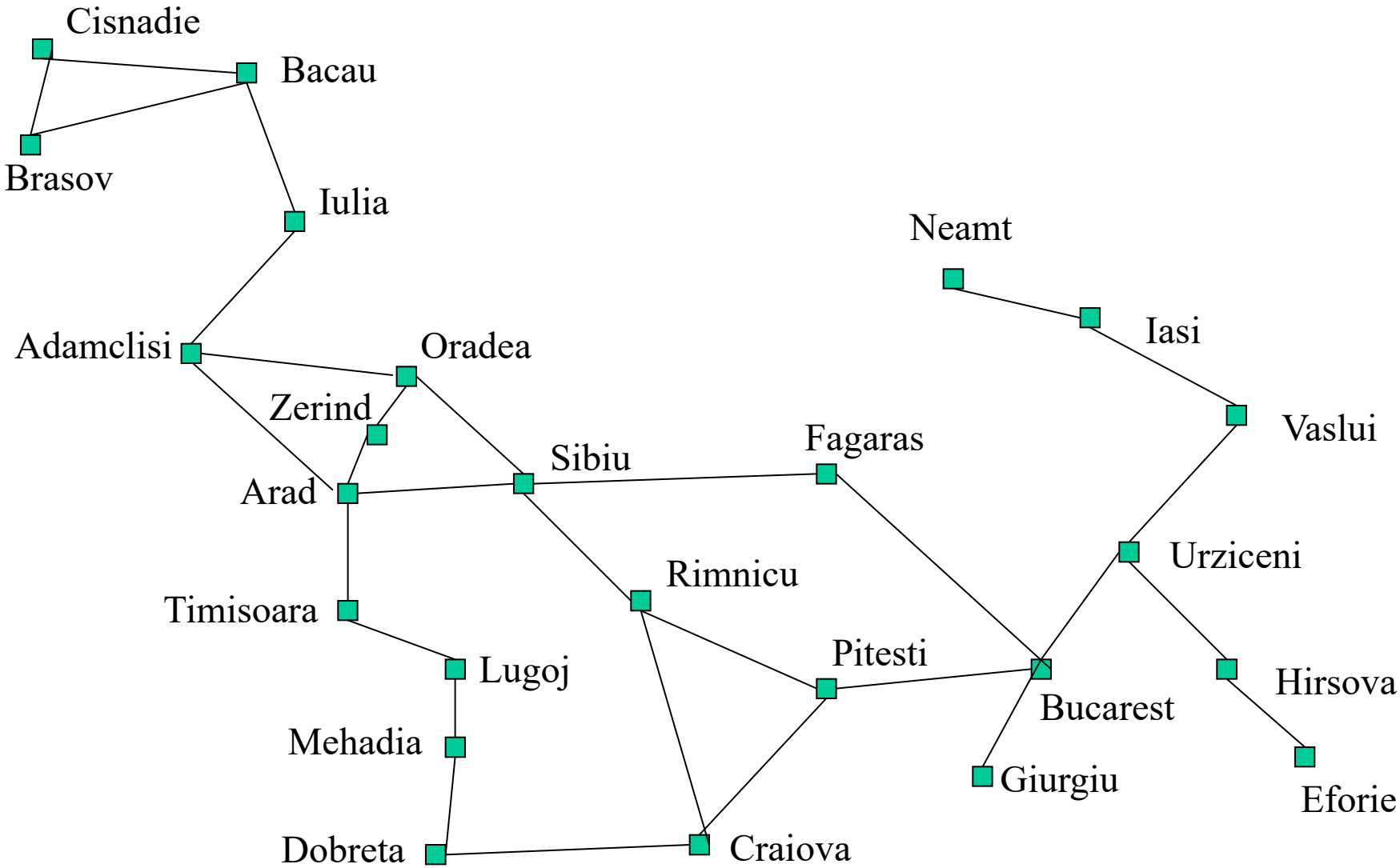
Búsqueda informada

- **Búsqueda no informada:**
  - Costo en tiempo
  - Costo en espacio

Búsqueda informada



Búsqueda informada



Búsqueda informada

¿Cómo mejorar la búsqueda?

Seleccionar mejor entre los nodos de un árbol

Búsqueda informada

Función heurística

- Una heurística es una función que asigna a cada nodo un valor que corresponde al **costo estimado de llegar a la meta** estando en dicho nodo
- Se denota por **h(n)**

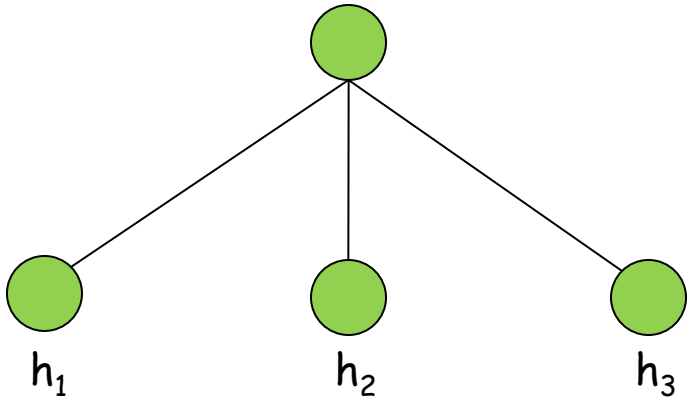
Búsqueda informada

Función heurística

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- Se denota por **h(n)**

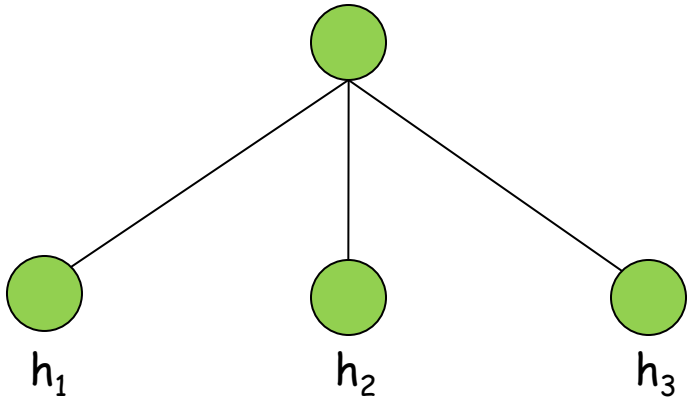
La heurística estima el valor real, no es 100% seguro

Búsqueda informada

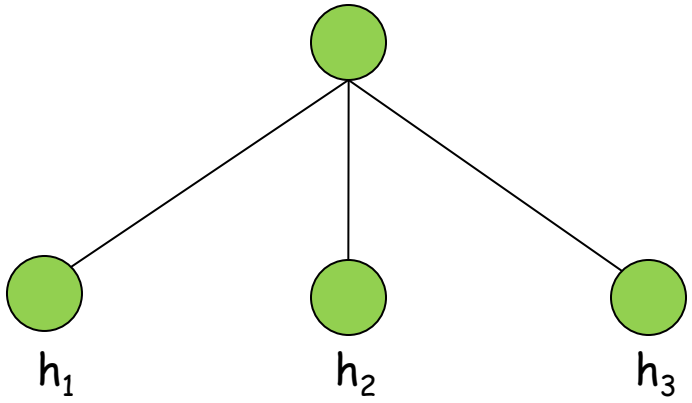


Búsqueda informada

h_1: costo estimado de llegar a la meta desde este nodo

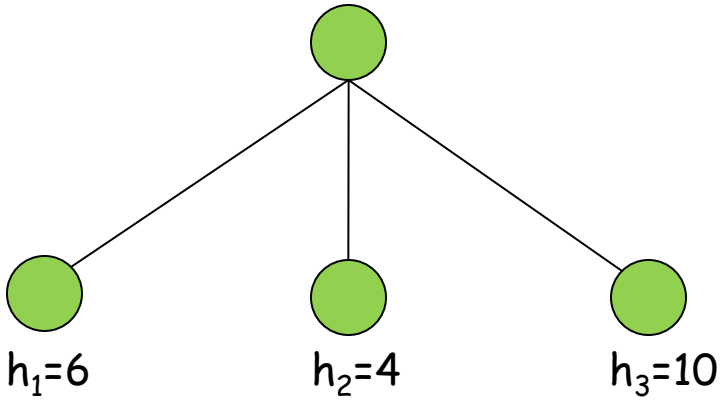


Búsqueda informada



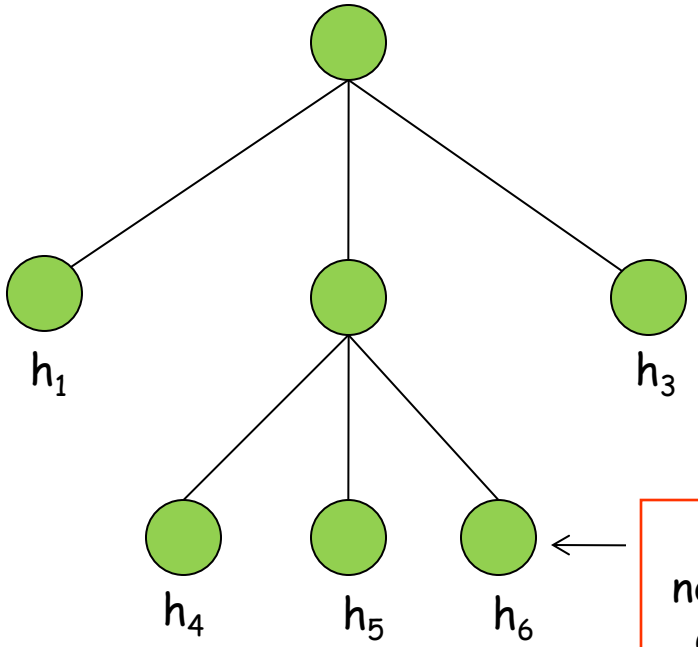
Si conociera los valores de h, cuál de los tres nodos escogería

Búsqueda informada



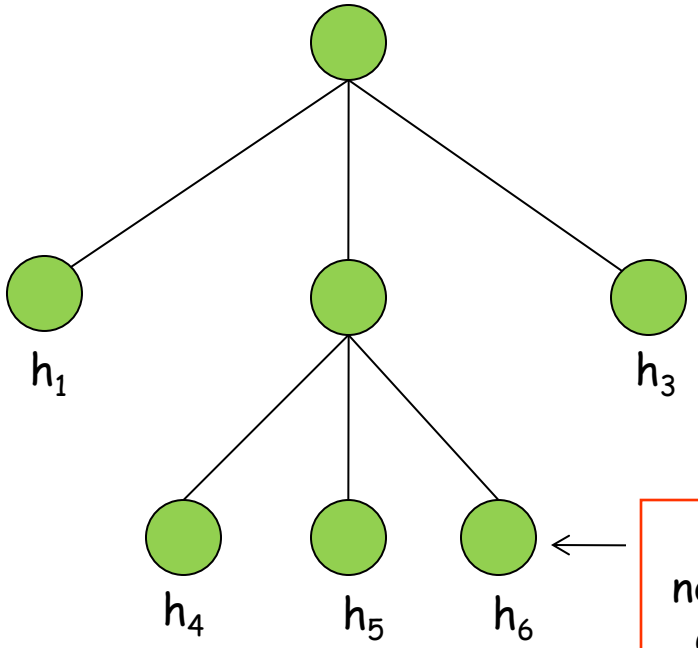
Si conociera los valores de h, cuál de los tres nodos escogería

Búsqueda informada



Suponga que se llega a un nodo meta, ¿cuál debería ser el valor de la heurística en este nodo?

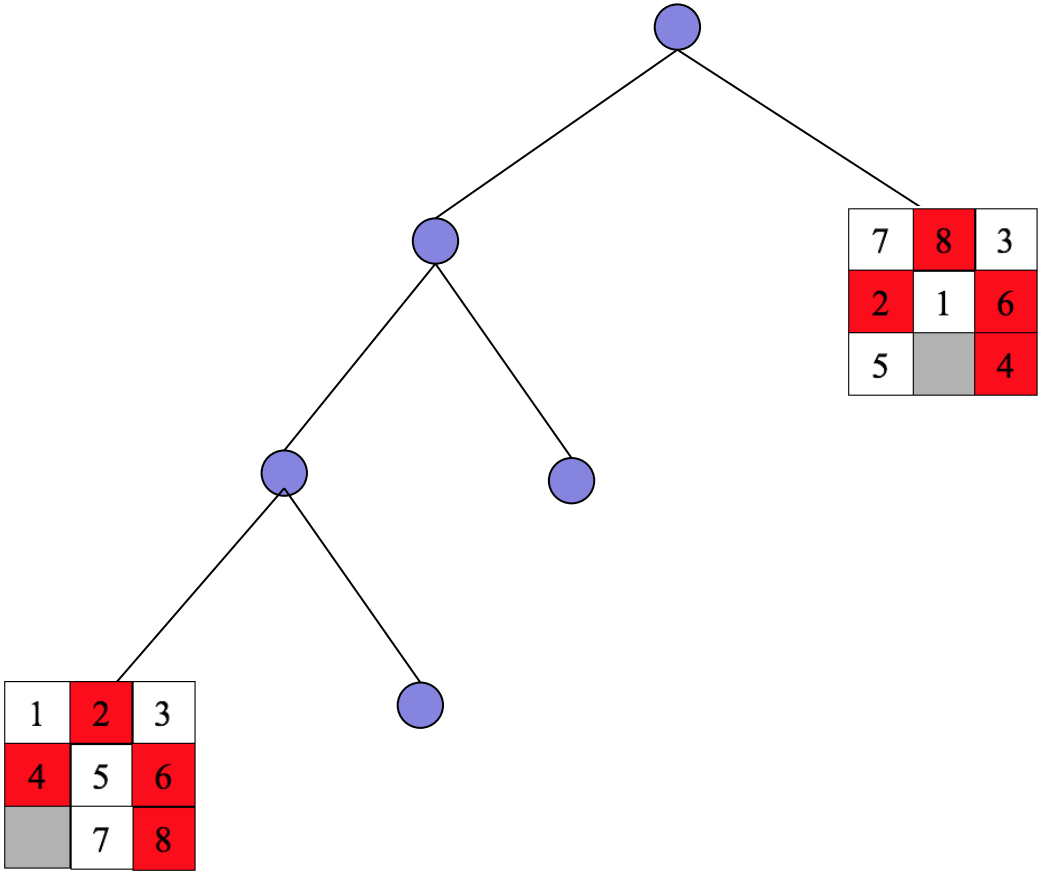
Búsqueda informada



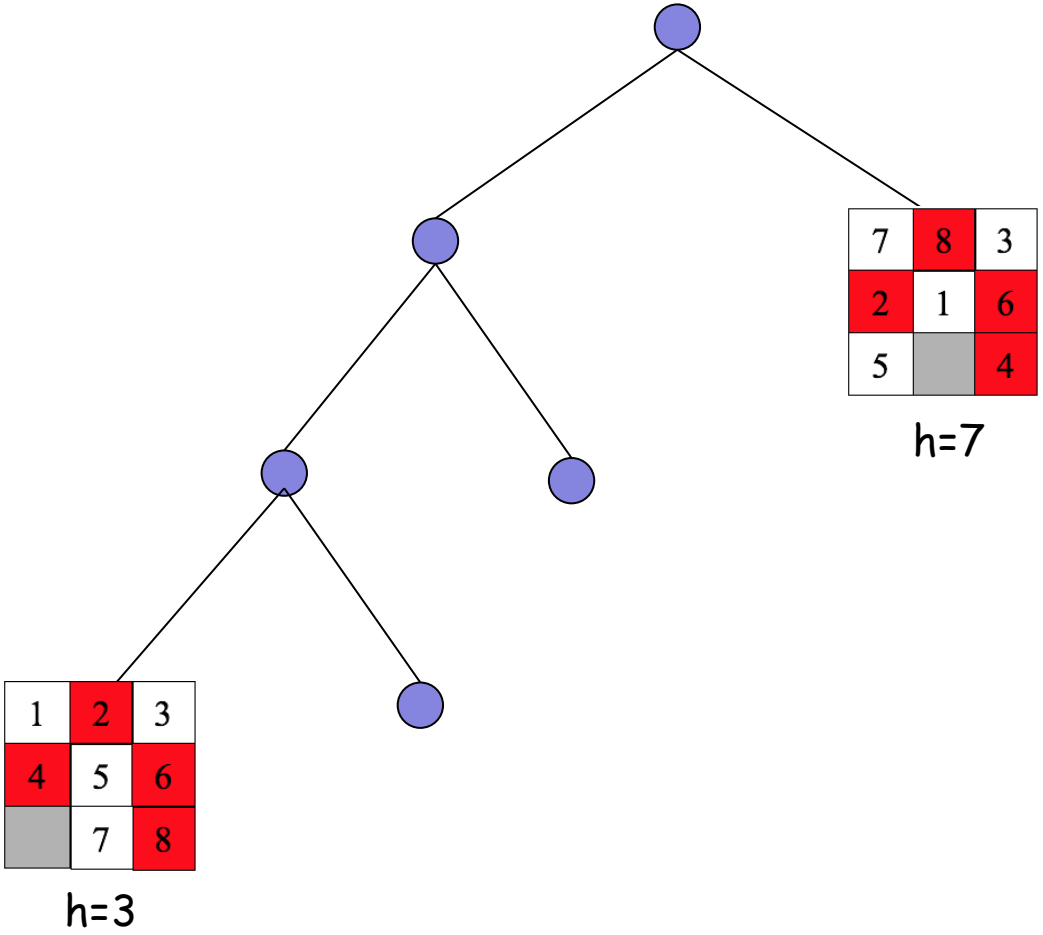
Suponga que se llega a un nodo meta, ¿cuál debería ser el valor de la heurística en este nodo?

La heurística h(n) debe ser 0 cuando n es un nodo meta

Búsqueda informada

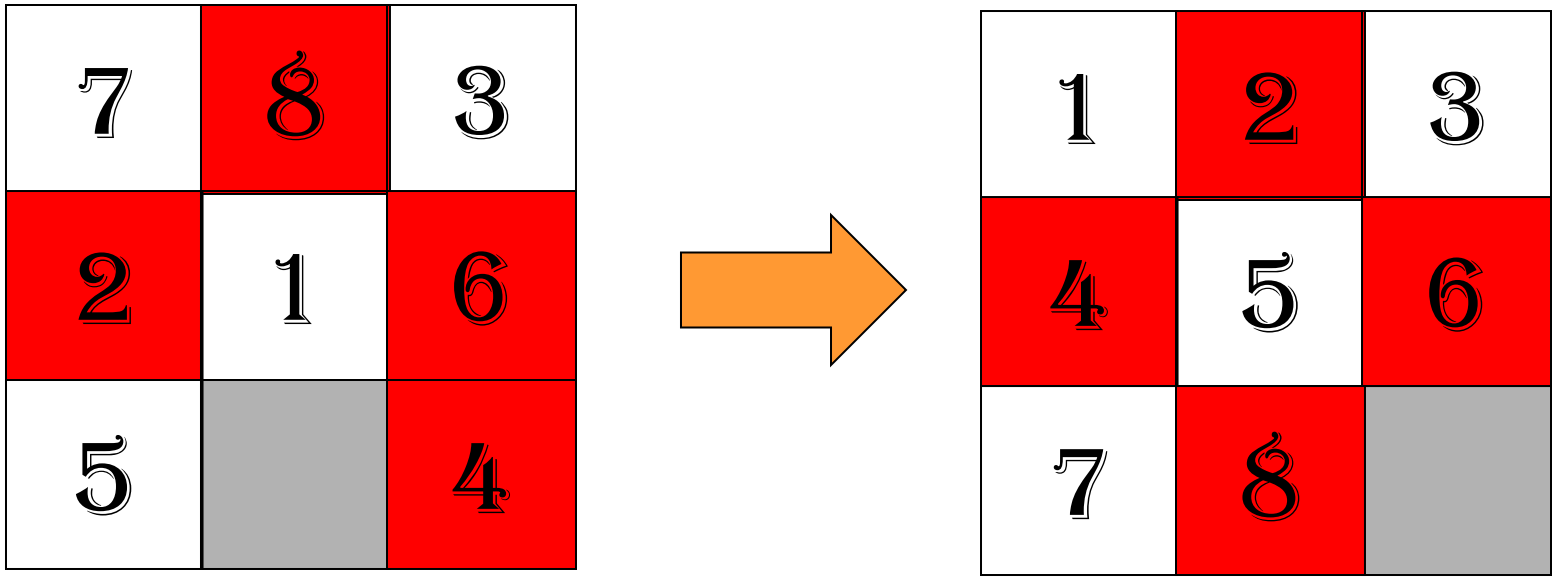


Búsqueda informada



Búsqueda informada

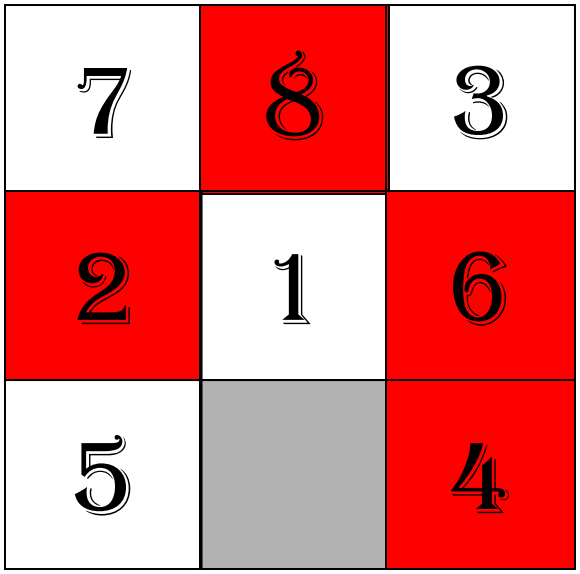
8-puzzle

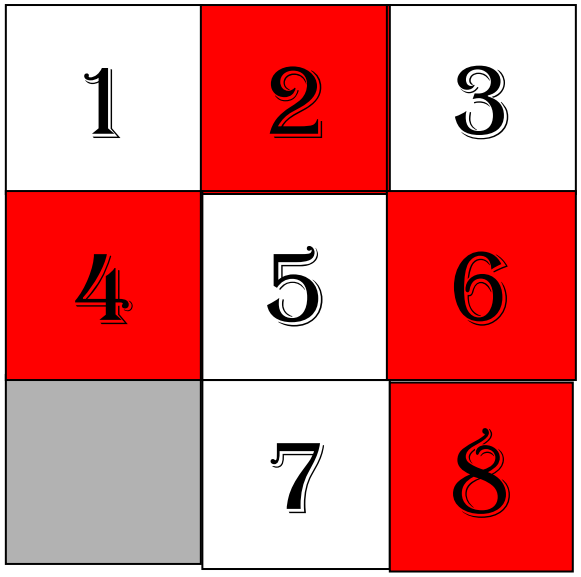


¿Cuál podría ser una heurística?

Búsqueda informada

8-puzzle



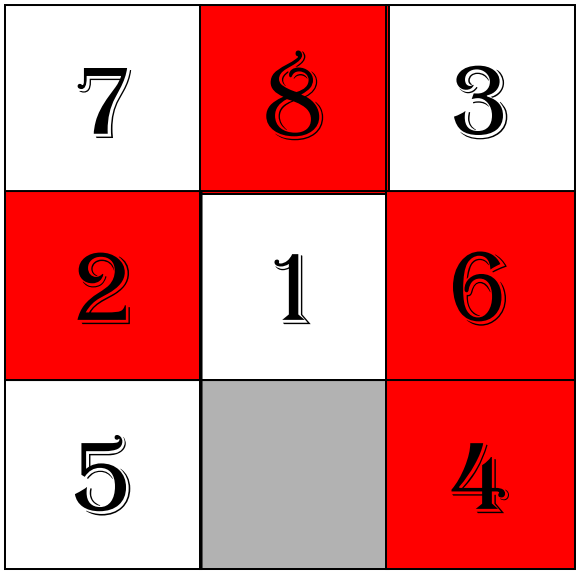


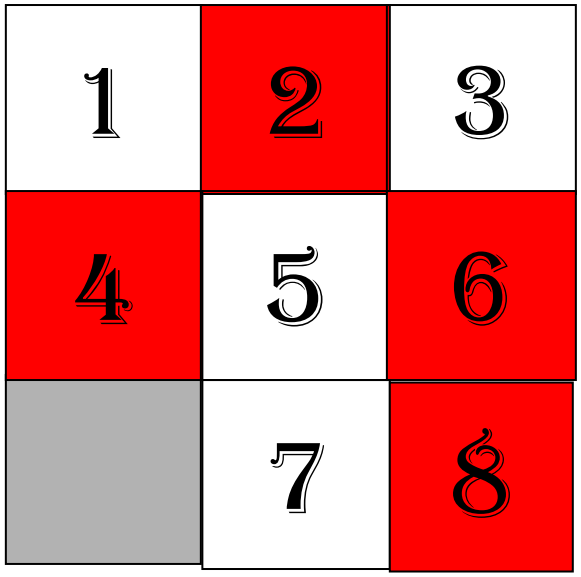
¿Cuál está más distante de la meta?

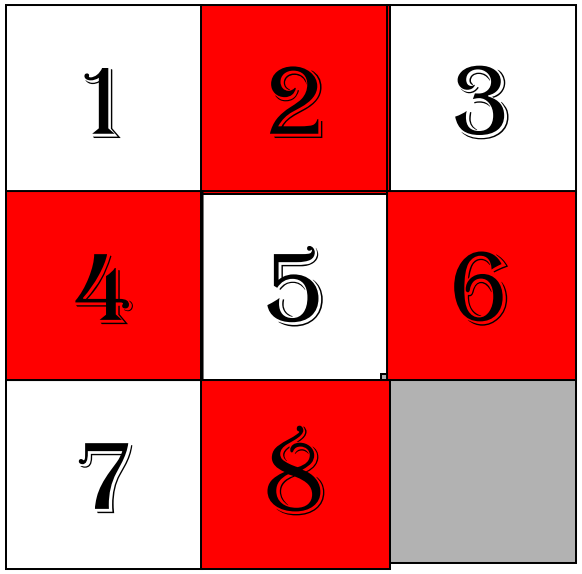
¿Cómo lo sabe?

Búsqueda informada

8-puzzle





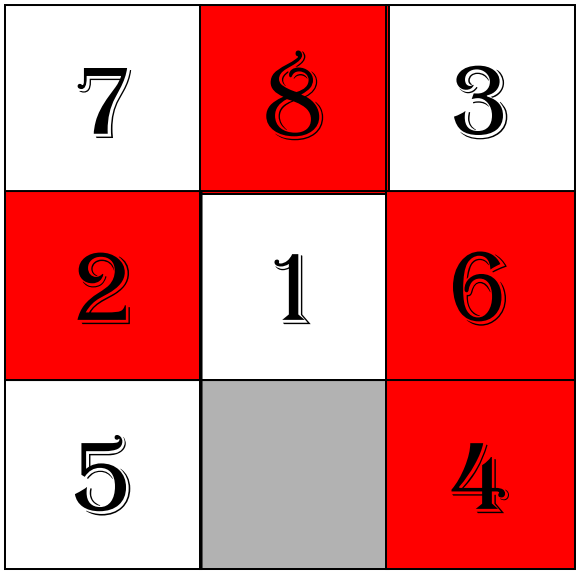


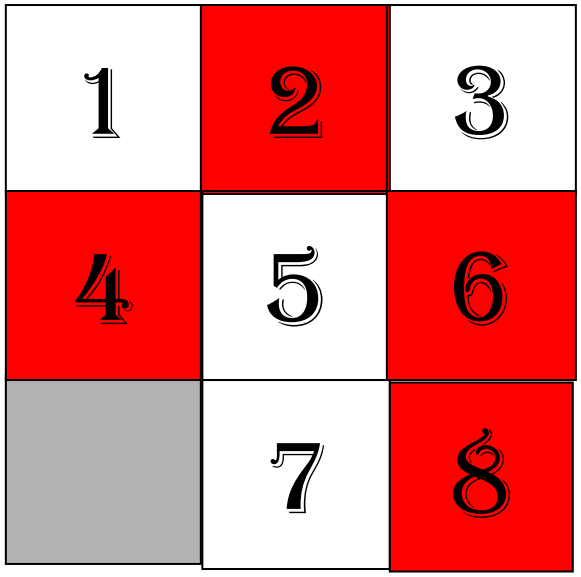
¿Cuál está más distante de la meta? ¿Cómo lo sabe?

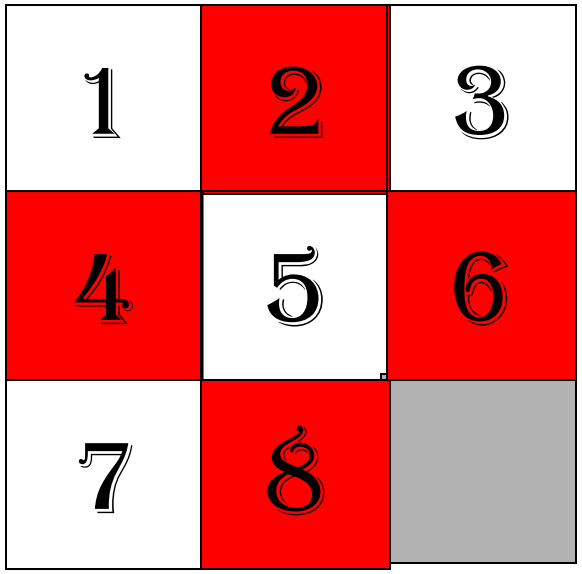
¿Cuál es el valor de la heurística en la meta?

Búsqueda informada

8-puzzle



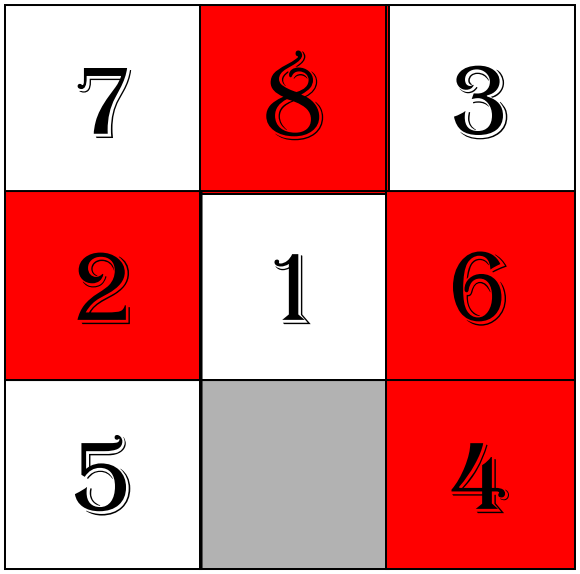




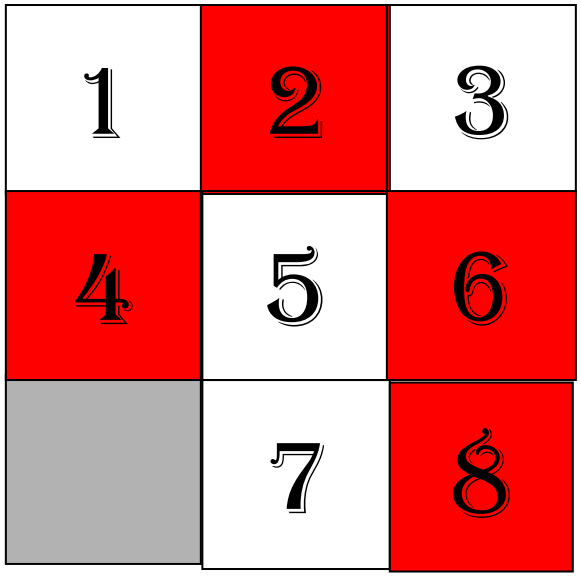
h(n): cantidad de placas en la posición incorrecta

Búsqueda informada

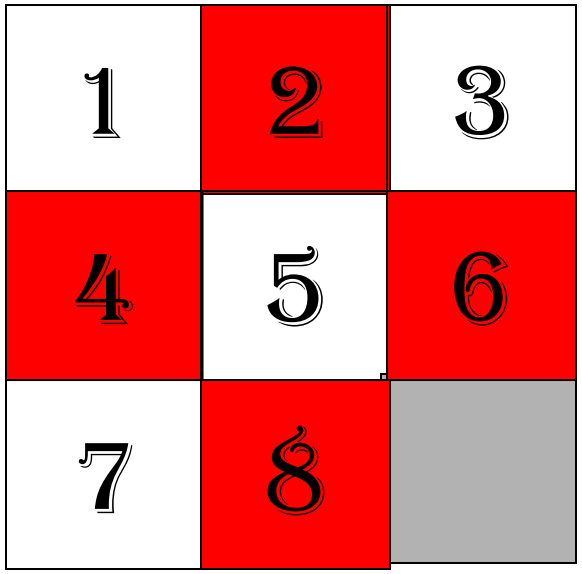
8-puzzle



h(n)=?



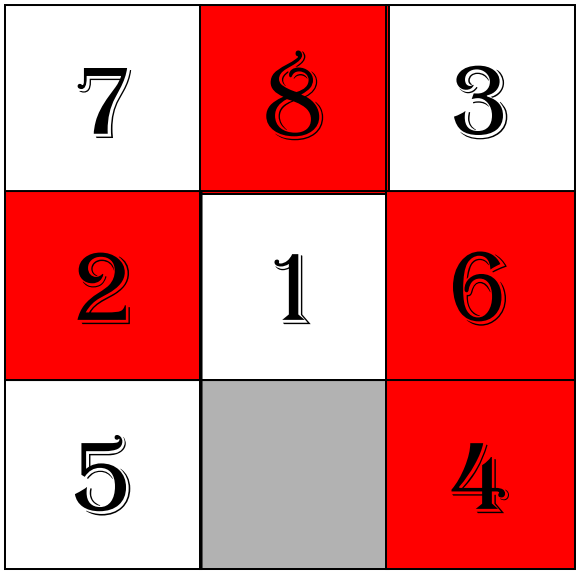
h(n)=?



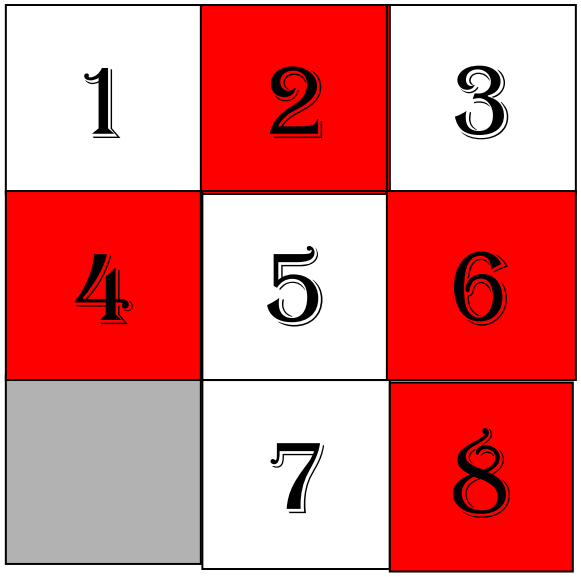
h(n)=?

Búsqueda informada

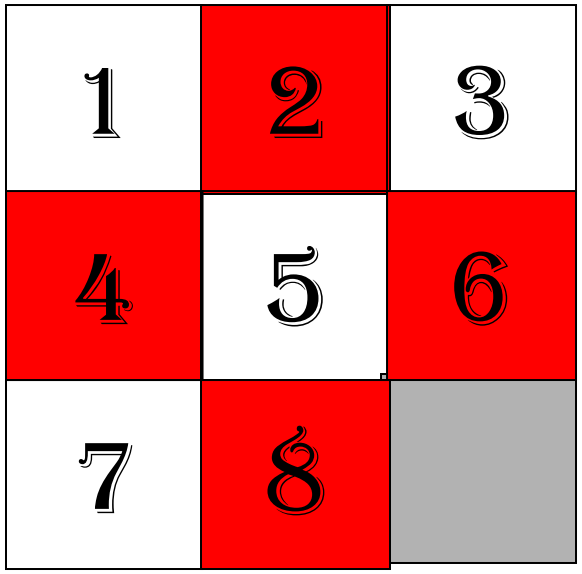
8-puzzle



h(n)=7



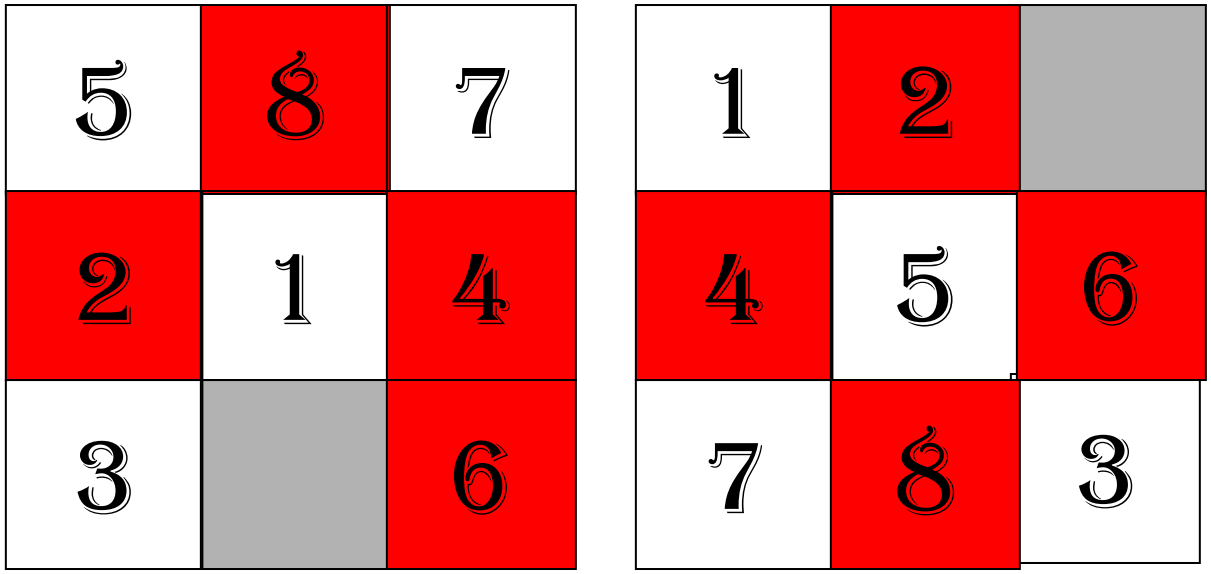
h(n)=3



h(n)=0

Búsqueda informada

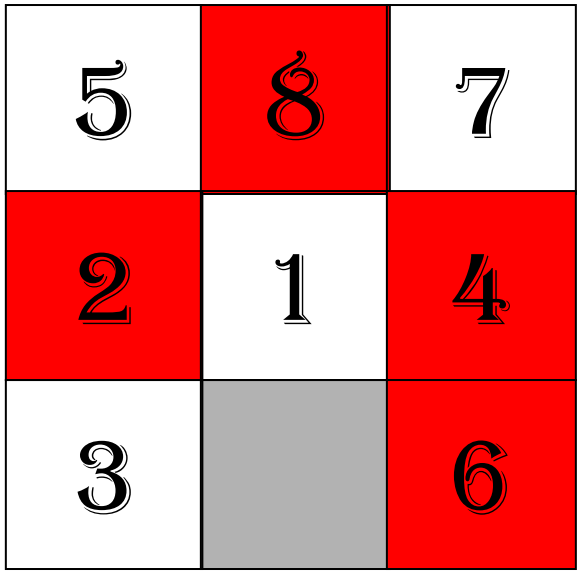
8-puzzle



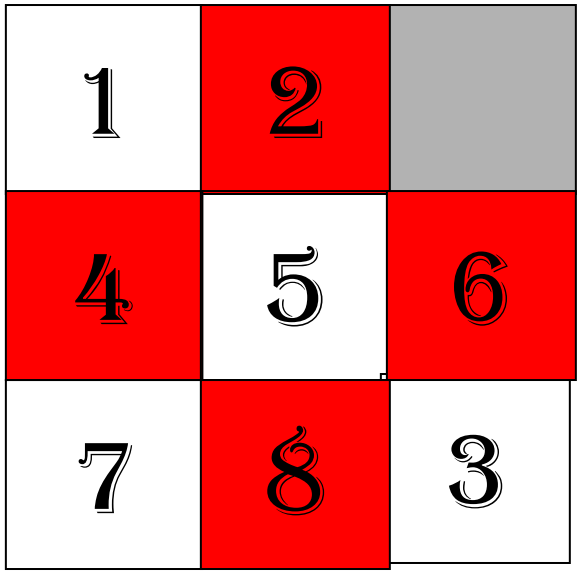
¿Cuál es de estos nodos escogería para expandir en un árbol de búsqueda?

Búsqueda informada

8-puzzle



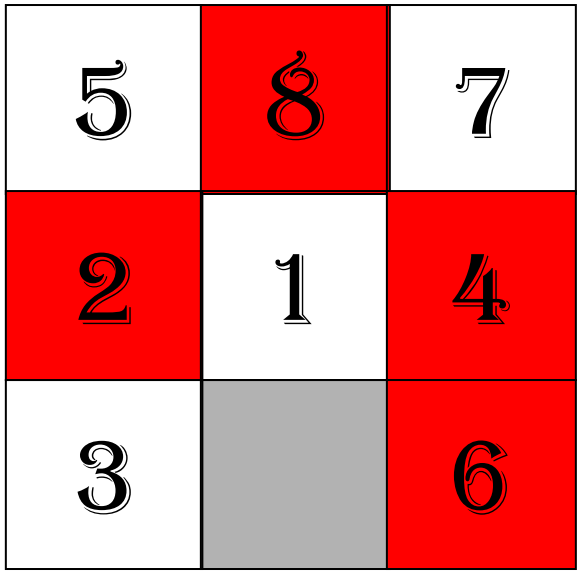
h(n)=?



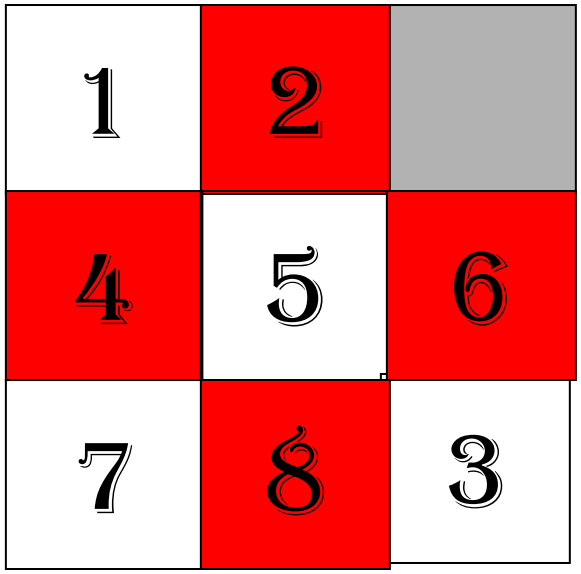
h(n)=?

Búsqueda informada

8-puzzle



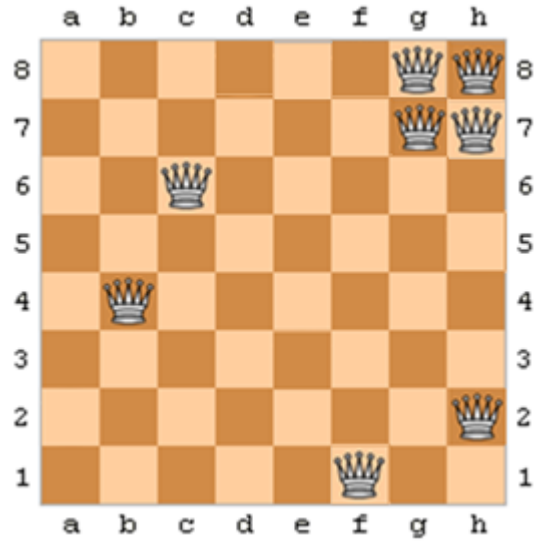
h(n)=9



h(n)=2

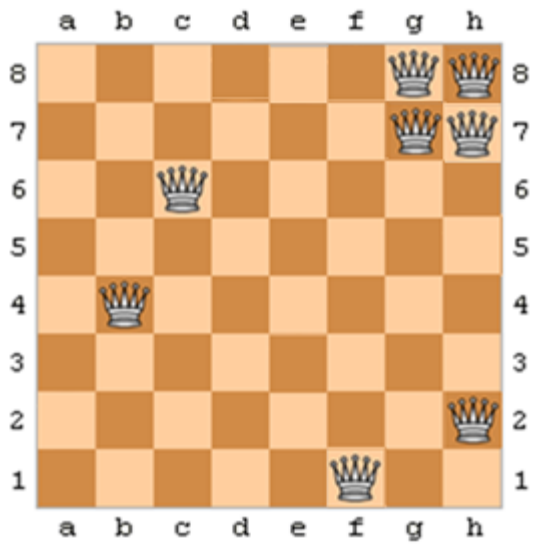
Búsqueda informada

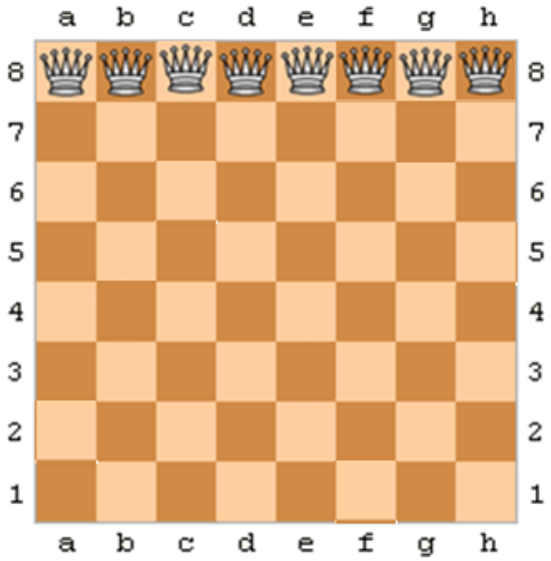
- Defina una **heurística**



El problema de las n reinas

Búsqueda informada

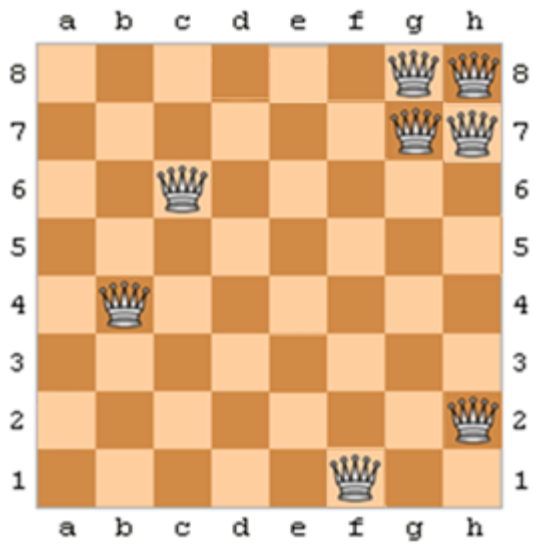


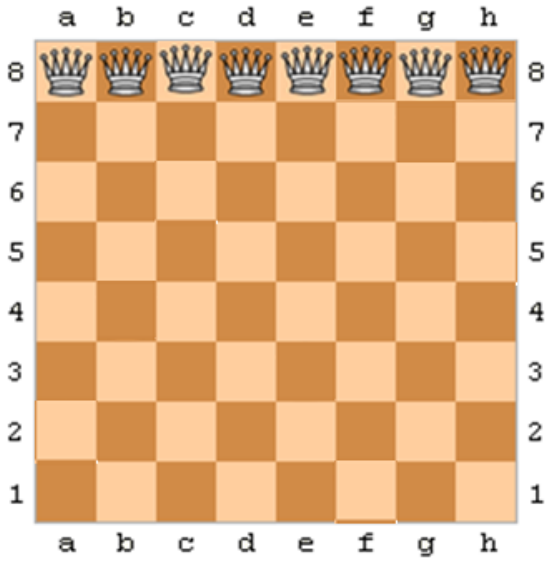


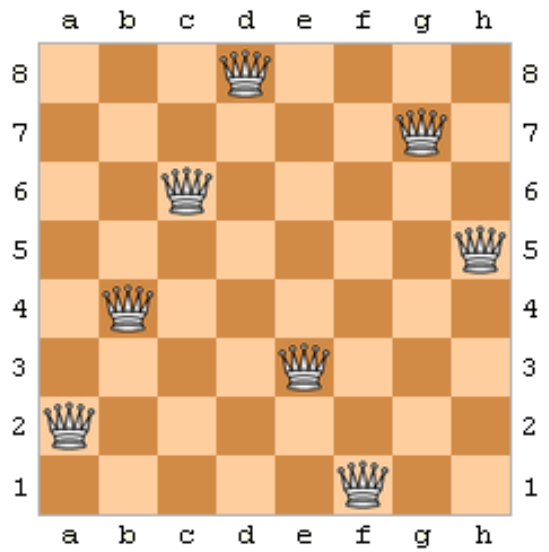
¿Cuál está más distante de la meta?

¿Cómo lo sabe?

Búsqueda informada

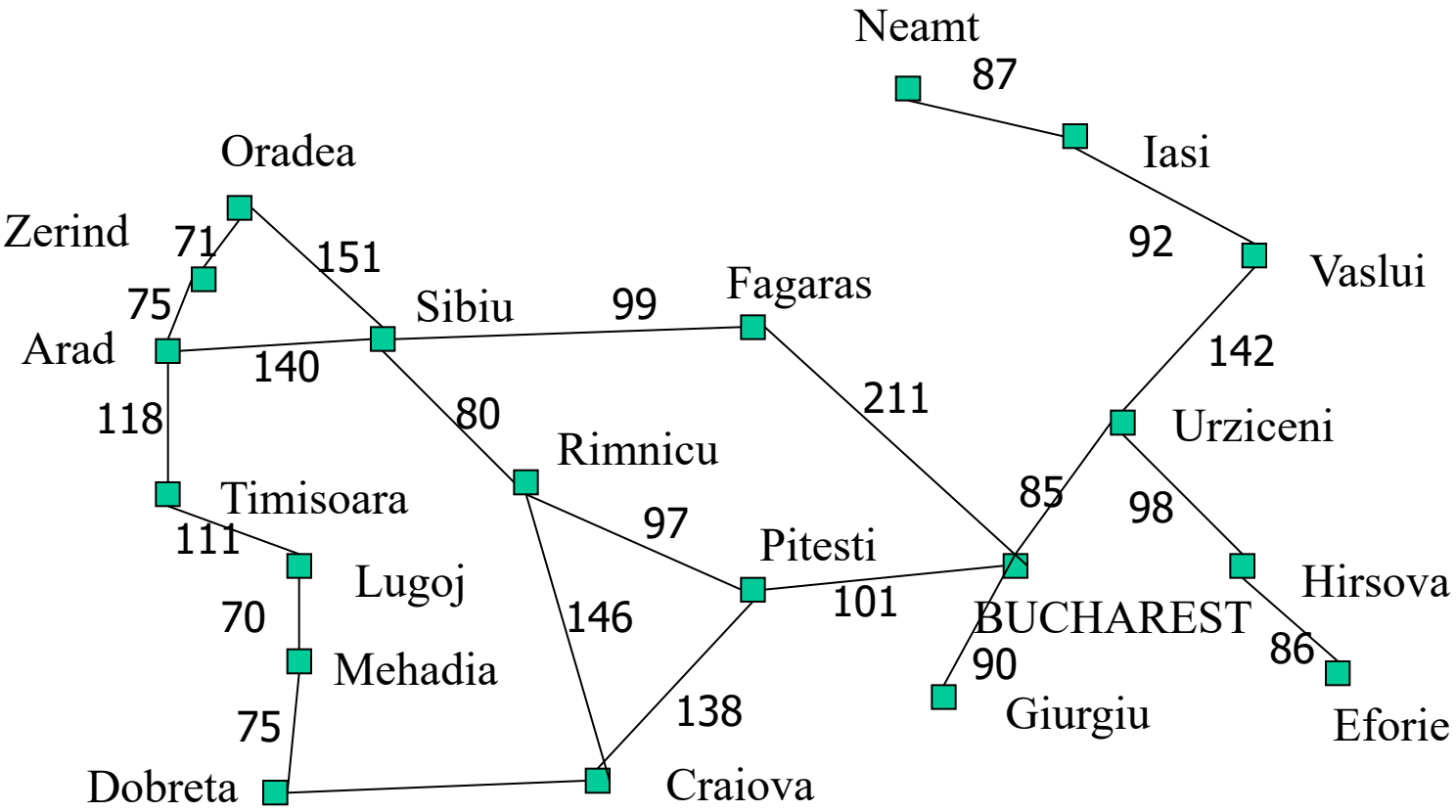






h(n): cantidad de pares de reinas que se atacan

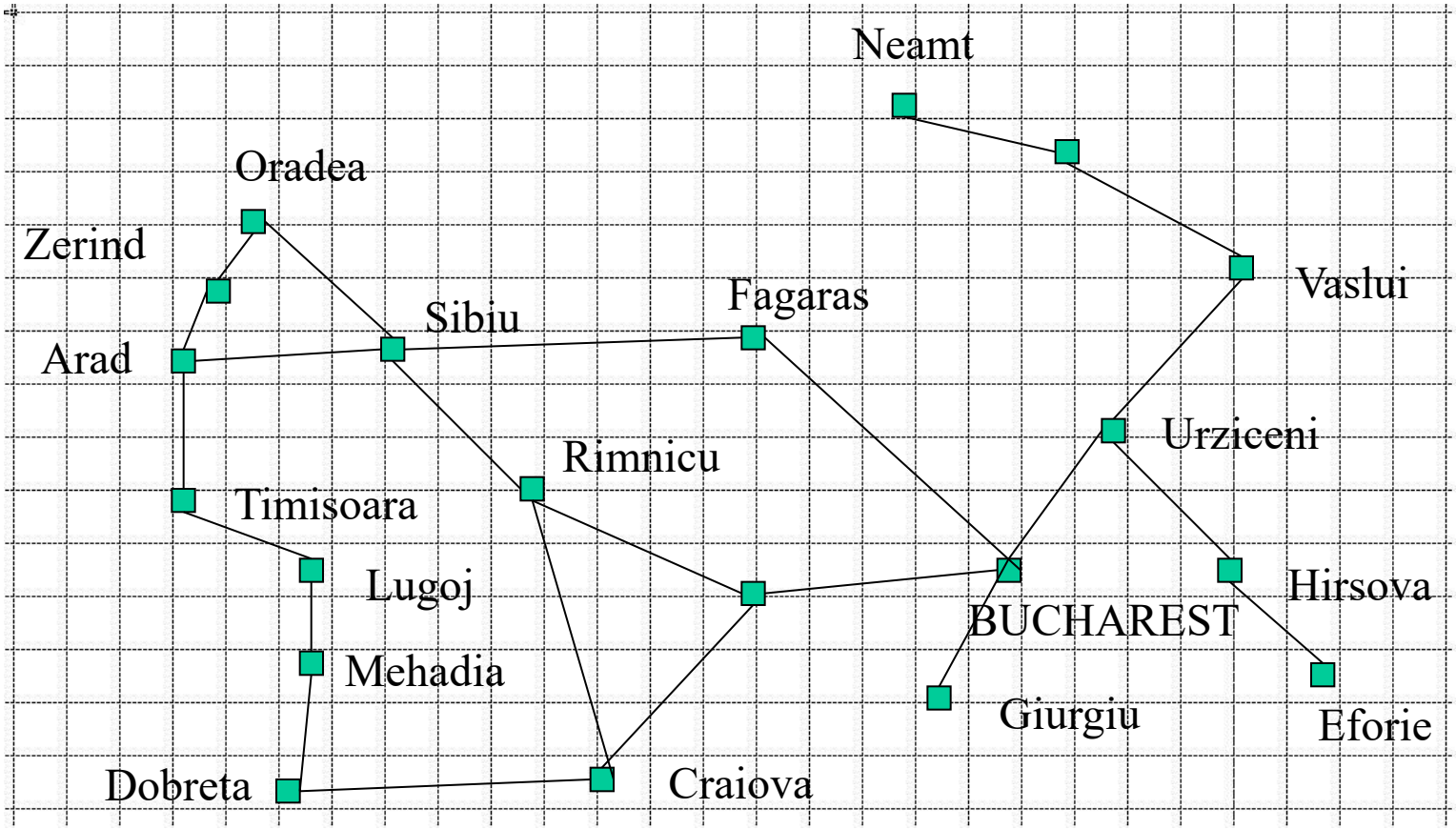
Búsqueda informada



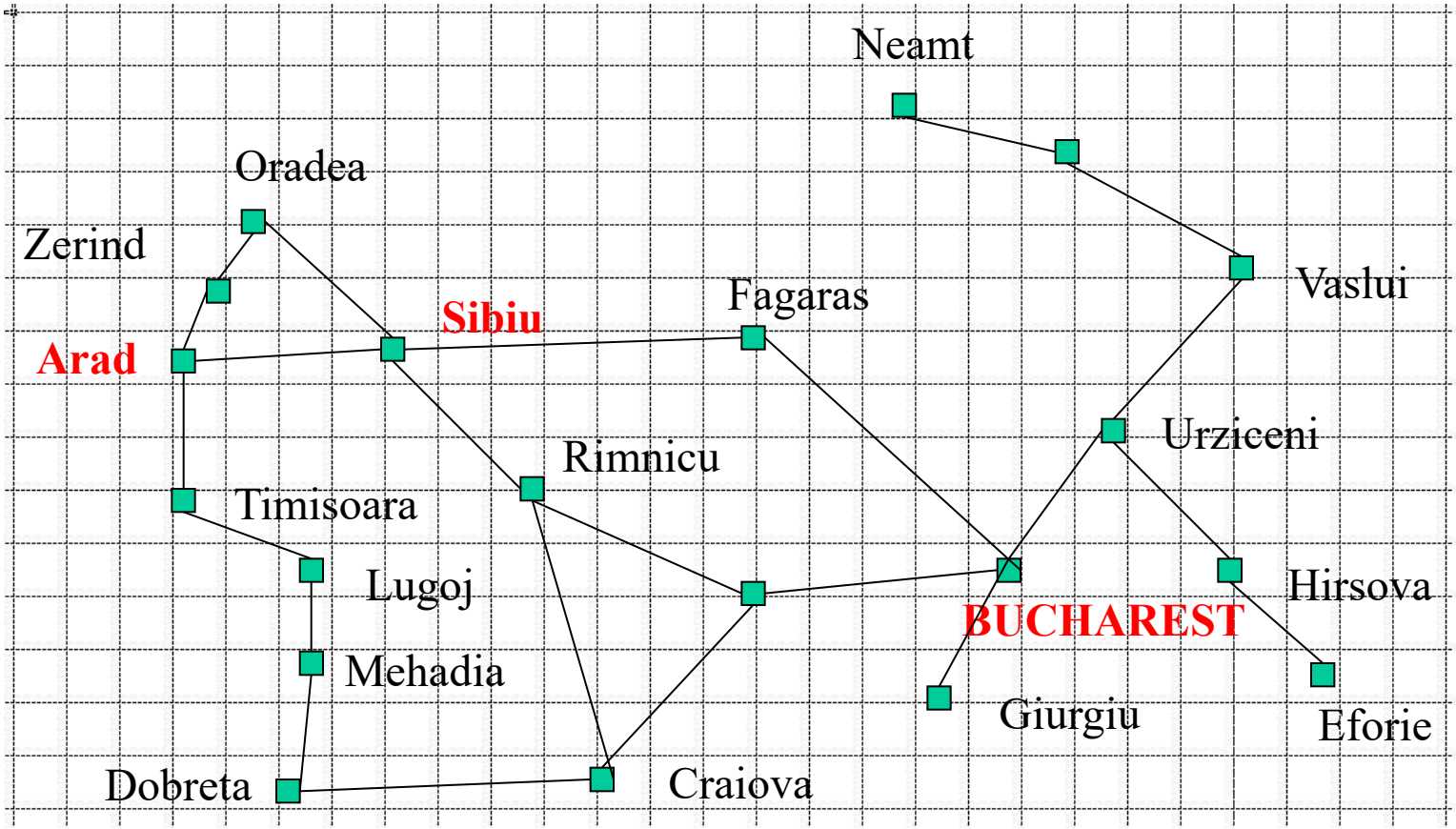
¿Cuál podría ser una heurística?

Se cuenta con las coordenadas de las ciudades

Búsqueda informada



Búsqueda informada



Ciudad	Coordenadas
Arad	(3,7)
Bucarest	(19,11)
Sibiu	(7,6)

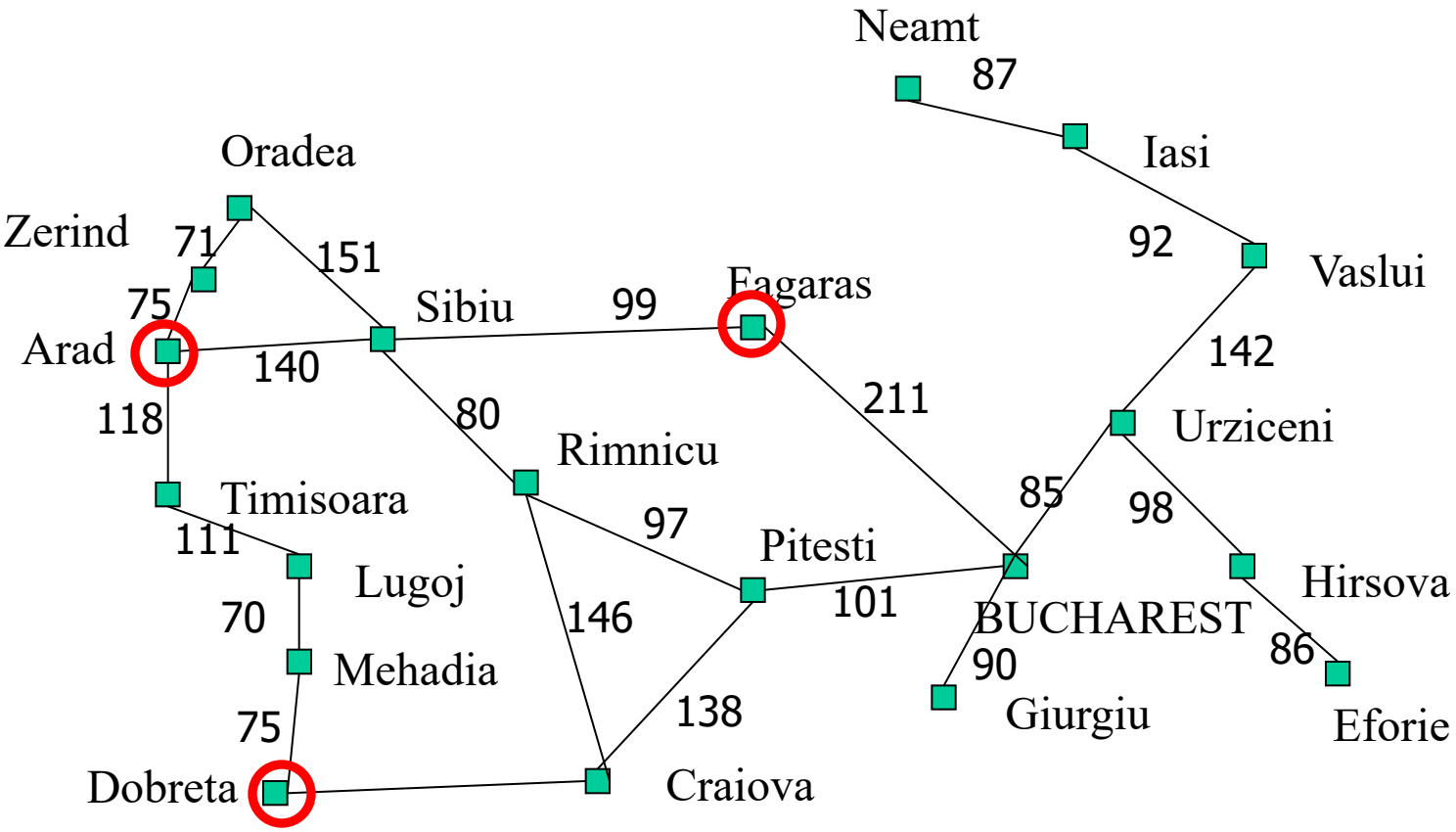
Para llegar a Bucarest, qué es mejor,
estar en Arad o en Sibiu?

Búsqueda informada

Ciudad	Coordenadas
Arad	(3,7)
Bucarest	(19,11)
Calih	(18,11)
Sibiu	(7,6)

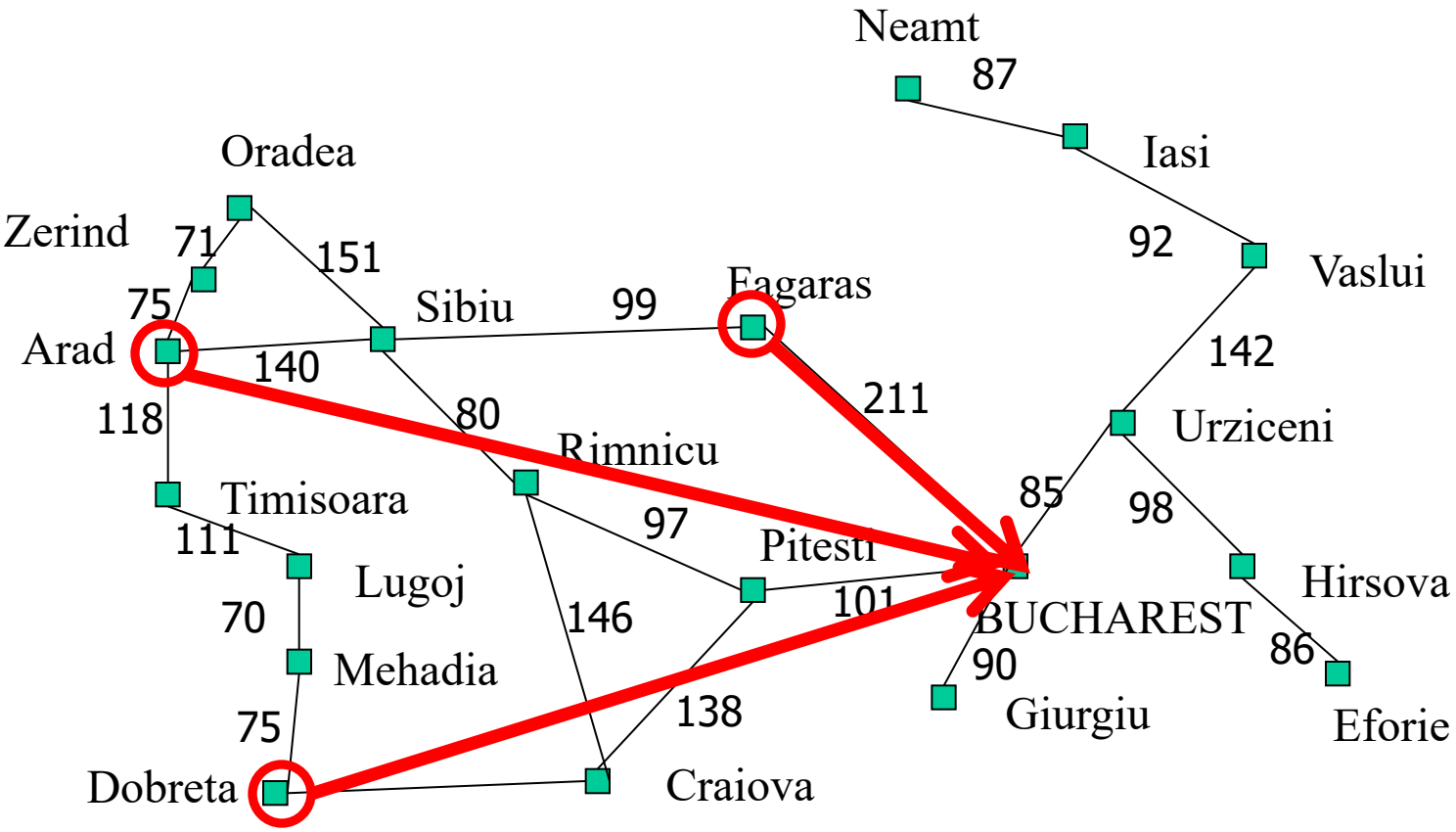
Para llegar a Bucarest, qué es mejor, estar en Arad, Calih o Sibiu?

Búsqueda informada



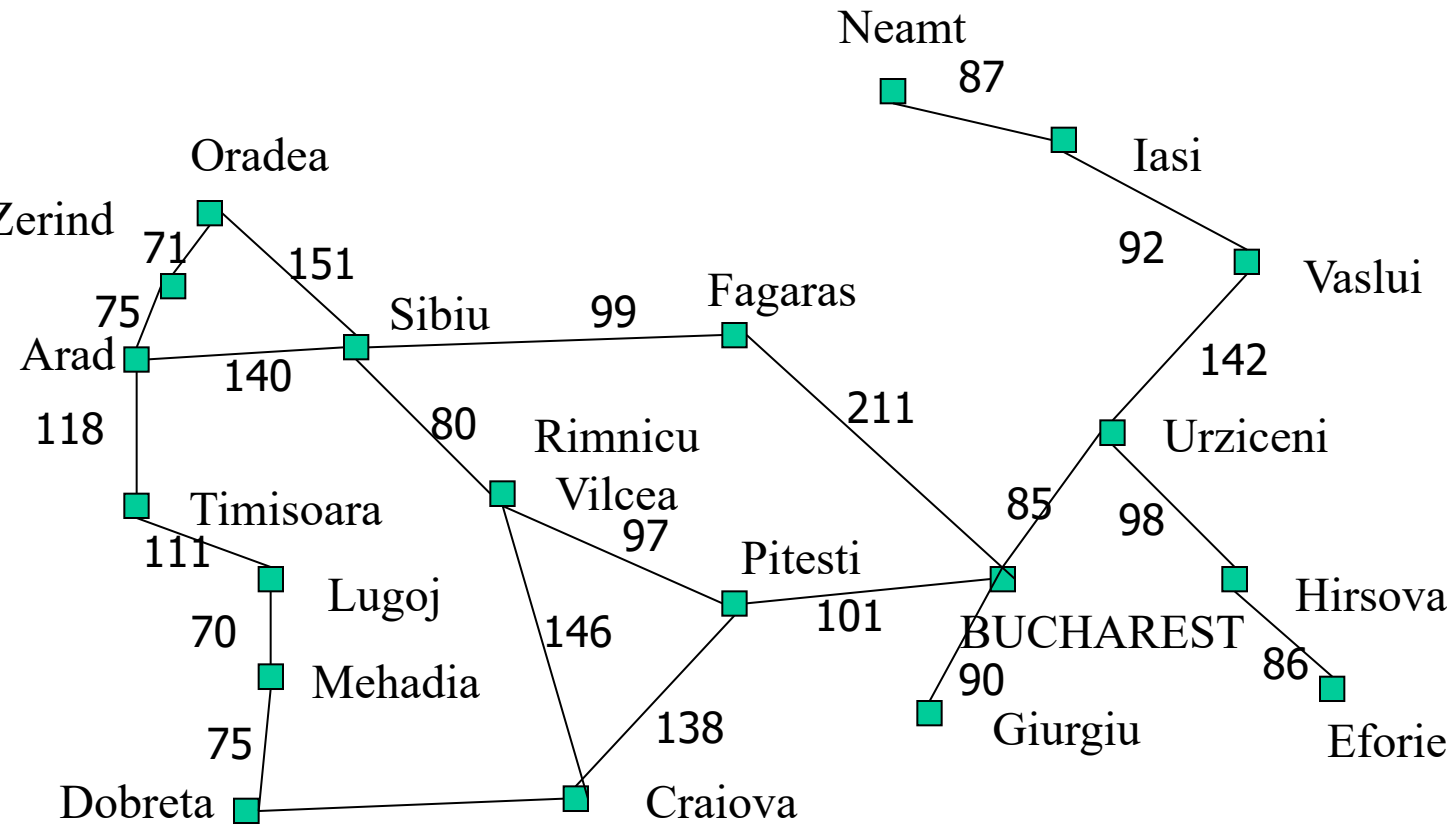
¿Cuál está más distante de la meta? ¿Cómo lo sabe?

Búsqueda informada



h(n): distancia en línea recta entre n y la ubicación de la meta

Búsqueda informada



Distancia en línea recta a Bucarest:

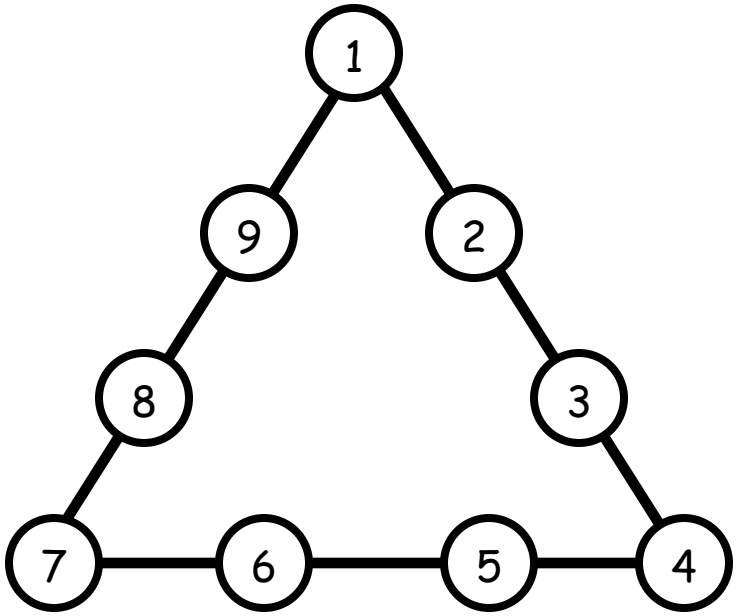
Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

h(n): distancia en línea recta entre n y la ubicación de la meta

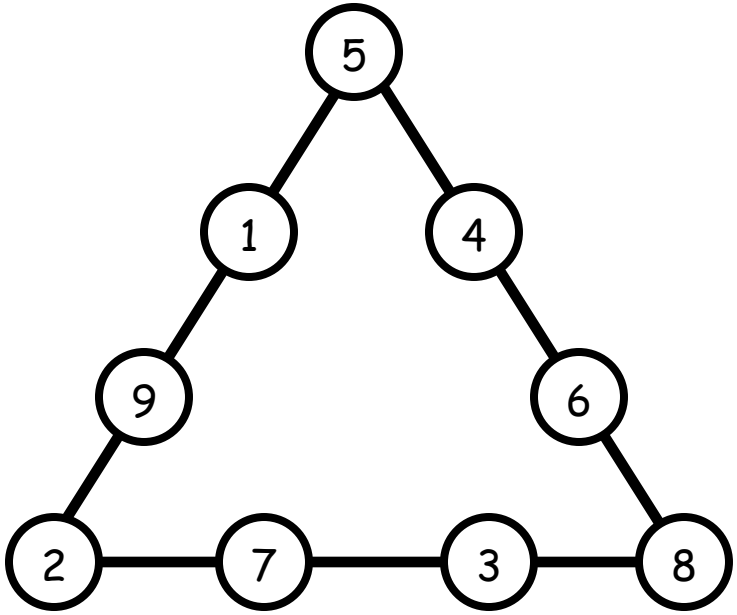
Búsqueda informada

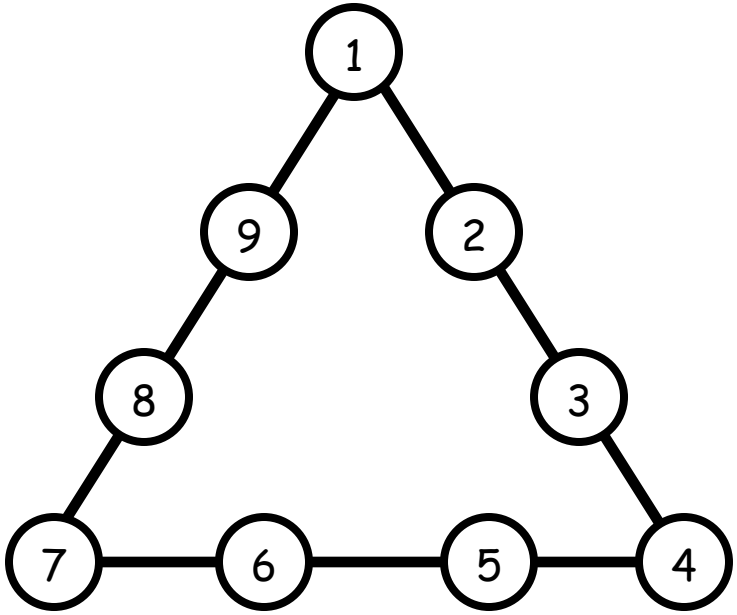
- Defina una **heurística**

El triángulo mágico. Utilizando el operador *intercambio(a,b)* que cambia de posición los números a y b, colocarlos de tal forma que la suma sobre cada lado sea 20.



Búsqueda informada

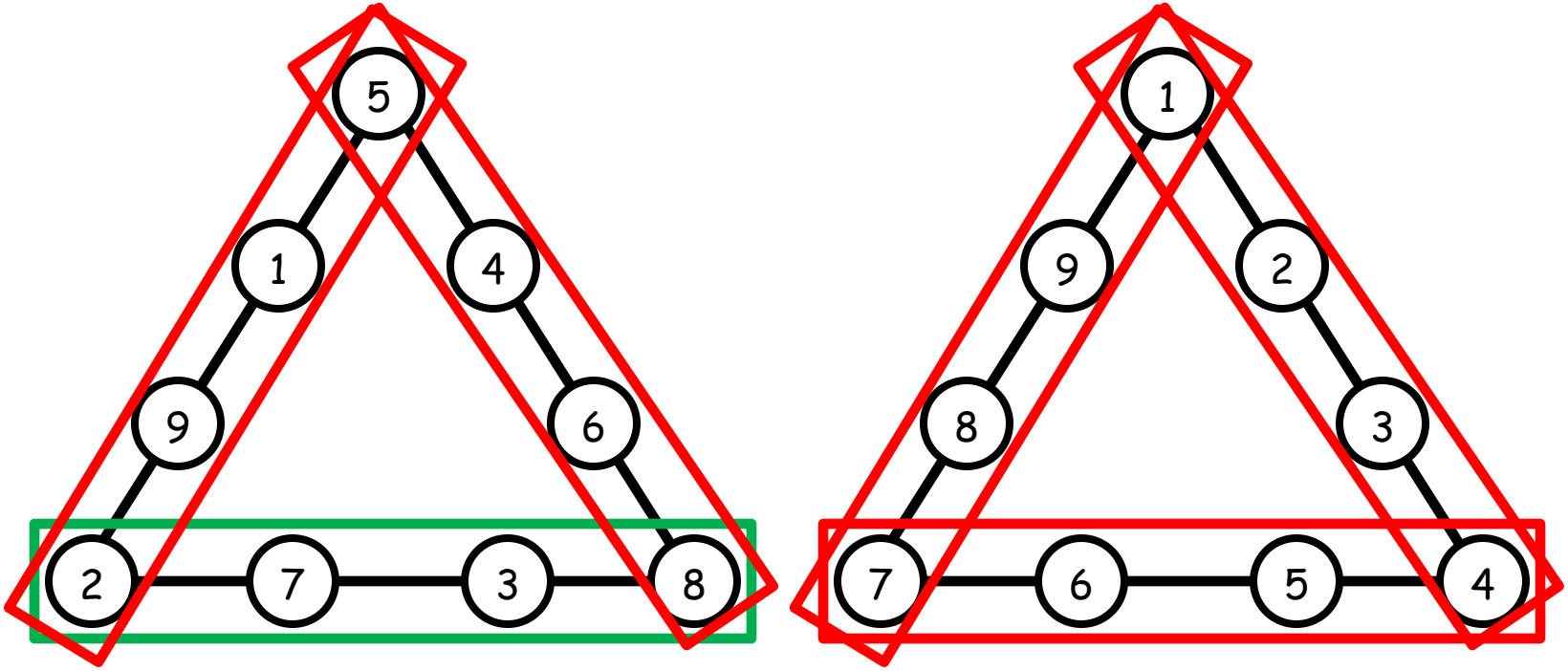




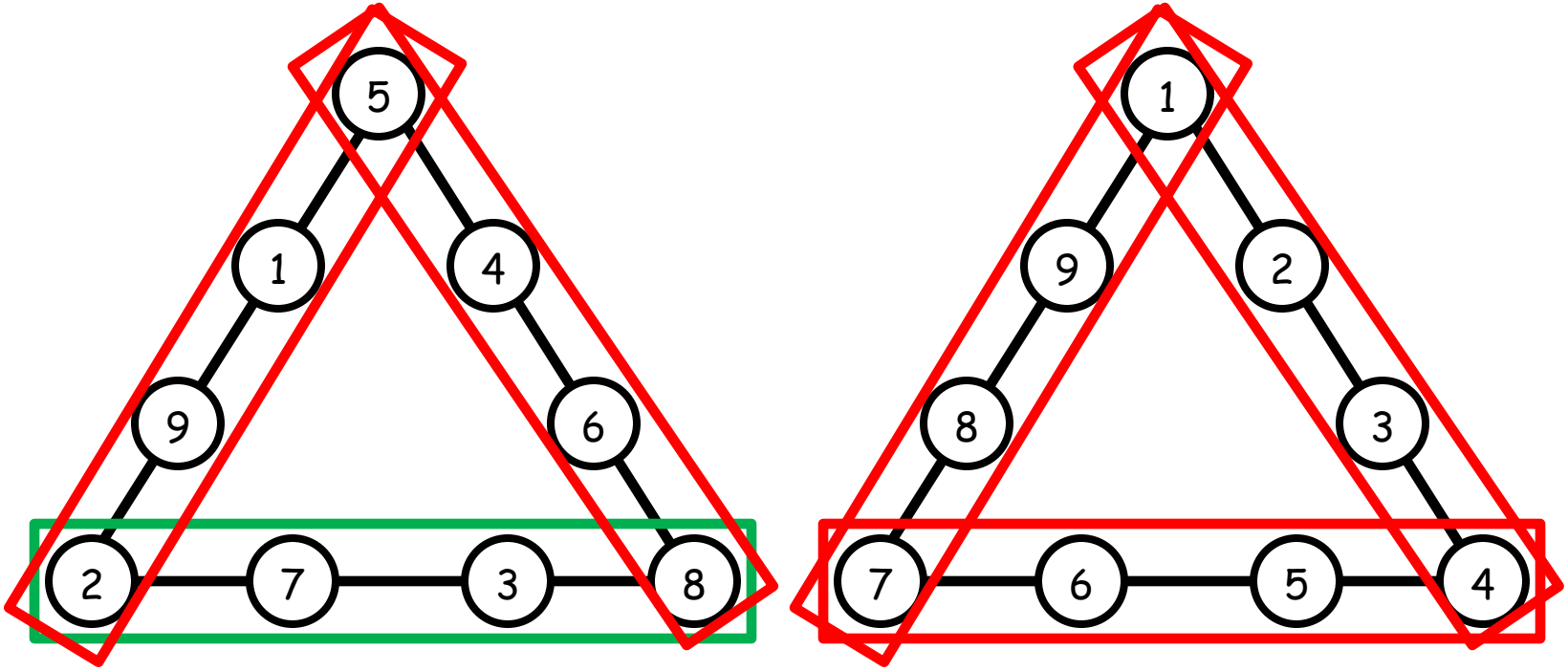
¿Cuál está más distante de la meta?

¿Cómo lo sabe?

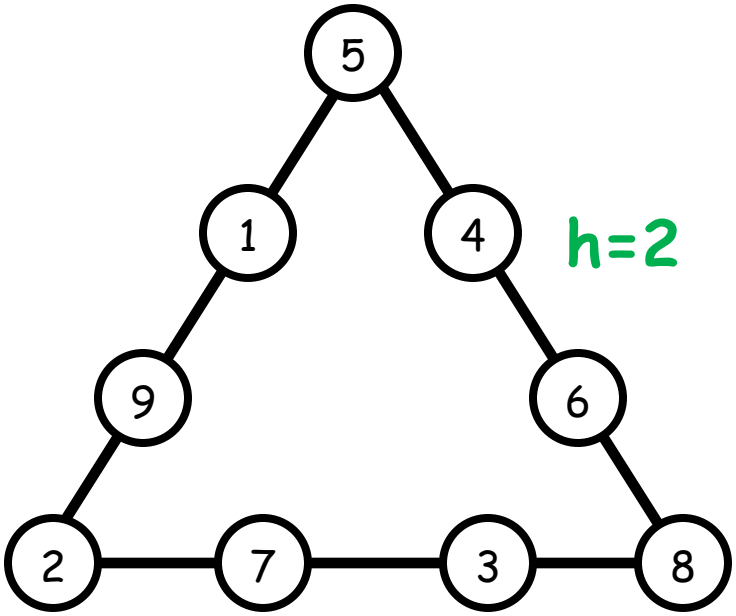
Búsqueda informada



Búsqueda informada

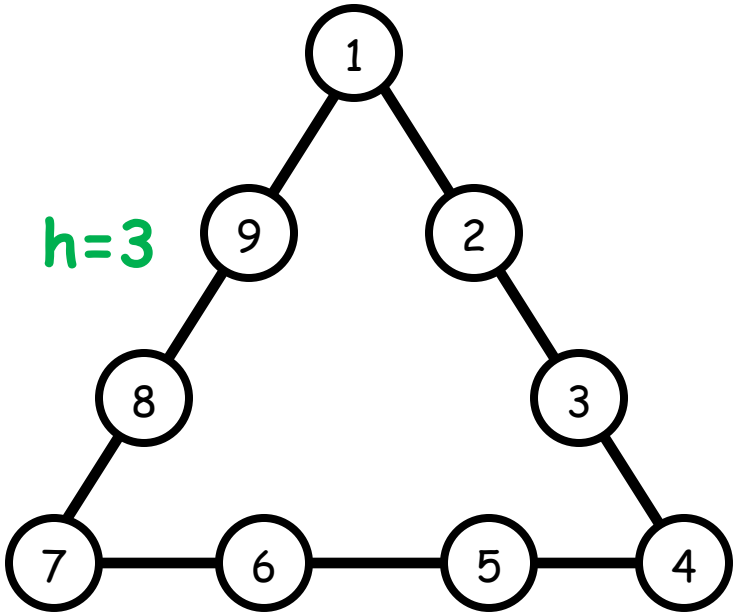


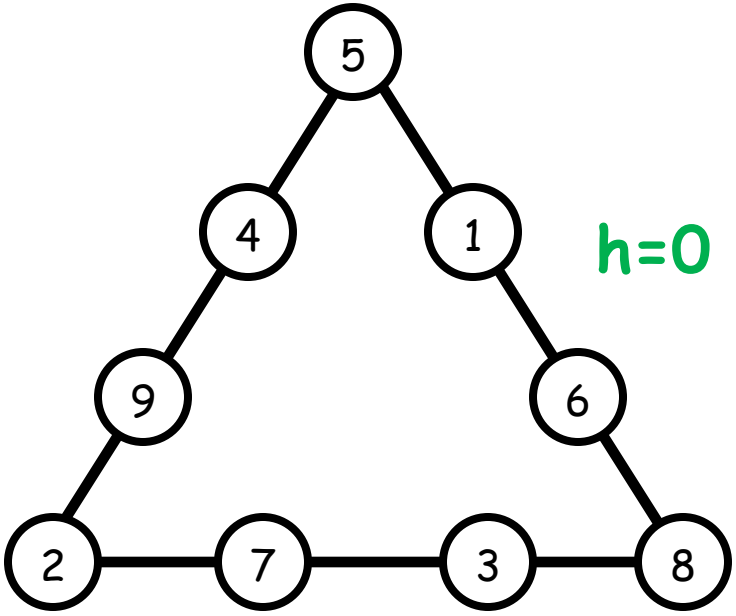
h(n): cantidad de lados que no suman 20



h=2

h=3





h=0

Búsqueda informada

Heurística

- La utilización de heurísticas en los métodos de búsqueda permiten reducir el espacio de estados a expandir en la construcción del árbol

Búsqueda informada

Estrategias de búsqueda informada

- Avara
- A*

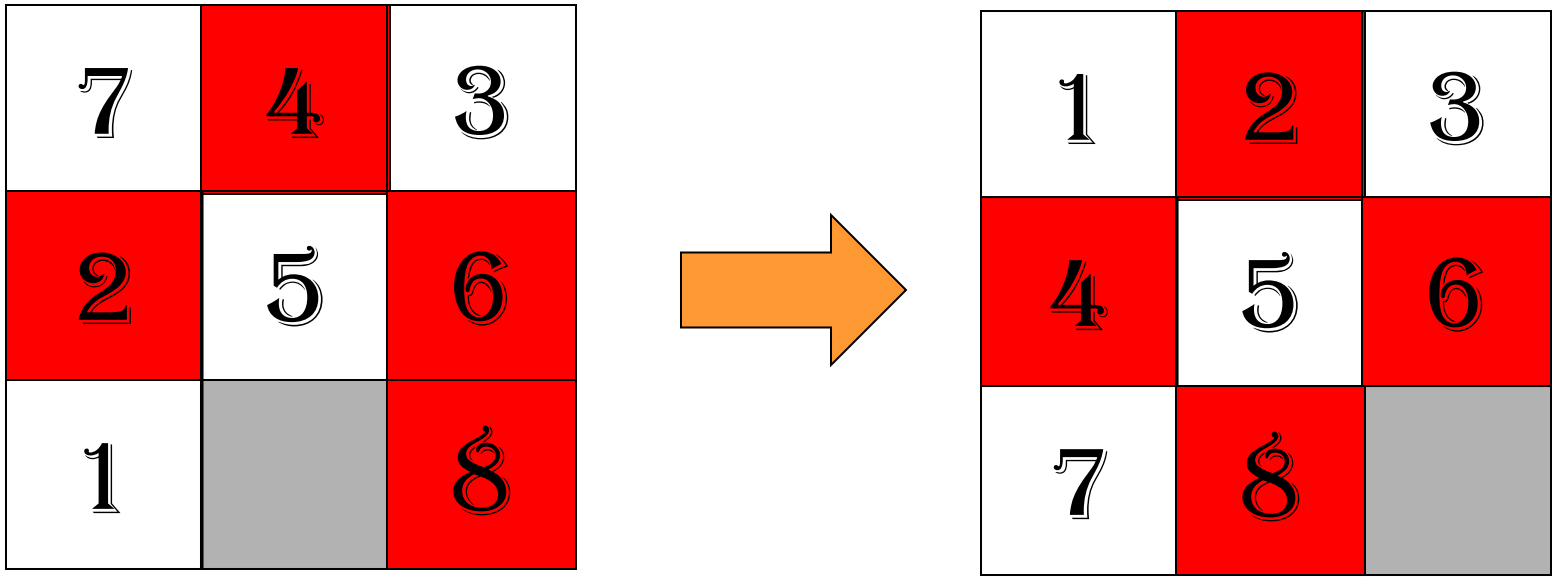
Búsqueda informada

Búsqueda Avara

- Expandir el nodo con **menor h(n)**, esto es, aquel que parece estar más cerca de la meta sin importar a qué profundidad esté

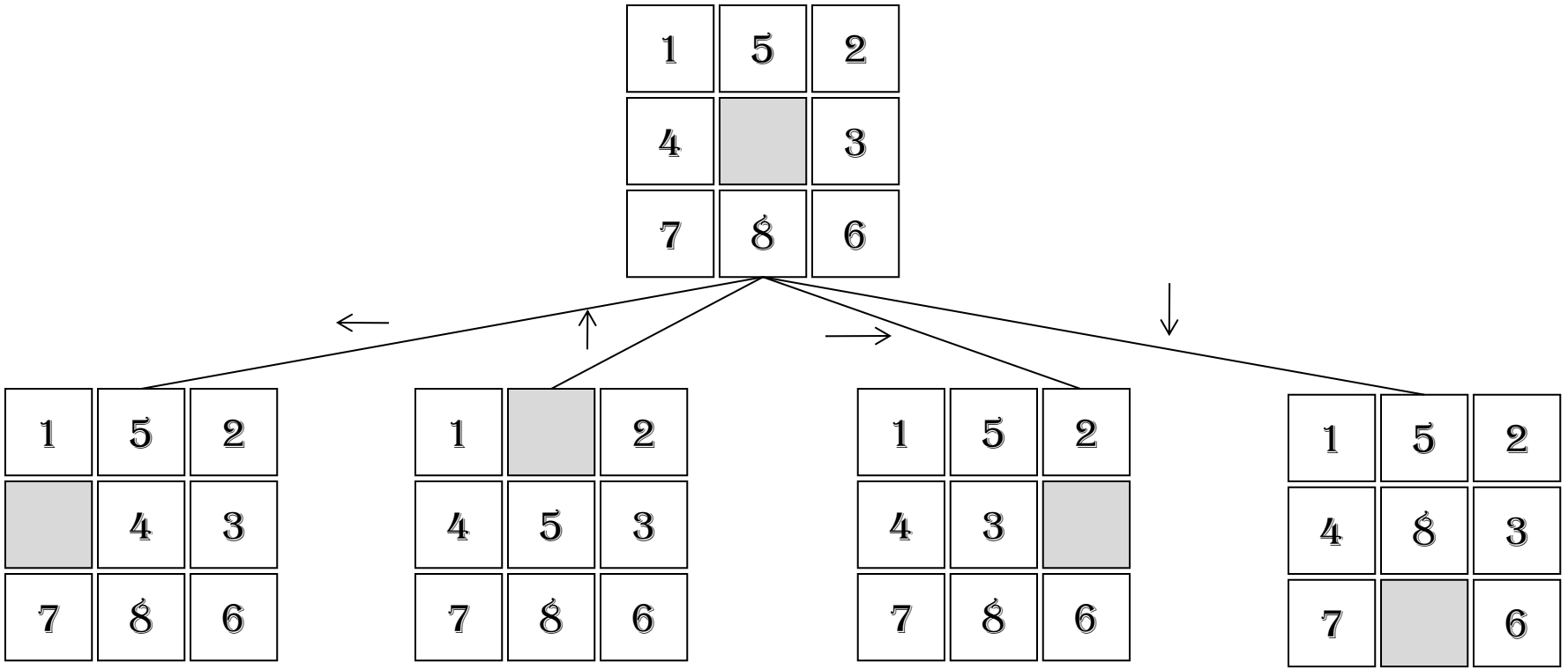
Búsqueda informada

8-puzzle

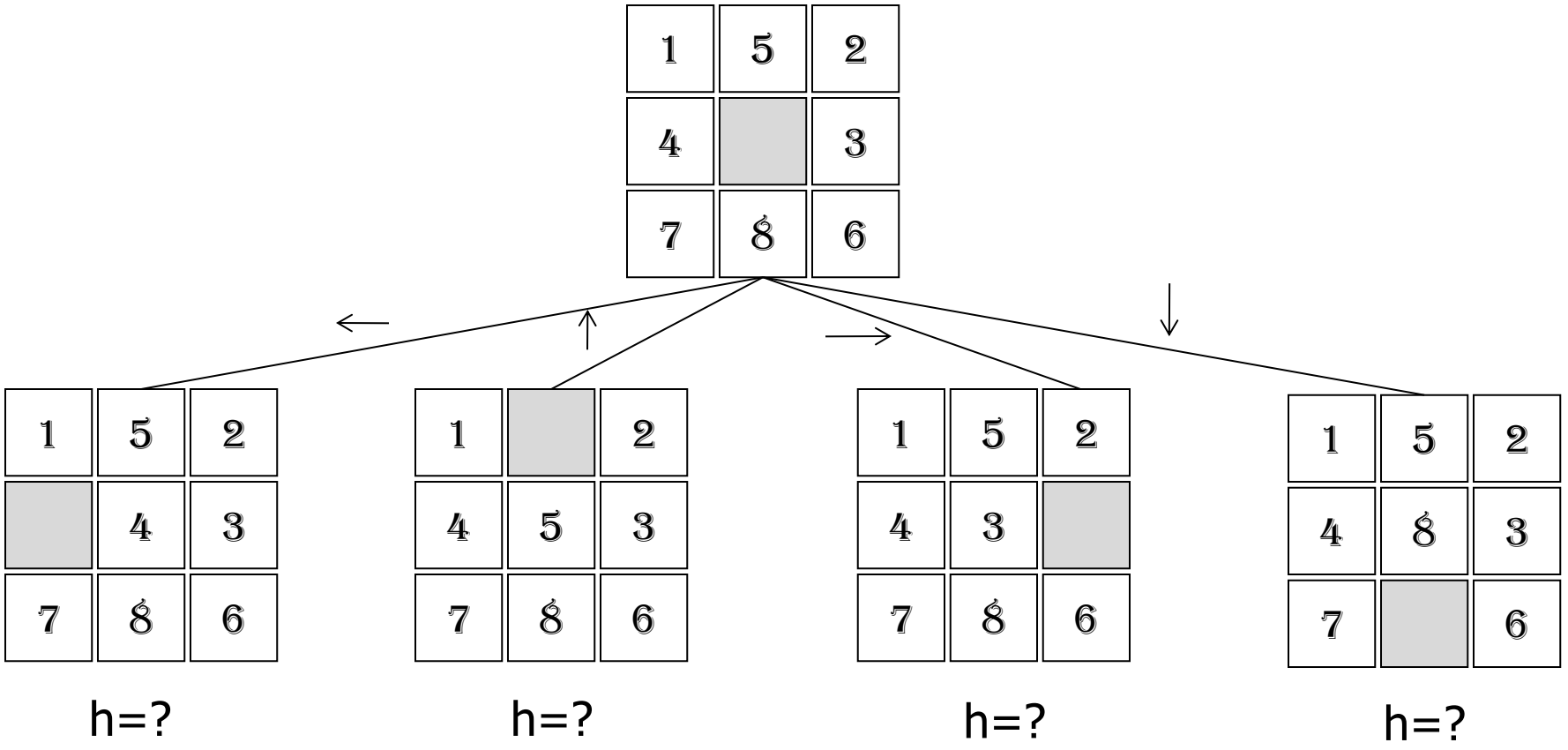


h(n): cantidad de piezas ubicadas en el lugar incorrecto (teniendo en cuenta la placa vacía)

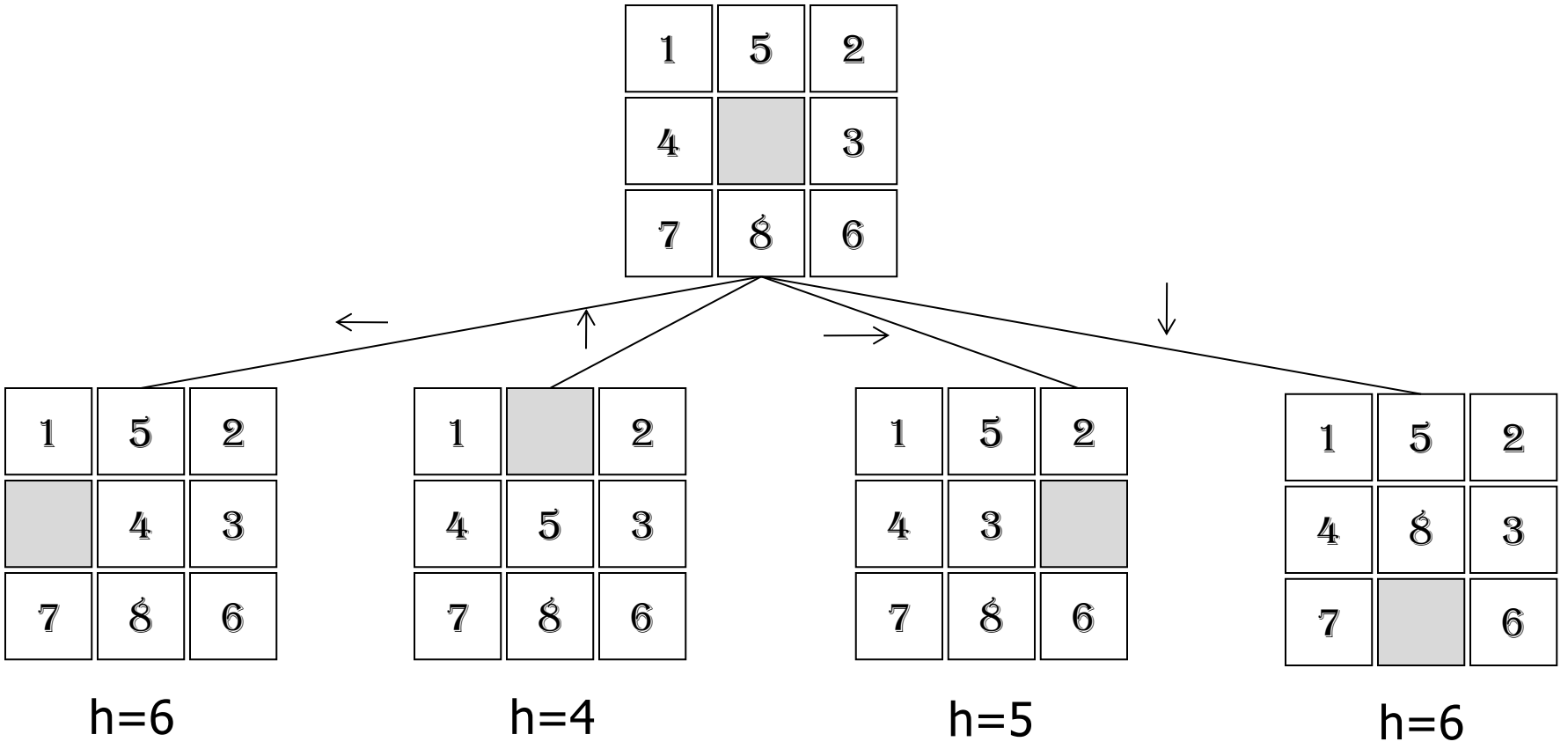
Búsqueda informada



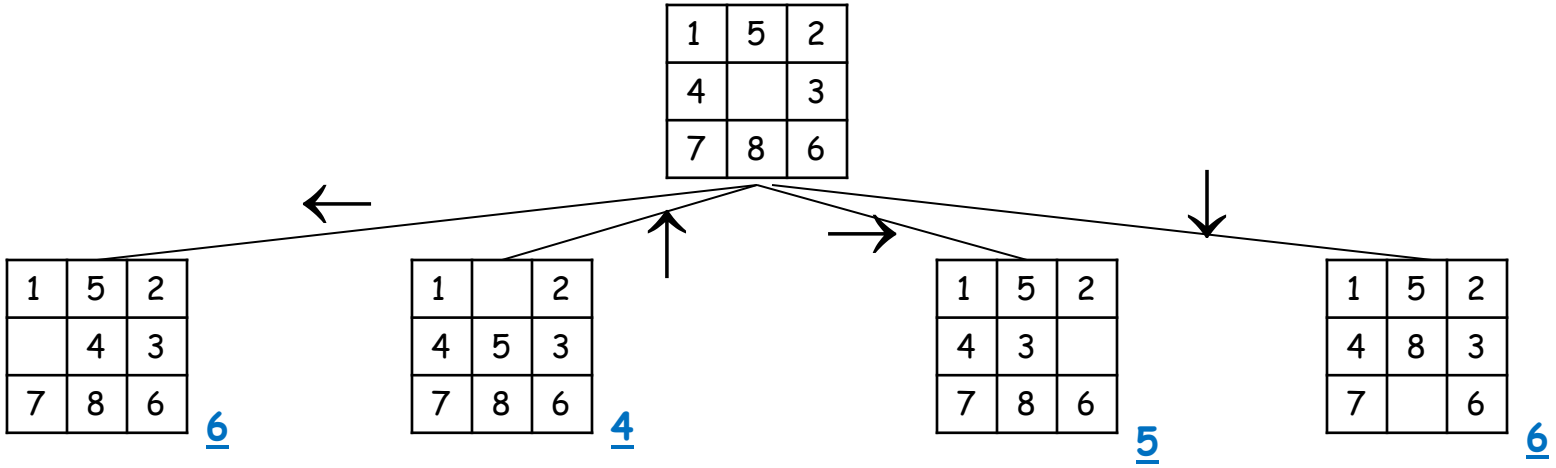
Búsqueda informada

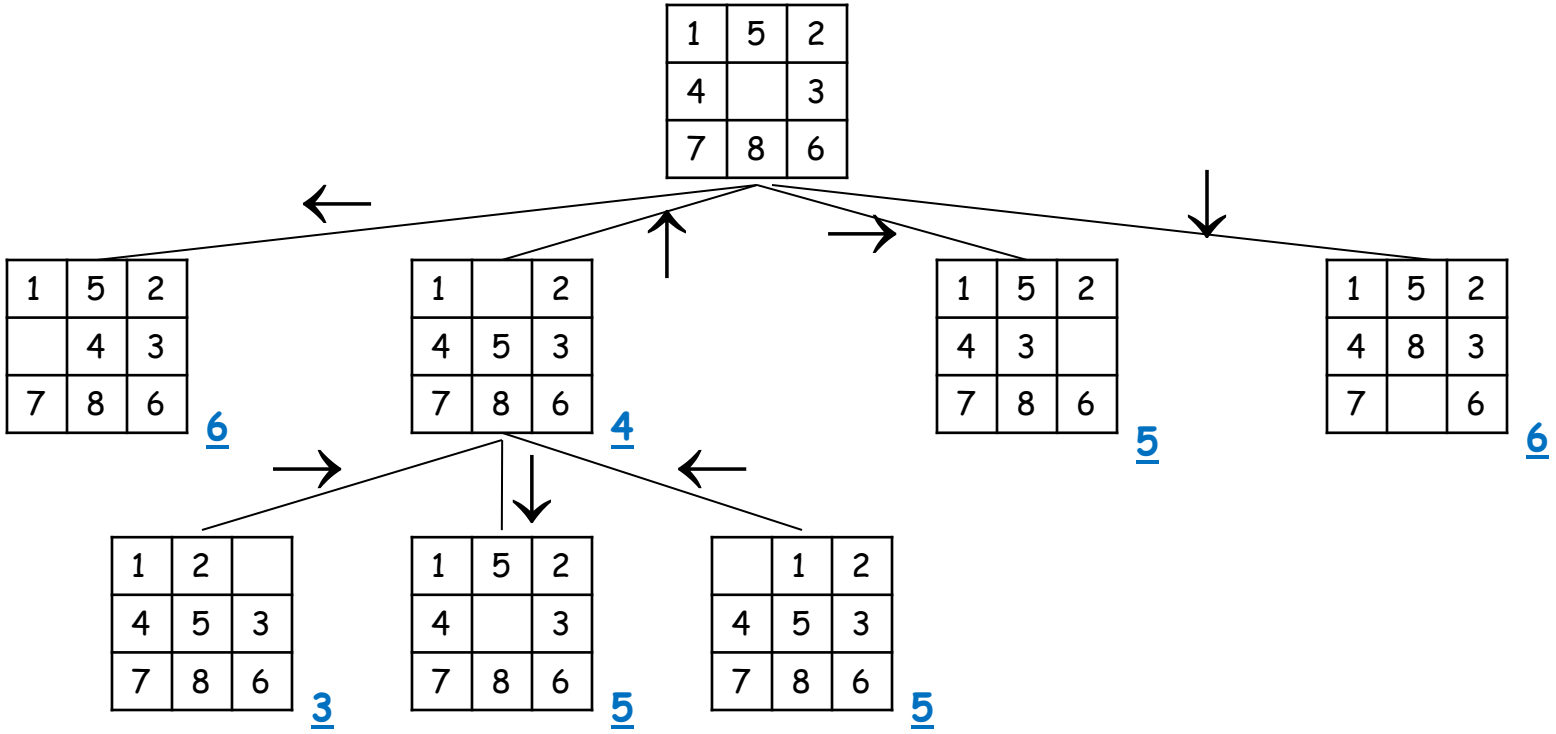


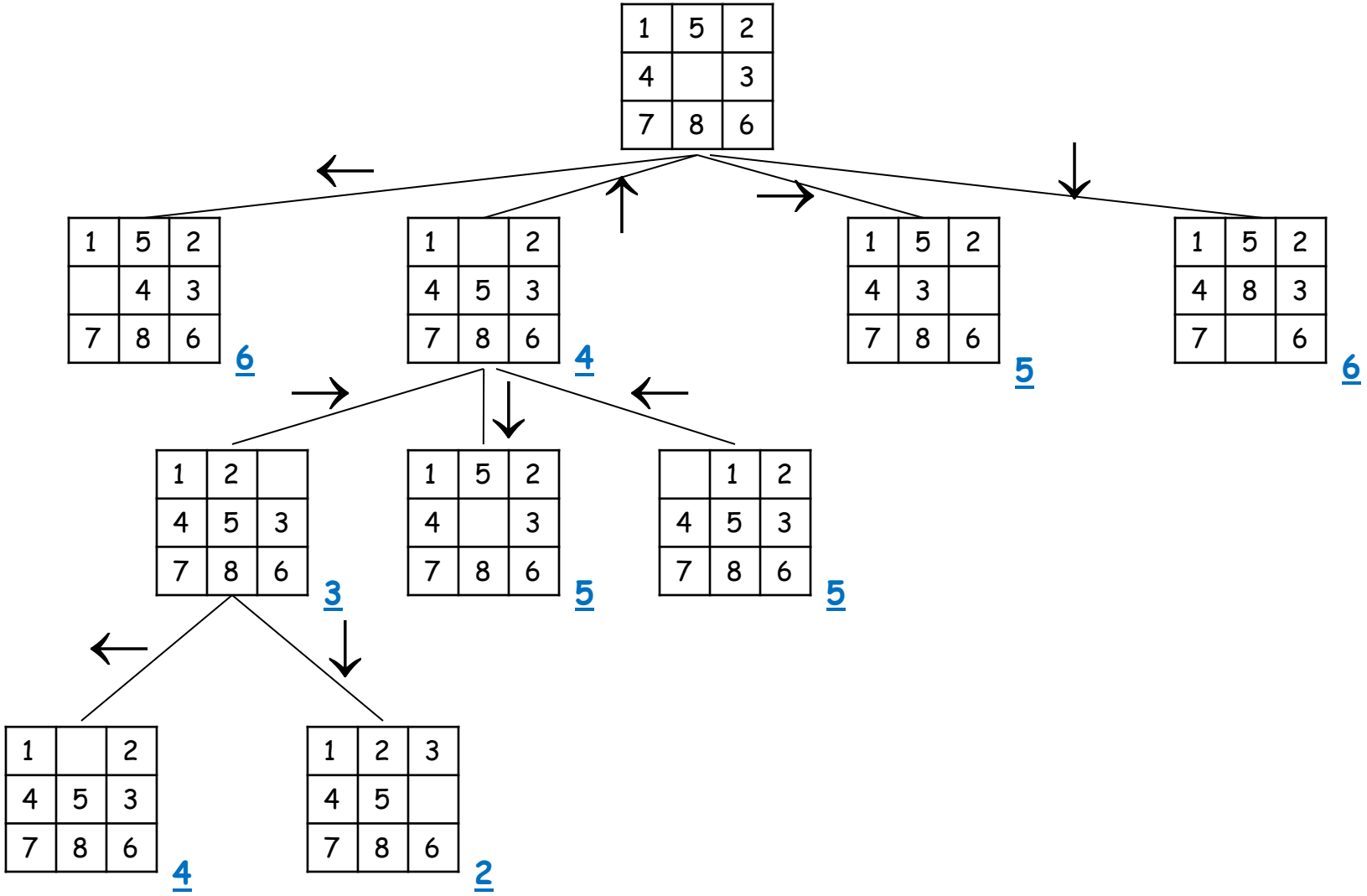
Búsqueda informada

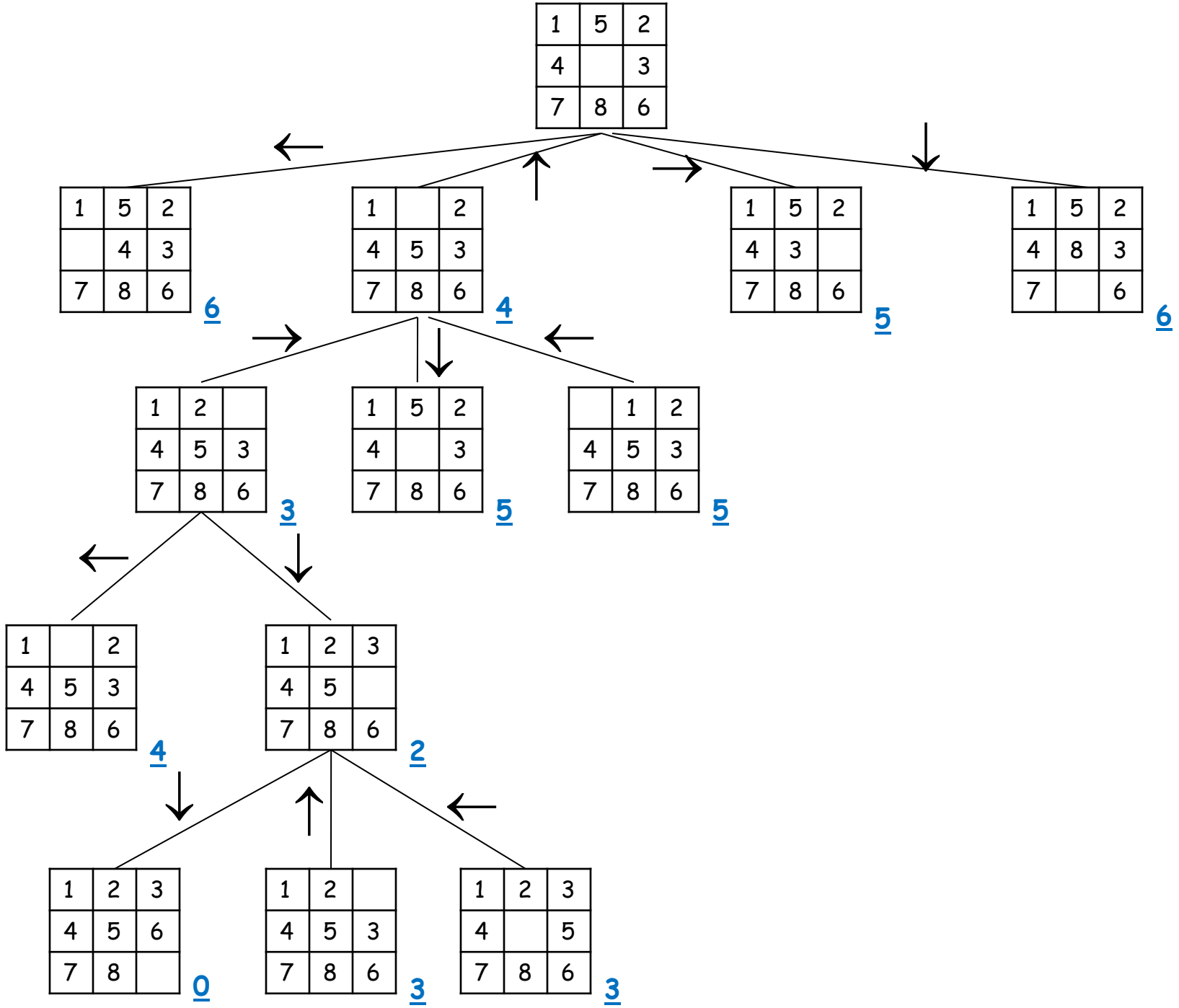


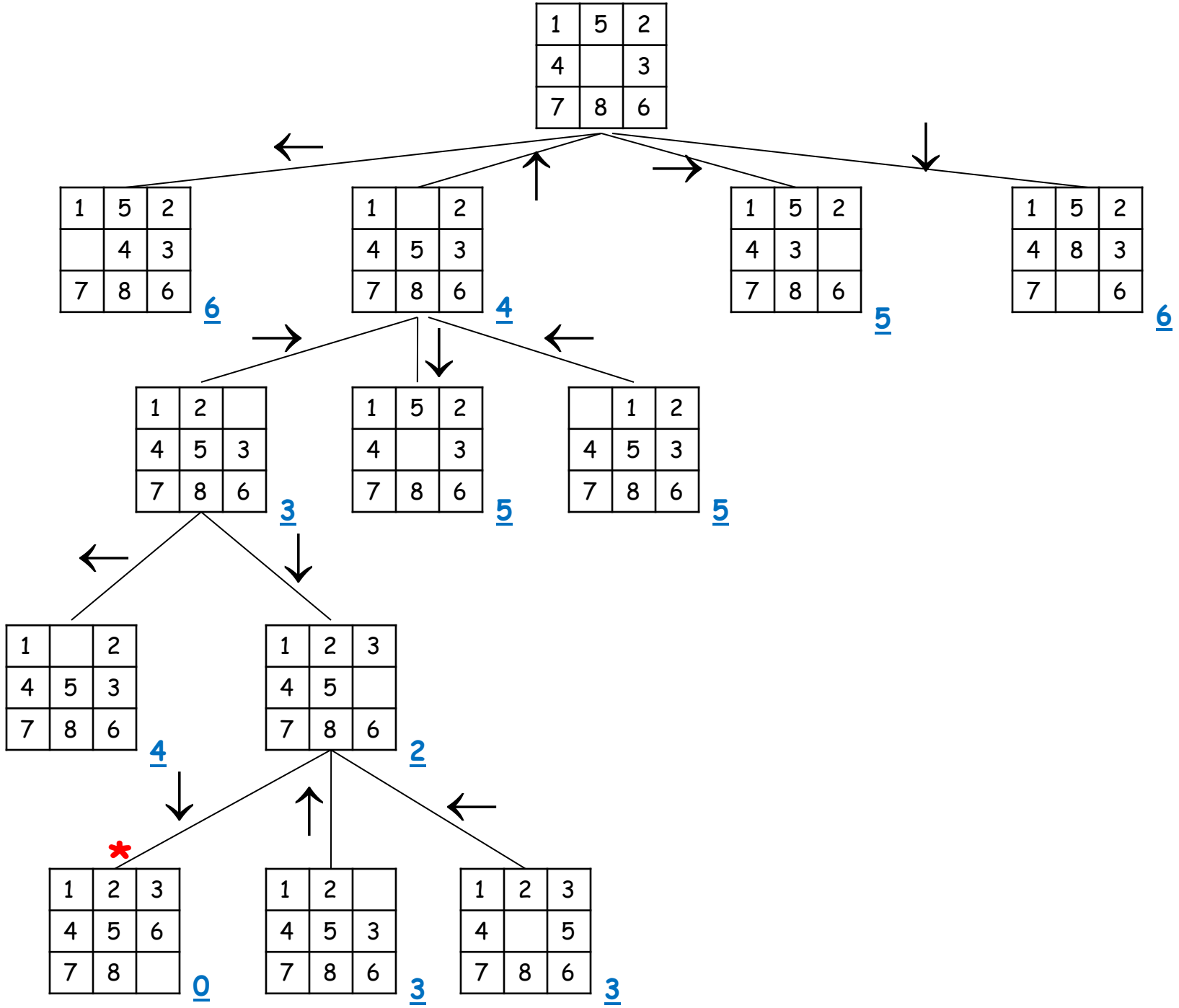
Complete el árbol de búsqueda
No evite devolverse


1 5 2
4 3
7 8 6
1 5 2
4 3
7 8 6
6
1 5 2
4 5 3
7 8 6
4
1 5 2
4 3
7 8 6
5
1 5 2
4 8 3
7 6
6

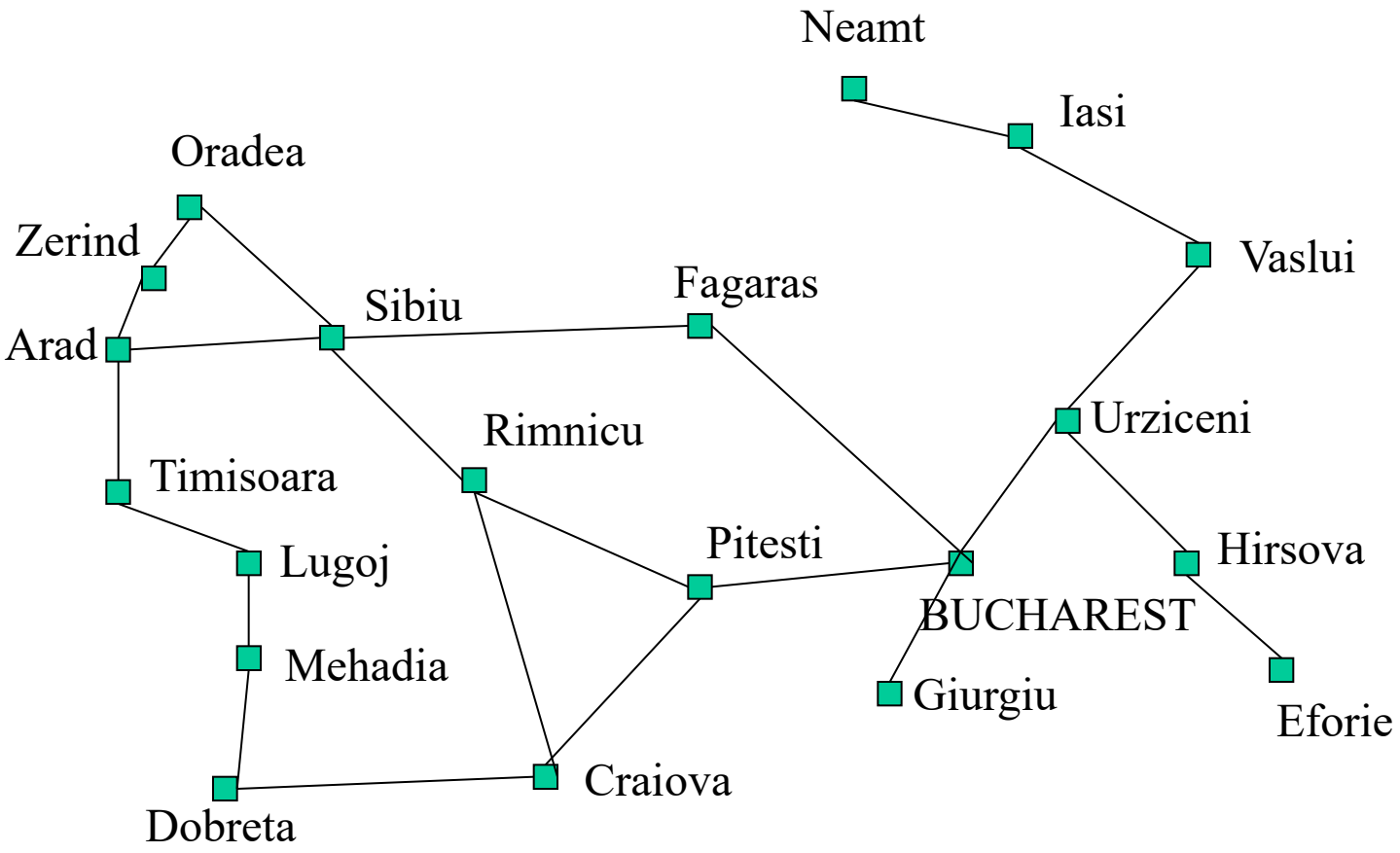

1 5 2
4 3
7 8 6
1 5 2
4 3
7 8 6
6
1 2
4 5 3
7 8 6
4
1 5 2
4 3
7 8 6
5
1 5 2
4 8 3
7 6
6
1 2
4 5 3
7 8 6
3
1 5 2
4 3
7 8 6
5
1 2
4 5 3
7 8 6
5


1 5 2
4 3
7 8 6
1 5 2
4 3
7 8 6
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7 8 6
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4 3
7 8 6
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7 6
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1 2
4 5 3
7 8 6
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7 8 6
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1 2
4 5 3
7 8 6
4
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4 5
7 8 6
2


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4 3
7 8 6
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4 3
7 8 6
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4 3
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7 6
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4 5 3
7 8 6
3
1 5 2
4 3
7 8 6
5
1 2
4 5 3
7 8 6
5
1 2 3
4 5 6
7 8
0
1 2
4 5 3
7 8 6
3
1 2 3
4 5
7 8 6
2
1 2 3
4 5 3
7 8 6
3


1 5 2
4 3
7 8 6
6
1 2
4 5 3
7 8 6
4
1 5 2
4 3
7 8 6
5
1 5 2
4 8 3
7 6
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7 8 6
3
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4 3
7 8 6
5
1 2
4 5 3
7 8 6
5
1 2
4 5 3
7 8 6
4
1 2 3
4 5
7 8 6
2
*
1 2 3
4 5 6
7 8
0
1 2
4 5 3
7 8 6
3
1 2 3
4 5
7 8 6
3

Búsqueda informada



Distancia en línea recta a Bucarest:

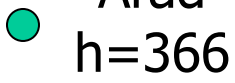
Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

h(n): distancia en línea recta entre n y la ubicación de la meta

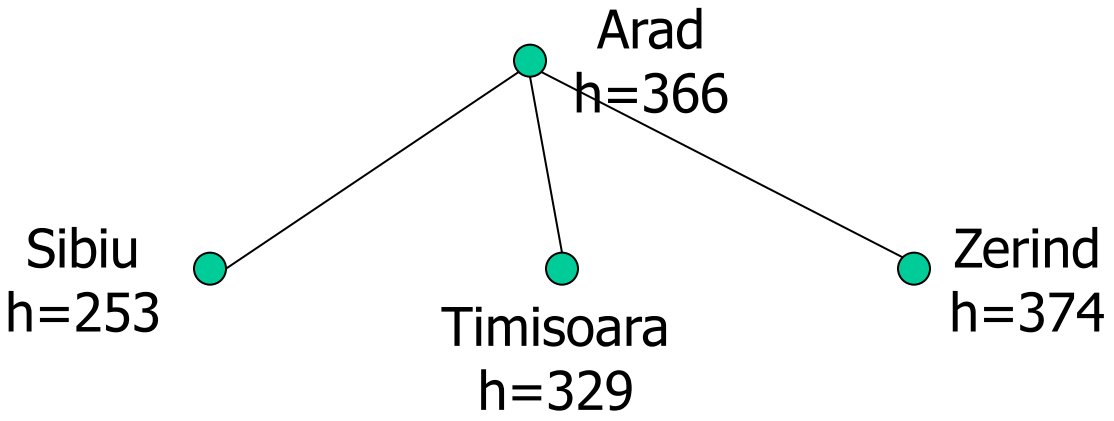
No evite devolverse

Búsqueda informada

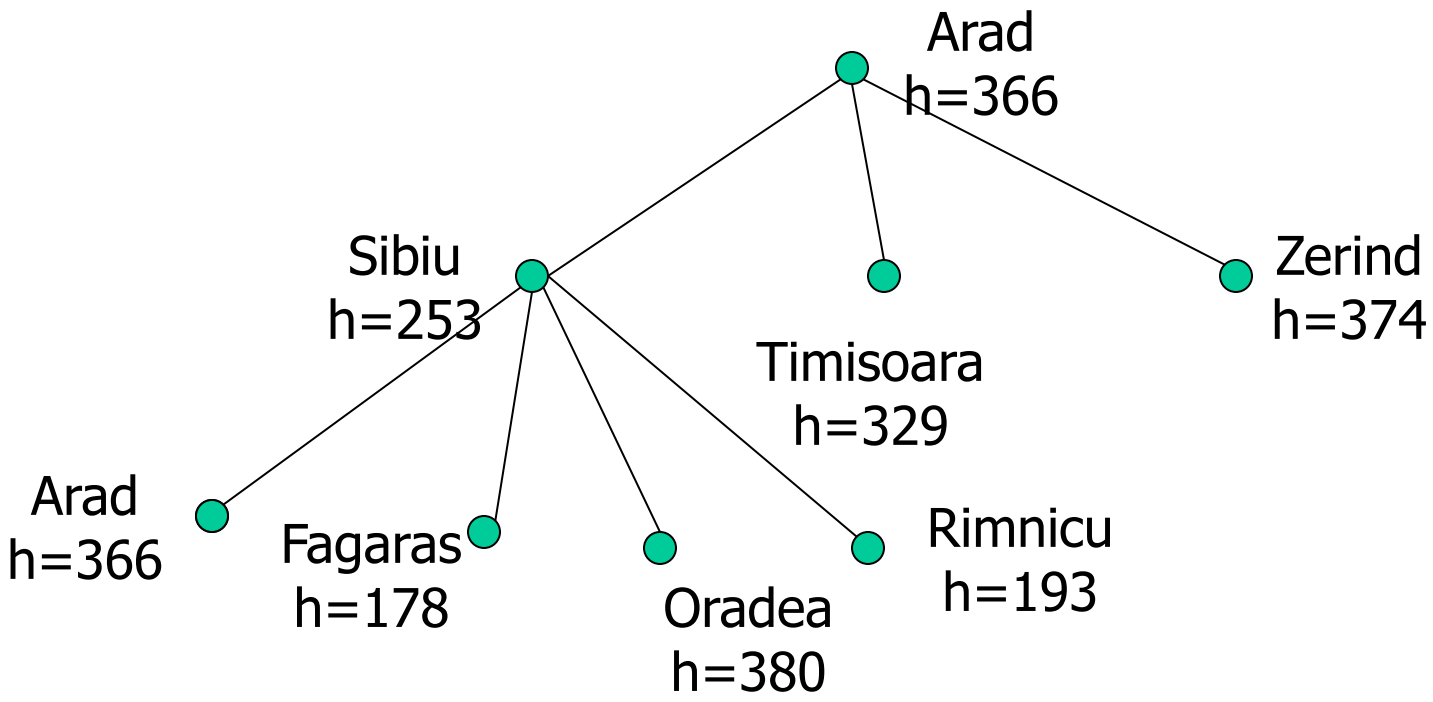
No evite devolverse



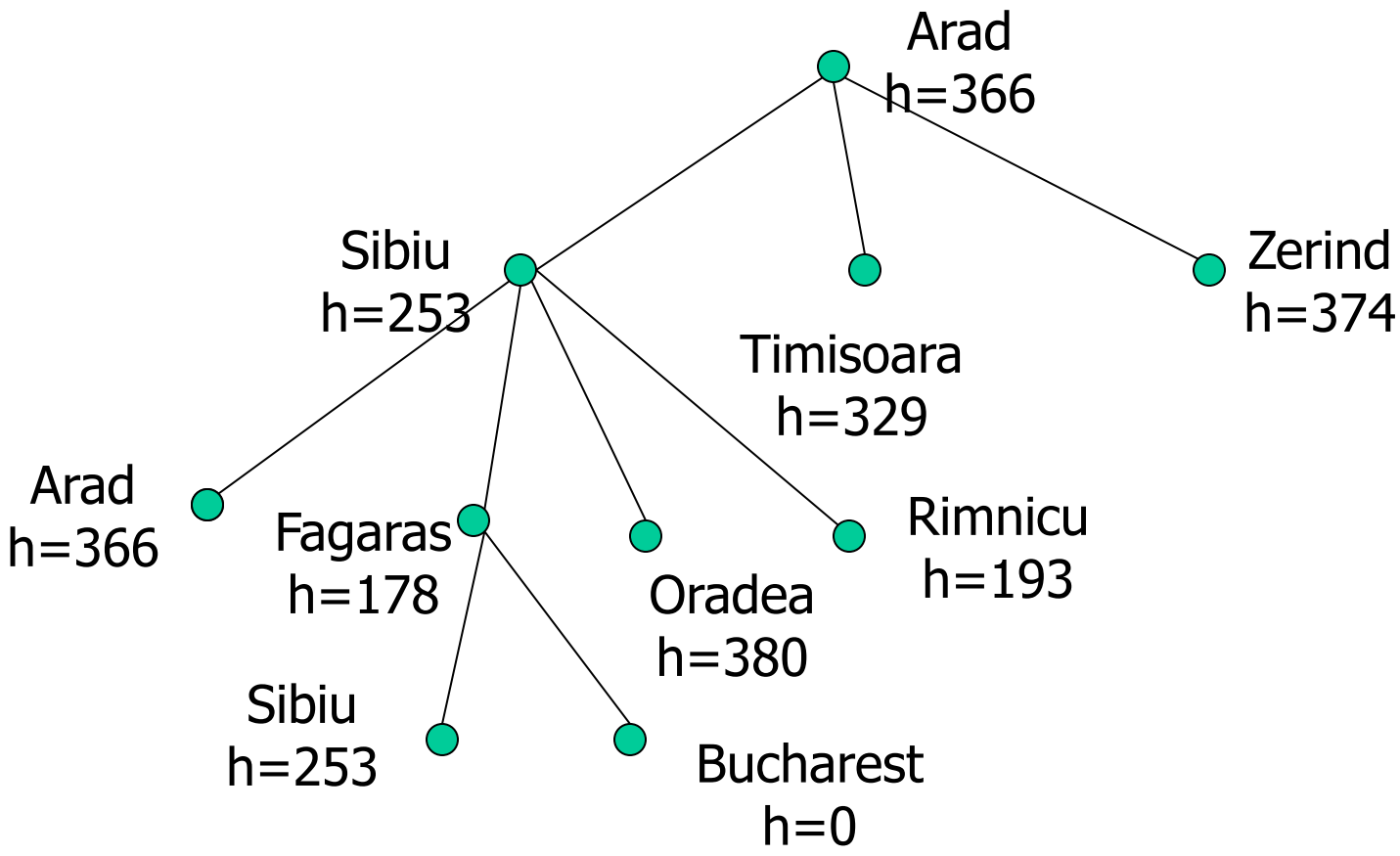
Búsqueda informada



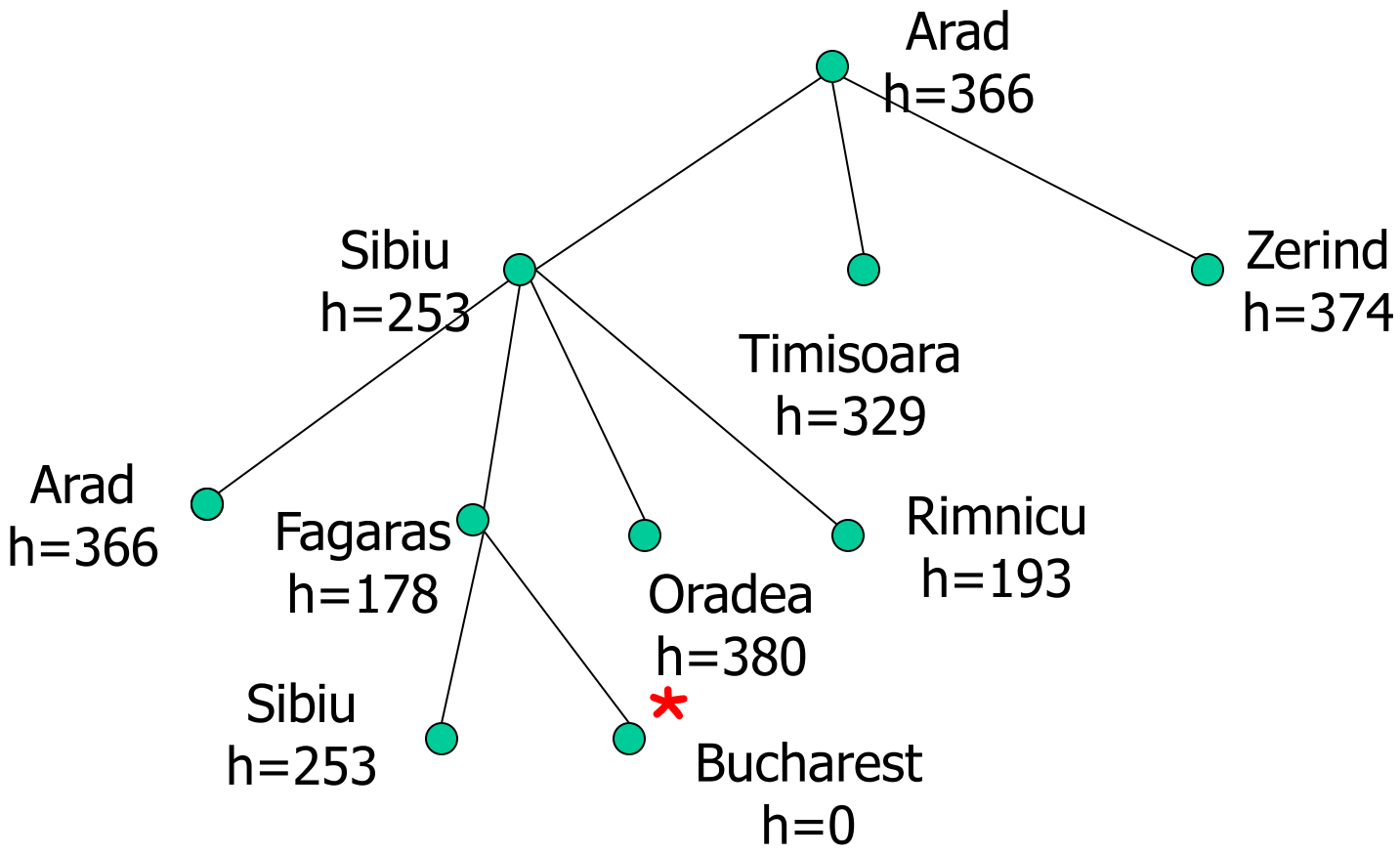
Búsqueda informada



Búsqueda informada

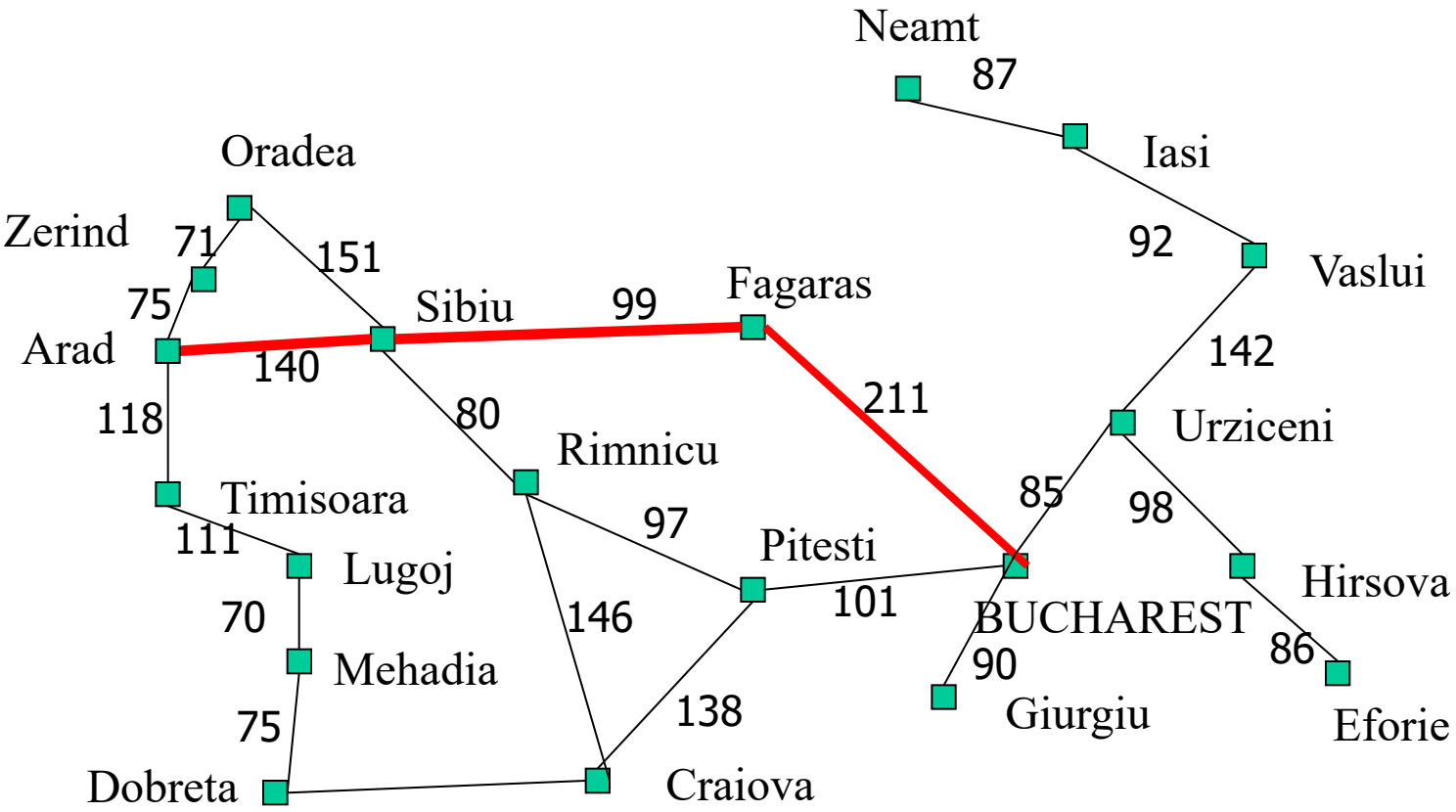


Búsqueda informada



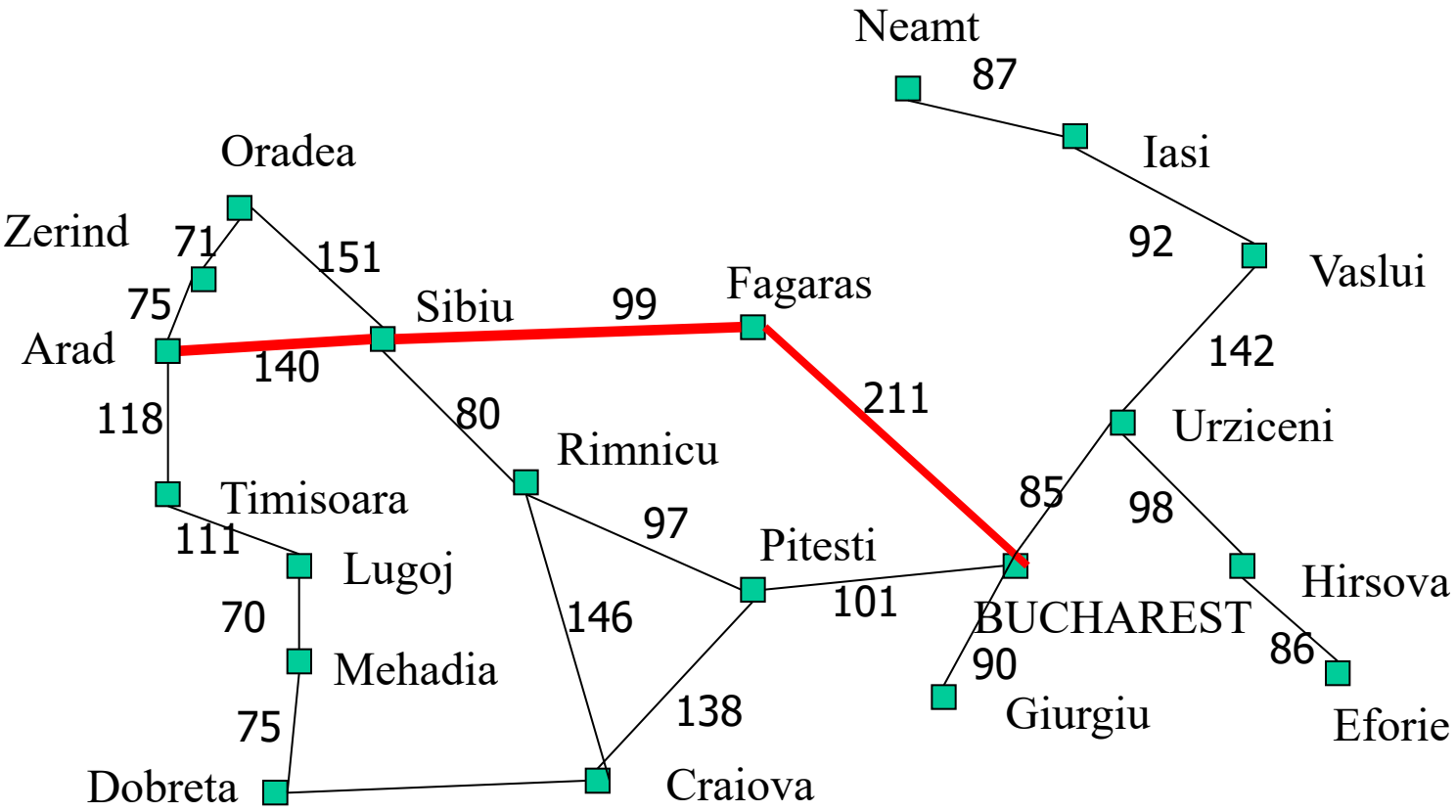
El camino obtenido es
Arad→Sibiu→*Fagaras*→*Bucarest*

Búsqueda informada



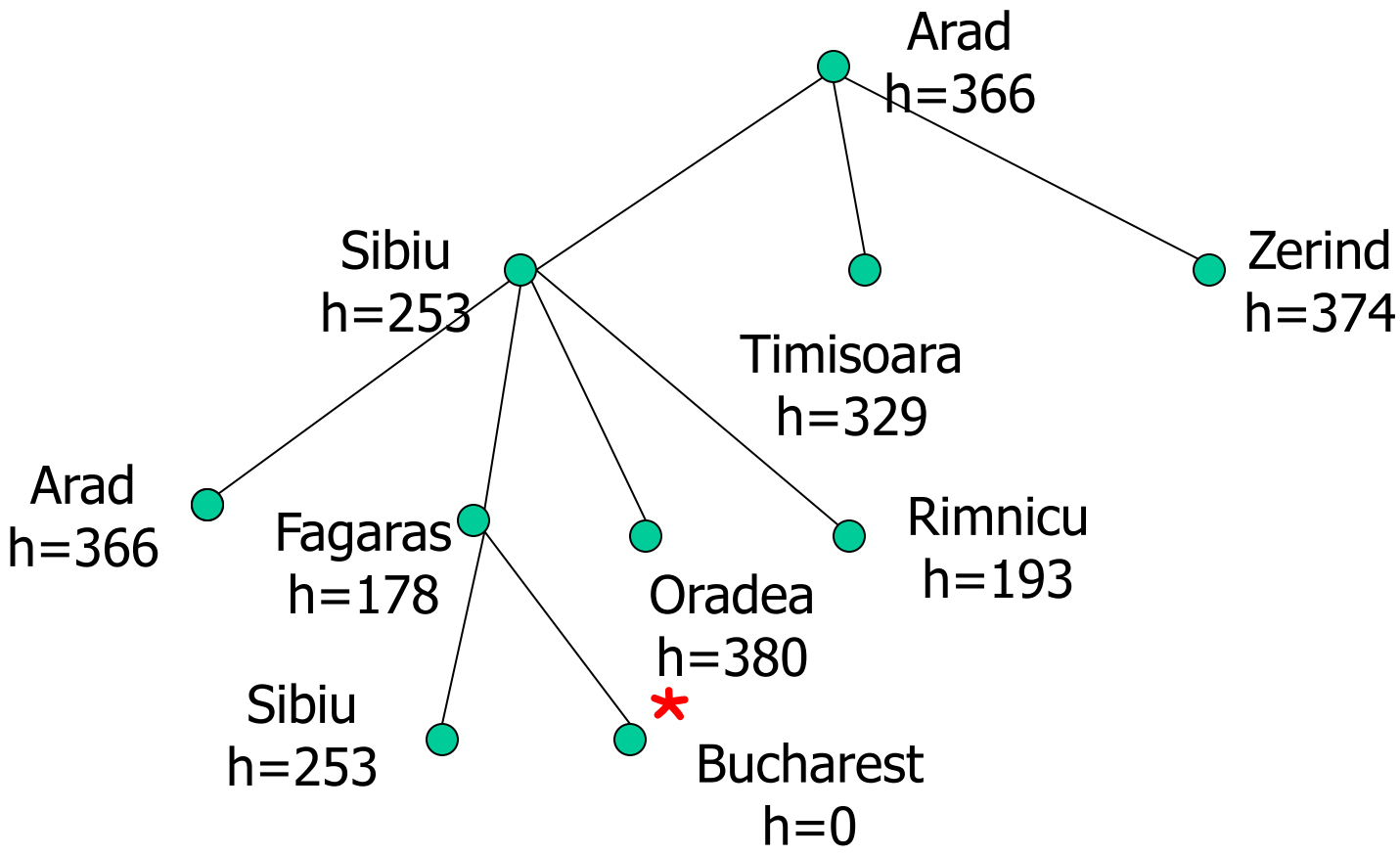
El camino obtenido es
Arad→Sibiu→*Fagaras*→*Bucarest*

Búsqueda informada



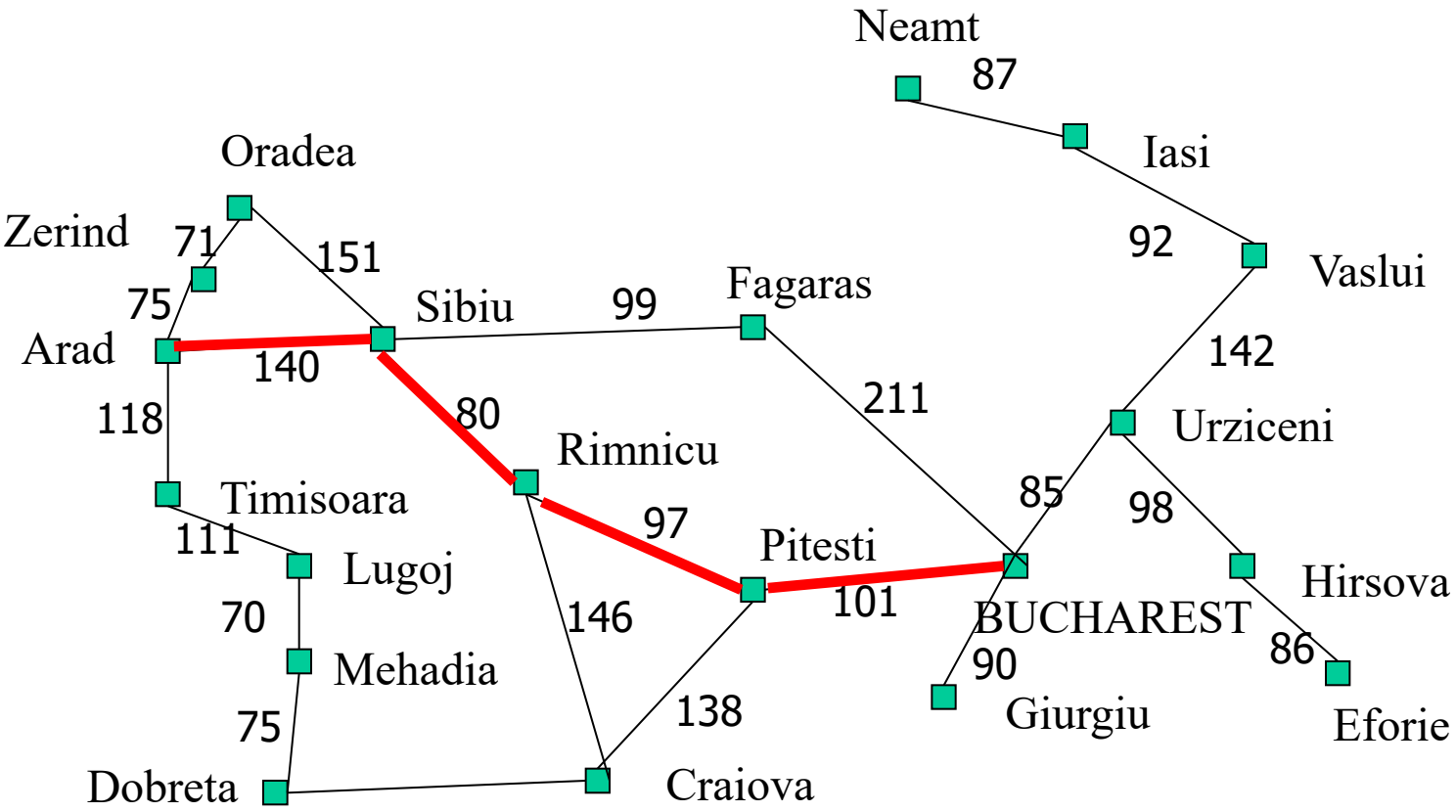
El **costo real** de la solución
Arad→Sibiu→*Fagaras*→*Bucarest* es **450**

Búsqueda informada



¿Es ésta la solución óptima?

Búsqueda informada



El camino óptimo es
Arad→Sibiu→*Rimnicu*→*Pitesti*→*Bucarest*
porque tiene costo **418**

Búsqueda informada

- La búsqueda avara **no es óptima**

Búsqueda informada

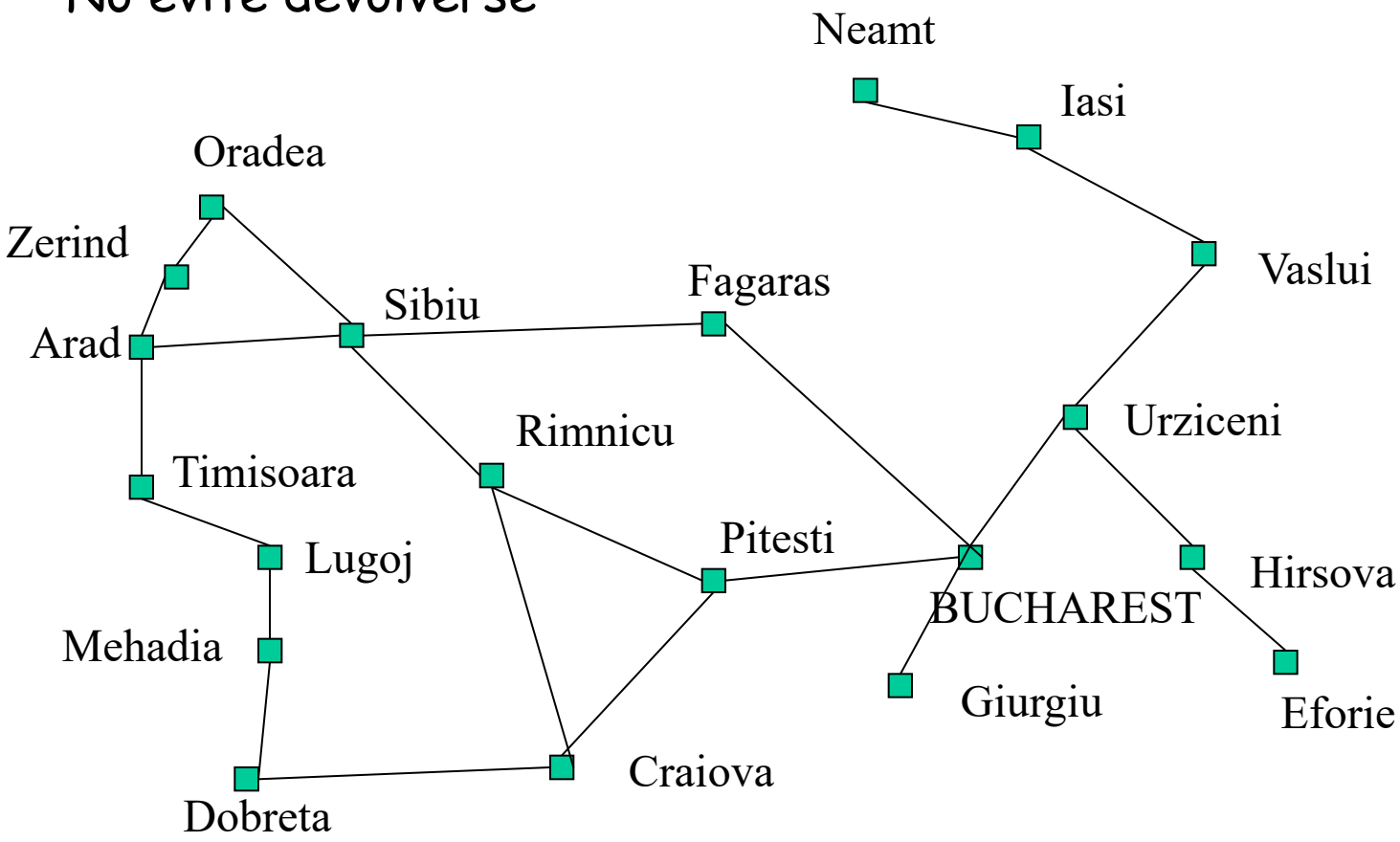
- Construya el árbol para ir de **Iasi** a **Fagaras**

  No evite devolverse

Búsqueda informada

- Construya el árbol para ir de **Iasi** a **Fagaras**

  No evite devolverse

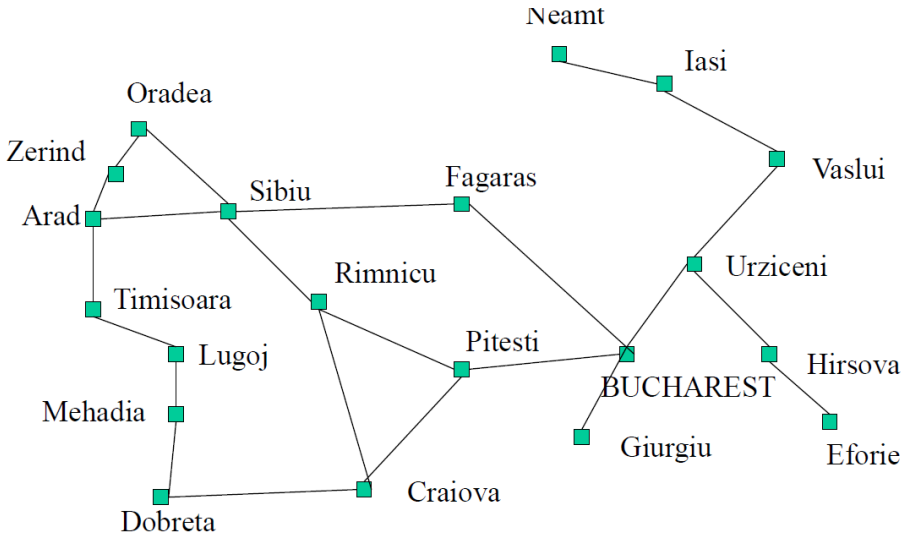


Distancia en línea recta a Fagaras:

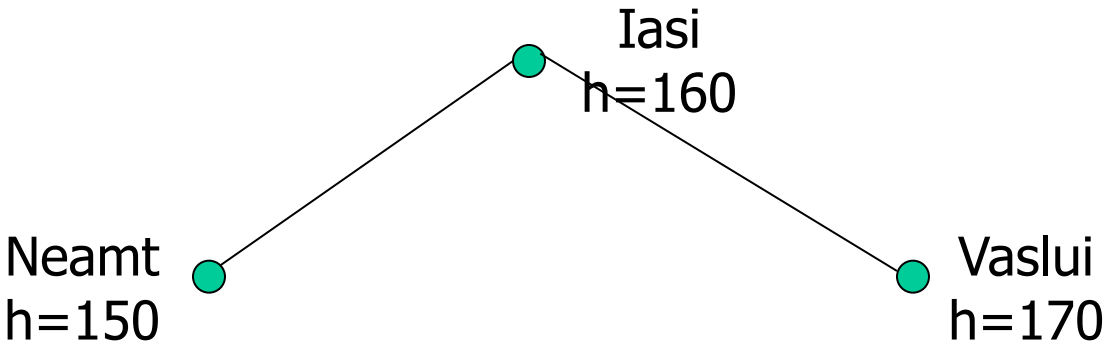
Fagaras	0
Iasi	160
Neamt	150
Urziceni	155
Vaslui	170

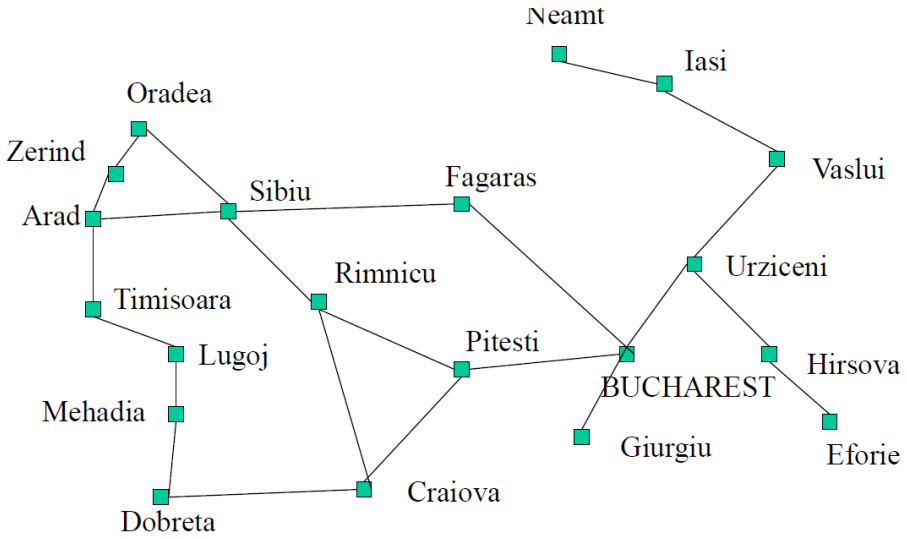
Búsqueda informada

Iasi
h=160

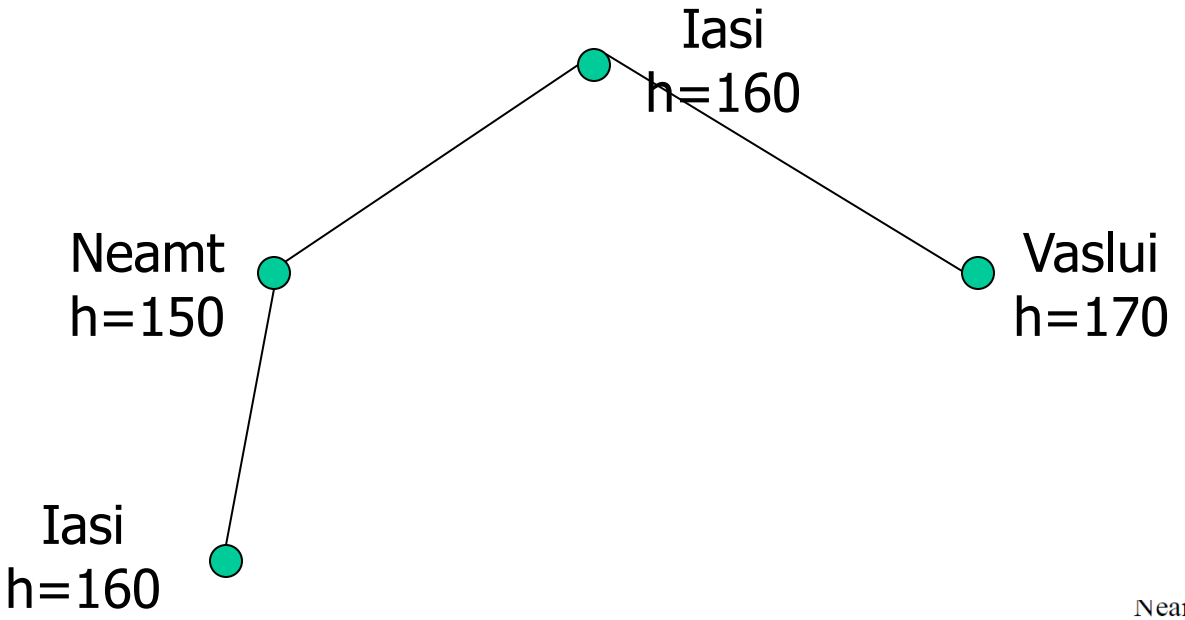


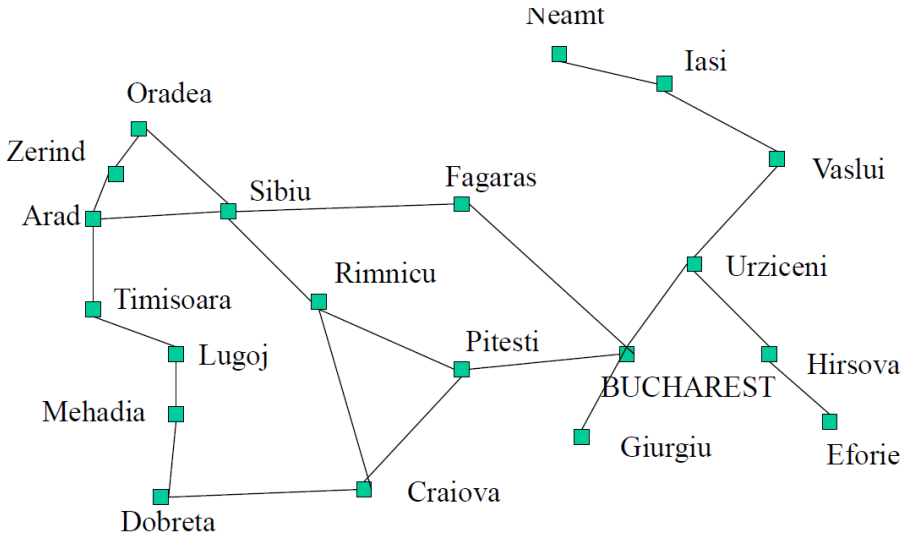
Búsqueda informada



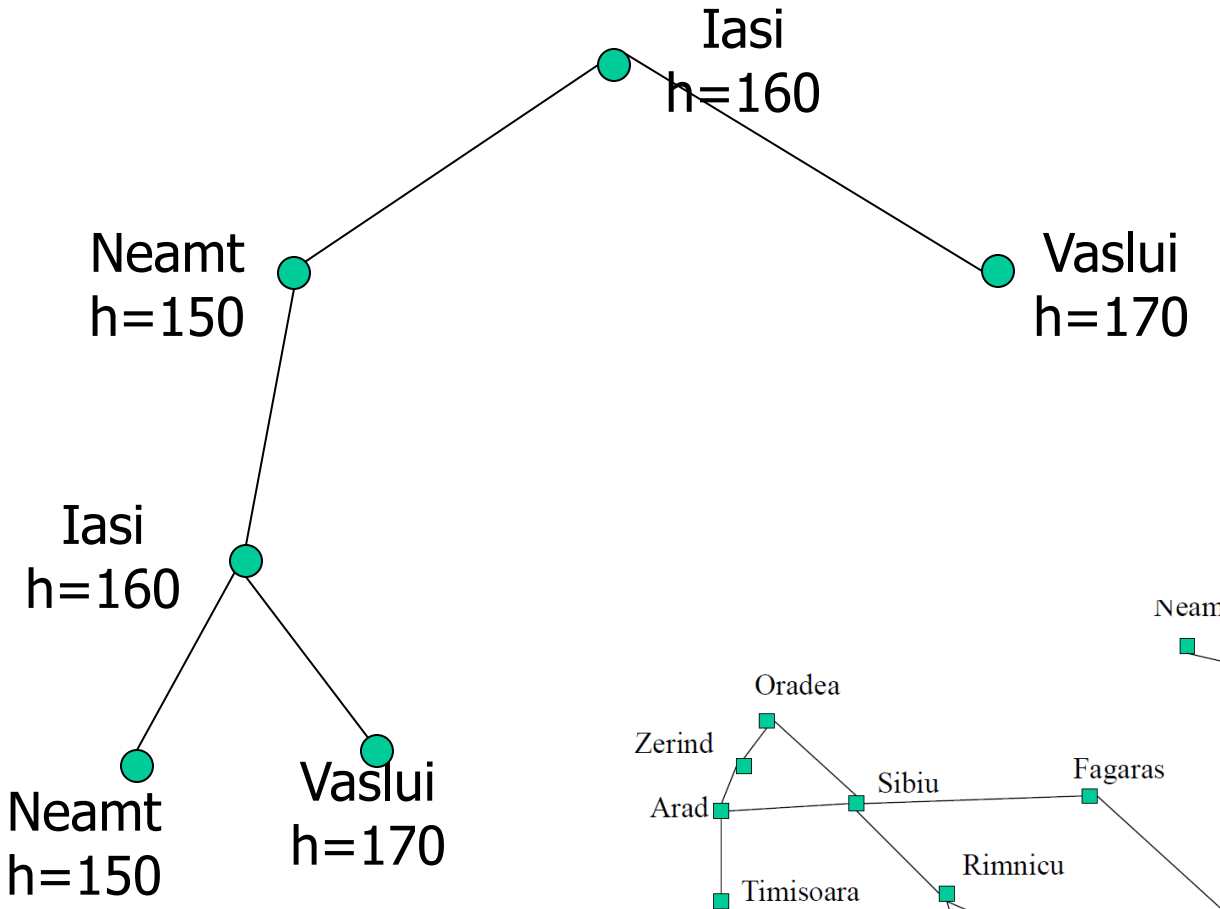


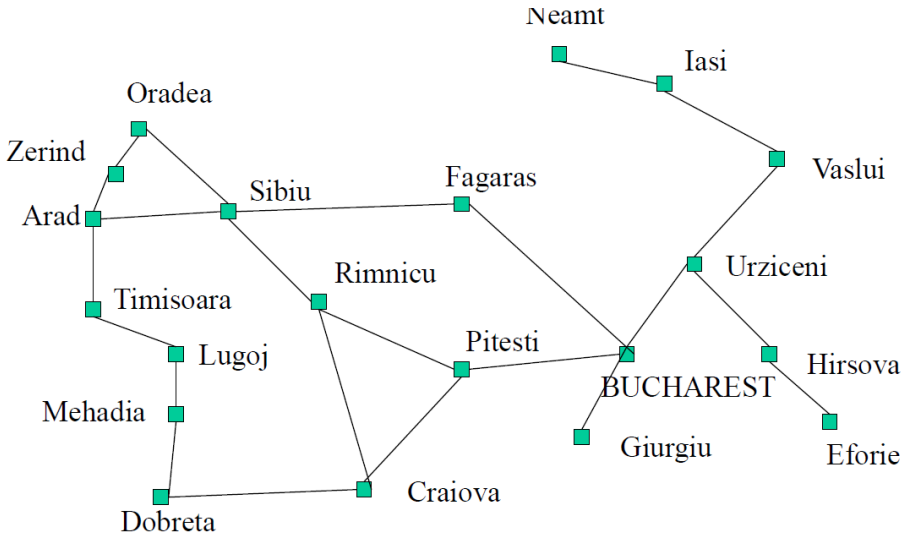
Búsqueda informada



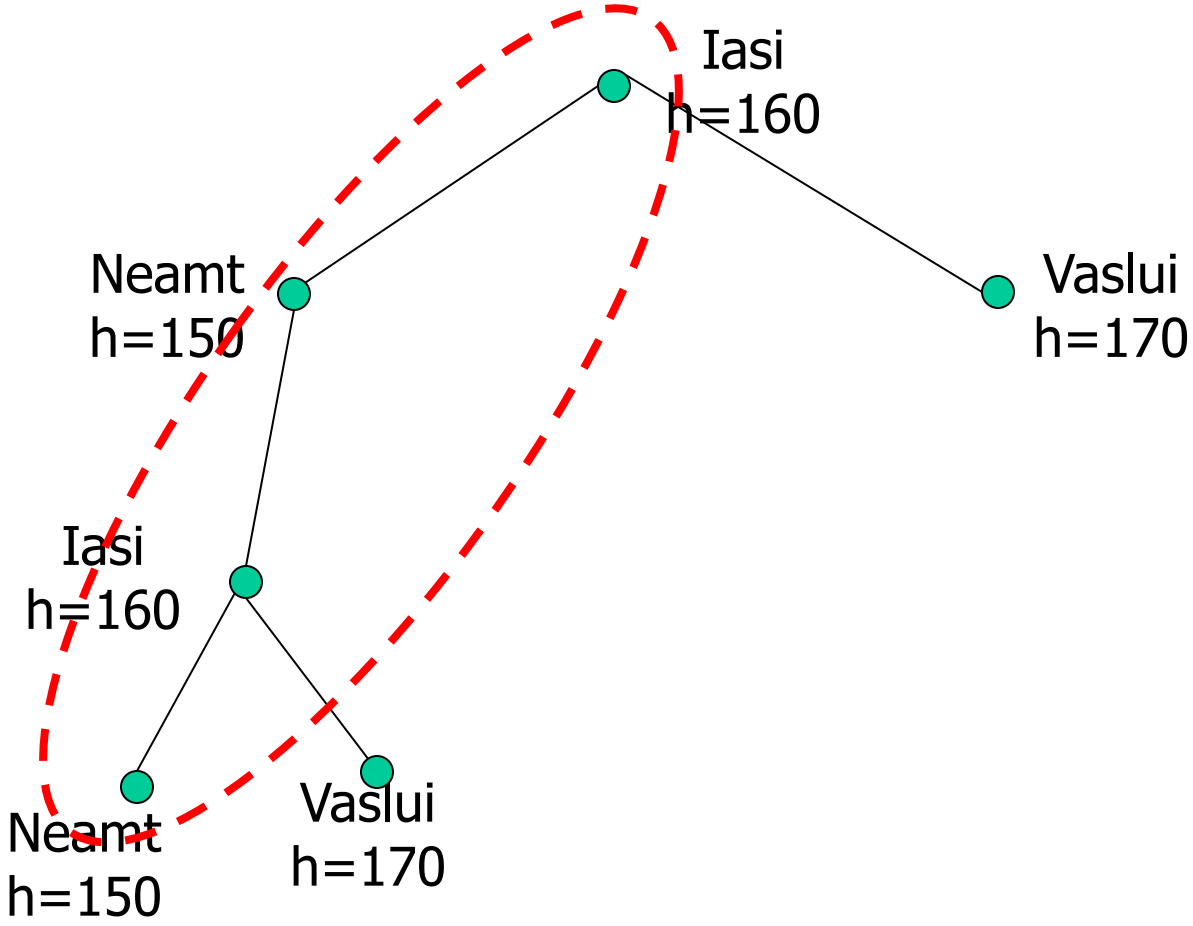


Búsqueda informada



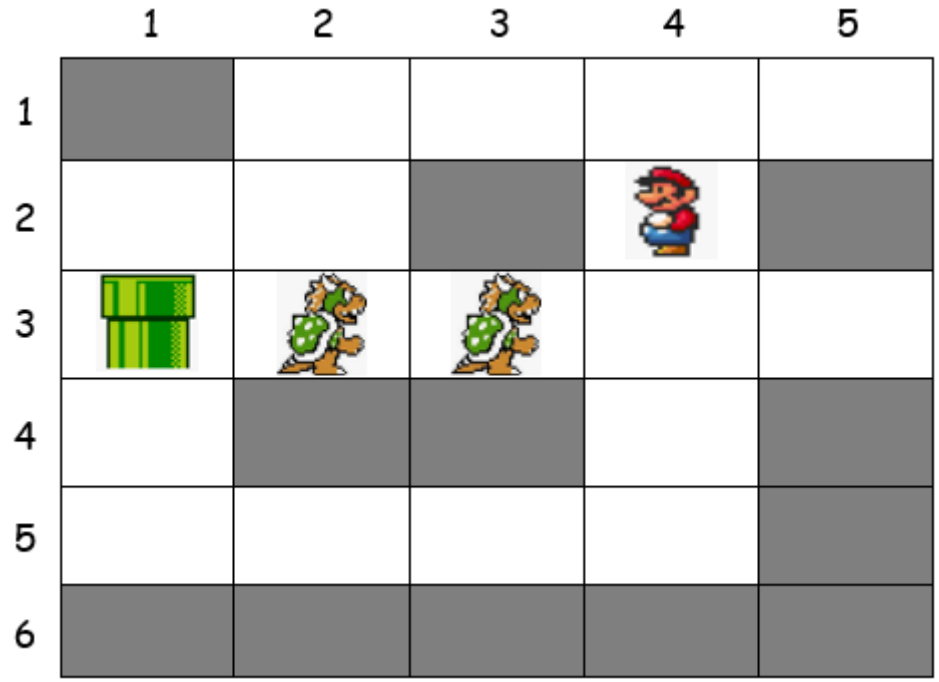


Búsqueda informada



Se queda en un **ciclo**

Búsqueda informada

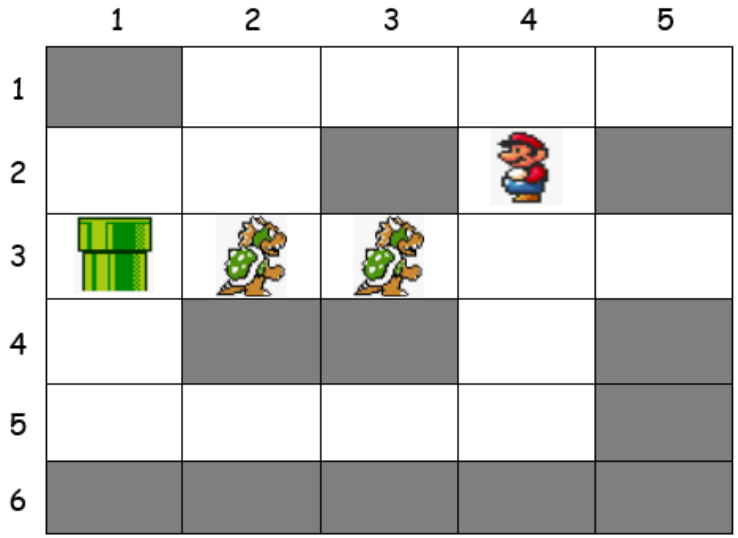


h(n): distancia de Manhattan a la posición (3,1)

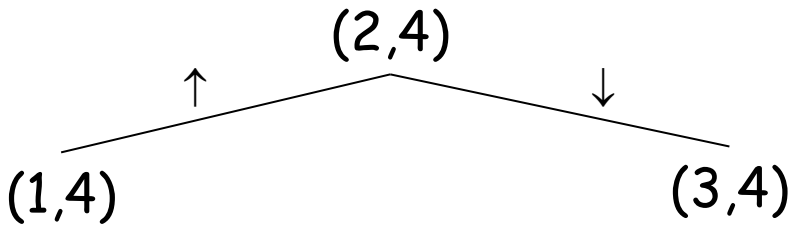
Evite devolverse

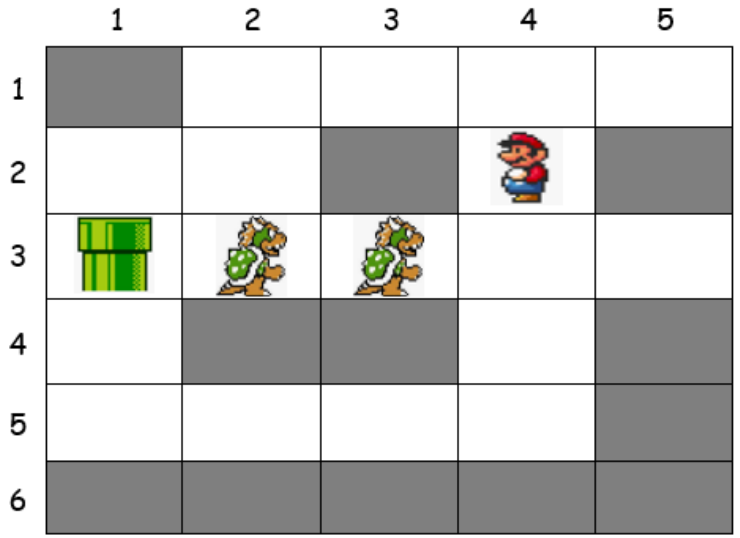
Búsqueda informada

(2,4)

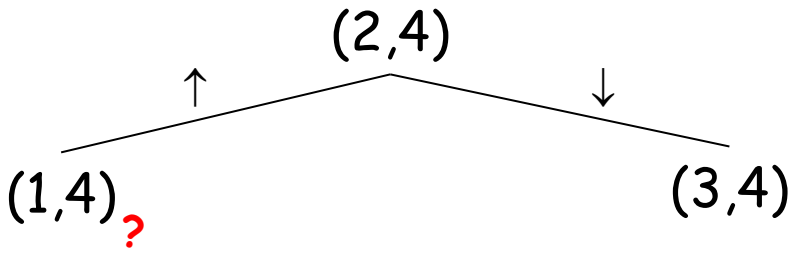


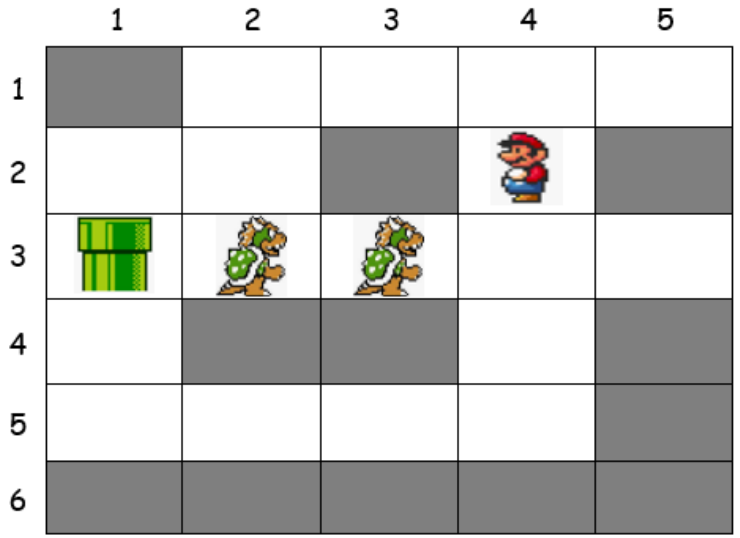
Búsqueda informada



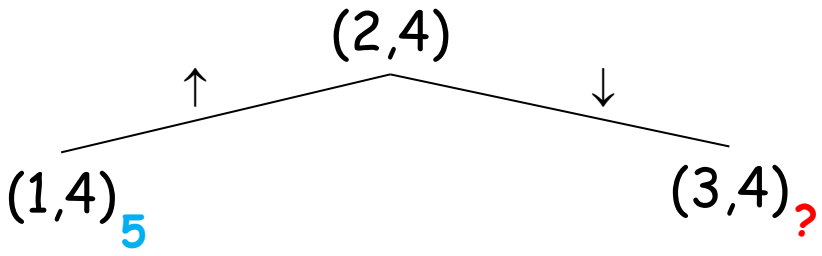


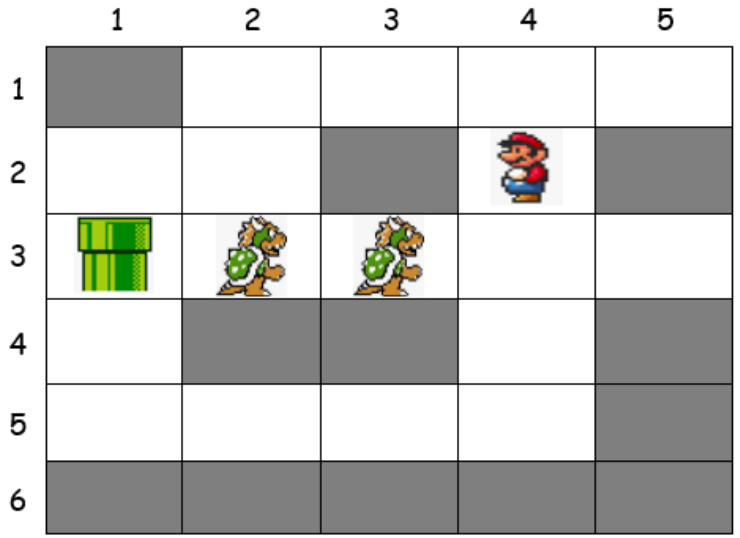
Búsqueda informada



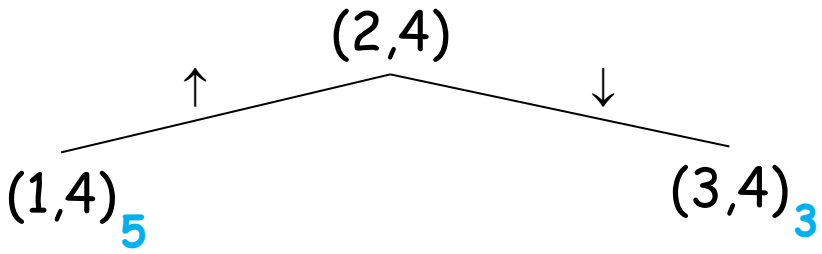


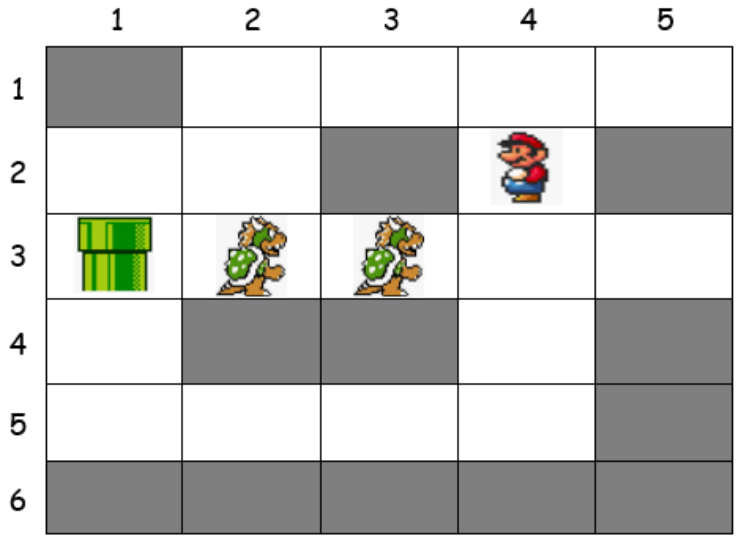
Búsqueda informada



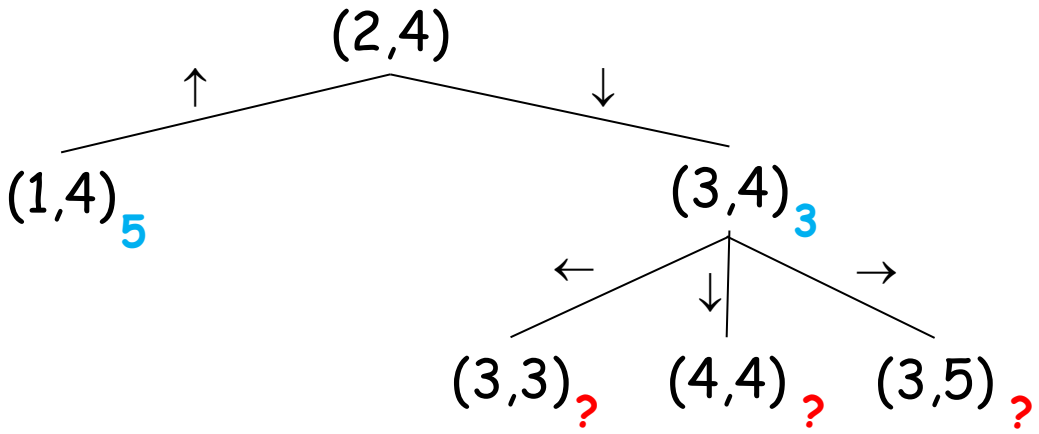


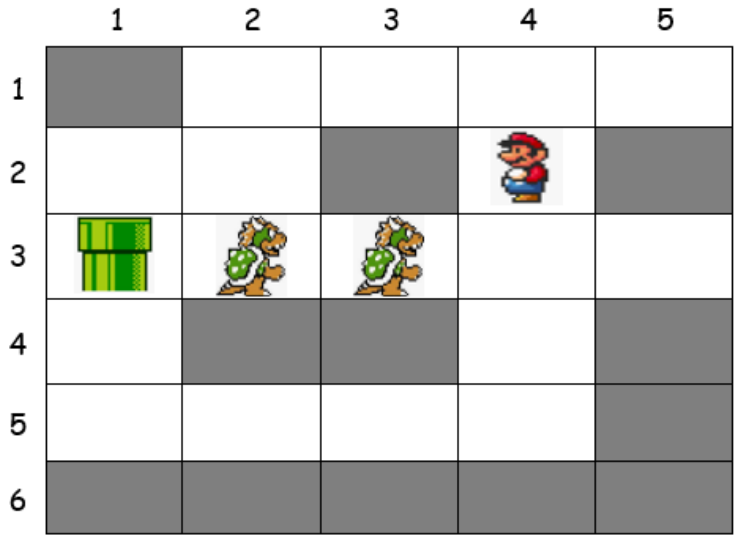
Búsqueda informada



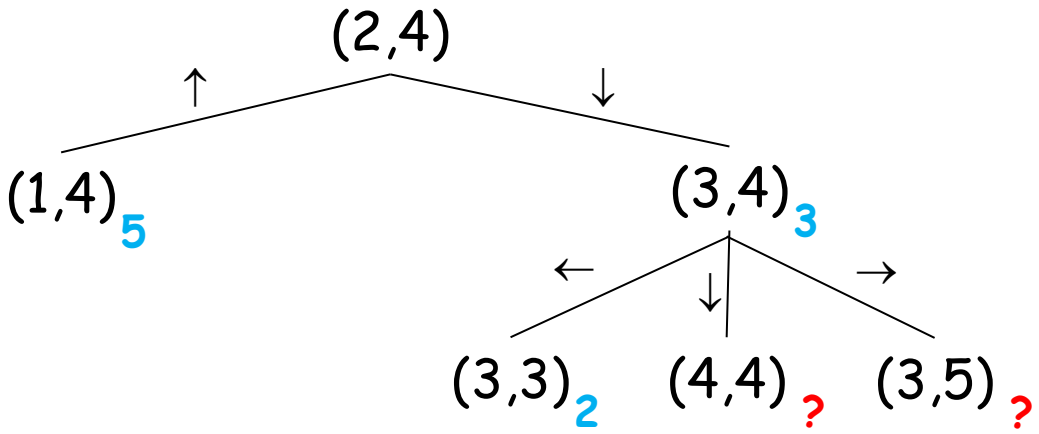


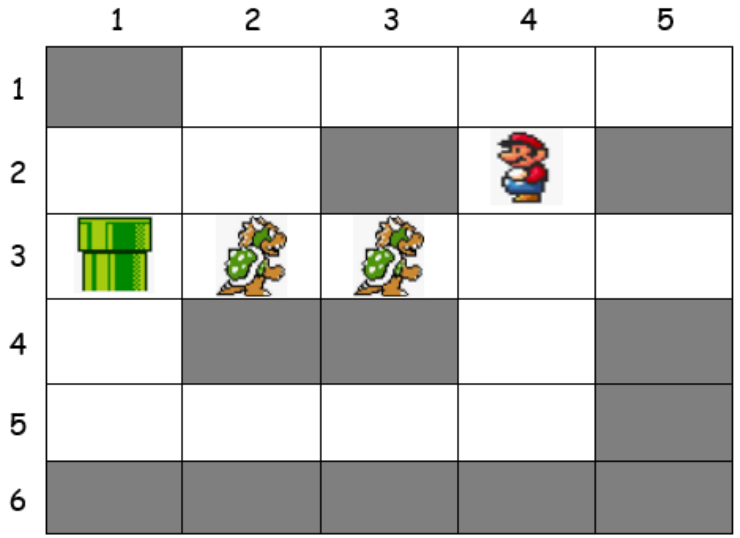
Búsqueda informada



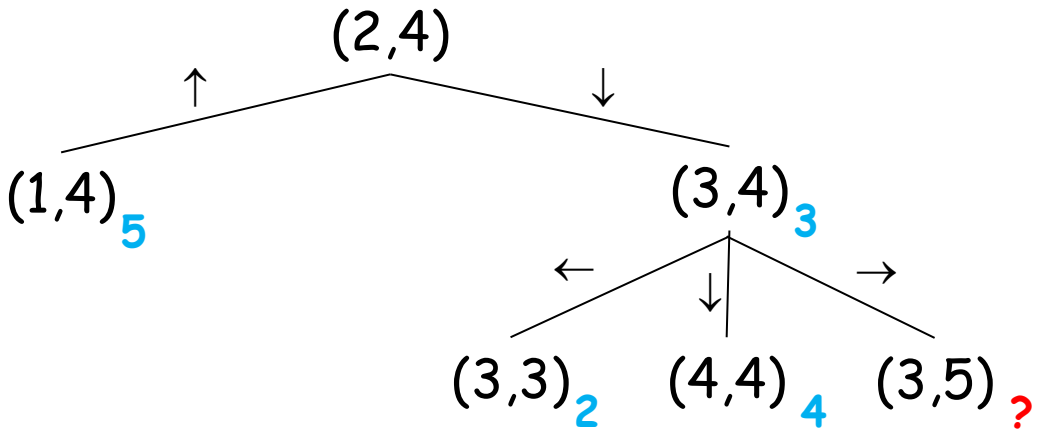


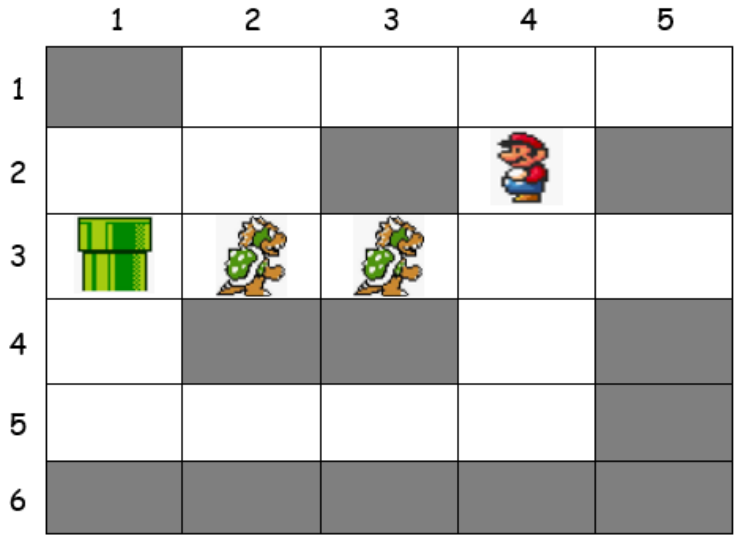
Búsqueda informada



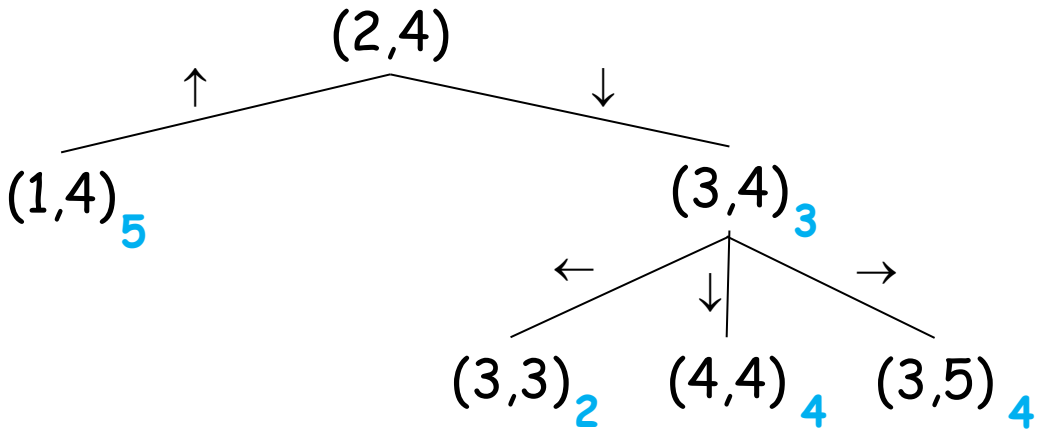


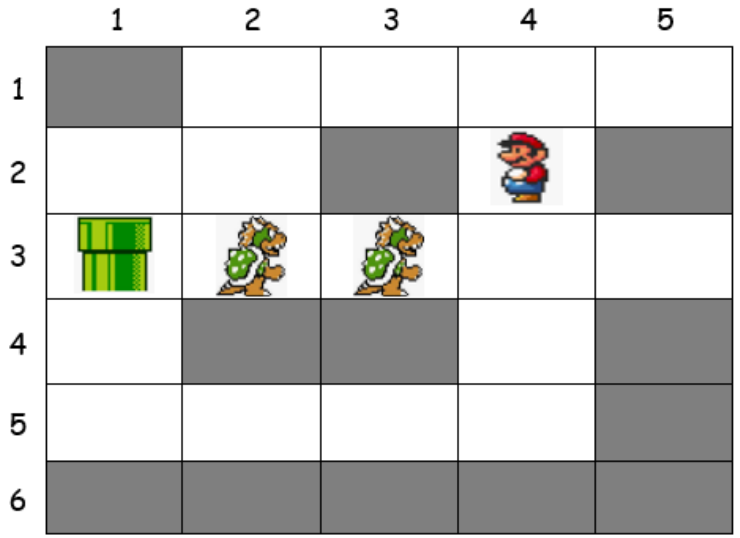
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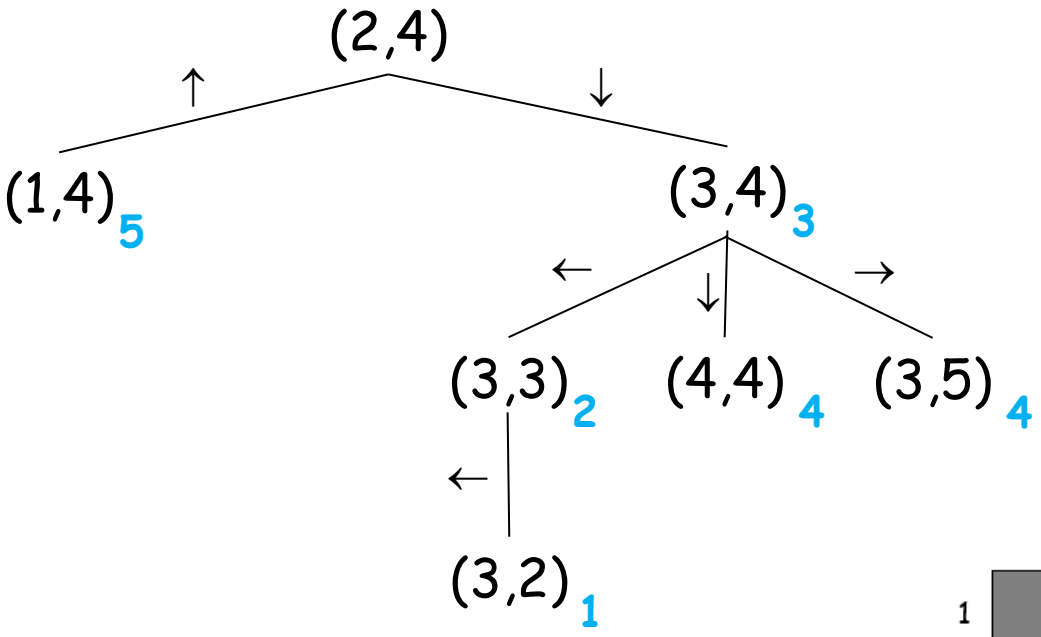


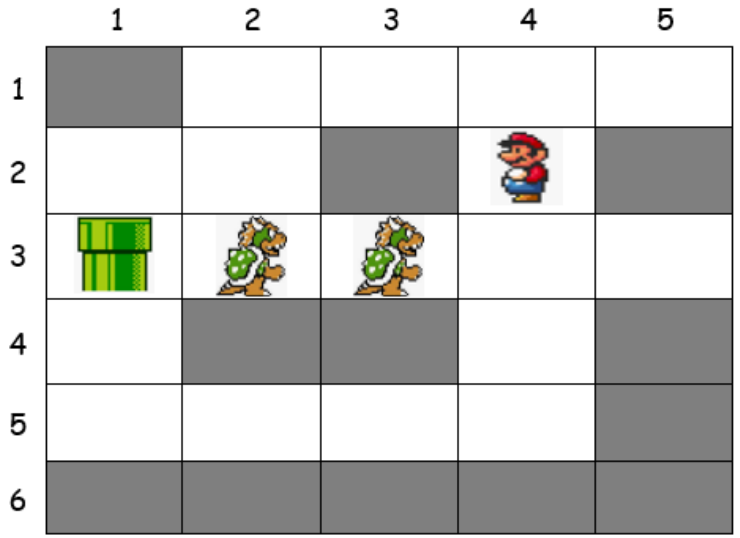
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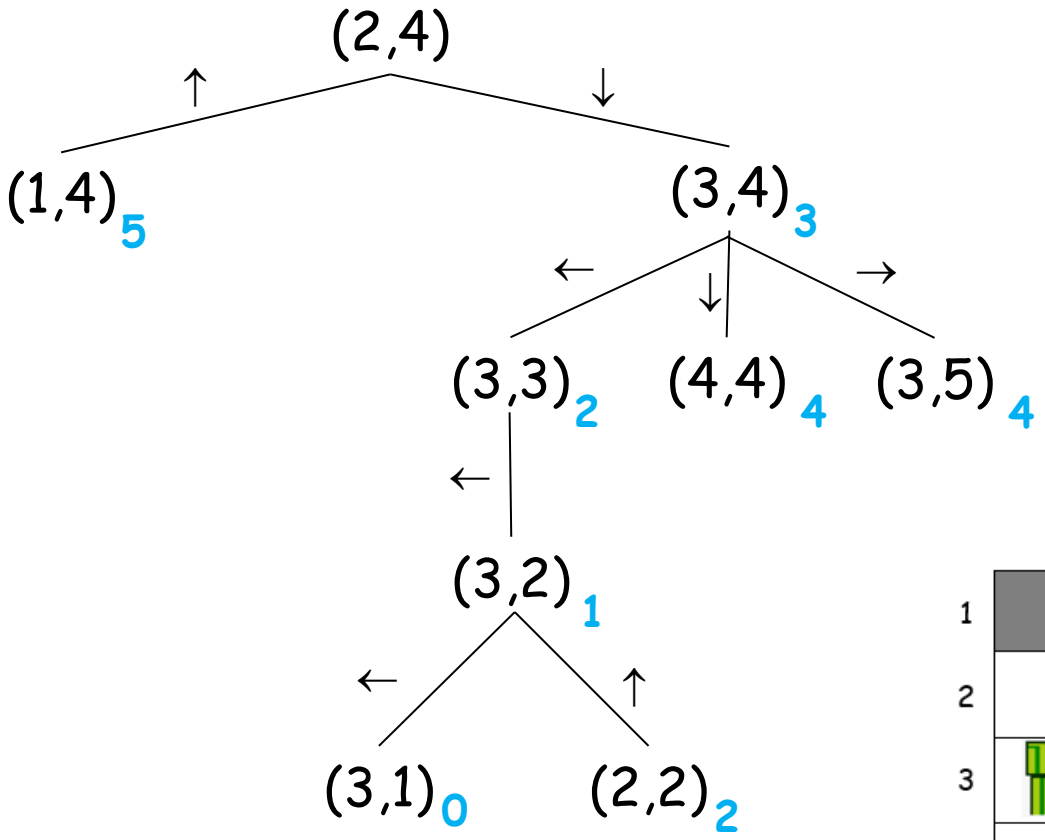


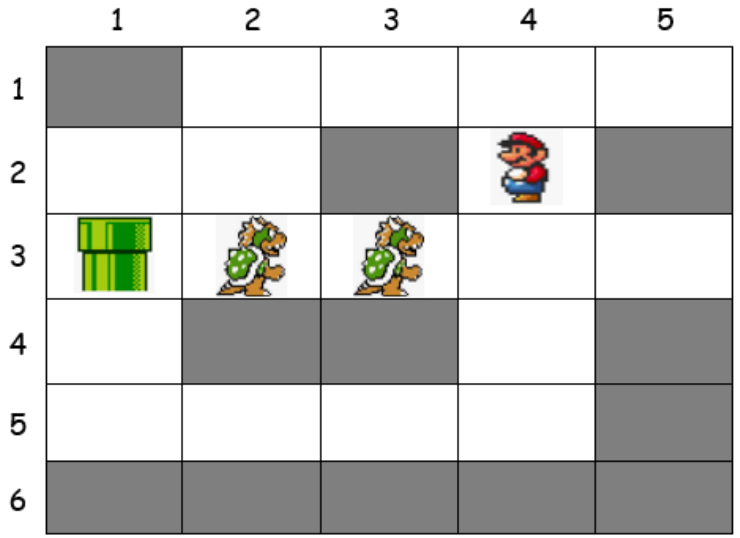
Búsqueda informada



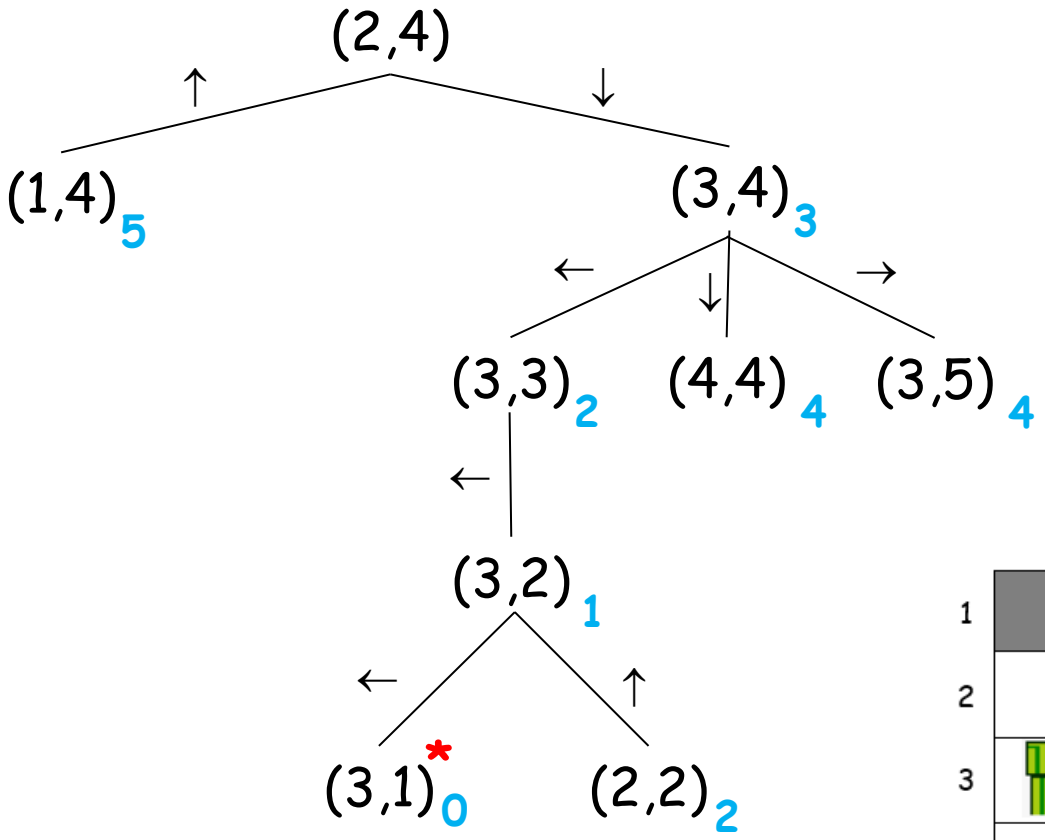


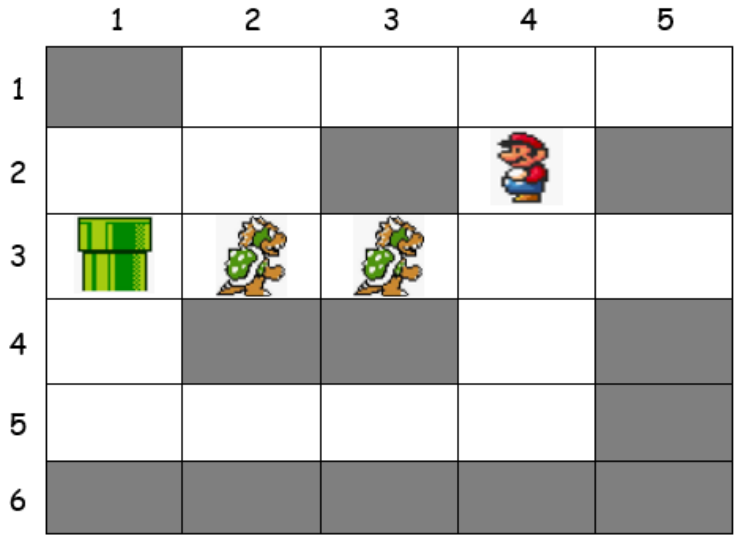
Búsqueda informada





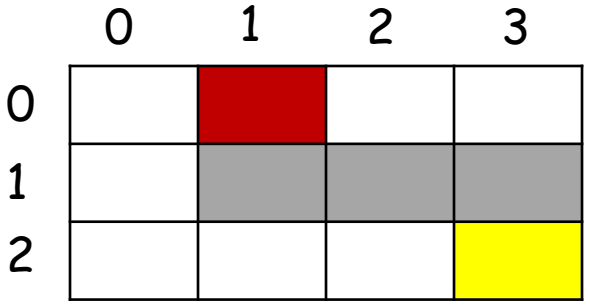
Búsqueda informada





Búsqueda informada

Aplicar búsqueda Avara

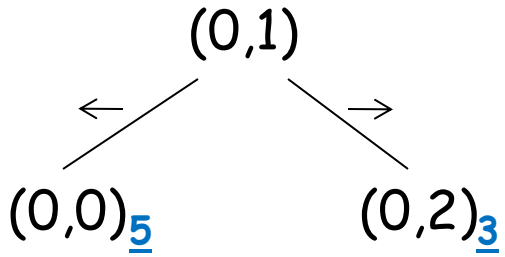


Agente Minero. Indique el camino que sigue el agente.

No evite devolverse

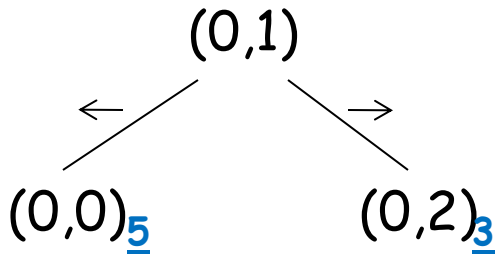
Punto de inicio

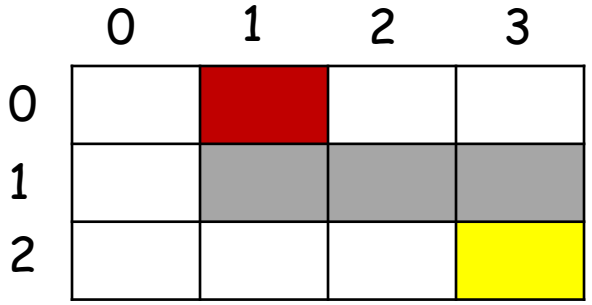
Lingote



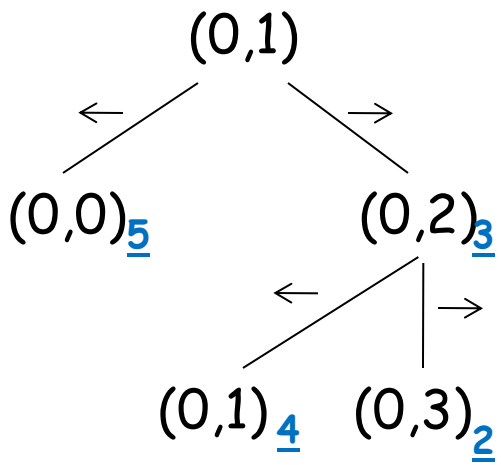
- Utilice **Manhattan**

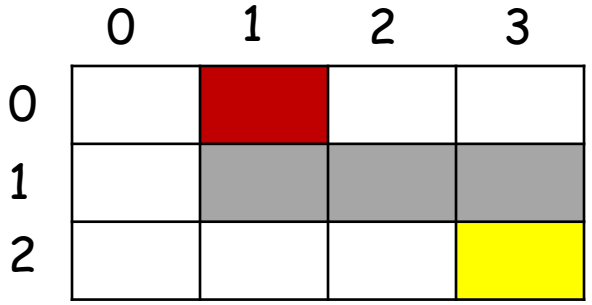
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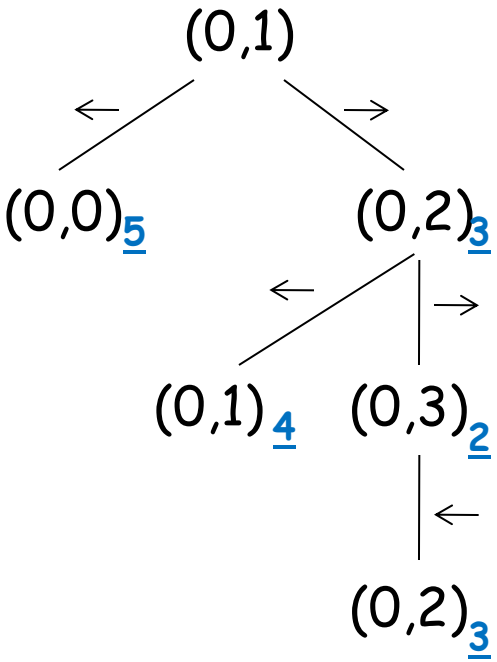


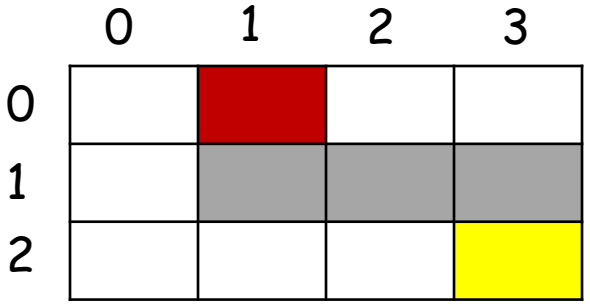
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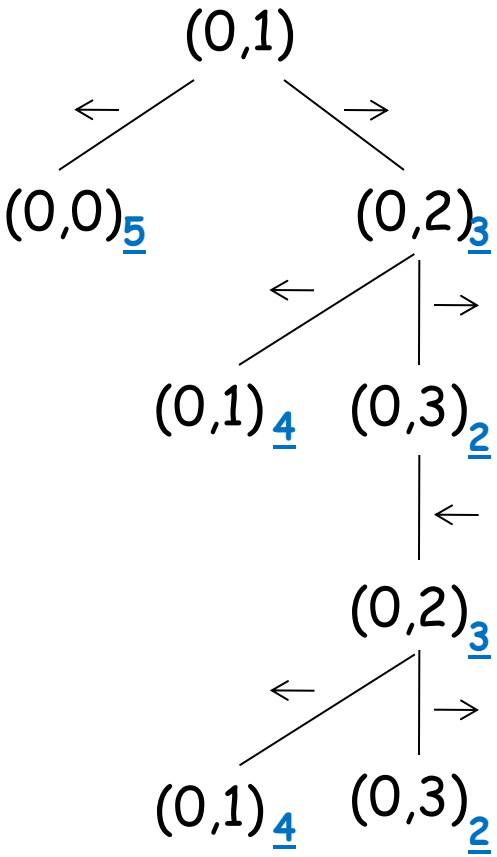


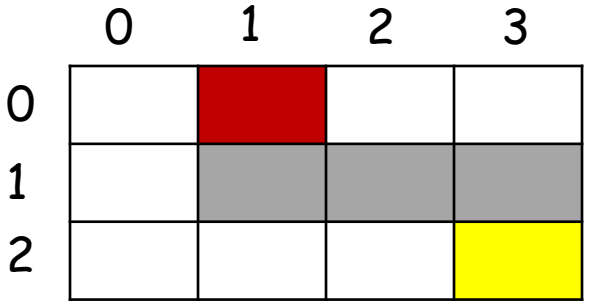
Búsqueda informada



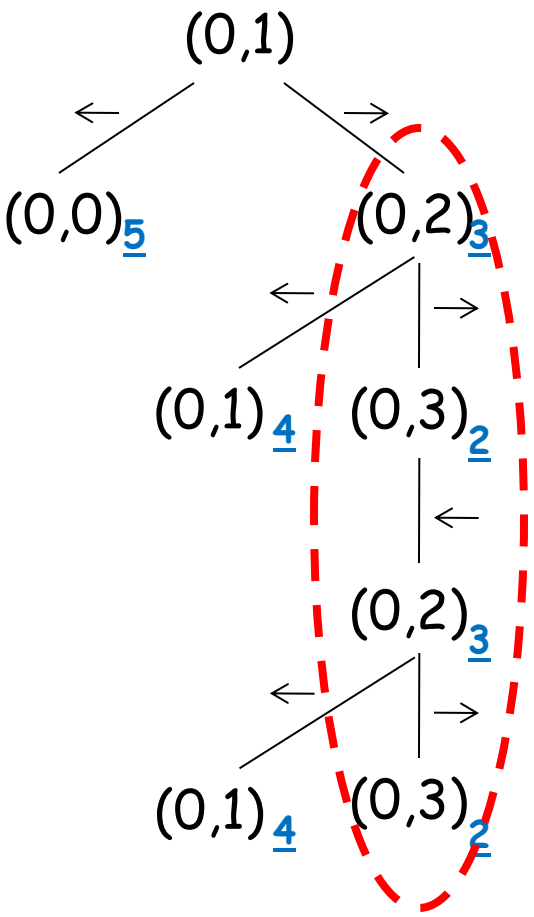


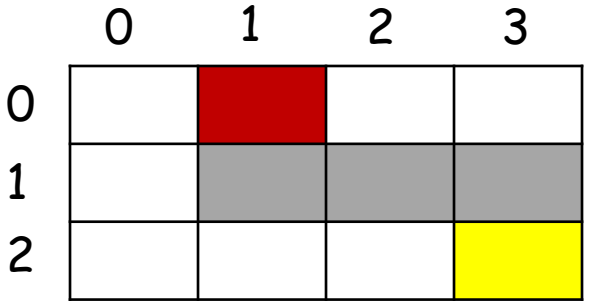
Búsqueda informada





Búsqueda informada





Búsqueda **avara** se queda en un ciclo

Búsqueda informada

Implementación

- Se puede implementar considerando la lista de nodos a expandir como una **cola de prioridad**, donde la prioridad es el valor de la **heurística**

Búsqueda informada

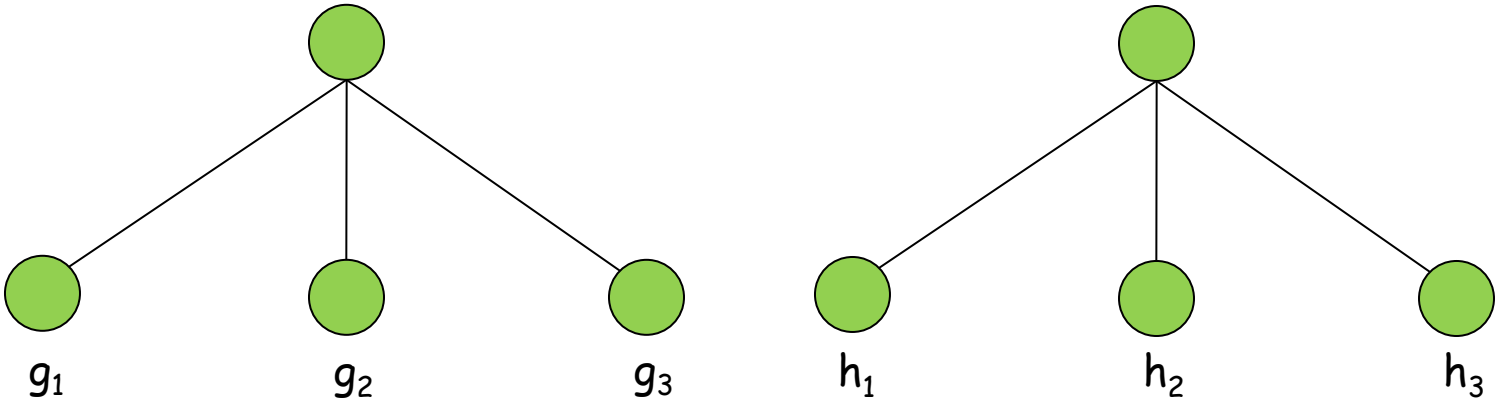
- **Completitud:** no, puede caer en ciclos
- **Complejidad temporal**: $O(b^d)$, exponencial
- **Complejidad espacial**: $O(b^d)$, exponencial
- **Solución óptima**: no, la función heurística no garantiza la solución óptima

Búsqueda informada

- ¿Cuál es la diferencia entre **costo uniforme** y **avara**?

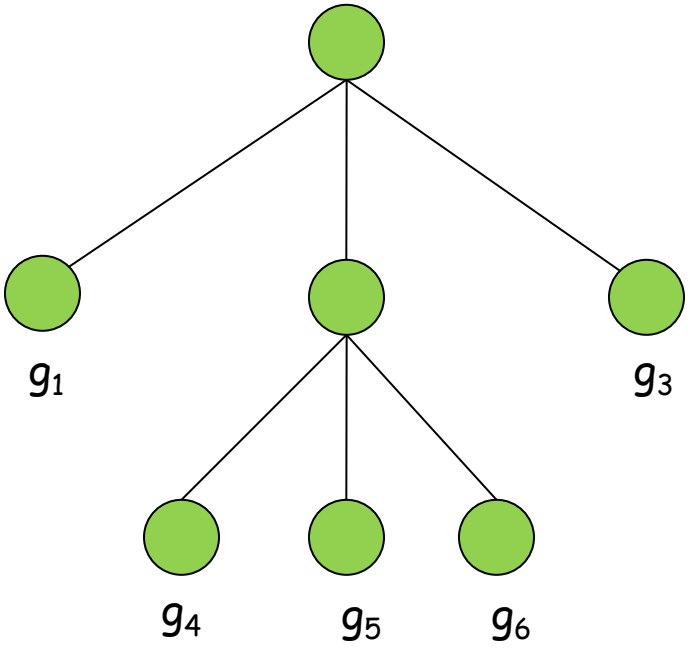
Búsqueda informada

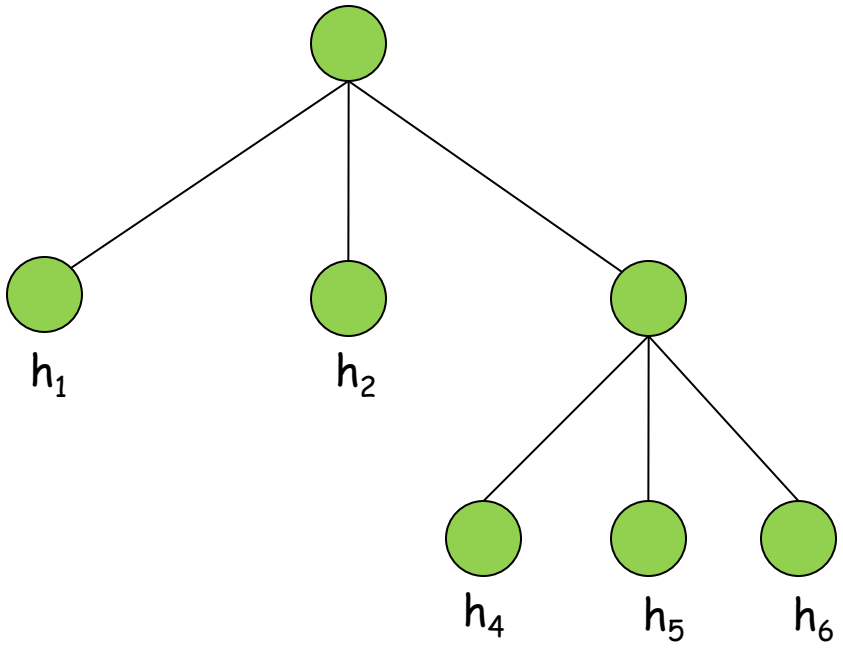
- ¿Cuál es la diferencia entre **costo uniforme** y **avara**?



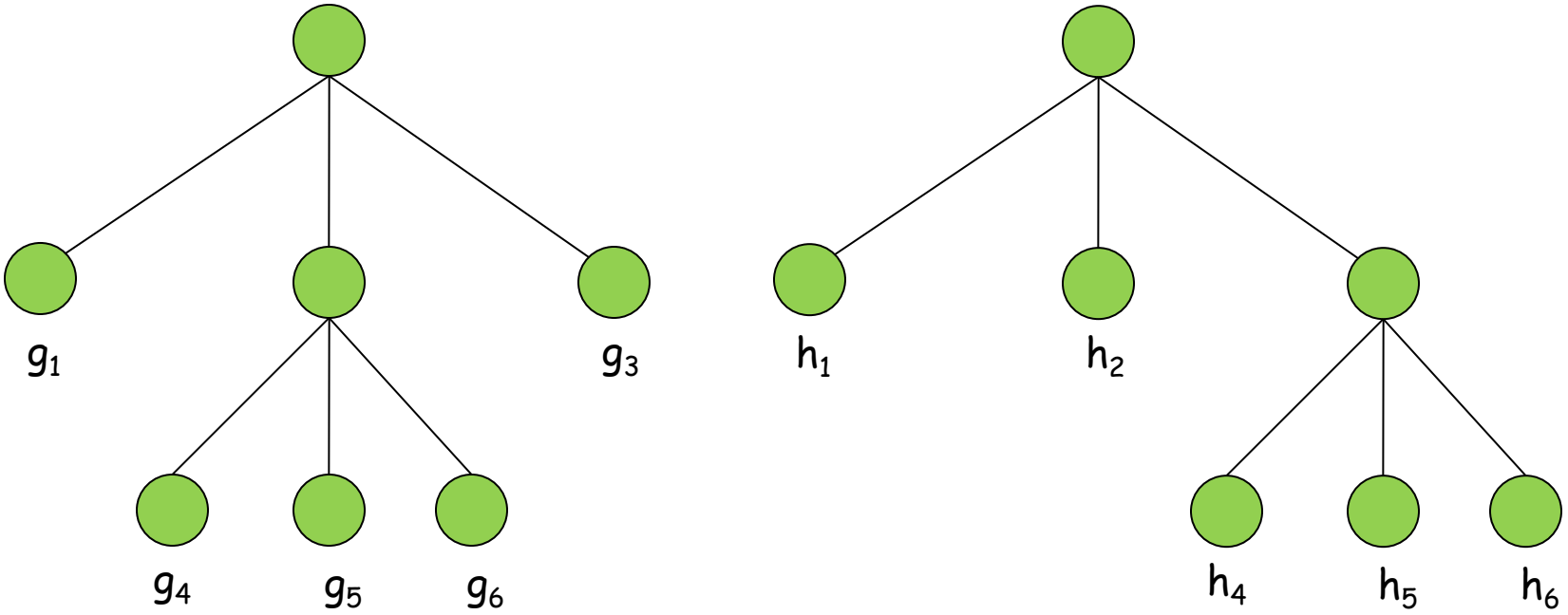
Búsqueda informada

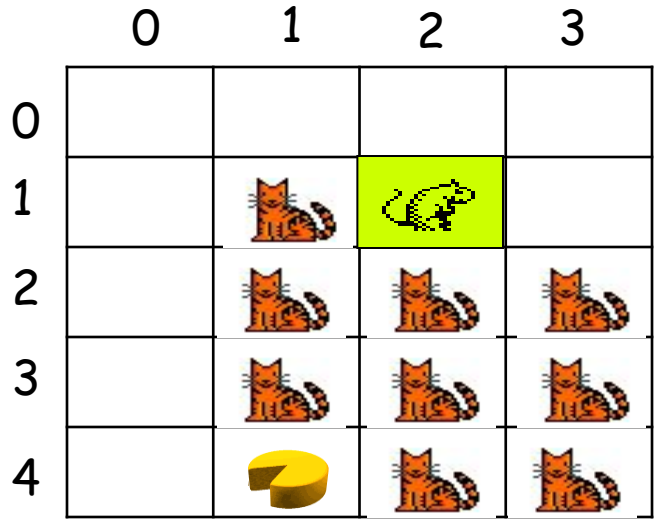
- ¿Cuál es la diferencia entre **costo uniforme** y **avara**?



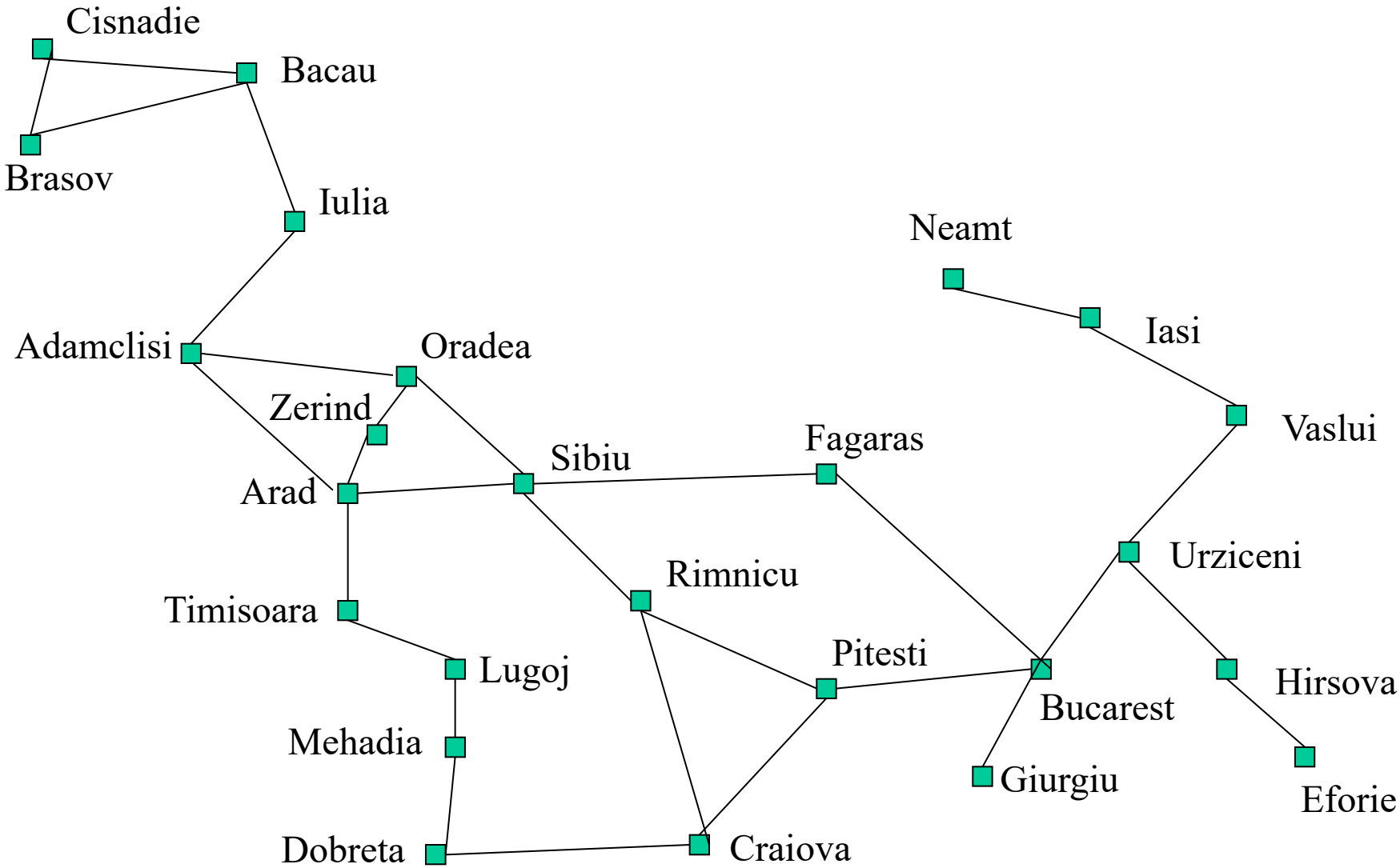


Búsqueda informada

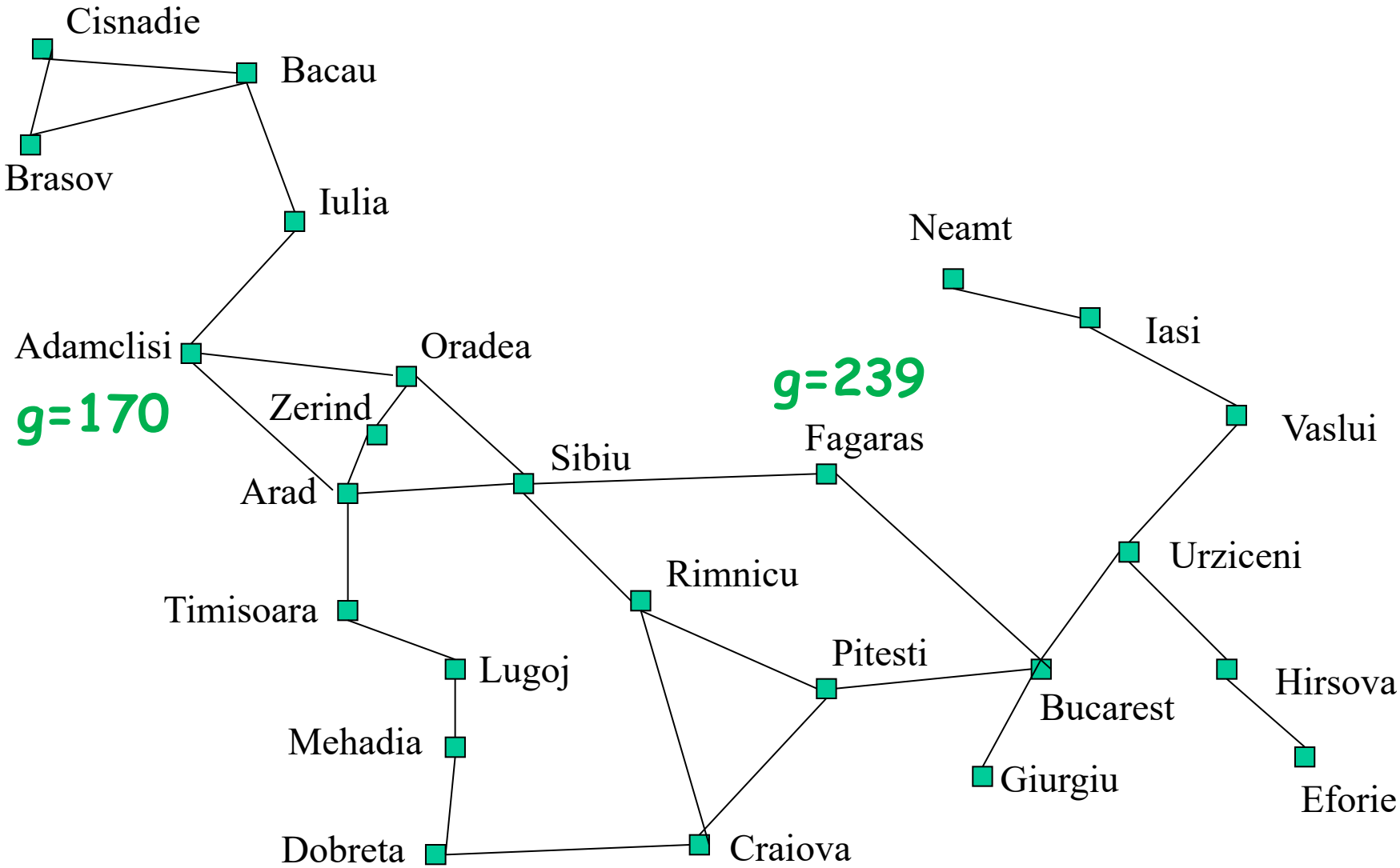




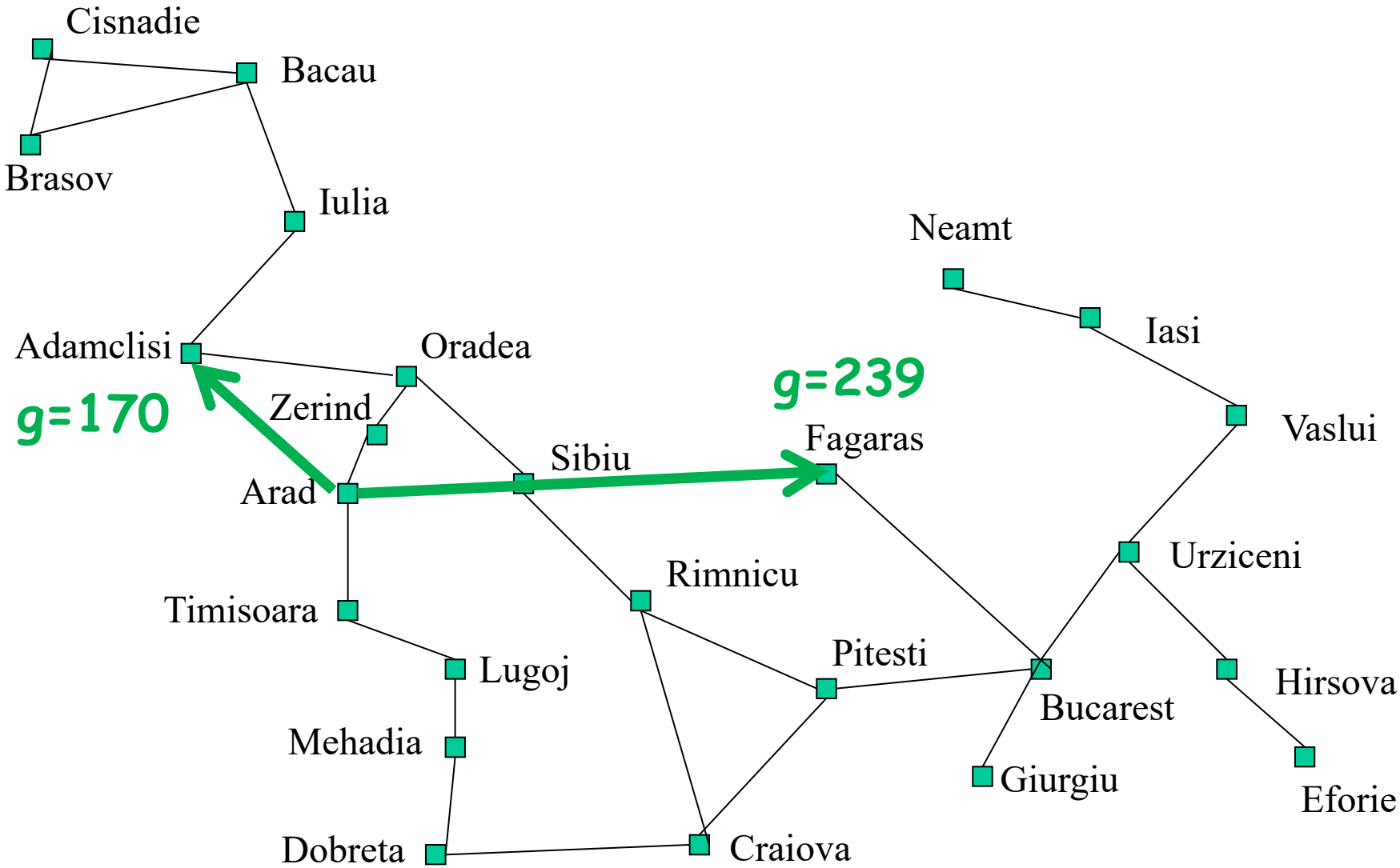
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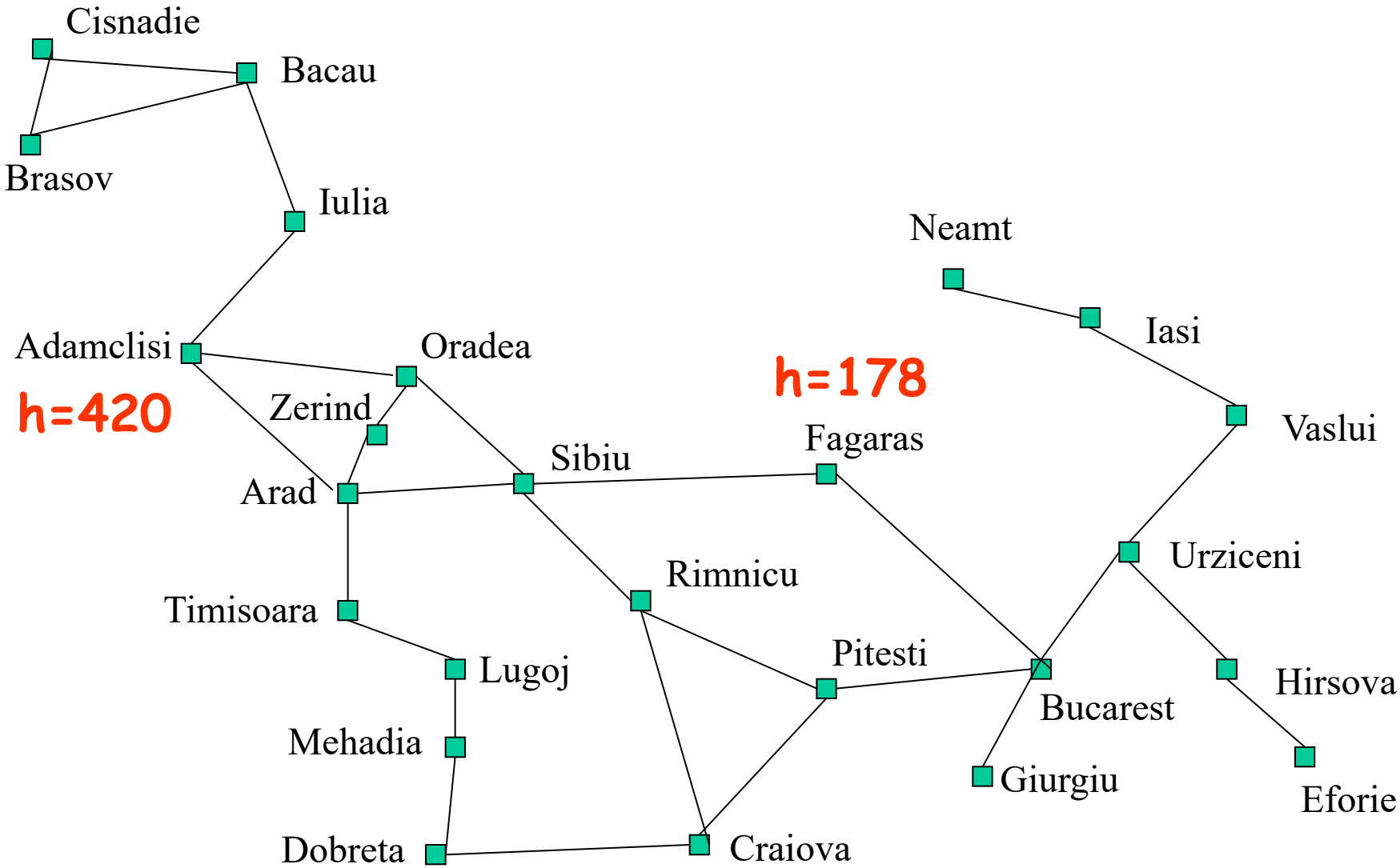
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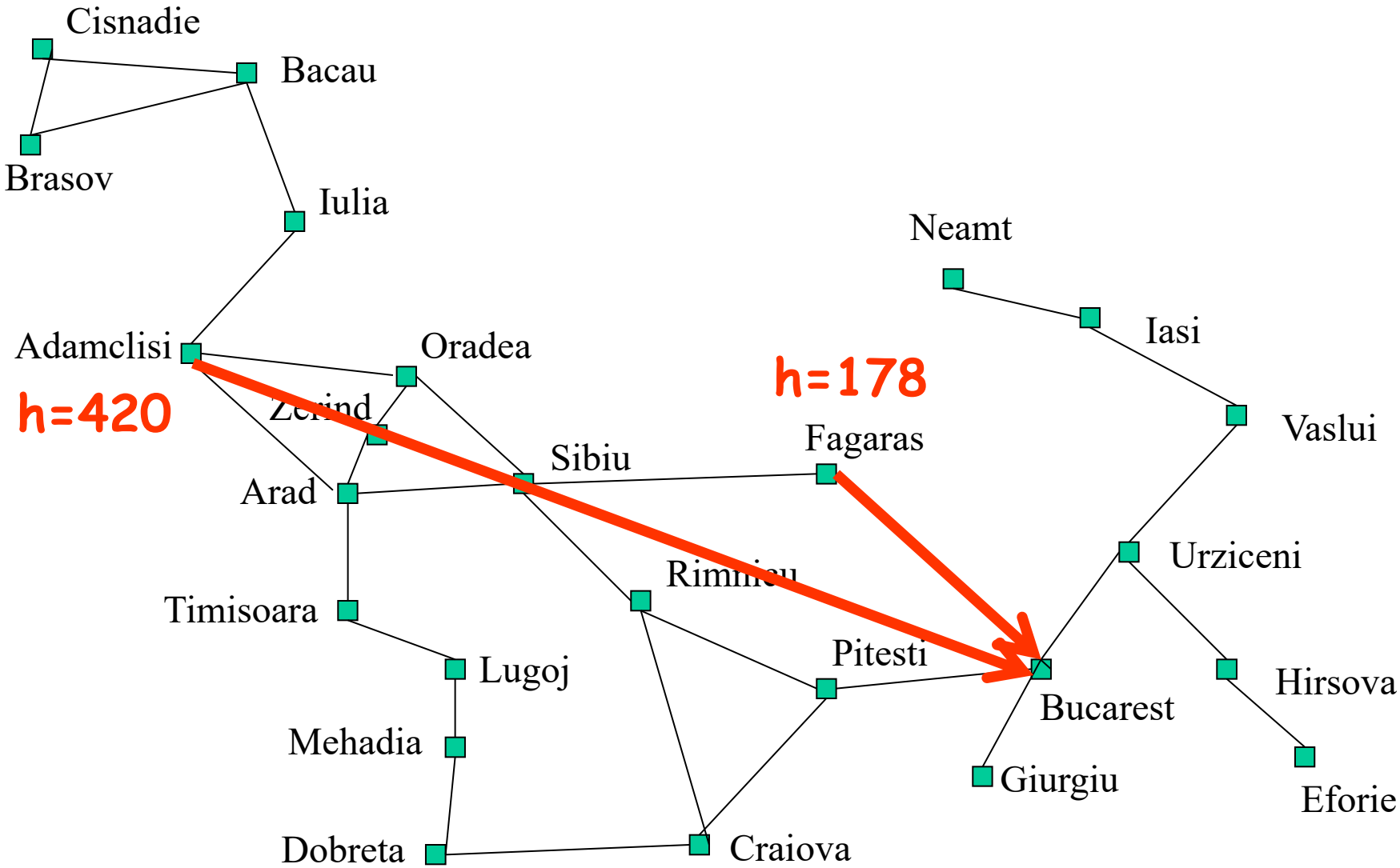
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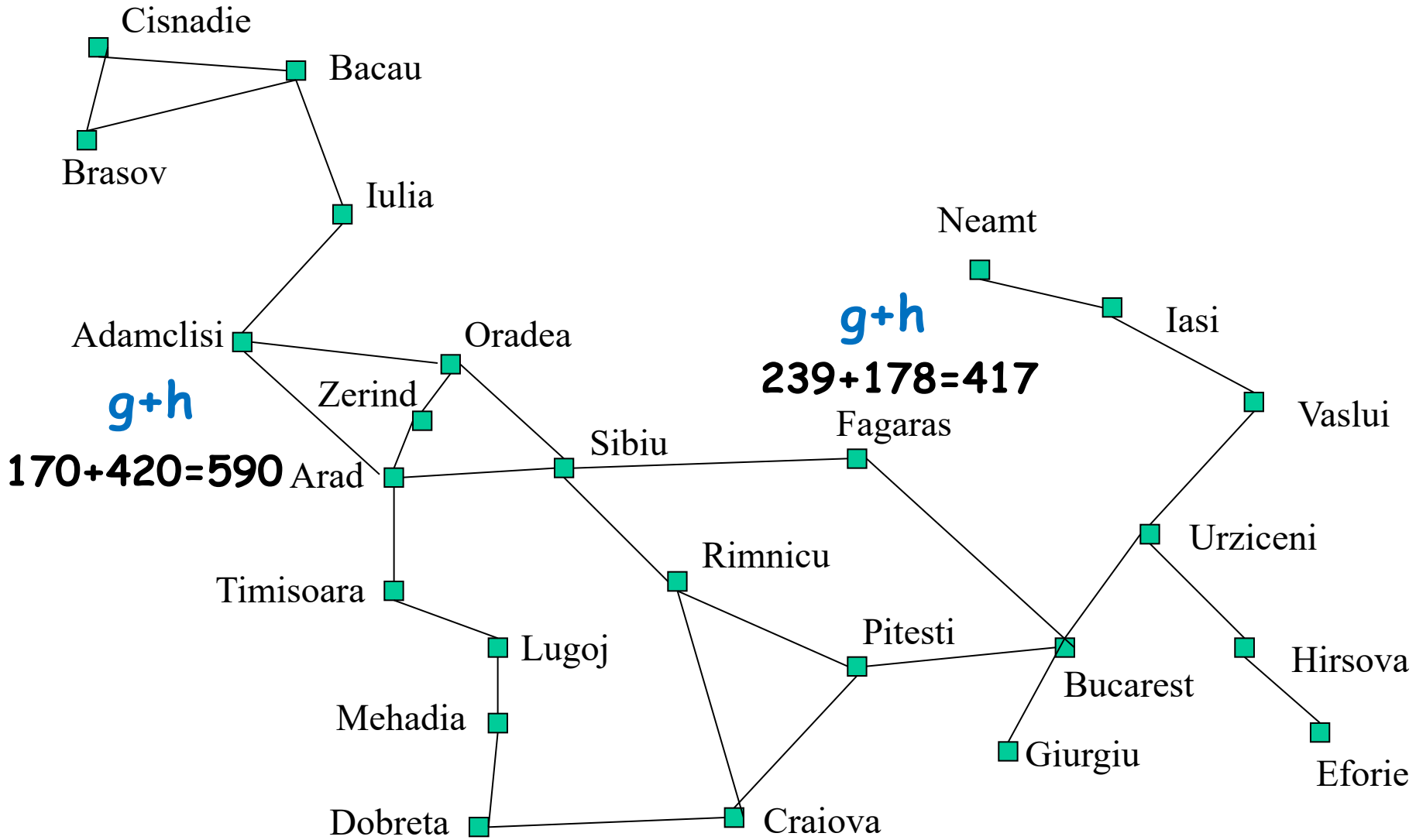
Búsqueda informada



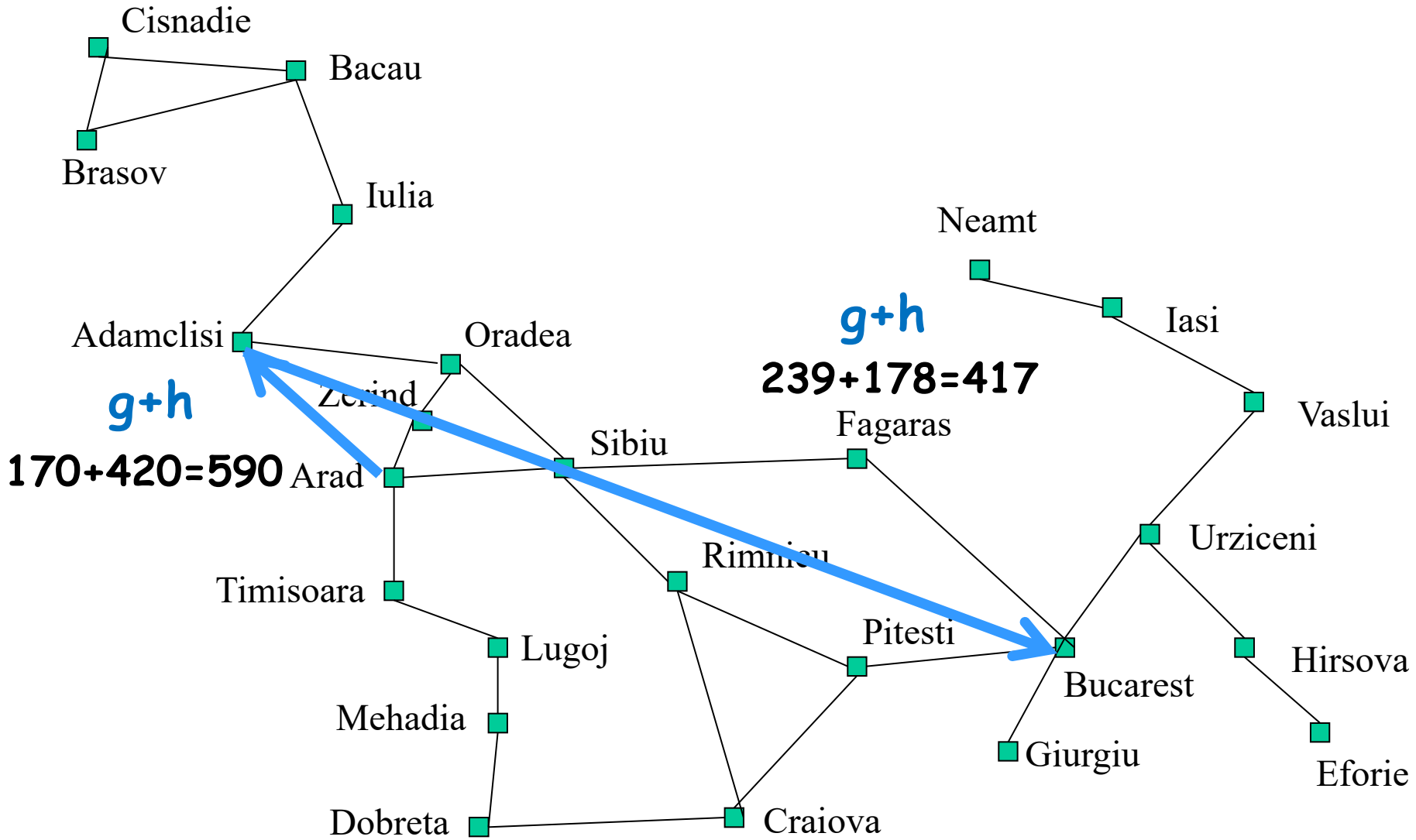
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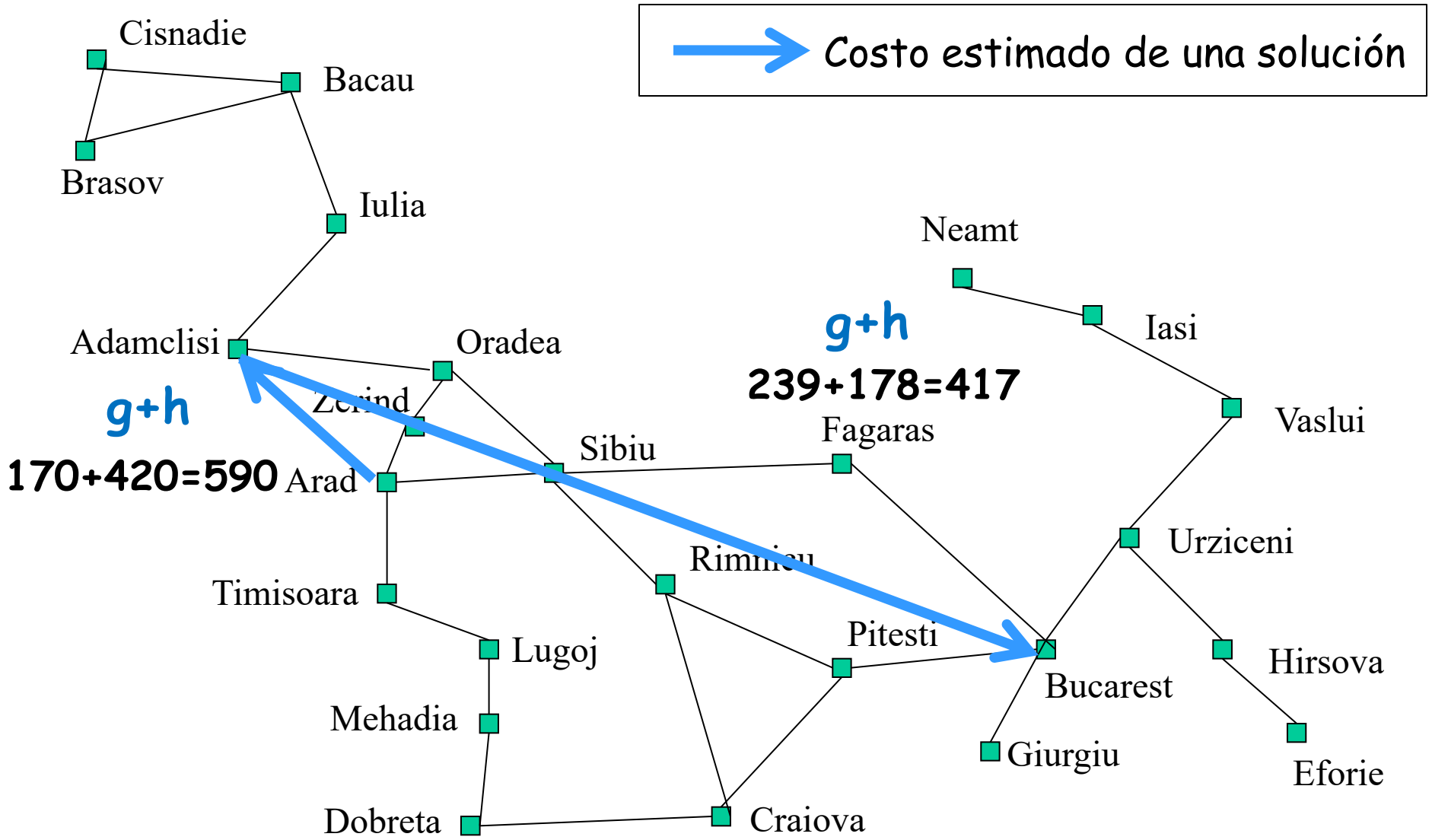
Búsqueda informada



Búsqueda informada



Búsqueda informada



Búsqueda informada

g =50.000, h =100.000



g =40.000, h =120.000



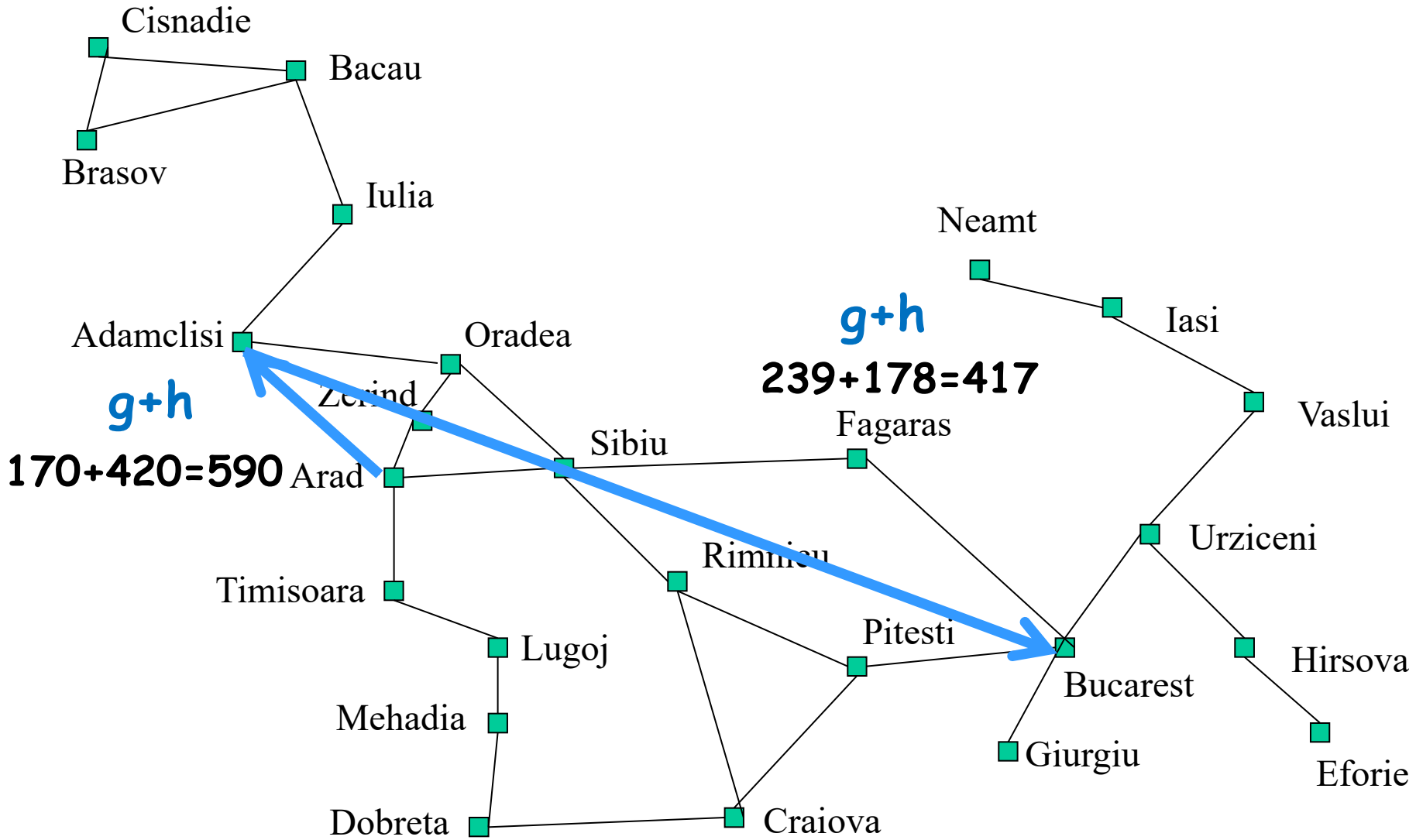
g =80.000, h =100.000



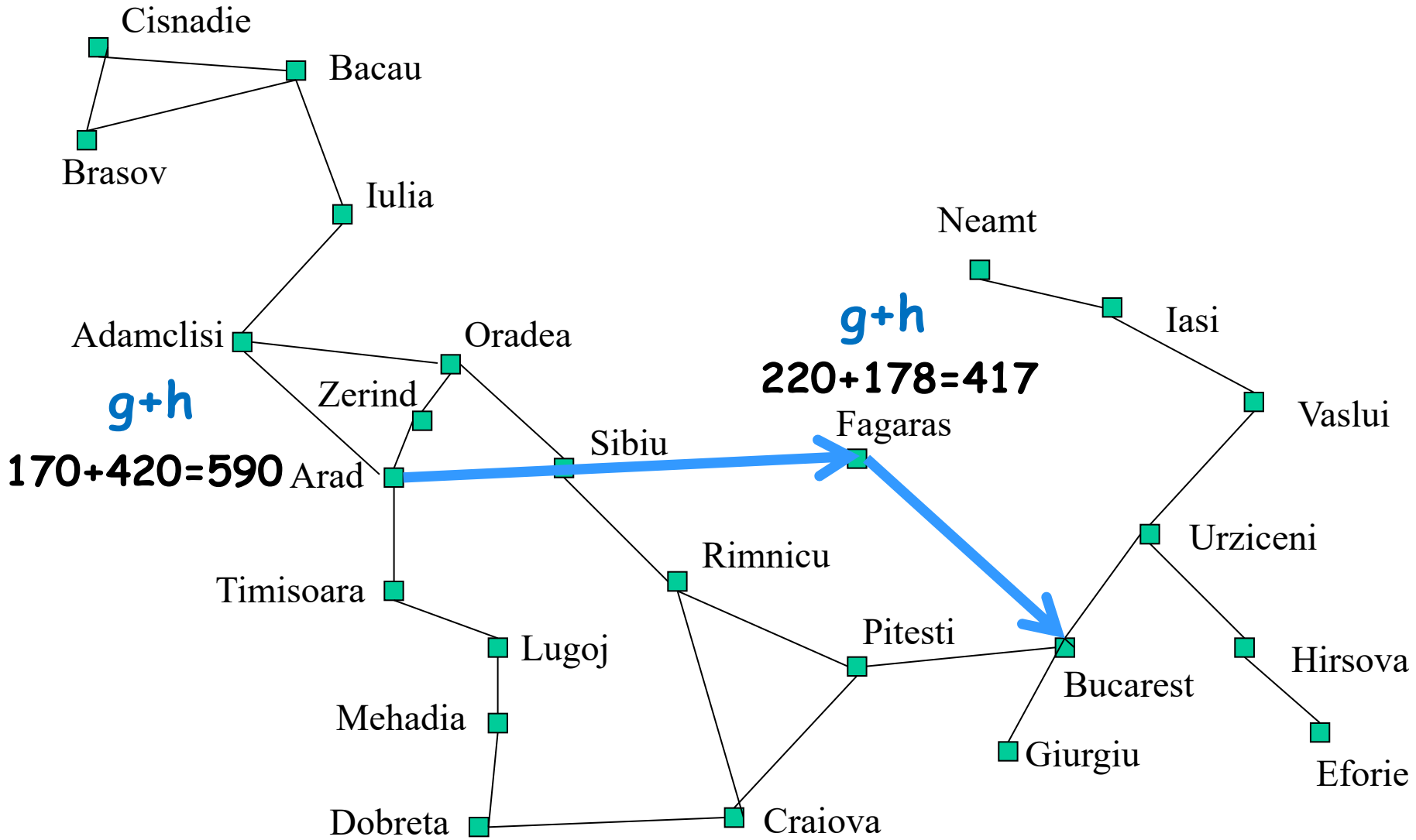
g =380.000, h =20.000



Búsqueda informada



Búsqueda informada



Búsqueda informada

Algoritmo A*

- Expandir el nodo con **menor f(n)**, esto es, aquel con el menor costo estimado para la solución

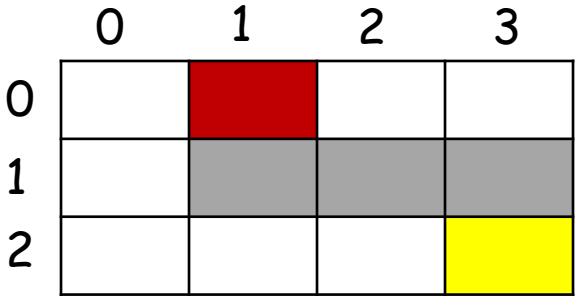
$$f(n) = g(n) + h(n)$$

Búsqueda informada

Aplicar A*

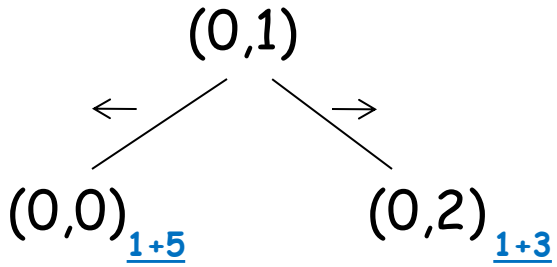
Agente Minero. Indique el camino que sigue el agente.

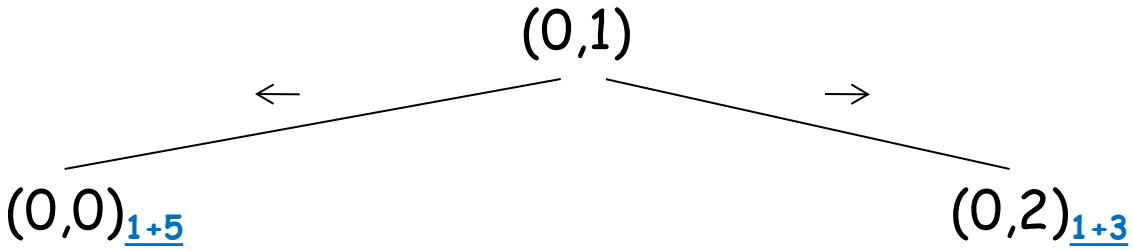
No evite devolverse

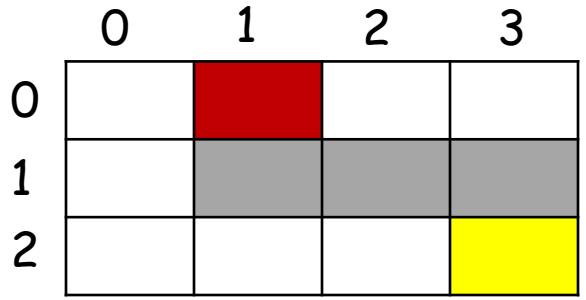


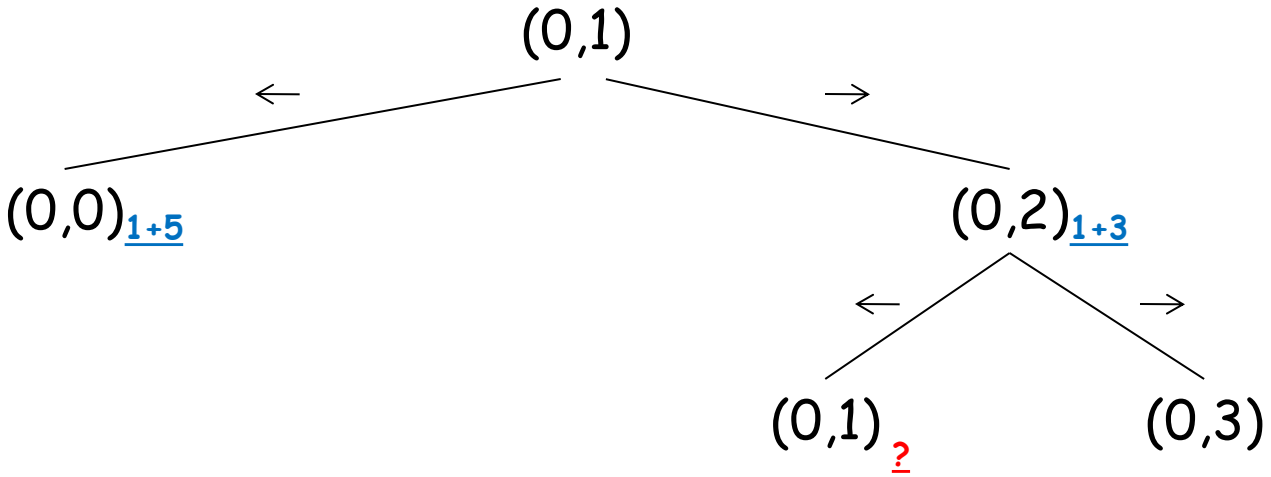
Punto de inicio

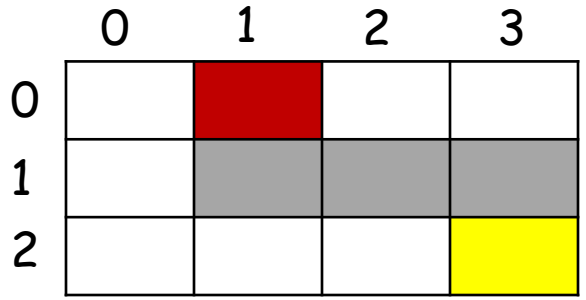
Lingote

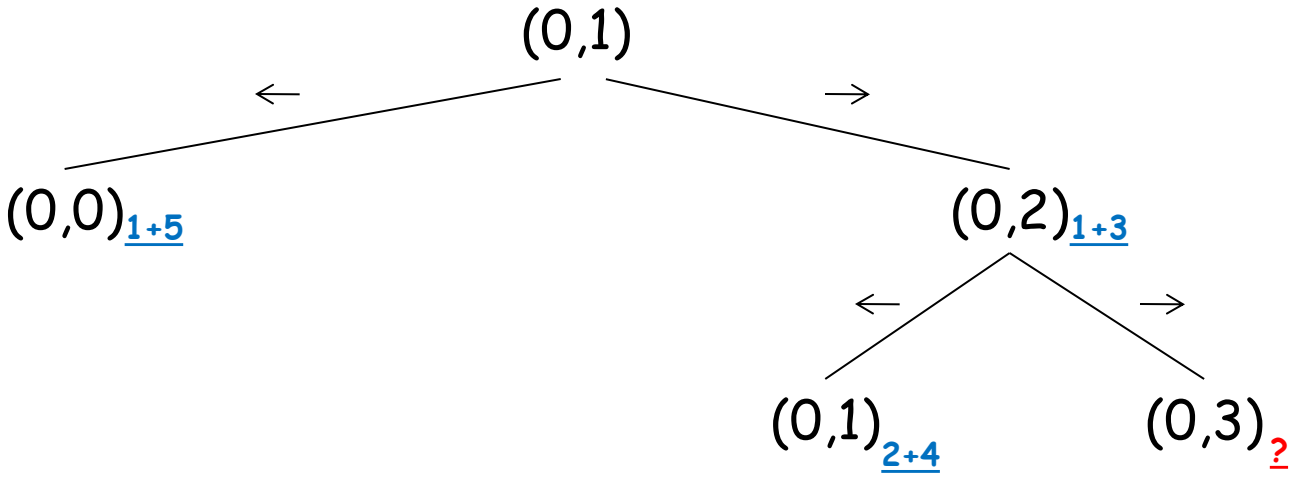


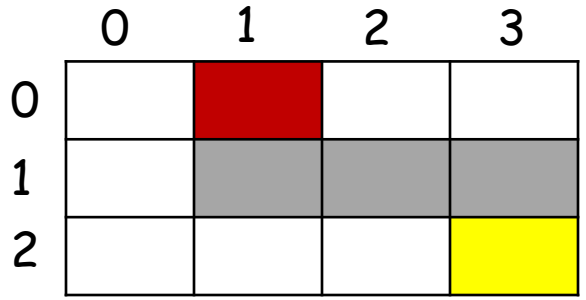

(0,1)
←
→
(0,0)1+5
(0,2)1+3

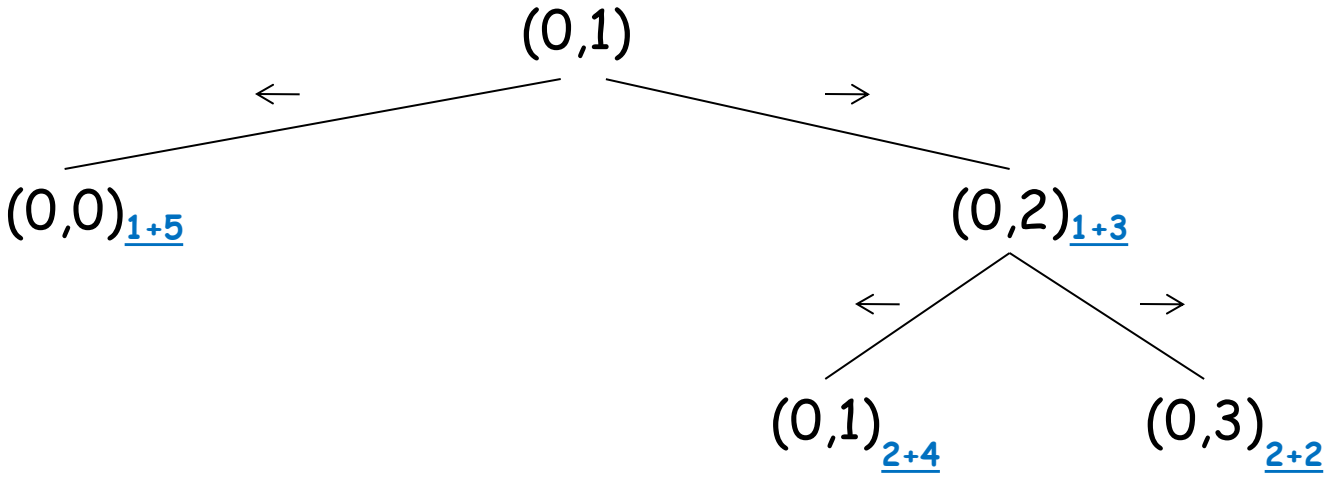

0
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1
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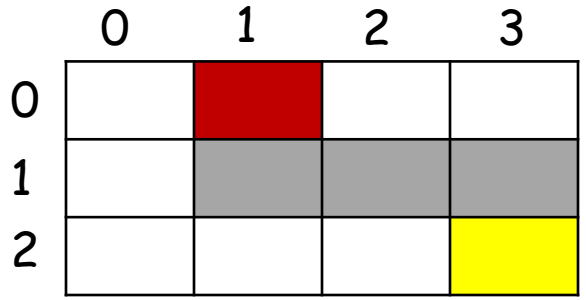

(0,1)
(0,0)1+5
(0,2)1+3
(0,1) ?
(0,3)

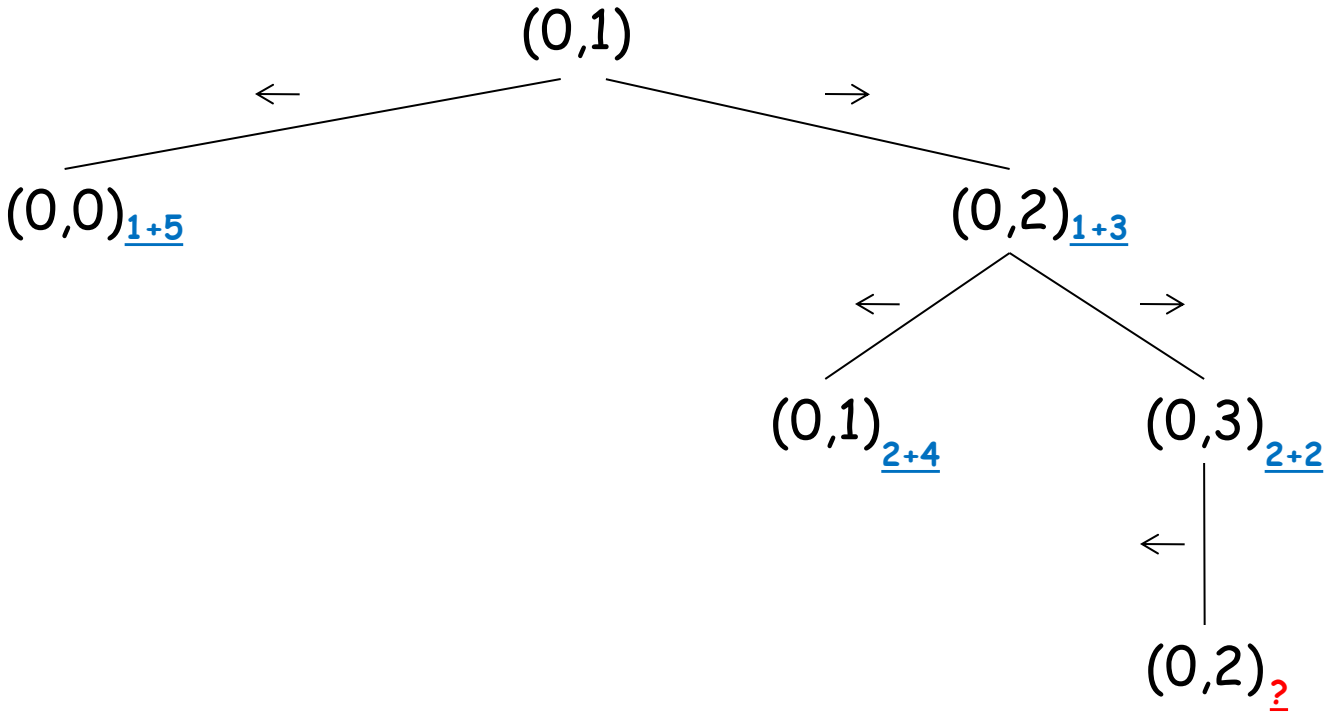

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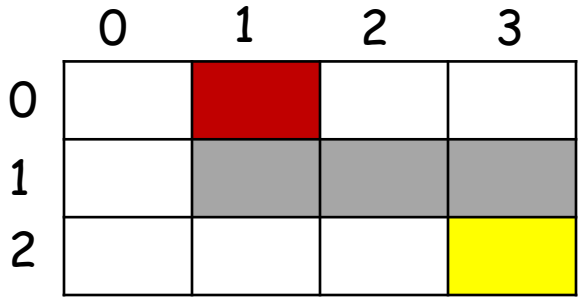

(0,1)
(0,0)1+5
(0,2)1+3
(0,1)2+4
(0,3)?

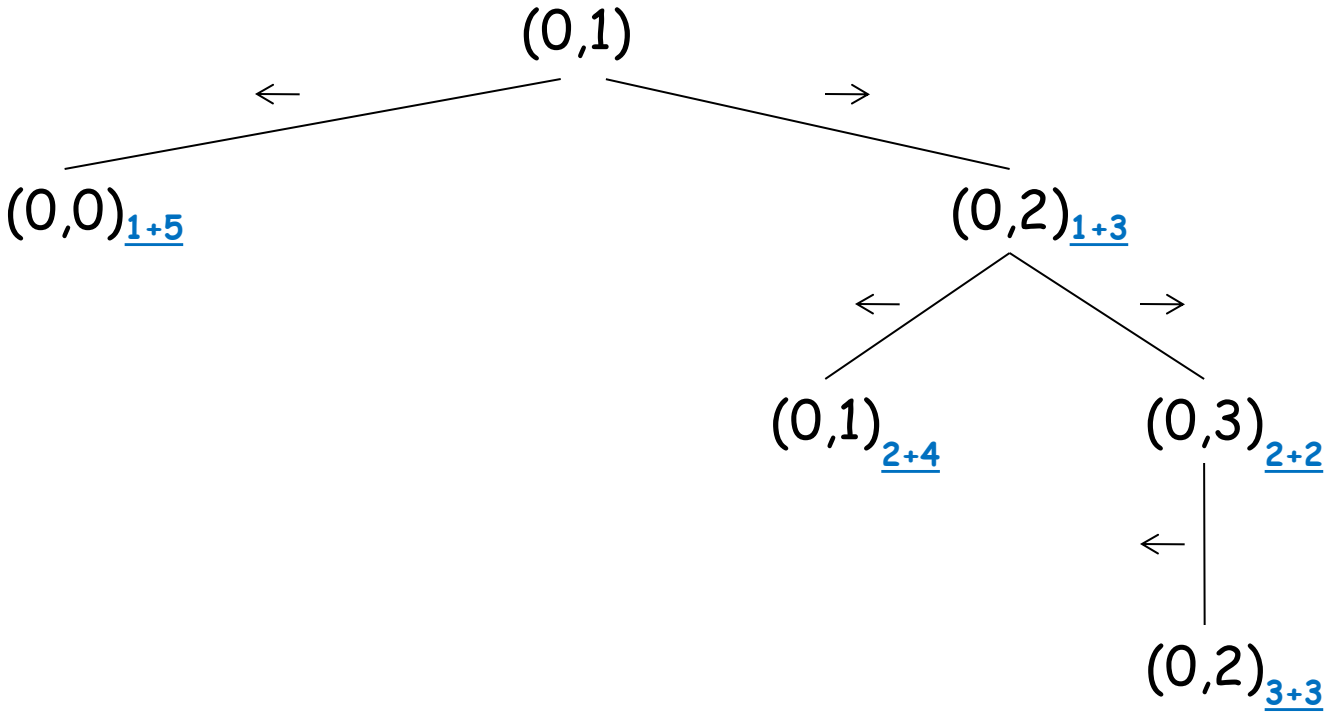

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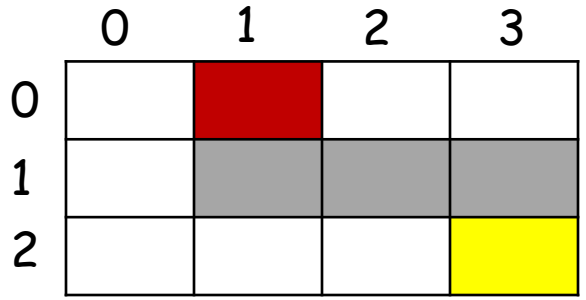

(0,1)
(0,0)1+5
(0,2)1+3
(0,1)2+4
(0,3)2+2

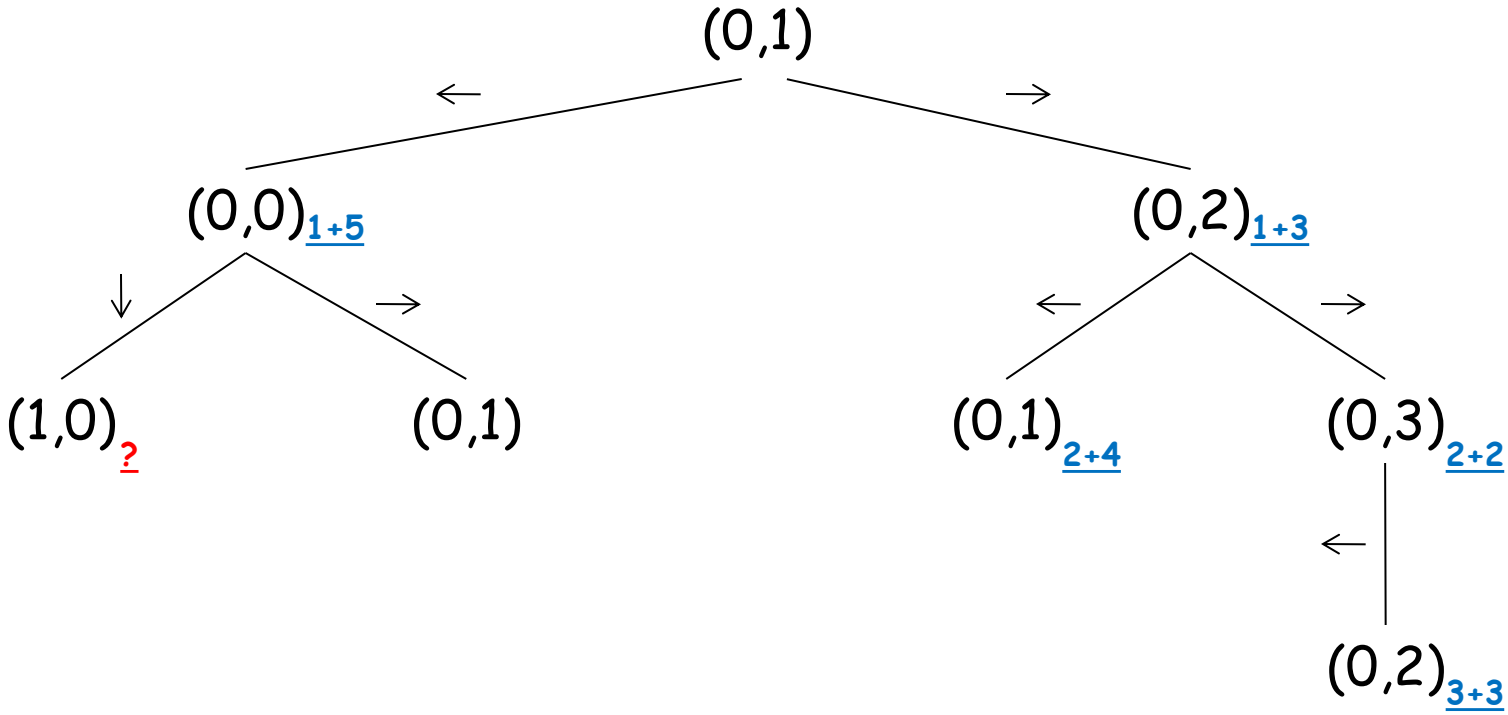

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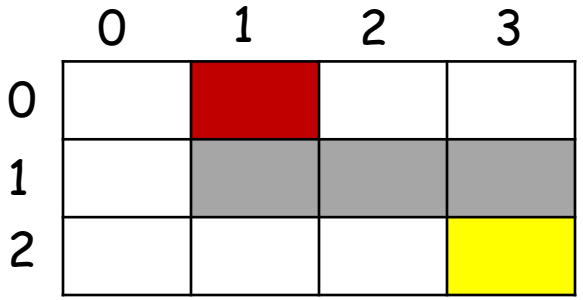

(0,1)
(0,0)1+5
(0,2)1+3
(0,1)2+4
(0,3)2+2
(0,2)?

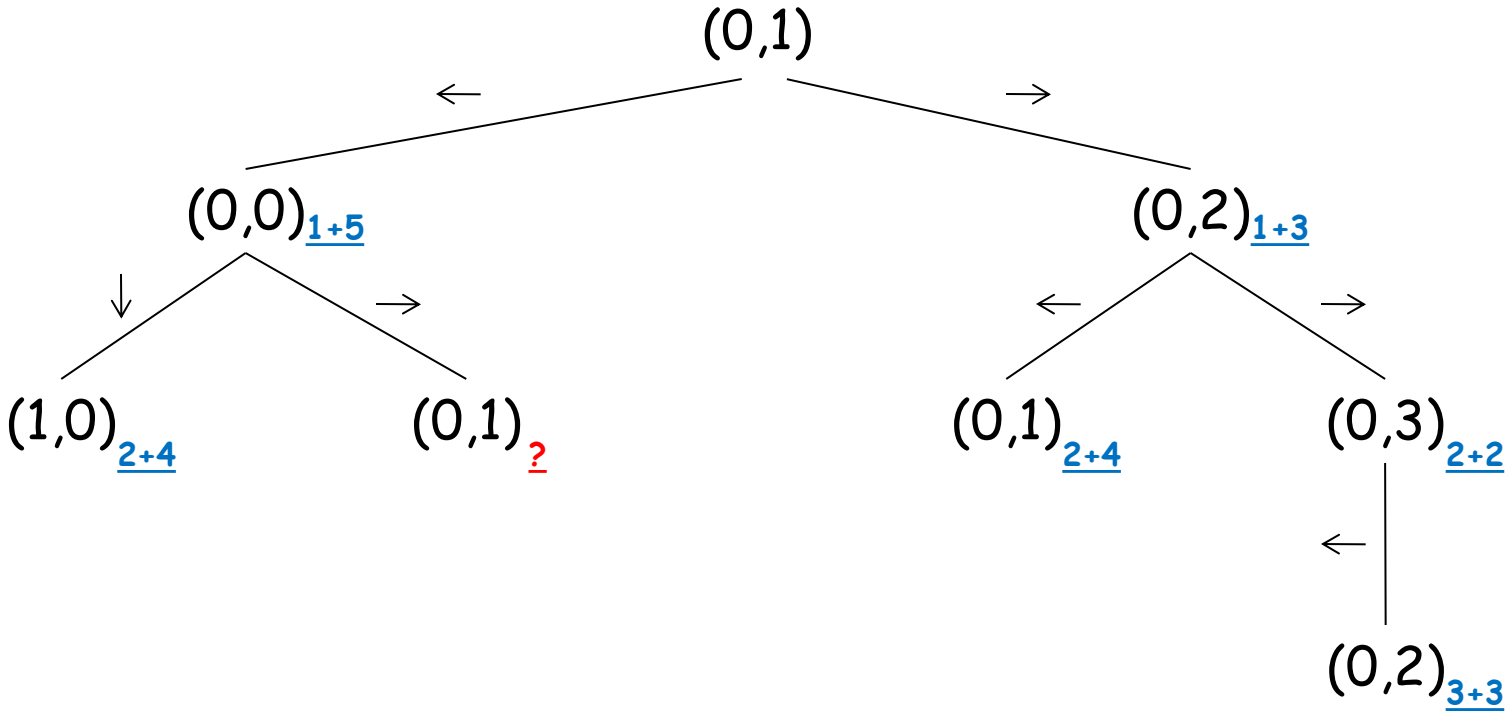

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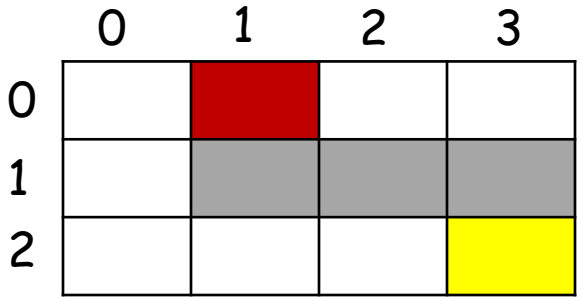

(0,1)
(0,0)1+5
(0,2)1+3
(0,1)2+4
(0,3)2+2
(0,2)3+3

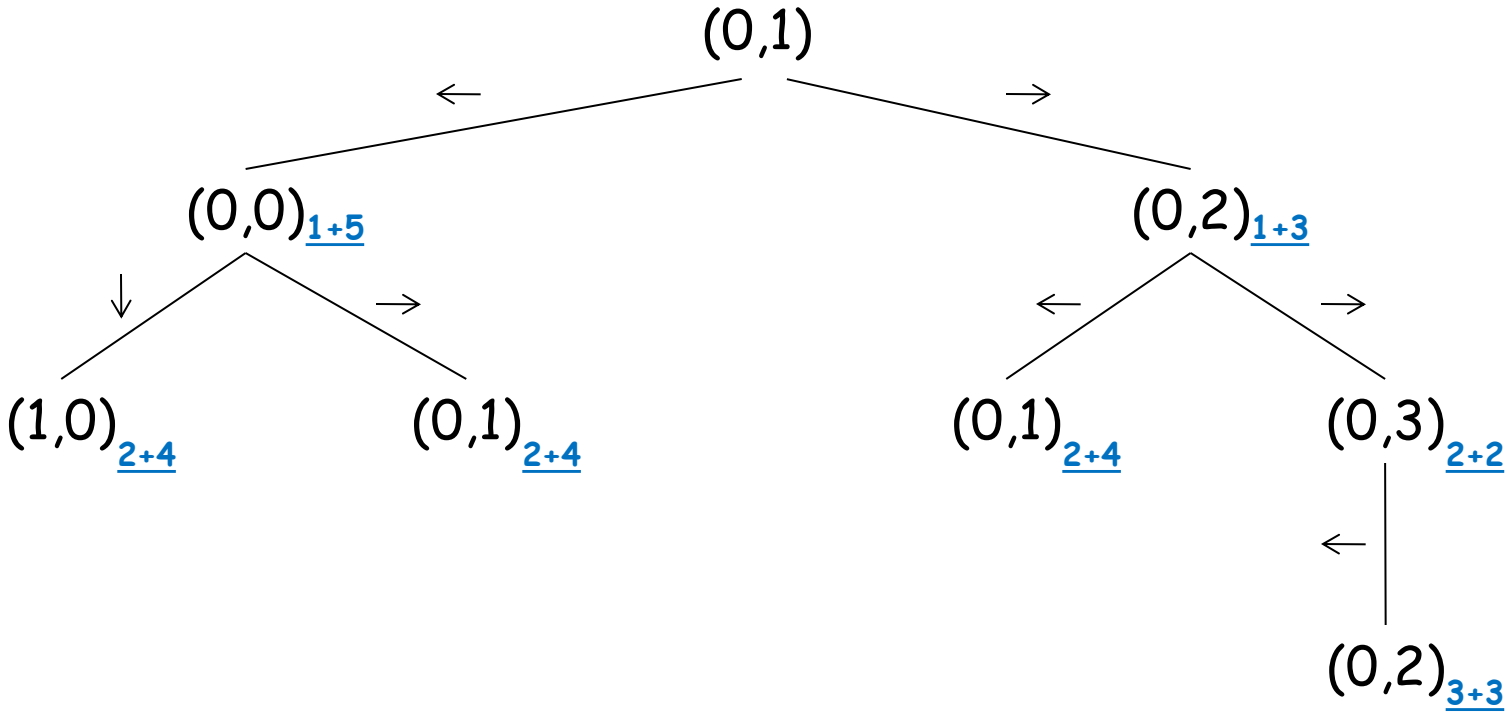

0
1
2
3
0
1
2

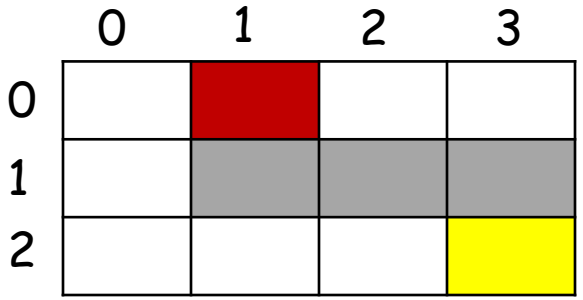

(0,1)
(0,0)1+5
(0,2)1+3
(1,0)?
(0,1)
(0,1)2+4
(0,3)2+2
(0,2)3+3

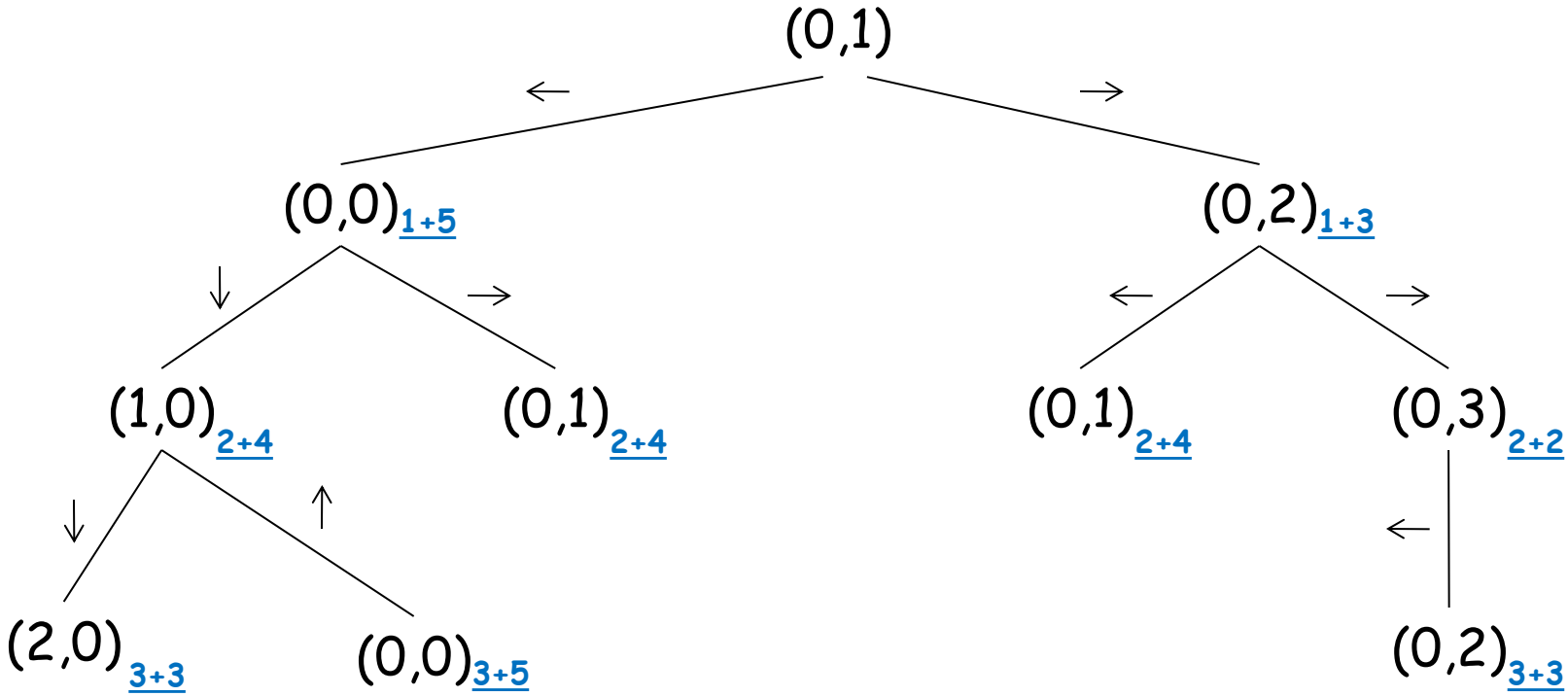

0
1
2
3
0
1
2

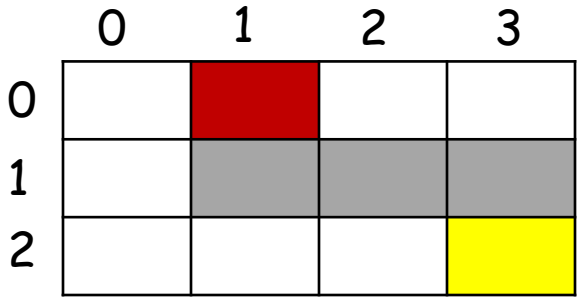

(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
(0,1) ?
(0,1) 2+4
(0,3) 2+2
(0,2) 3+3

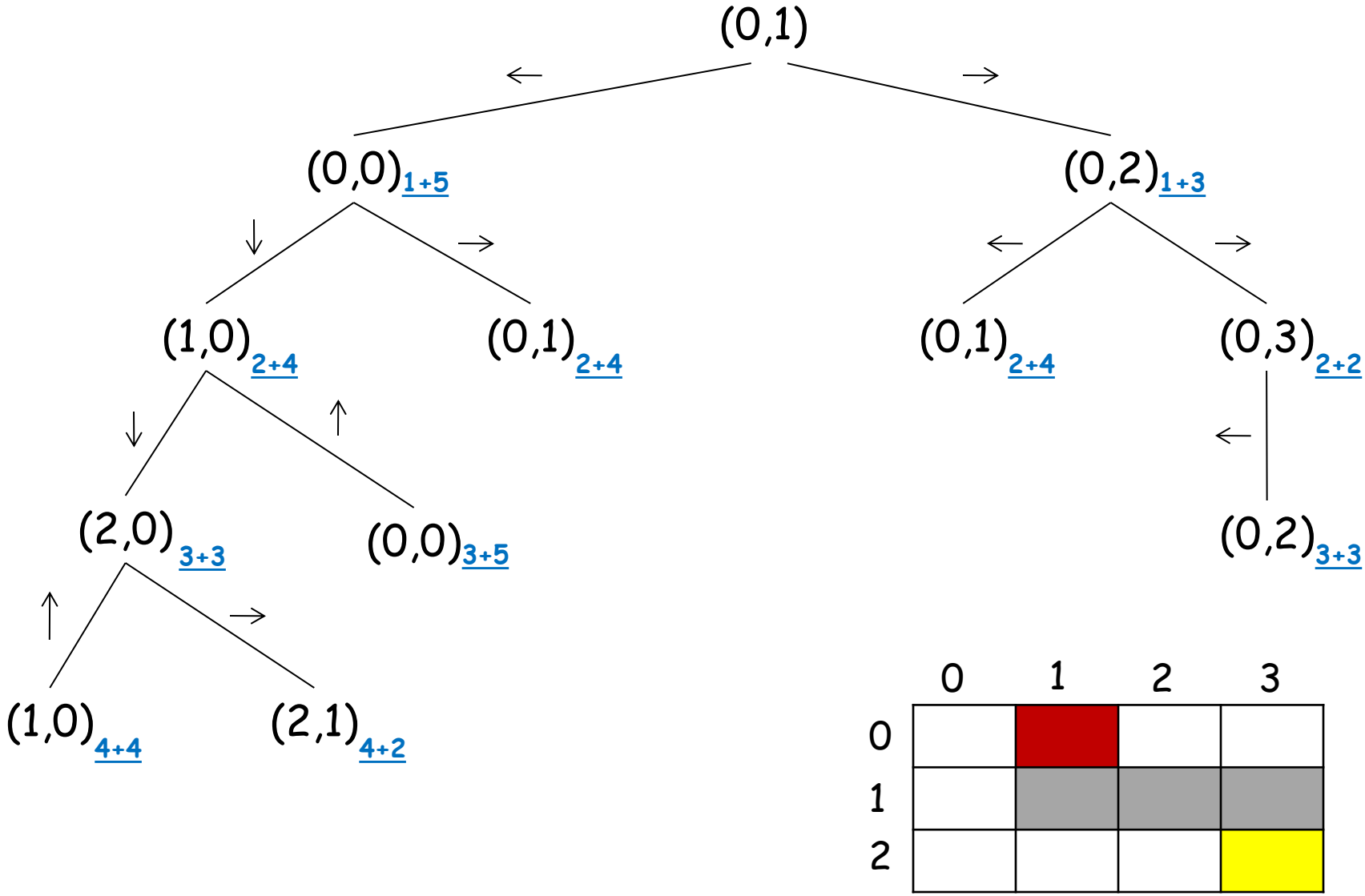

0
1
2
3
0
1
2


(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
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(0,3) 2+2
(0,2) 3+3

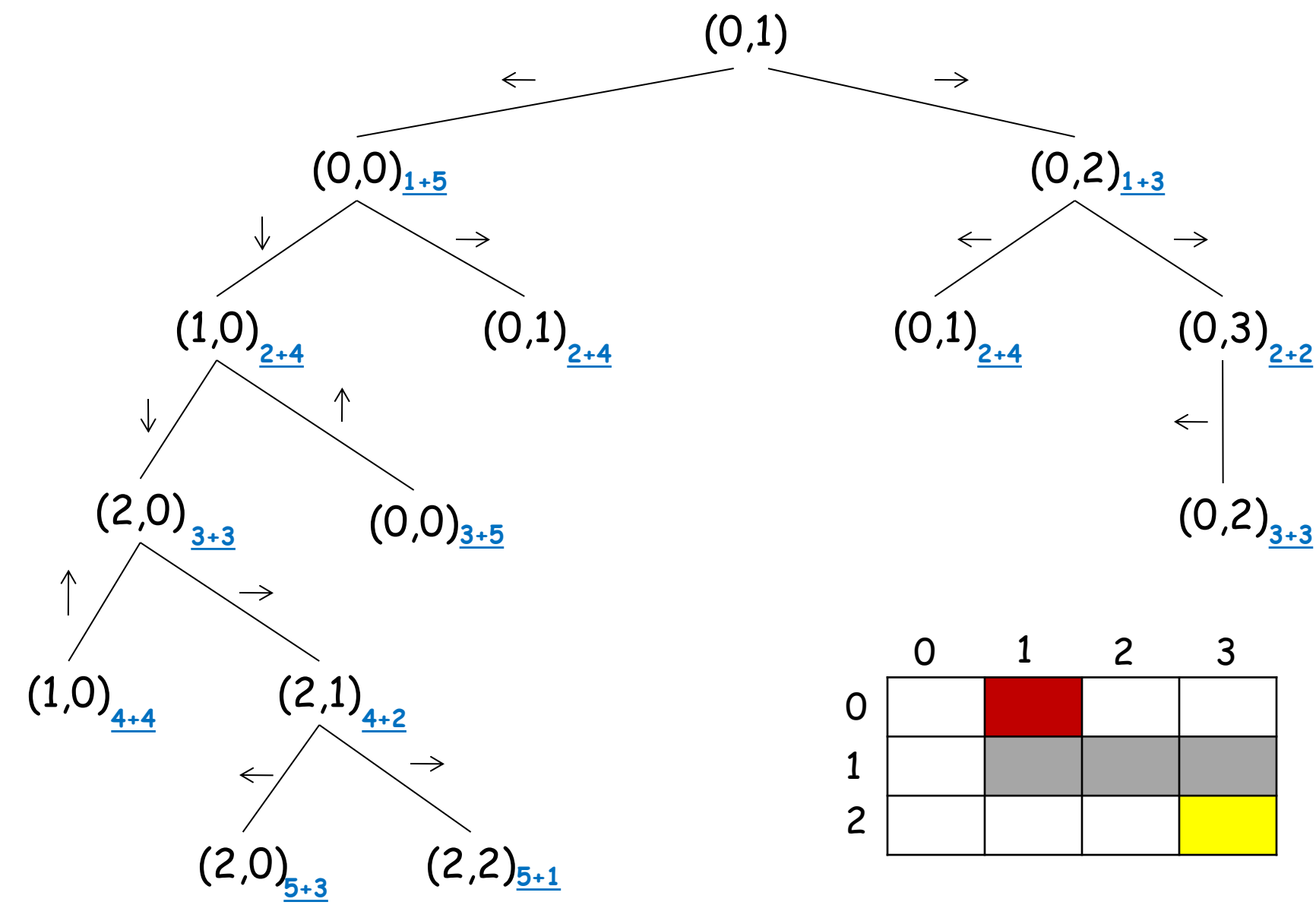

0
1
2
3
0
1
2


(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
(0,1) 2+4
(0,1) 2+4
(0,3) 2+2
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(0,0) 3+5
(0,2) 3+3

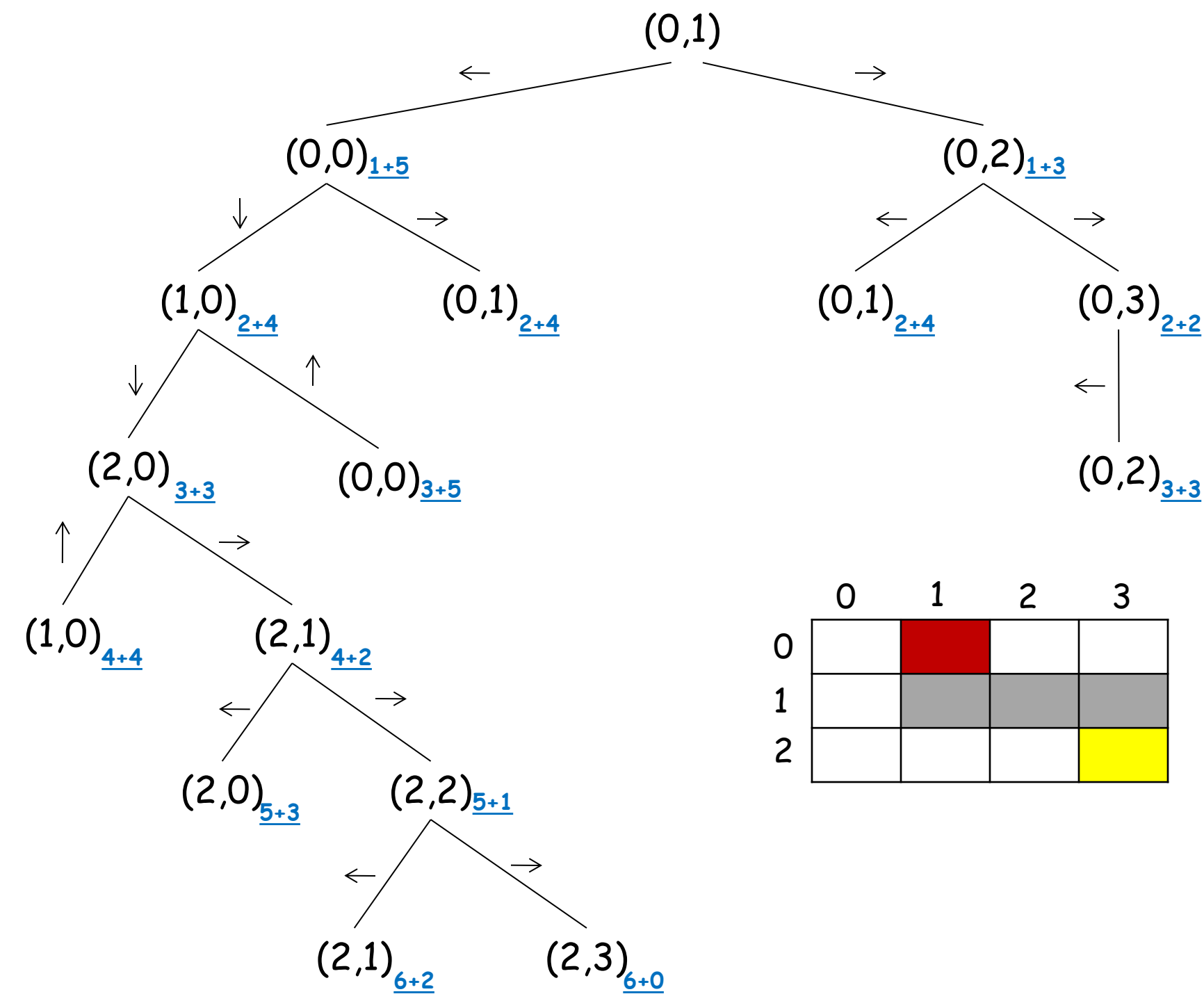

0
1
2
3
0
1
2

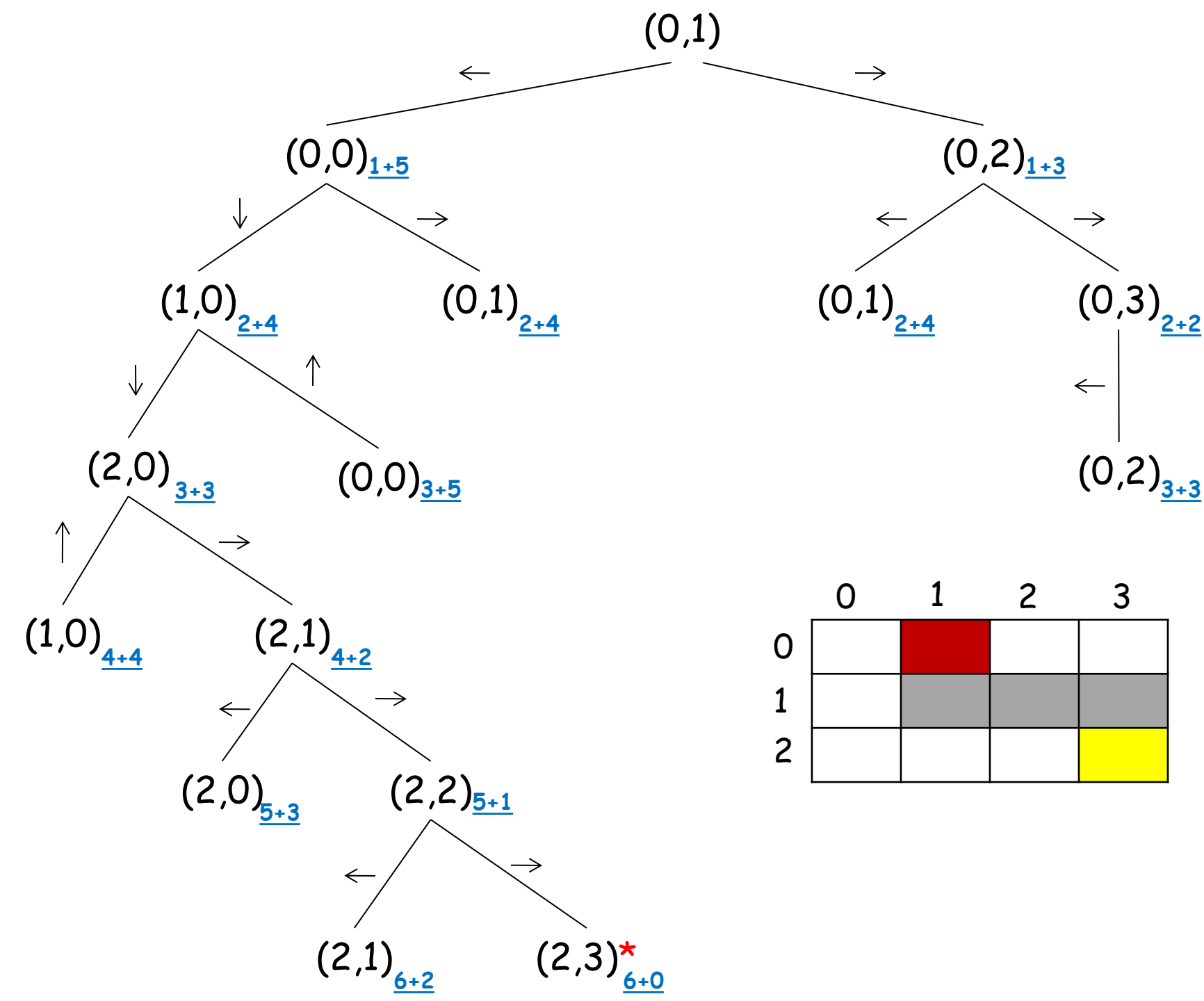

(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
(0,1) 2+4
(0,1) 2+4
(0,3) 2+2
(2,0) 3+3
(0,0) 3+5
(0,2) 3+3
(1,0) 4+4
(2,1) 4+2

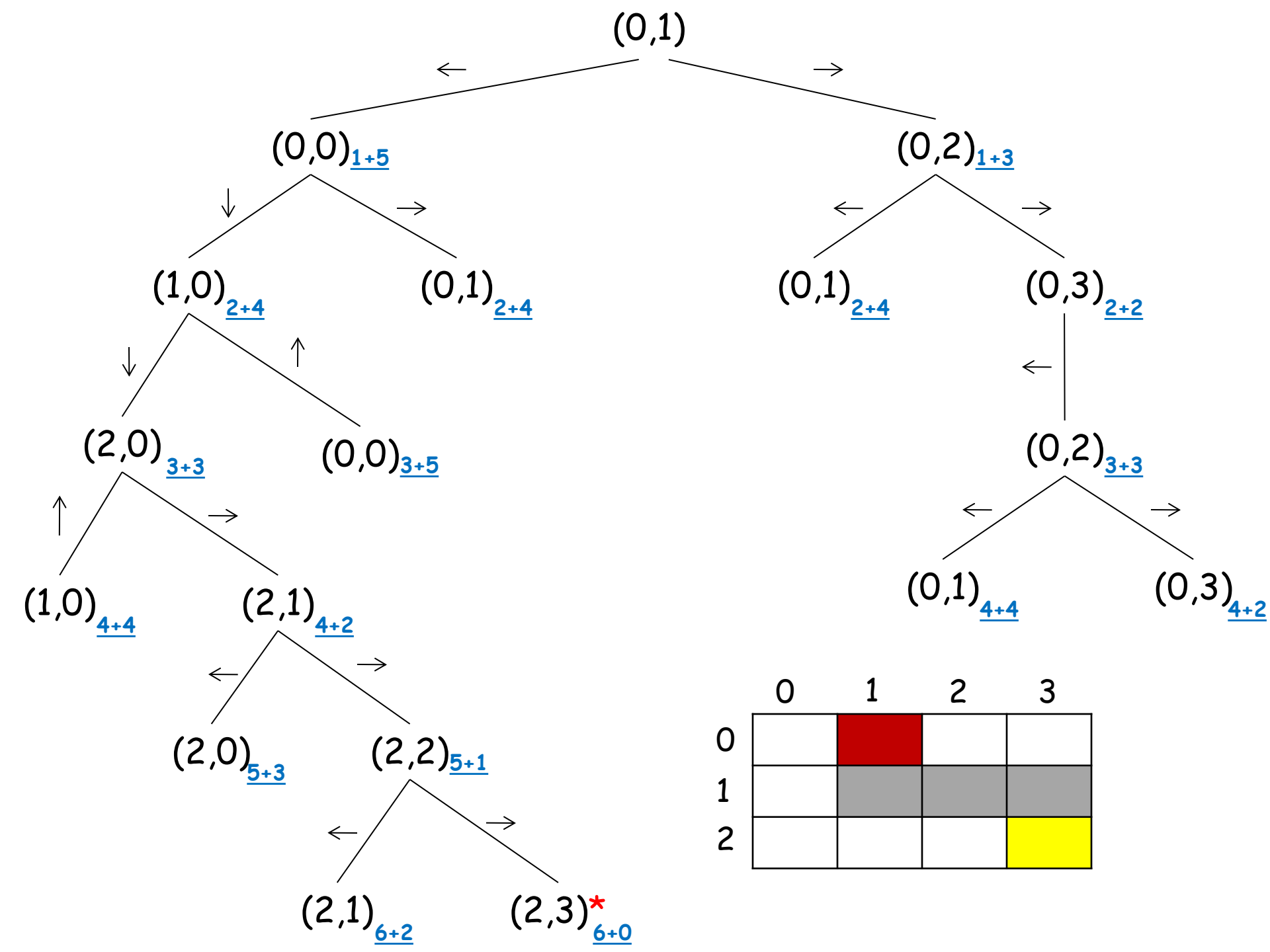
0
1
2
3
0
1
2

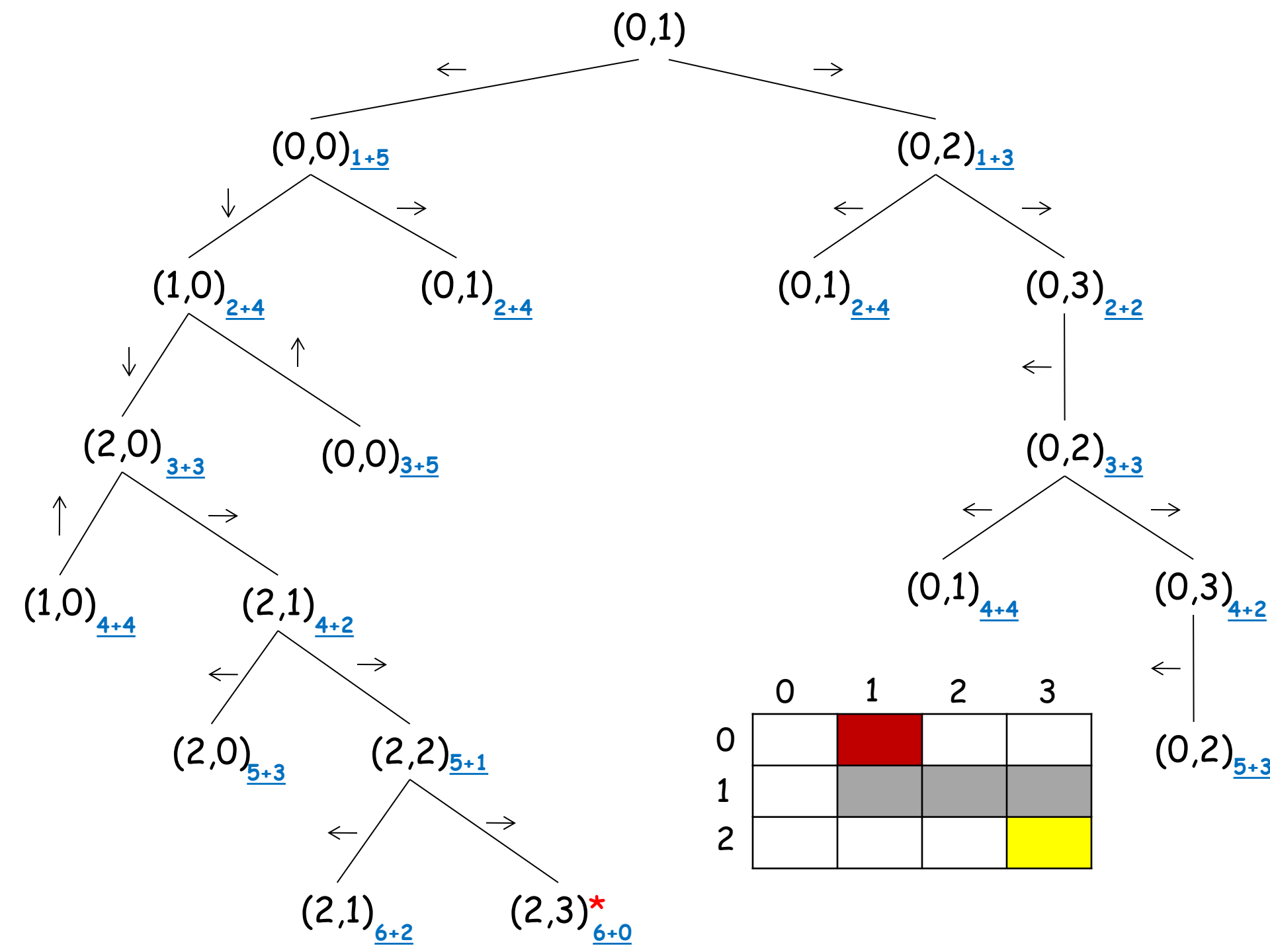

(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
(0,1) 2+4
(0,1) 2+4
(0,3) 2+2
(2,0) 3+3
(0,0) 3+5
(0,2) 3+3
(1,0) 4+4
(2,1) 4+2
(2,0) 5+3
(2,2) 5+1

0
1
2
3
0
1
2

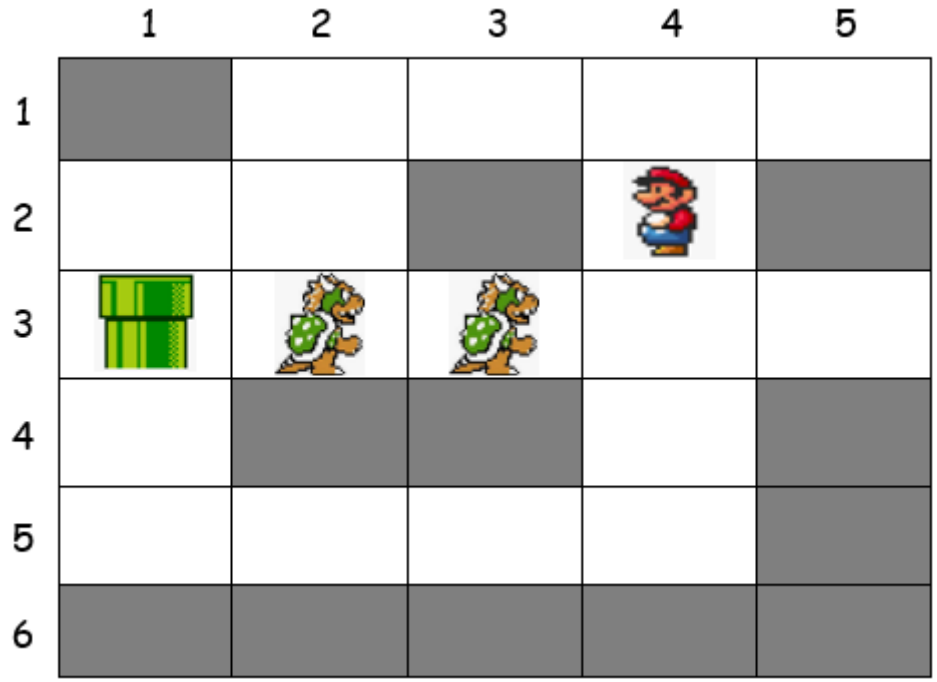

(0,1)
(0,0)1+5
(0,2)1+3
(1,0)2+4
(0,1)2+4
(0,1)2+4
(0,3)2+2
(2,0)3+3
(0,0)3+5
(0,2)3+3
(1,0)4+4
(2,1)4+2
(2,0)5+3
(2,2)5+1
(2,1)6+2
(2,3)6+0
0
1
2
3
0
1
2


(0,1)
(0,0)1+5
(0,2)1+3
(1,0)2+4
(0,1)2+4
(0,1)2+4
(0,3)2+2
(2,0)3+3
(0,0)3+5
(0,2)3+3
(1,0)4+4
(2,1)4+2
(2,0)5+3
(2,2)5+1
(2,1)6+2
(2,3)*6+0
0
1
2
3
0
1
2


(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
(0,1) 2+4
(0,1) 2+4
(0,3) 2+2
(2,0) 3+3
(0,0) 3+5
(0,2) 3+3
(1,0) 4+4
(2,1) 4+2
(0,1) 4+4
(0,3) 4+2
(2,0) 5+3
(2,2) 5+1
(2,1) 6+2
(2,3)* 6+0
0 1 2 3
0
1
2


(0,1)
(0,0) 1+5
(0,2) 1+3
(1,0) 2+4
(0,1) 2+4
(0,1) 2+4
(0,3) 2+2
(2,0) 3+3
(0,0) 3+5
(0,2) 3+3
(1,0) 4+4
(2,1) 4+2
(0,1) 4+4
(0,3) 4+2
(2,0) 5+3
(2,2) 5+1
(0,2) 5+3
(2,1) 6+2
(2,3)* 6+0
0 1 2 3
0
1
2

Búsqueda informada

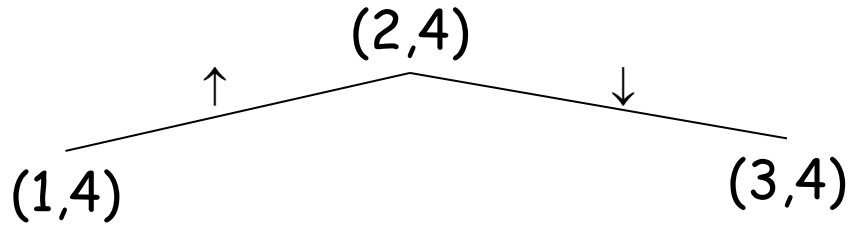


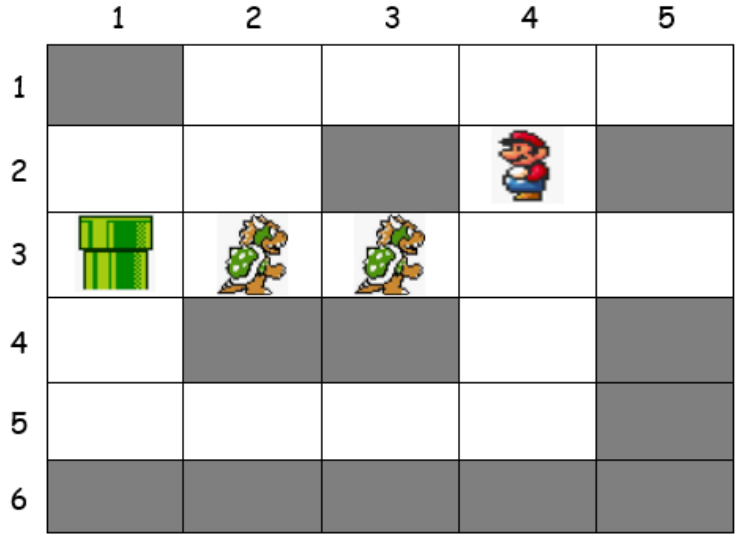
h(n): distancia de Manhattan a la posición (3,1)

Cada movimiento cuesta 1, llegar a una casilla con koopa cuesta 5 (allí está incluido el costo del movimiento)

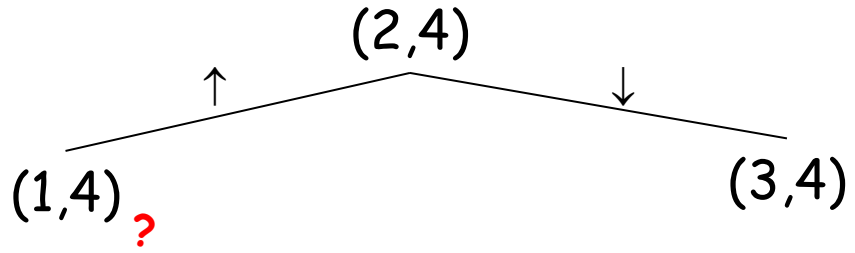
Evite devolverse

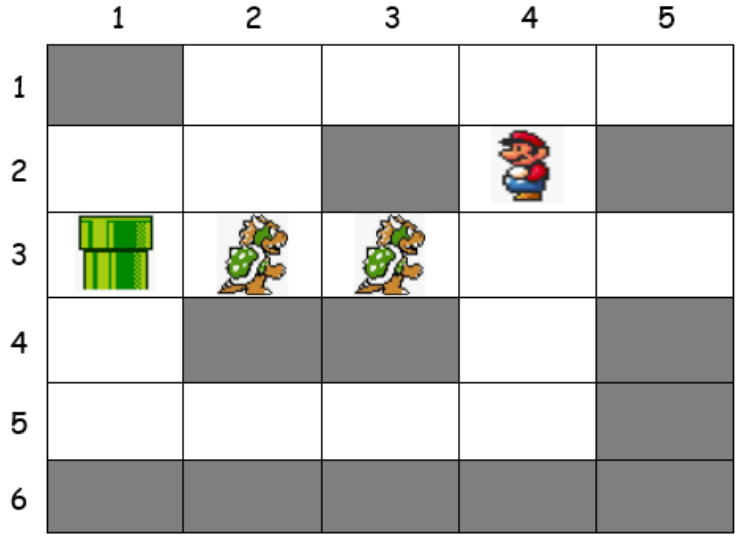
Búsqueda informada



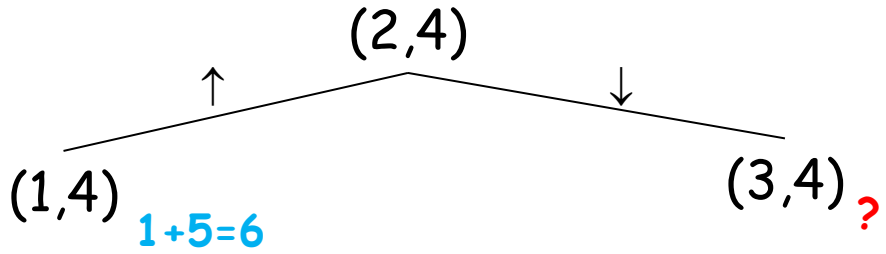


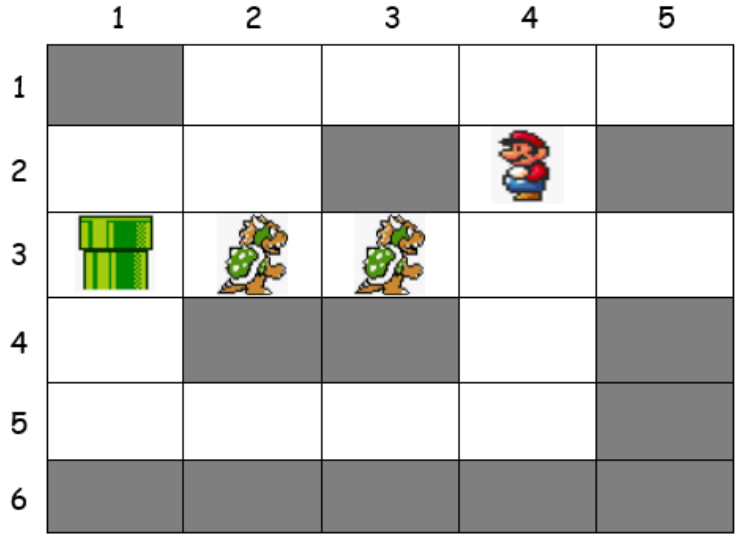
Búsqueda informada



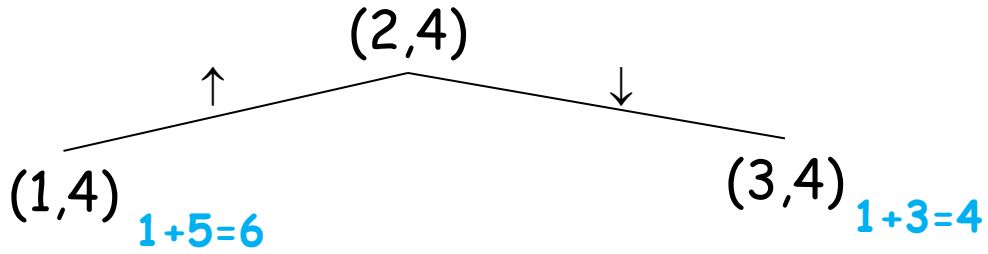


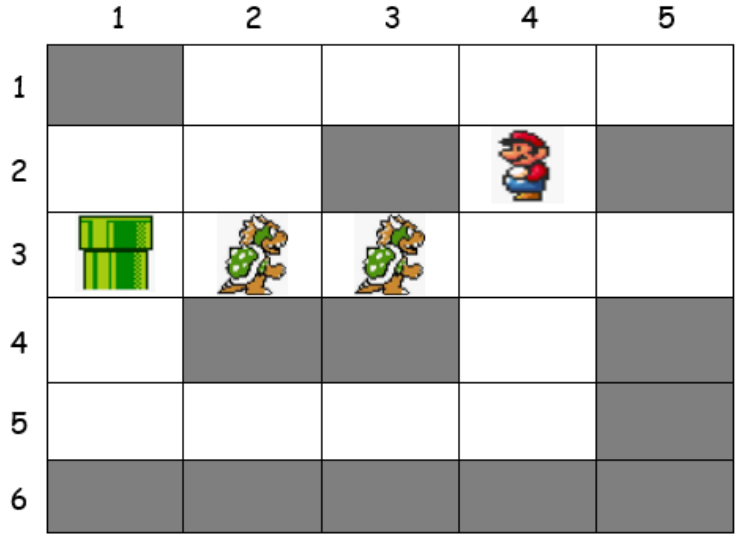
Búsqueda informada



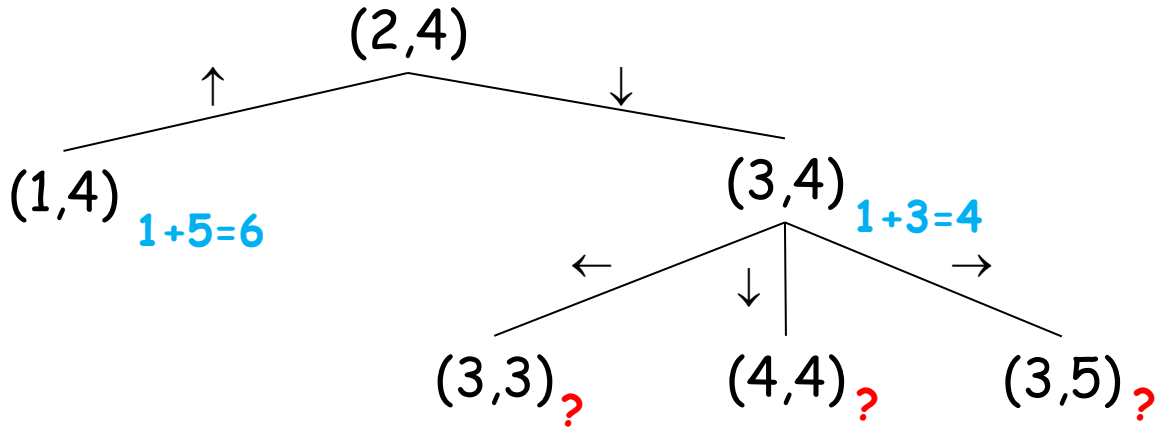


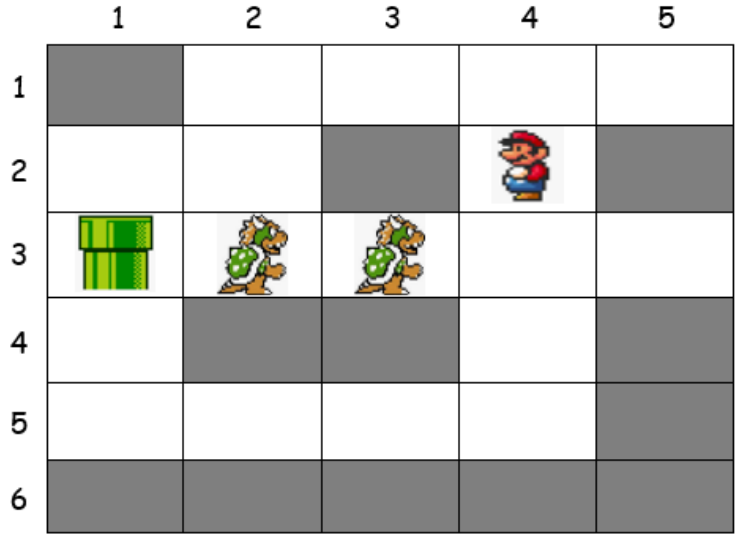
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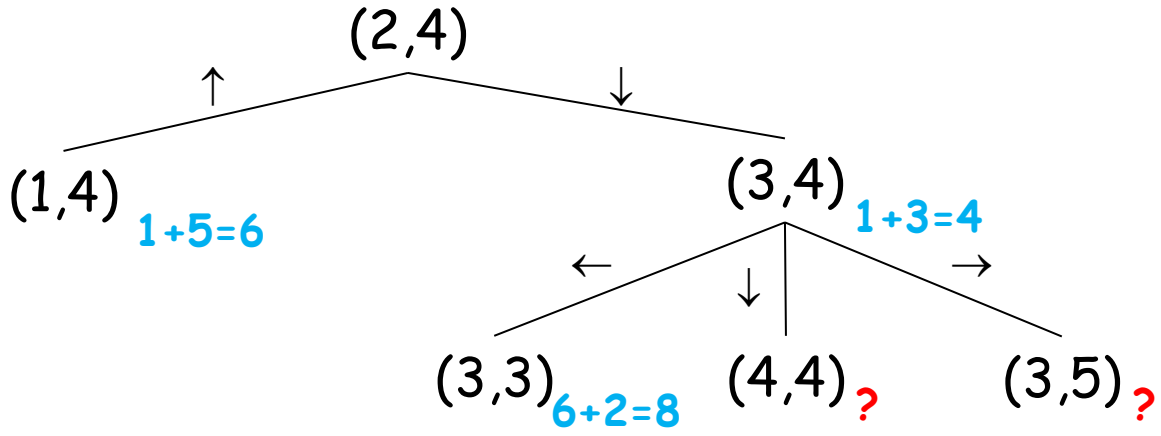


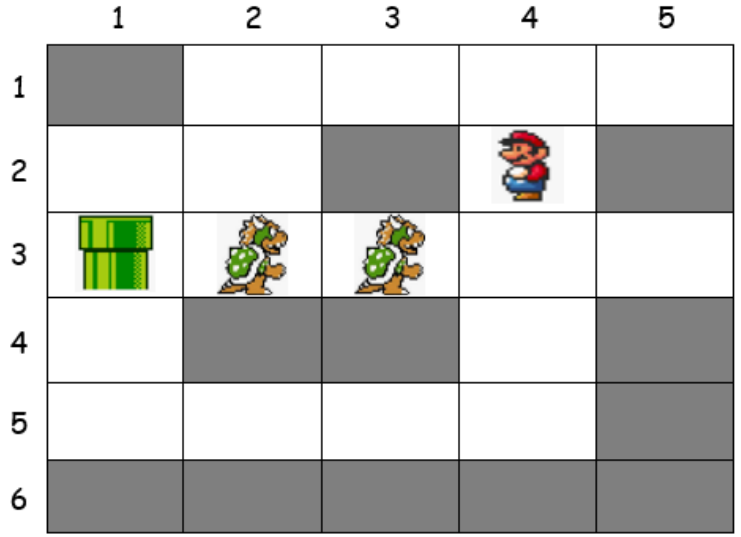
Búsqueda informada



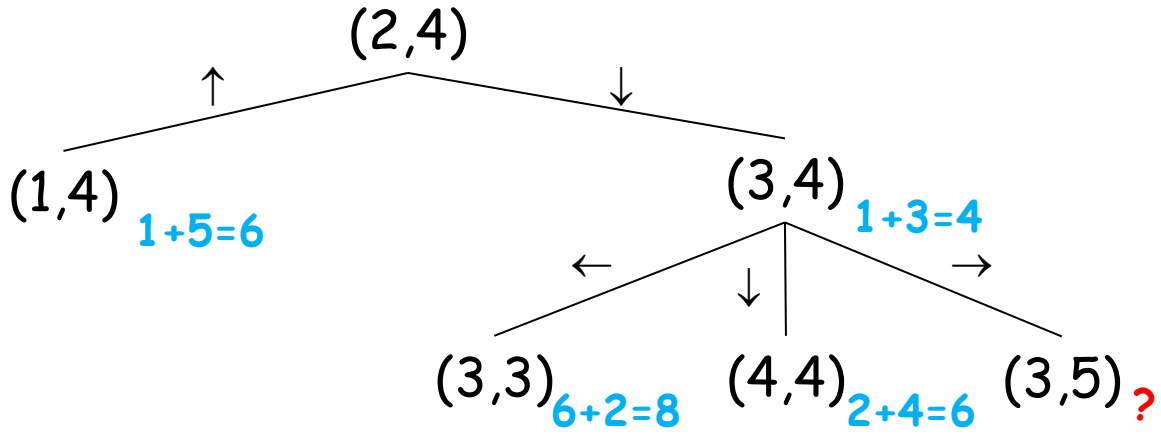


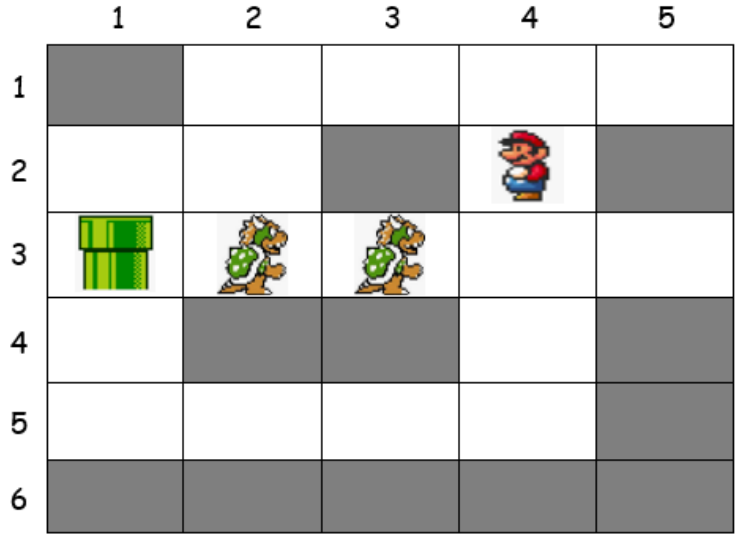
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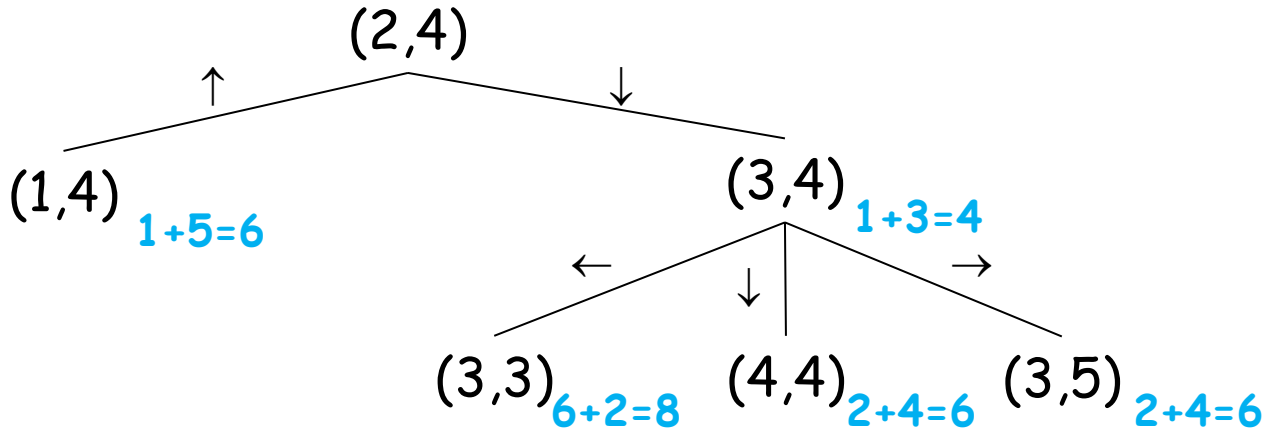


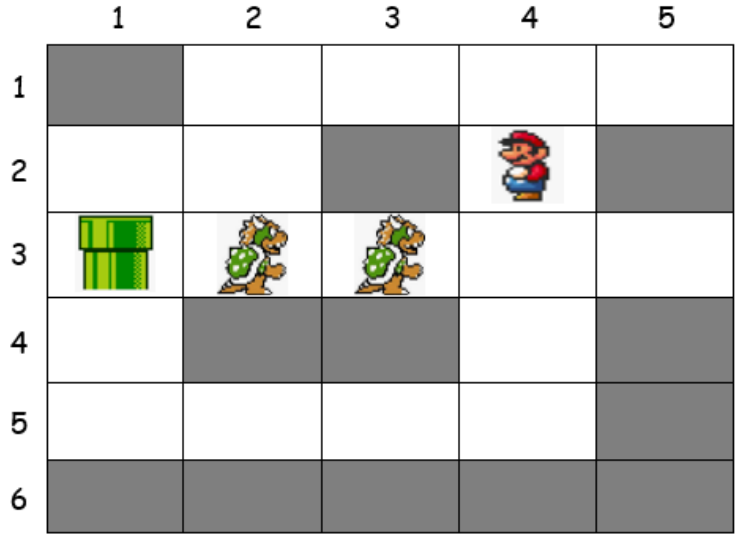
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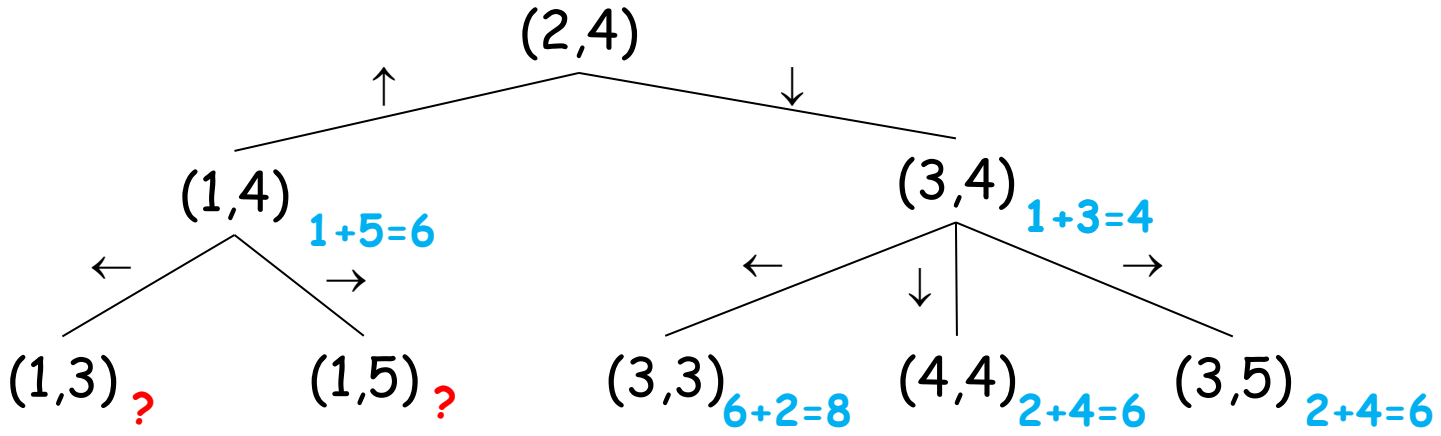


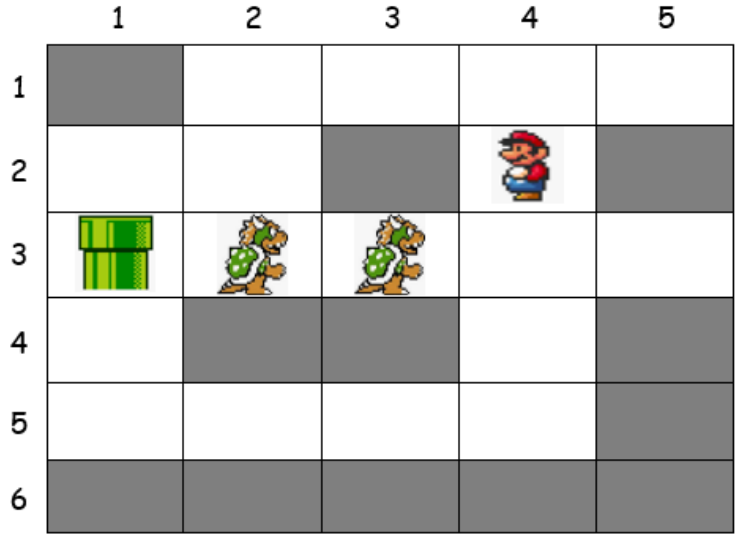
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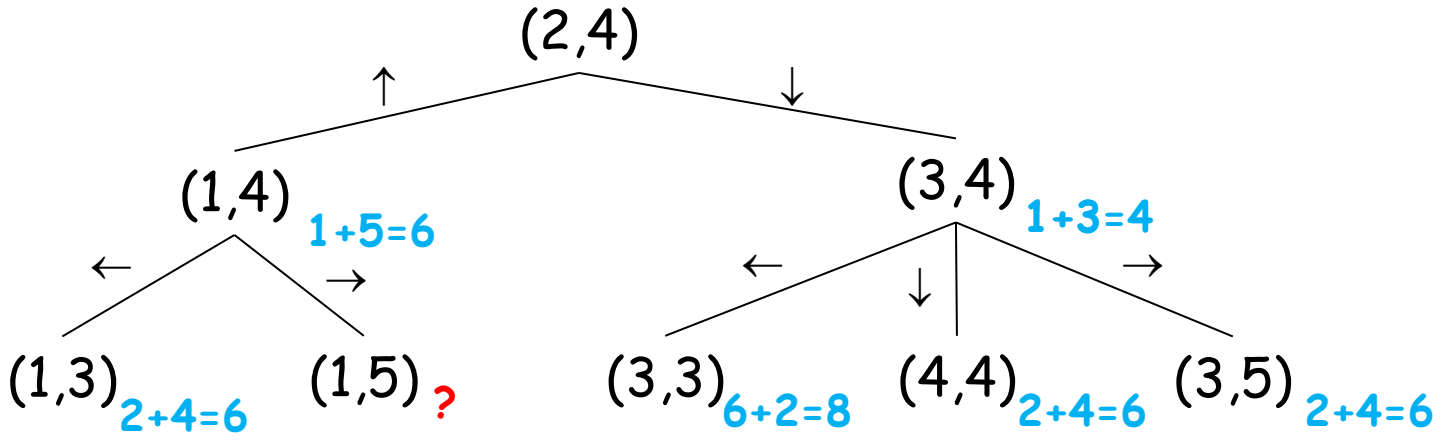


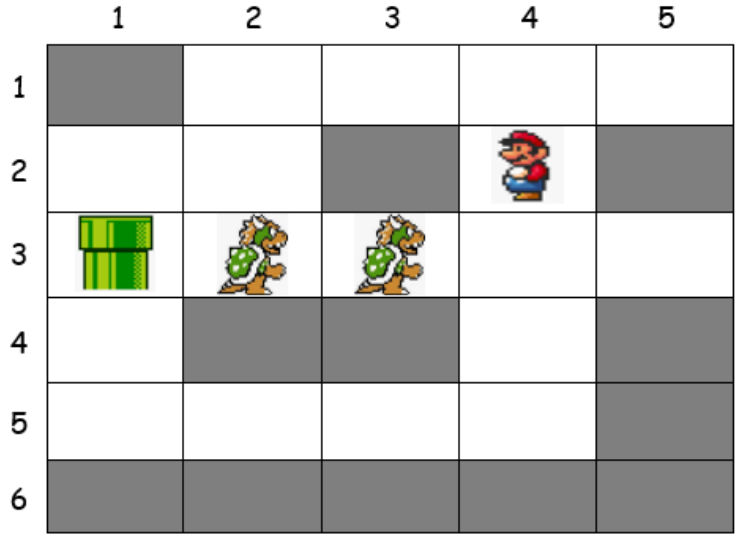
Búsqueda informada



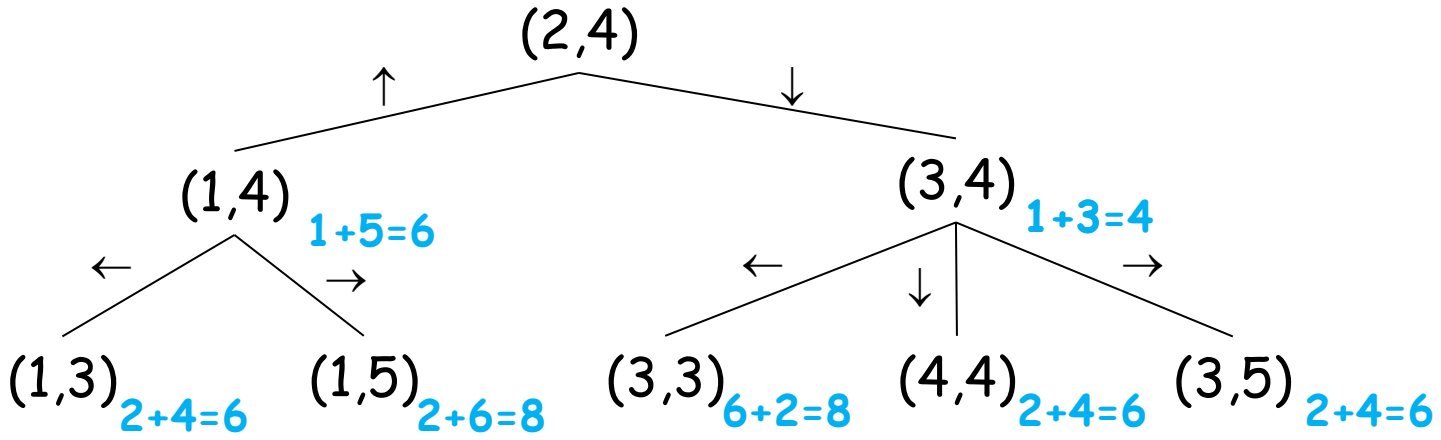


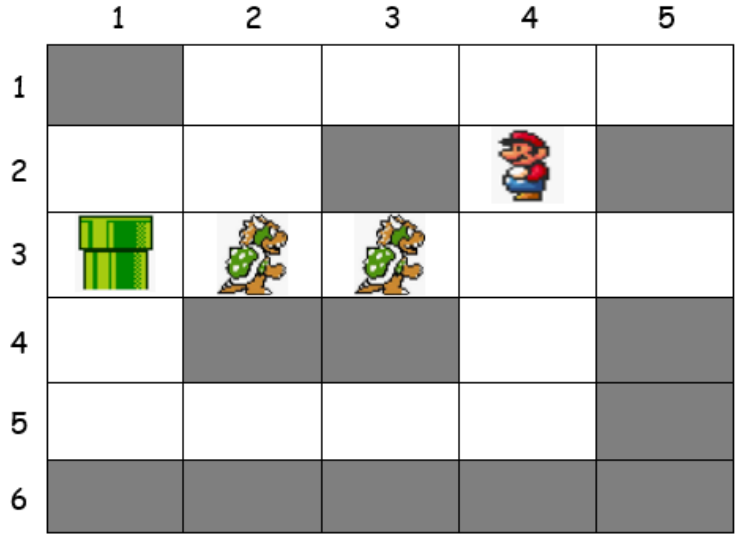
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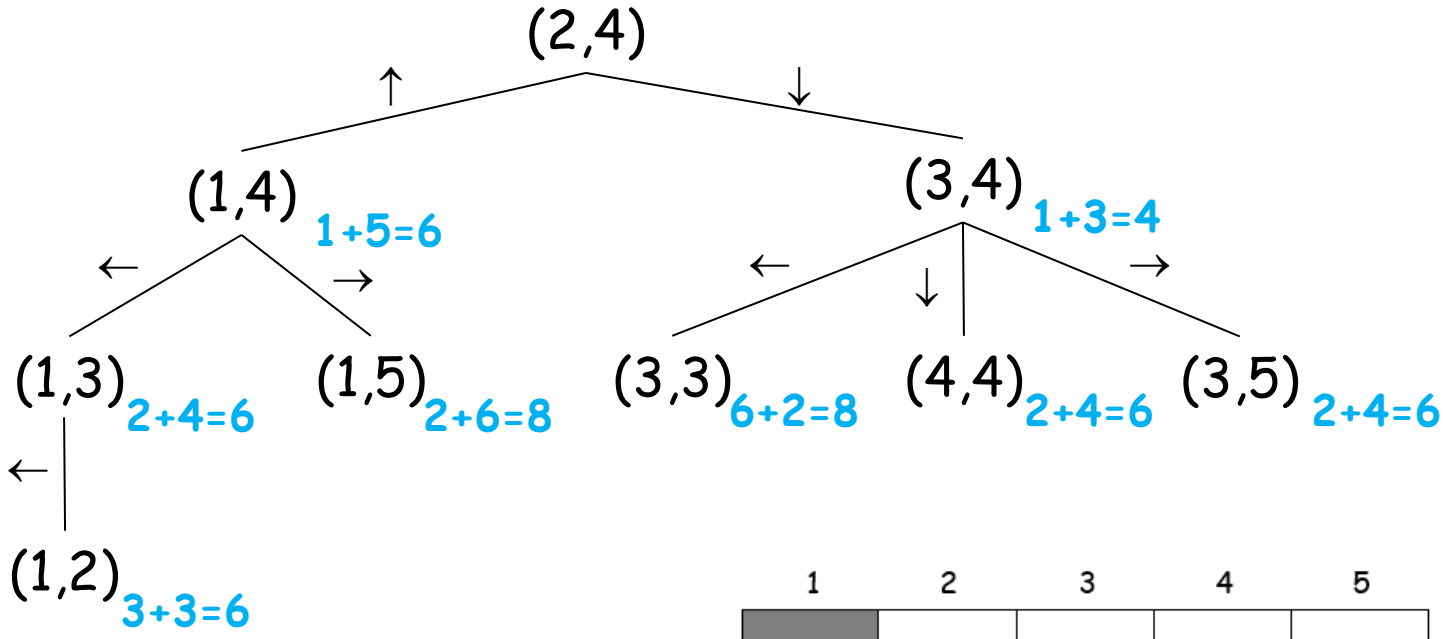


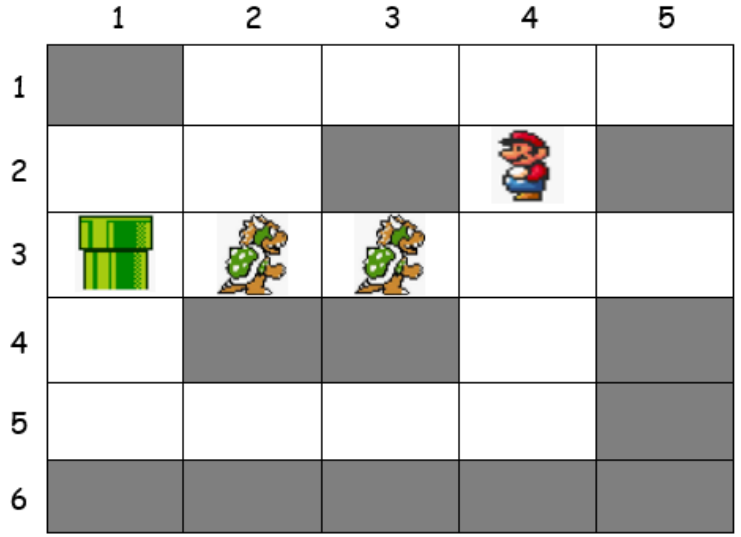
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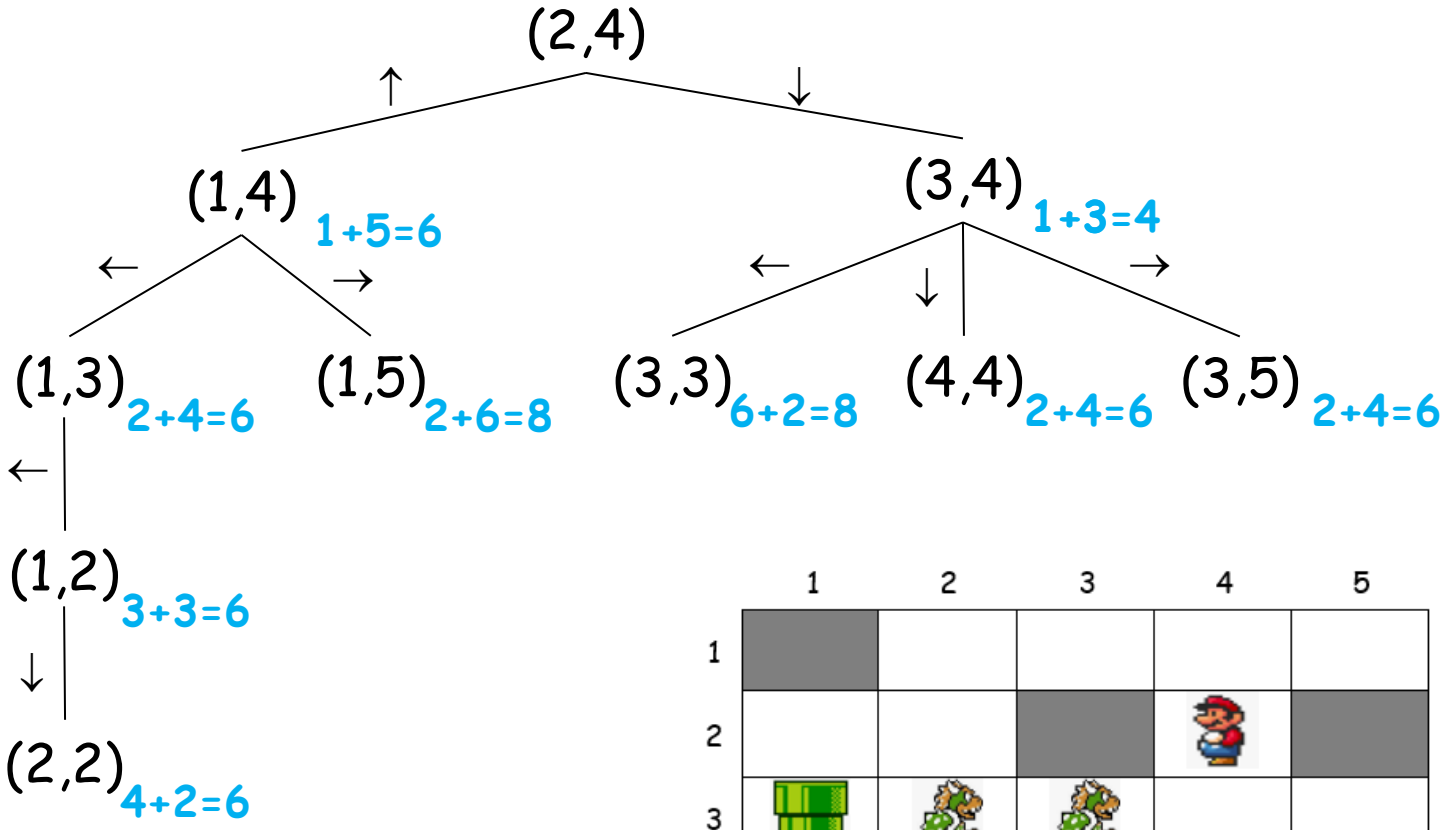


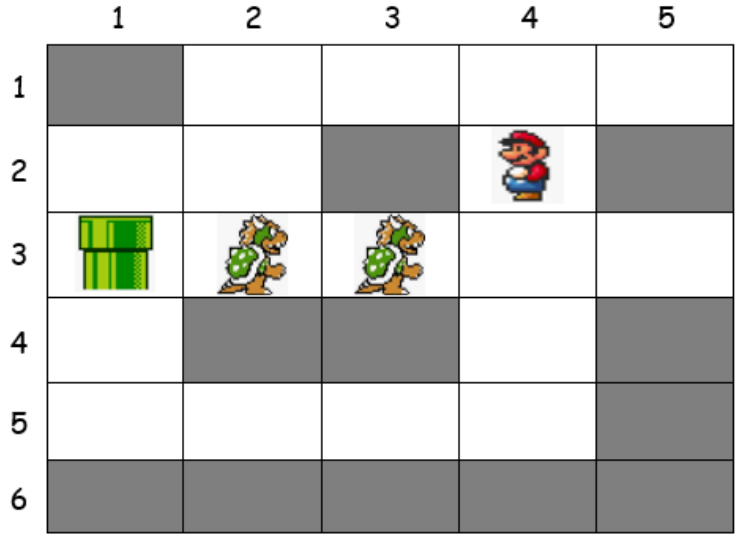
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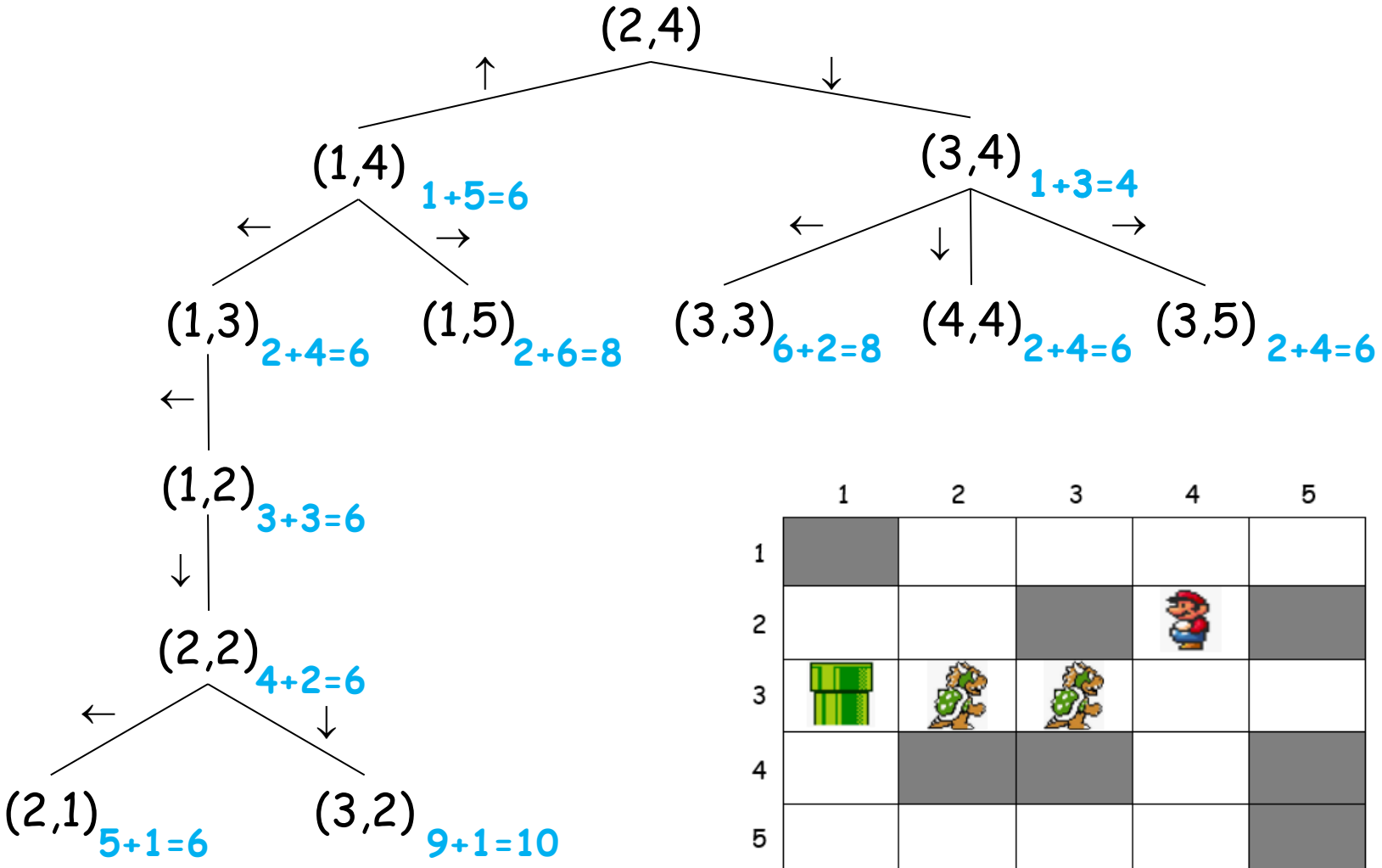


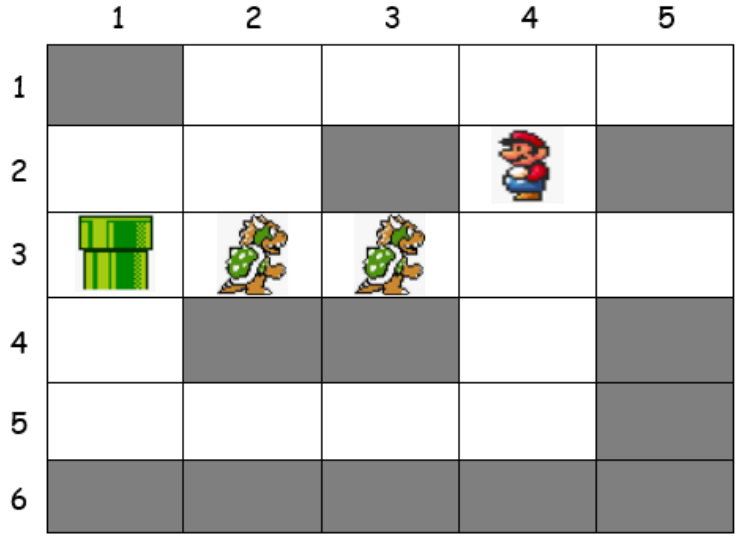
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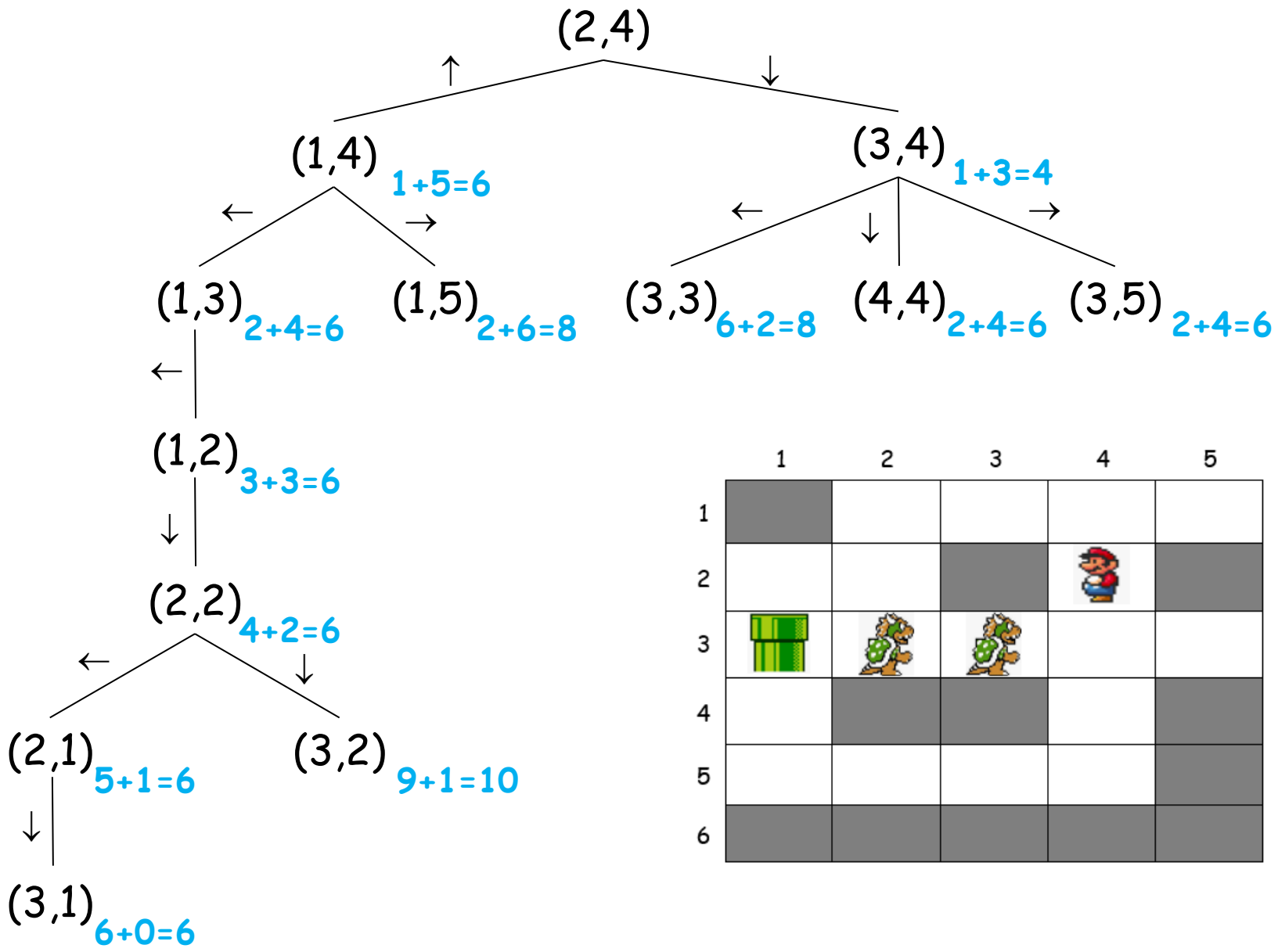


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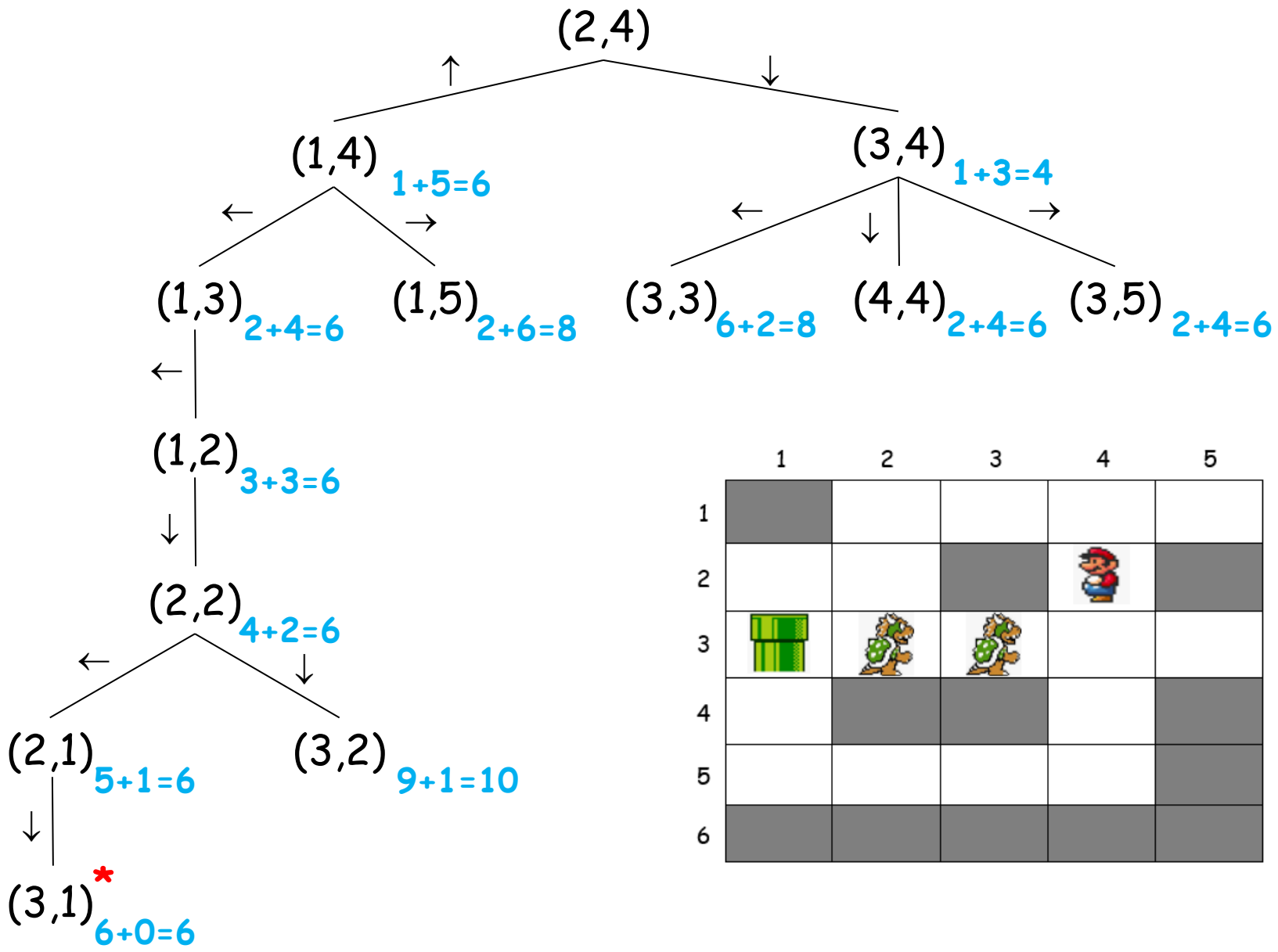




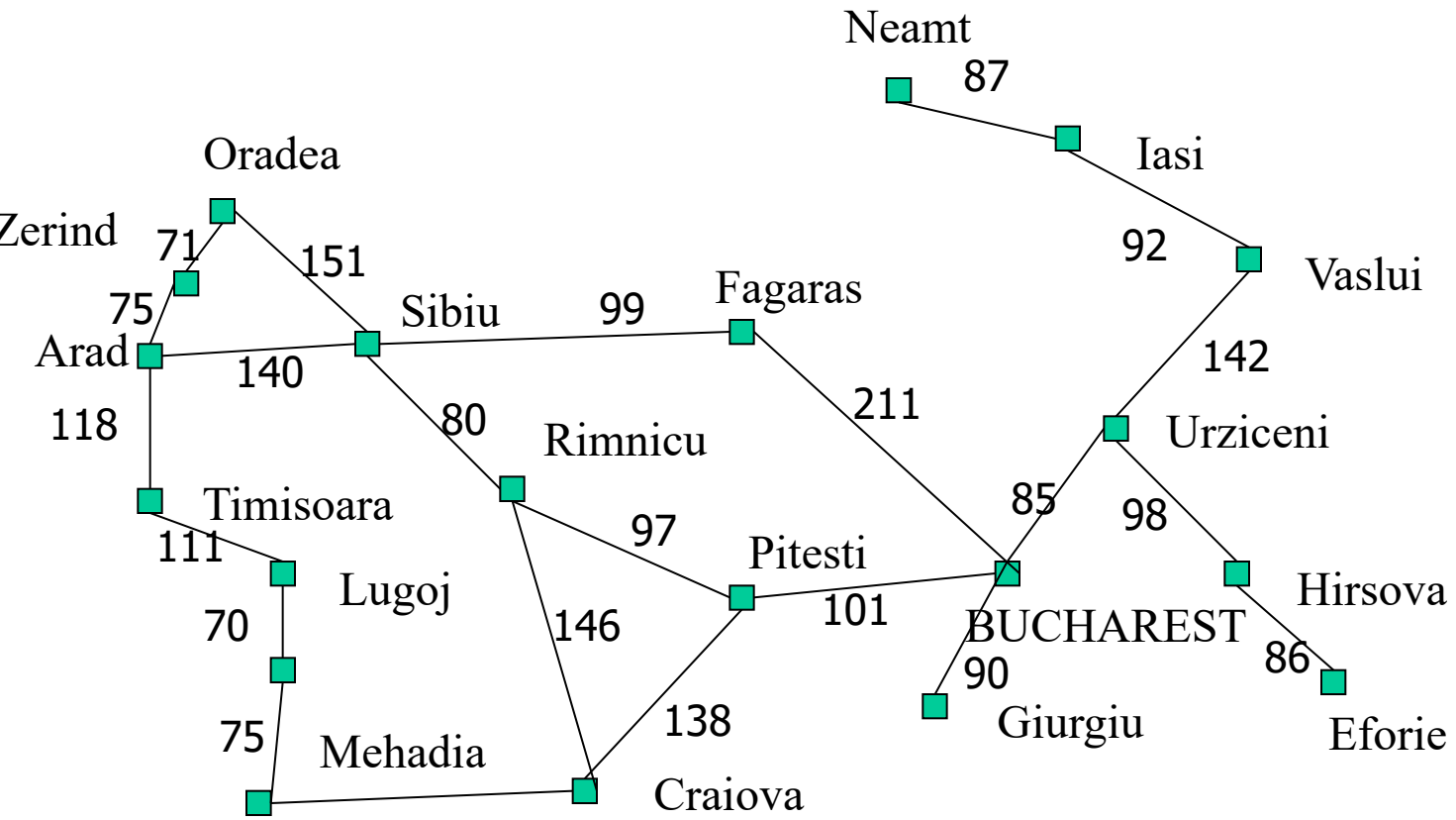
Búsqueda informada



Búsqueda informada



Búsqueda informada



Distancia en línea recta a Bucarest:

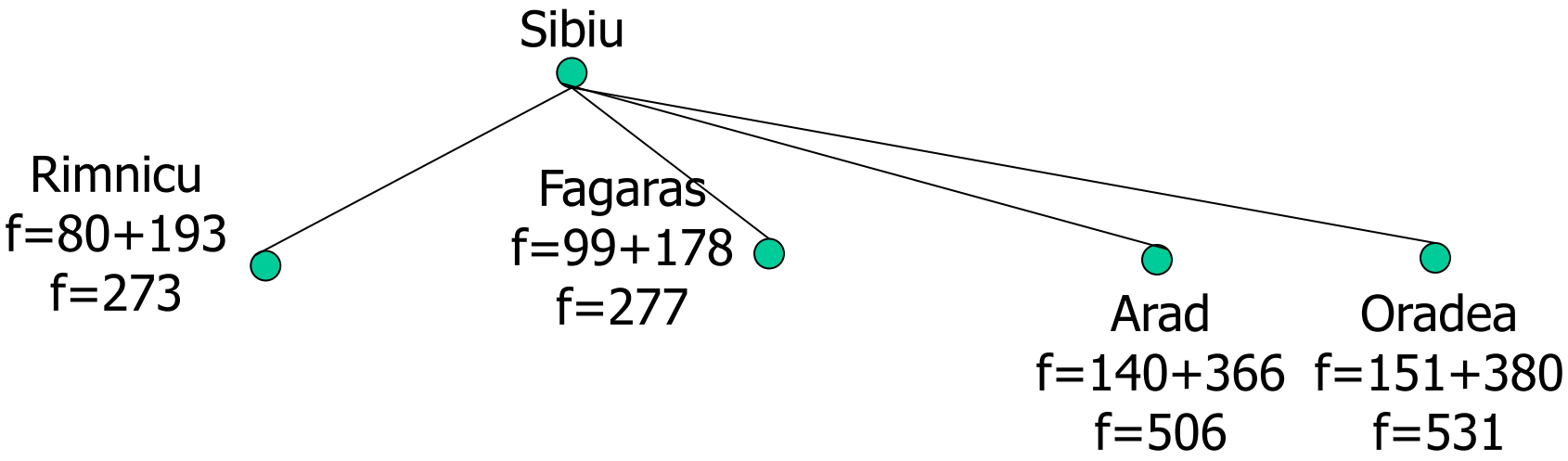
Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Problema: ir de Sibiu a Bucarest

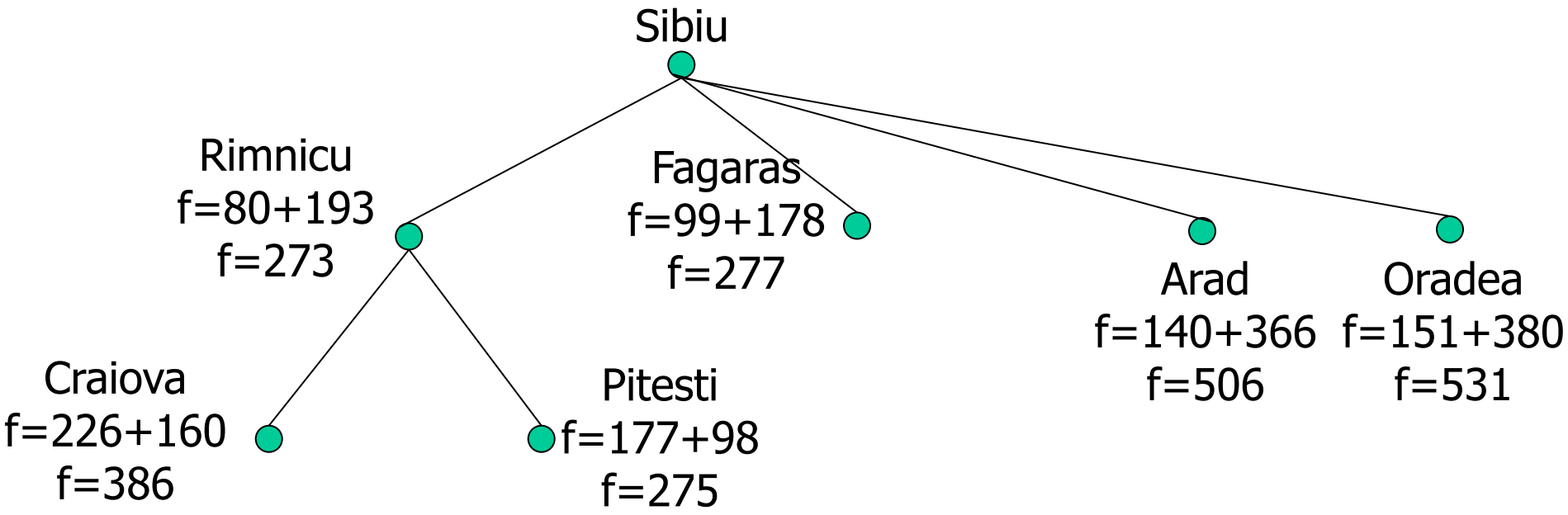
Evite devolverse

Aplique A*

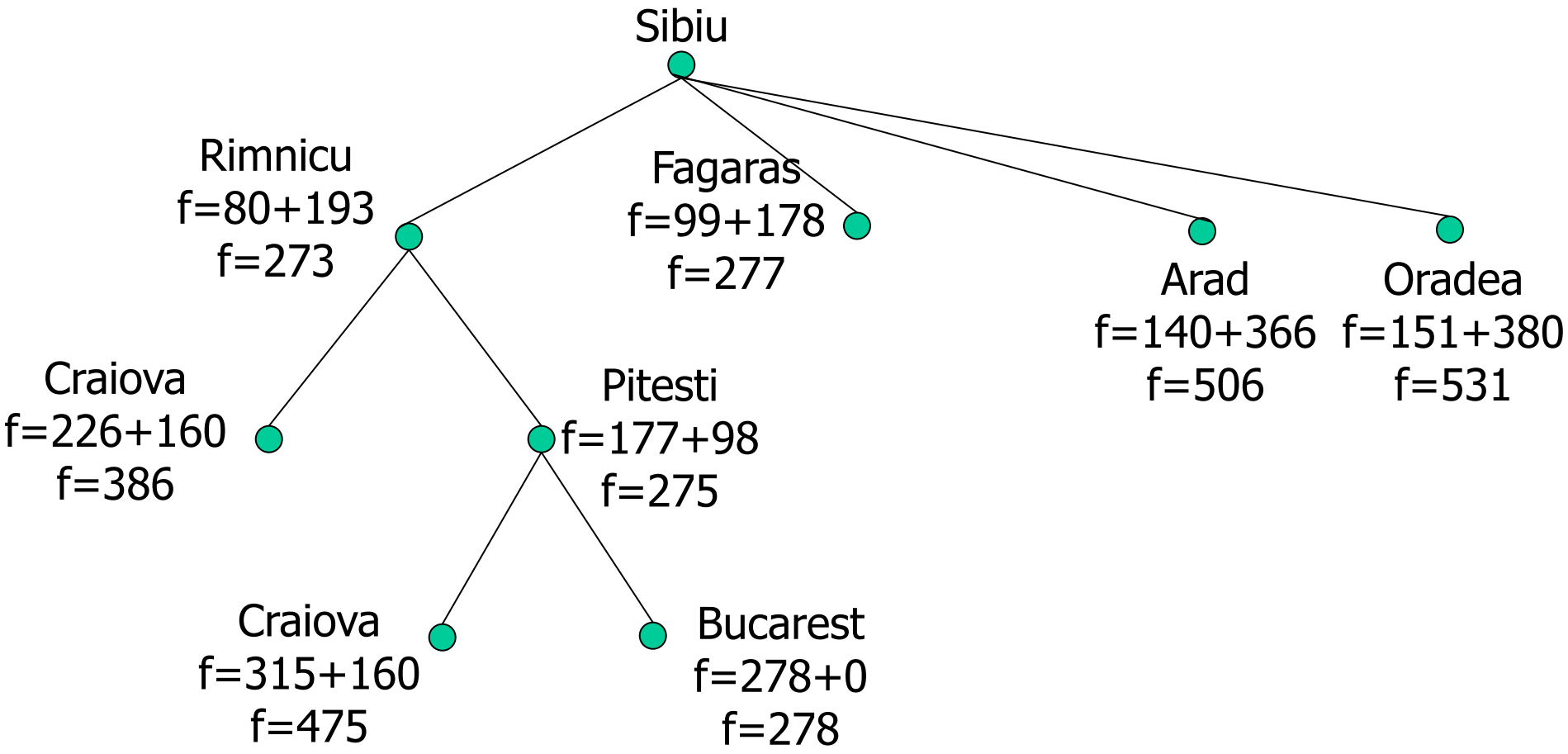
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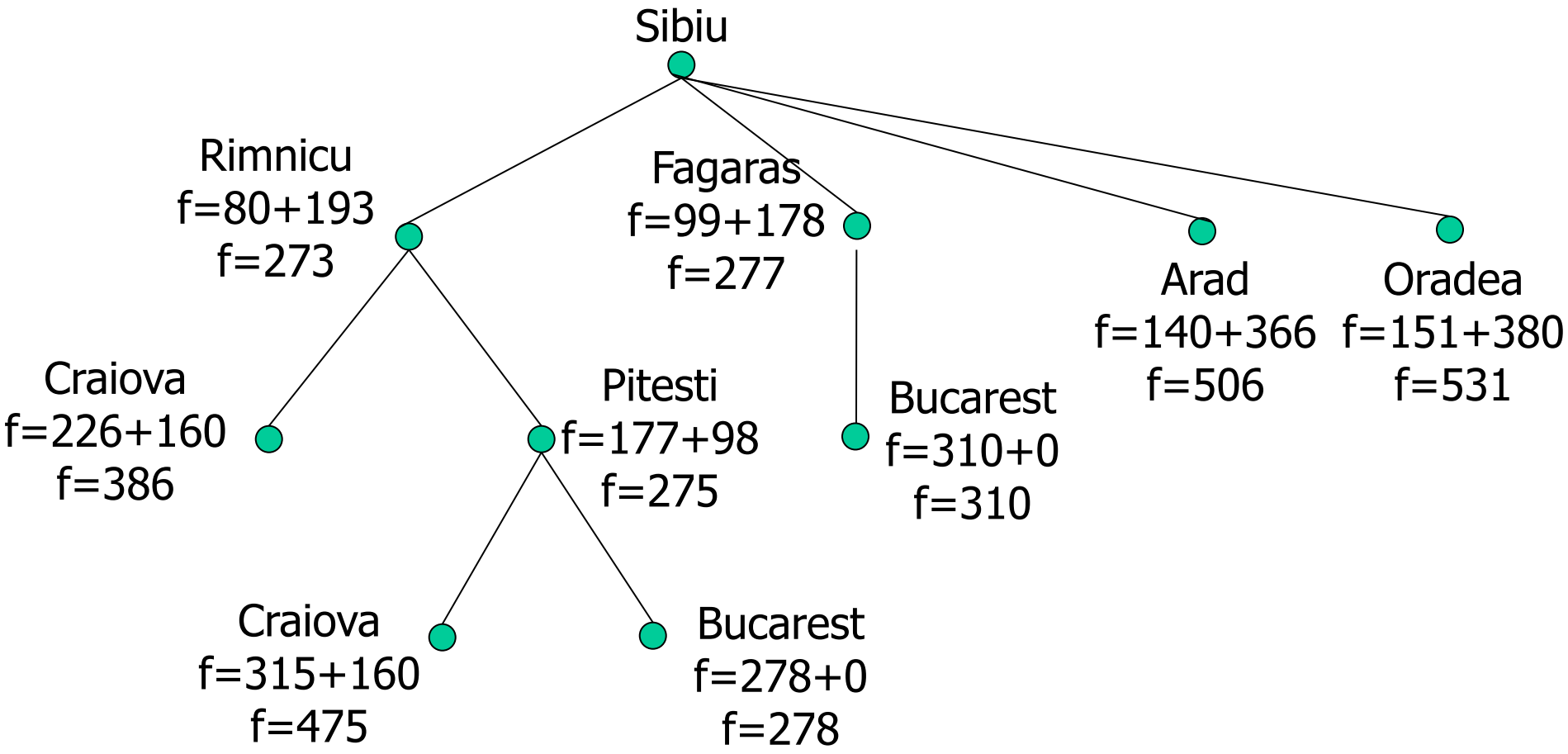
Búsqueda informada



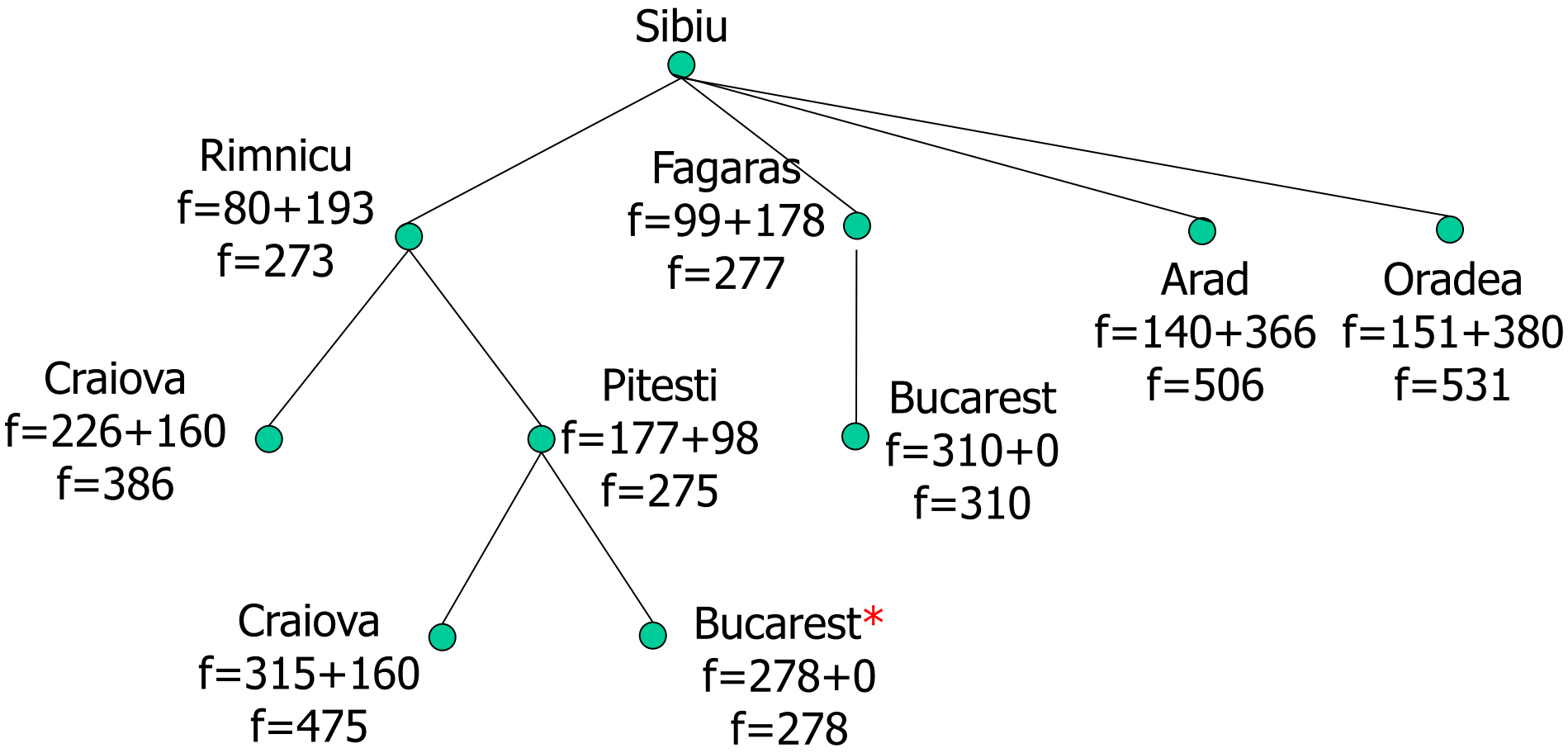
Búsqueda informada



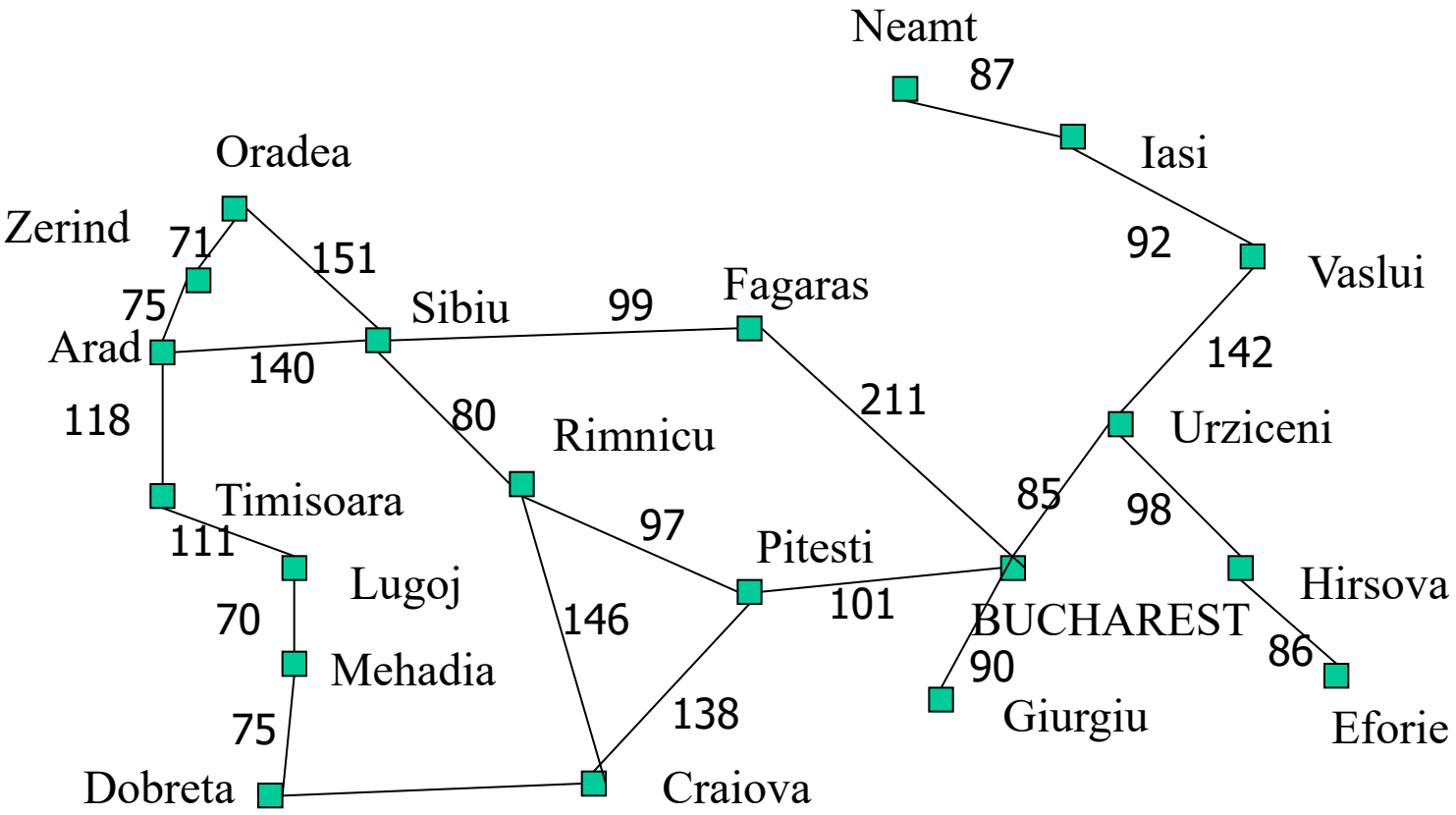
Búsqueda informada



Búsqueda informada



Búsqueda informada



La solución obtenida es
Sibiu→*Rimnicu→Pitesti→Bucarest que tiene*
costo **278** *y es la óptima*

Búsqueda informada

Implementación

- Se puede implementar considerando la lista de nodos a expandir como una **cola de prioridad**, donde la prioridad es el valor de **f(n)** y se selecciona aquel con menor prioridad

Búsqueda informada

Algoritmo A*

- La estrategia es **completa y óptima** si se cumple que la heurística h(n) sea **admisible**

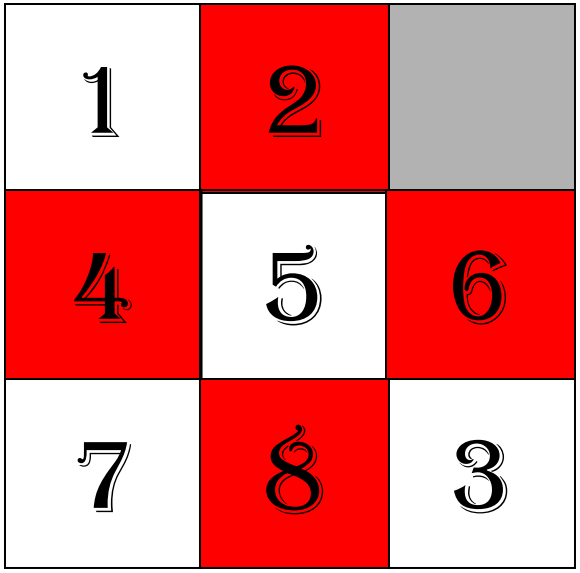
Búsqueda informada

- Una **heurística es admisible** si nunca asigna un valor mayor al real, es decir, se cumple que

$$h(n) \leq \text{costo_real}(n)$$

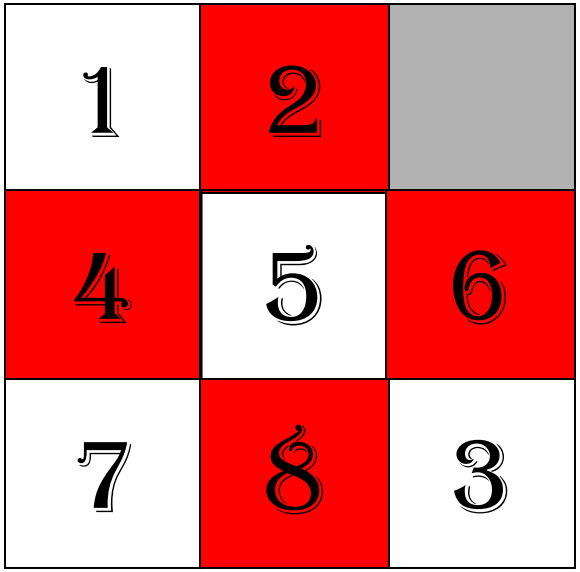
- Las heurísticas admisibles se conocen como **heurísticas optimistas**

Búsqueda informada



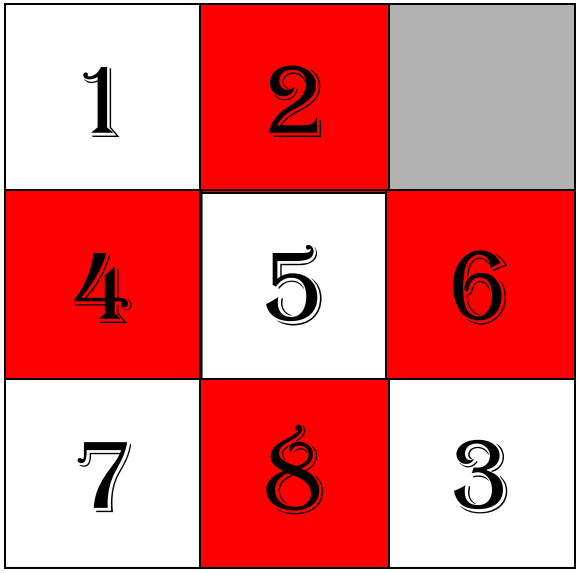
h=?

Búsqueda informada



h=2. ¿Qué significa el valor de la heurística?

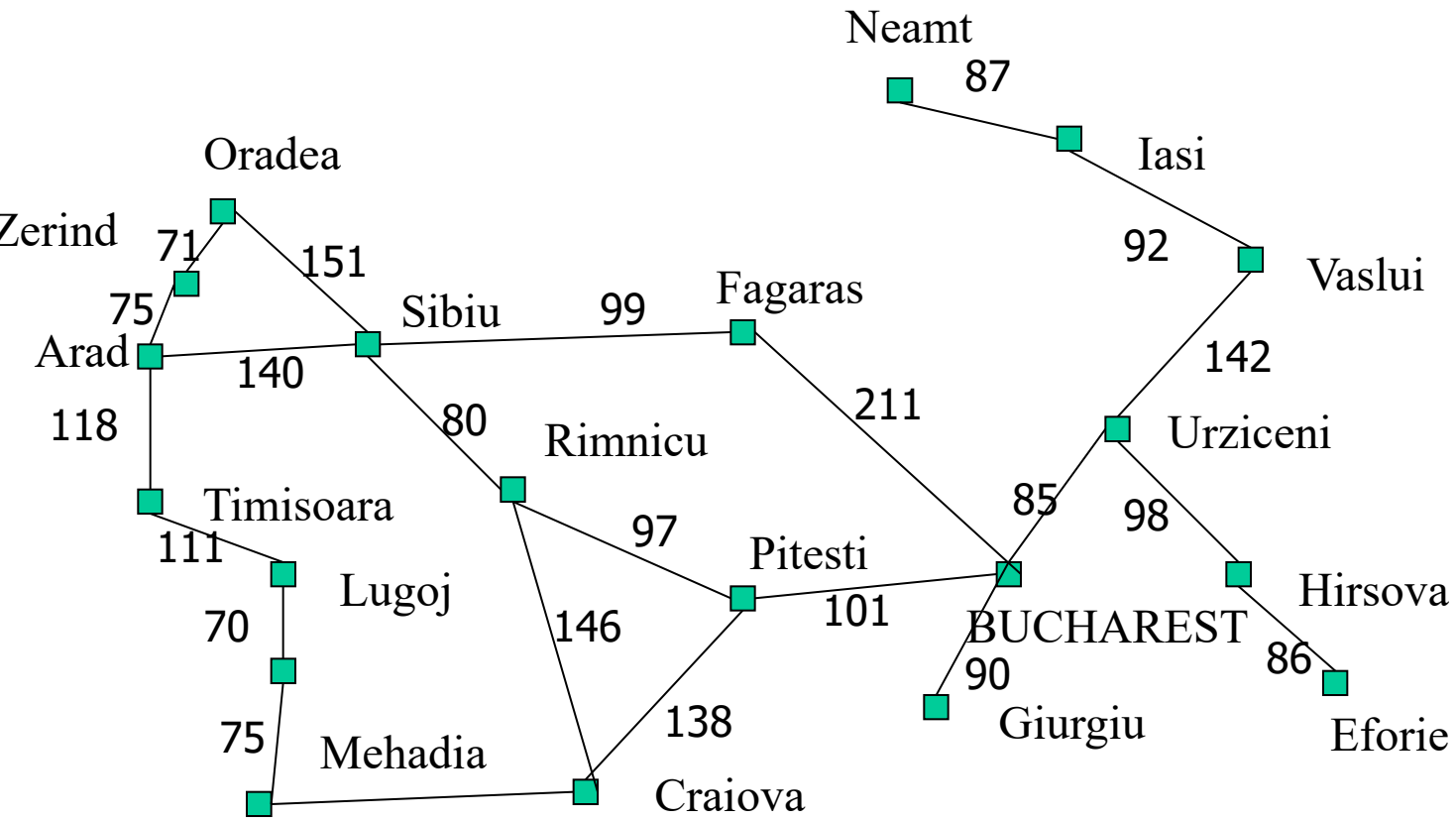
Búsqueda informada



h=2. Estima que en 2 movimientos puede encontrar una solución

¿h(n) ≤ costo_real(n)?

Búsqueda informada

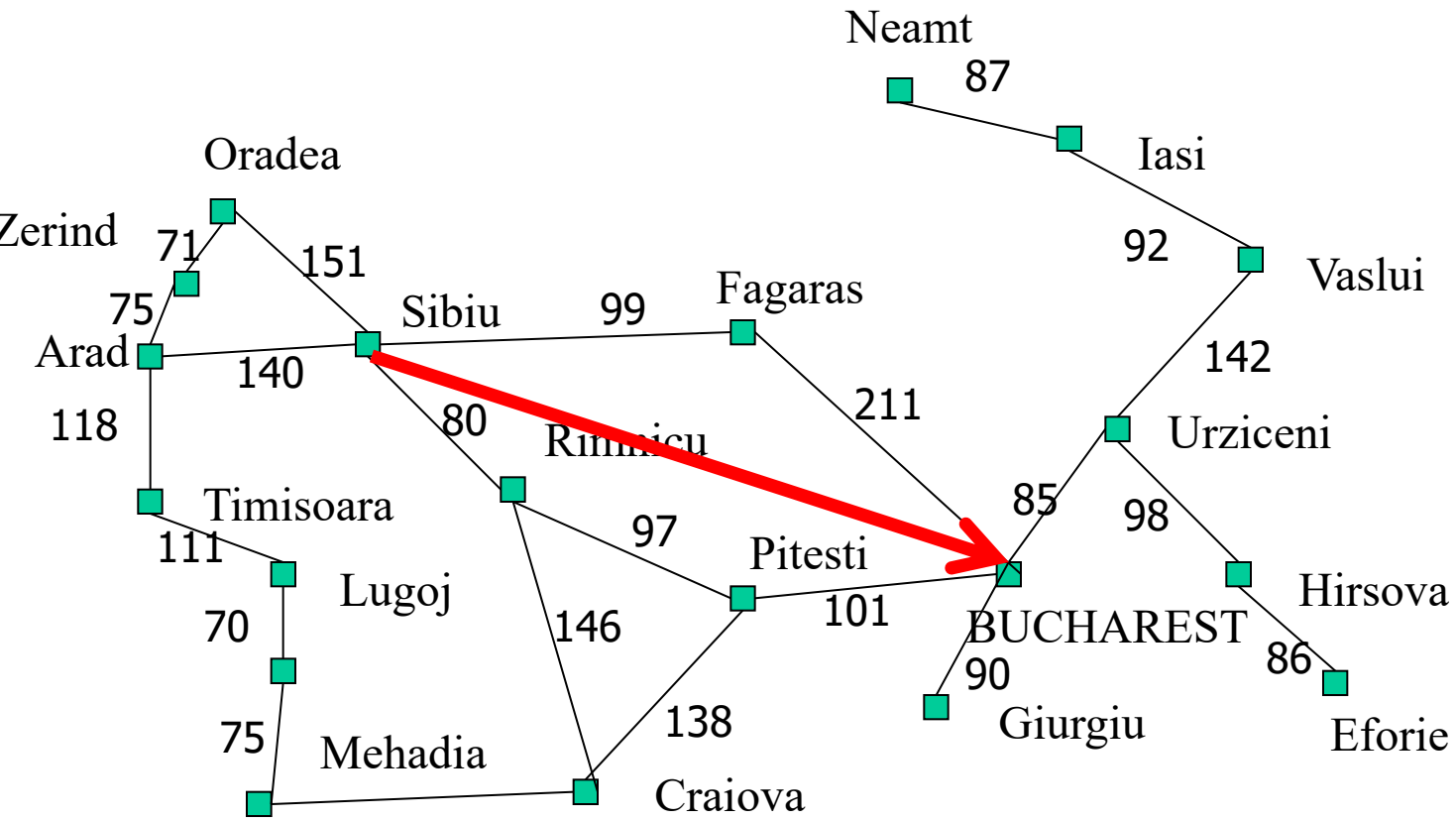


Distancia en línea recta a Bucarest:

Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

¿La distancia en línea recta es admisible?

Búsqueda informada

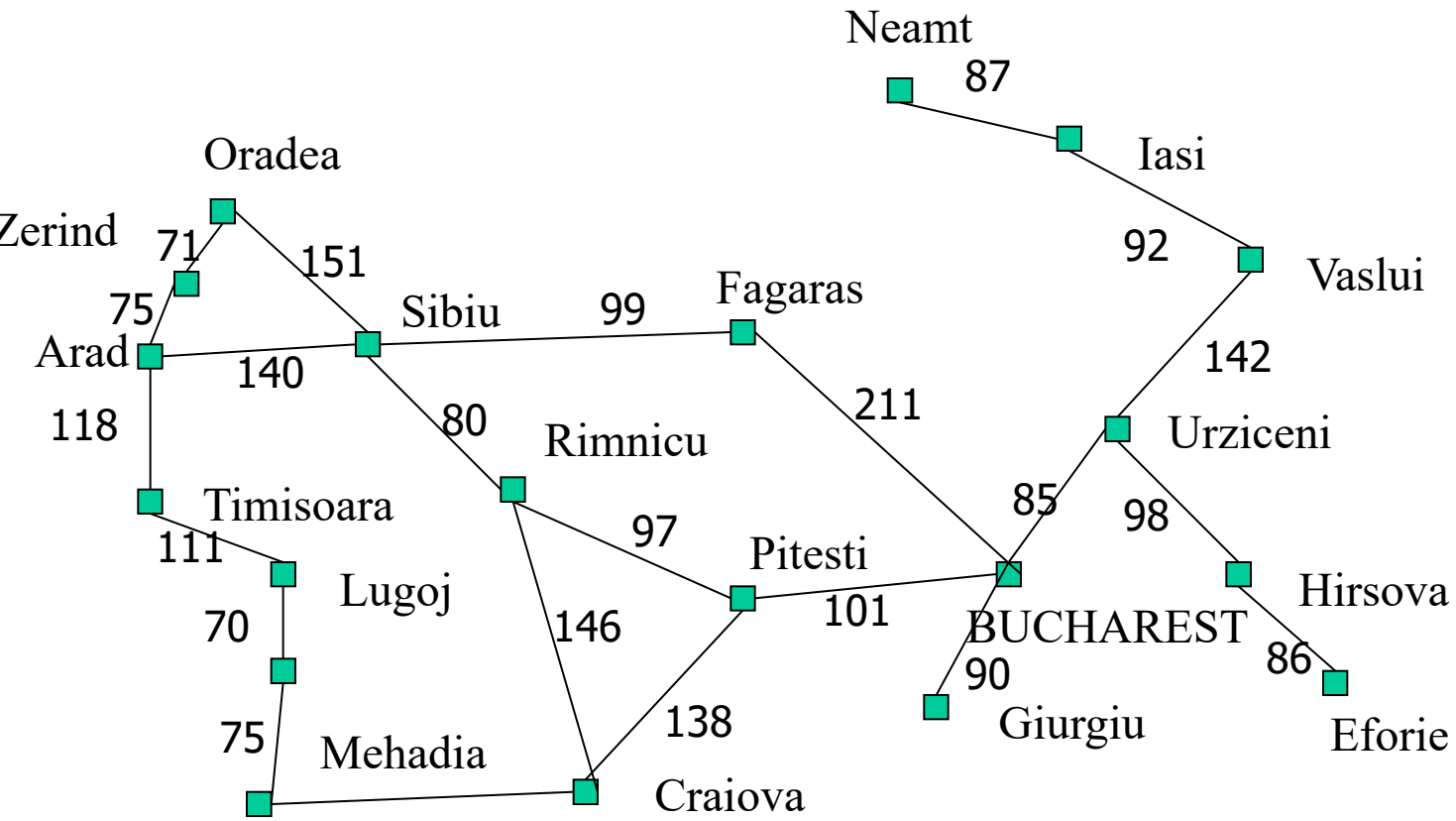


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¿La distancia en línea recta es admisible?

Búsqueda informada



Distancia en línea recta a Bucarest:

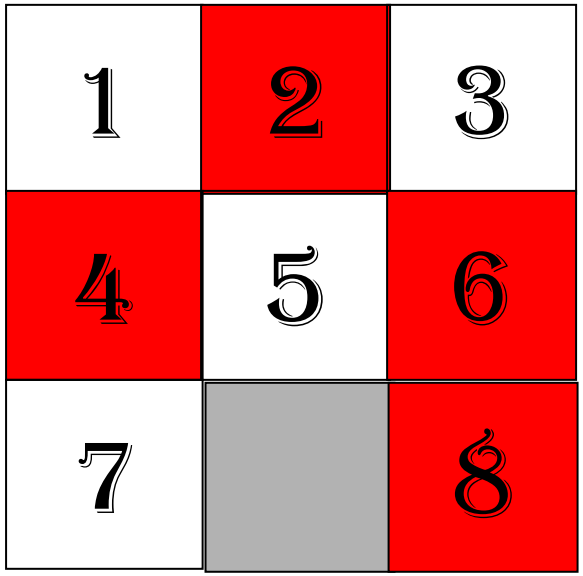
Arad	366
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Pitesti	98
Rimnicu	193
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Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

La heurística es menor al valor real

Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

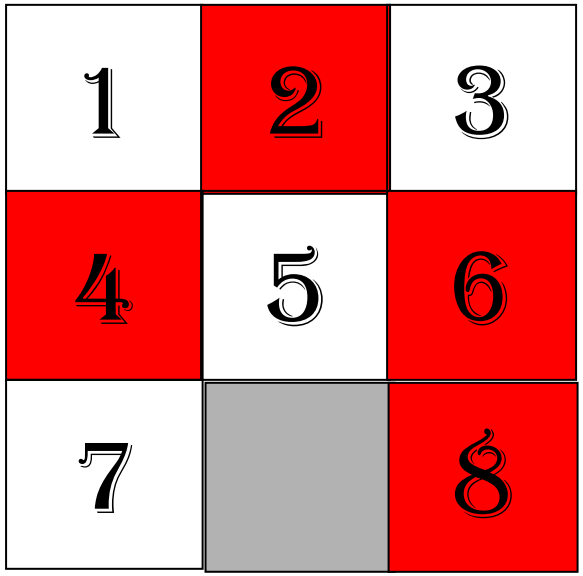
Cada movimiento de la placa vacía cuesta 1



Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



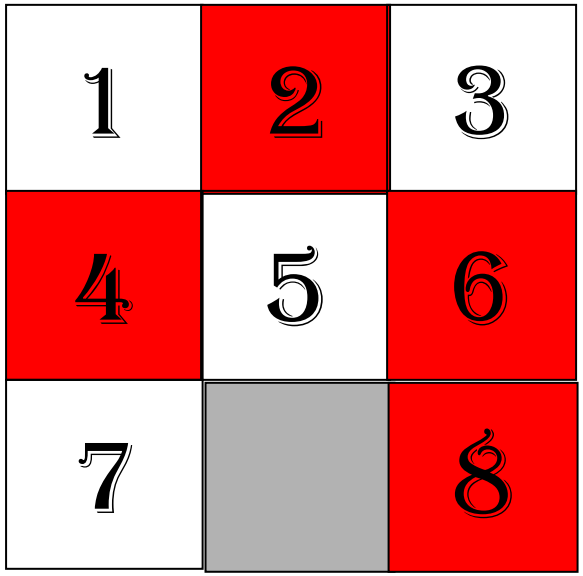
h = ?

VR = ?

Búsqueda informada

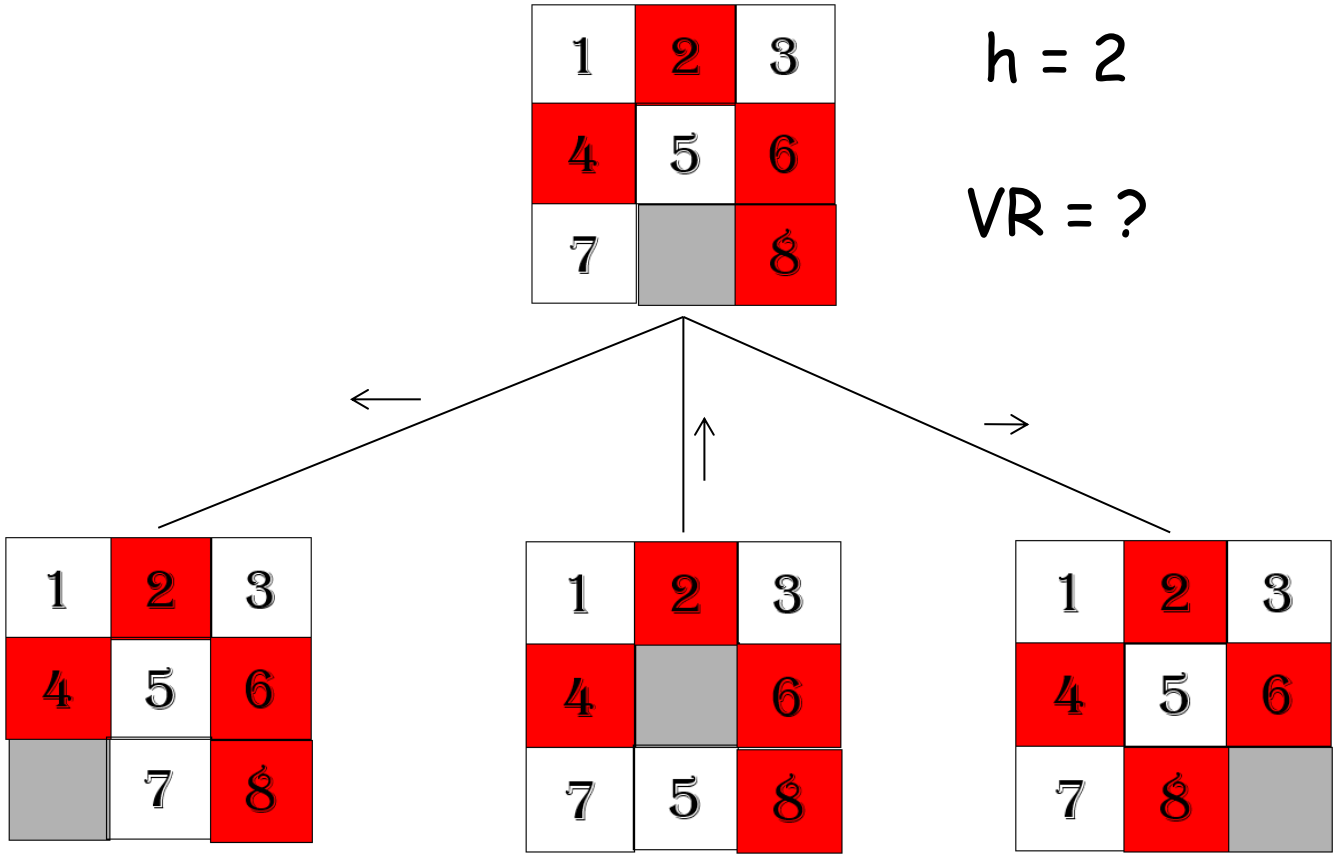
h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

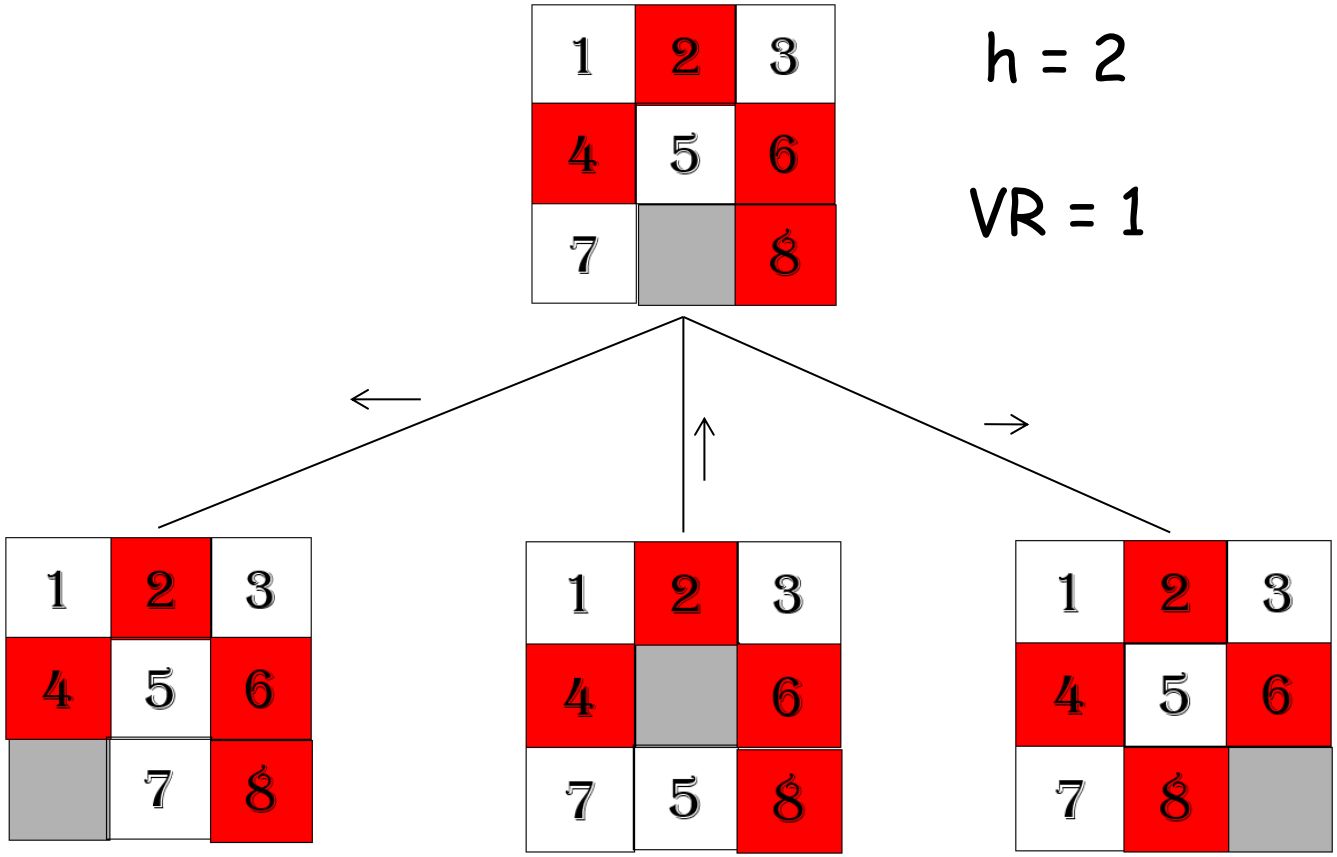
Cada movimiento de la placa vacía cuesta 1



h = 2

VR = ?

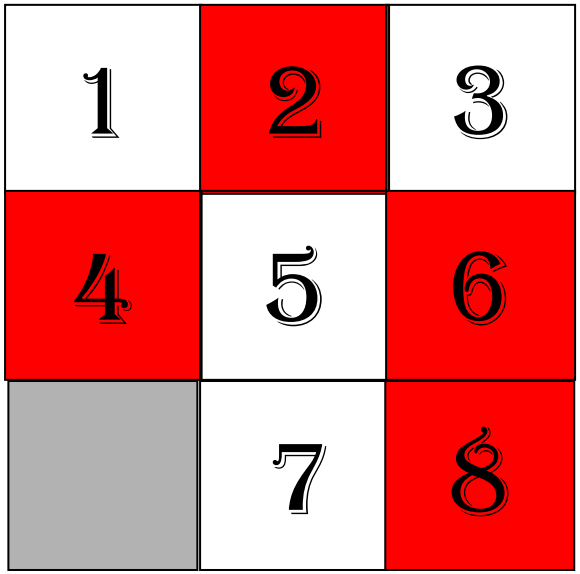

1 2 3
4 5 6
7 8
h = 2
VR = ?
1 2 3
4 5 6
7 8
1 2 3
4 6
7 5 8
1 2 3
4 5 6
7 8


1 2 3
4 5 6
7 8
h = 2
VR = 1
1 2 3
4 5 6
7 8
1 2 3
4 6
7 5 8
1 2 3
4 5 6
7 8

Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



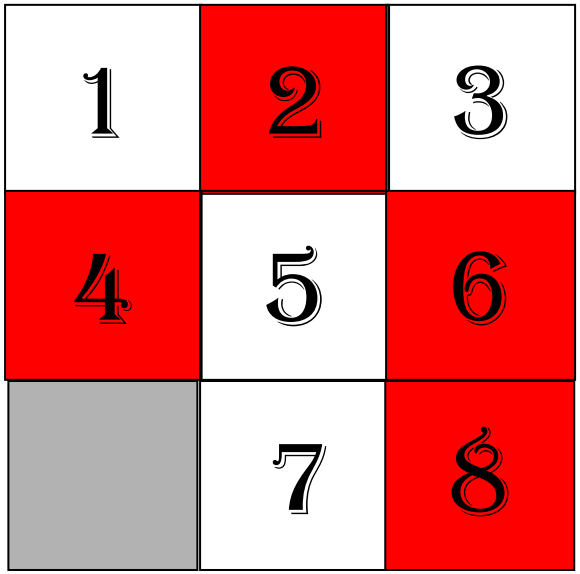
h = ?

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



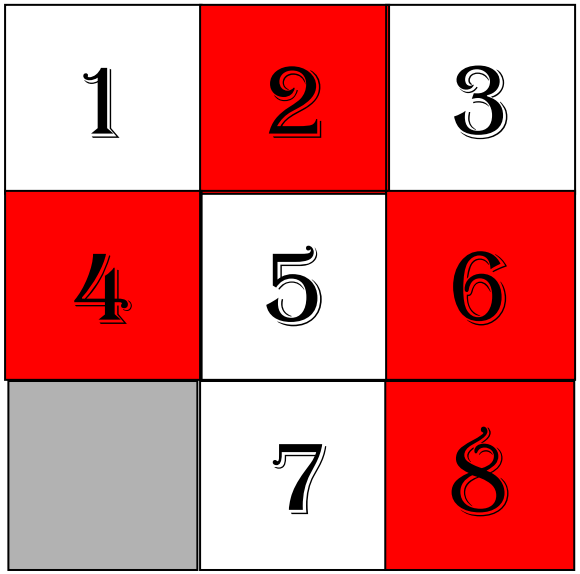
h = 3

VR = ?

Búsqueda informada

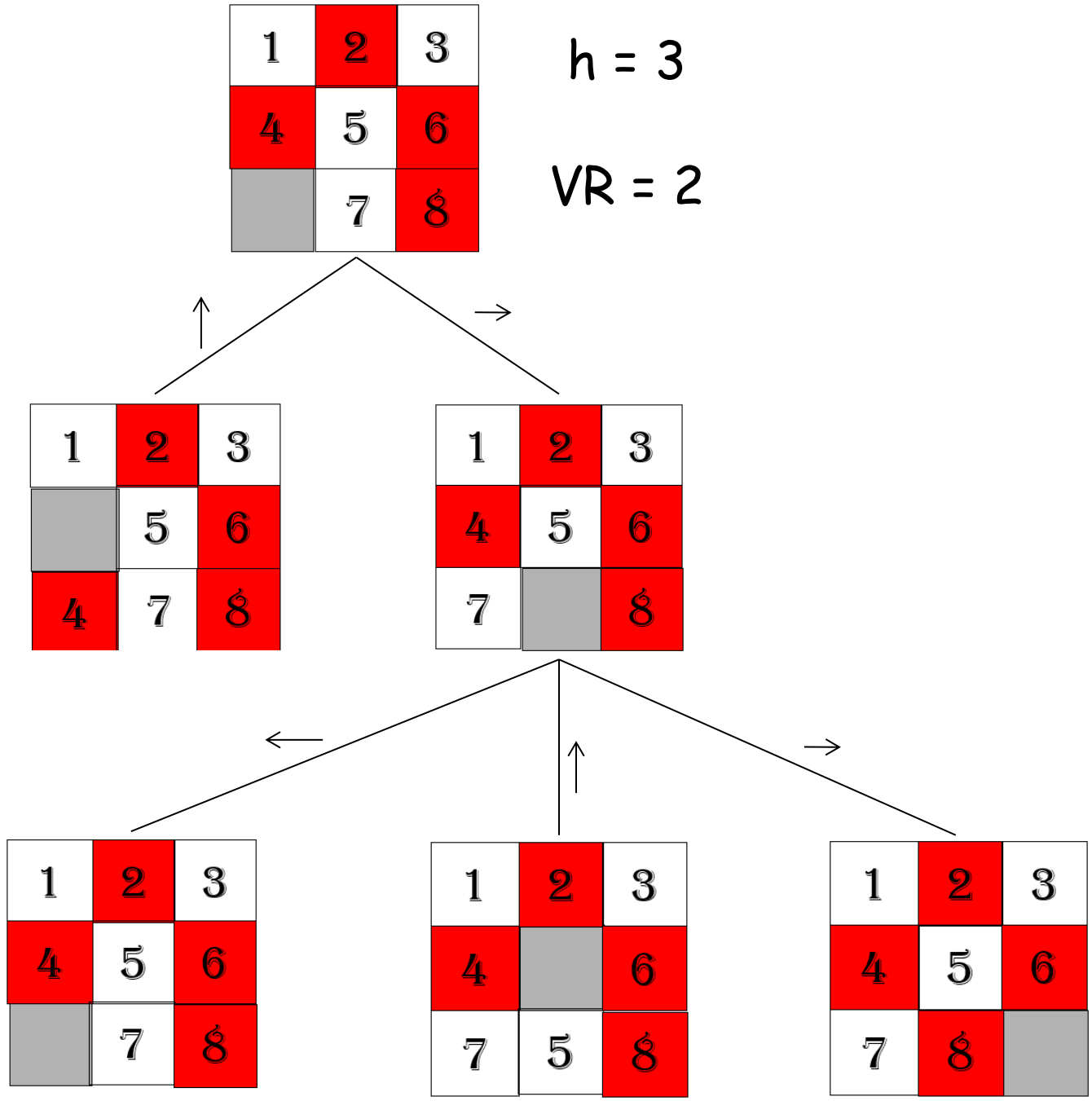
h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



h = 3

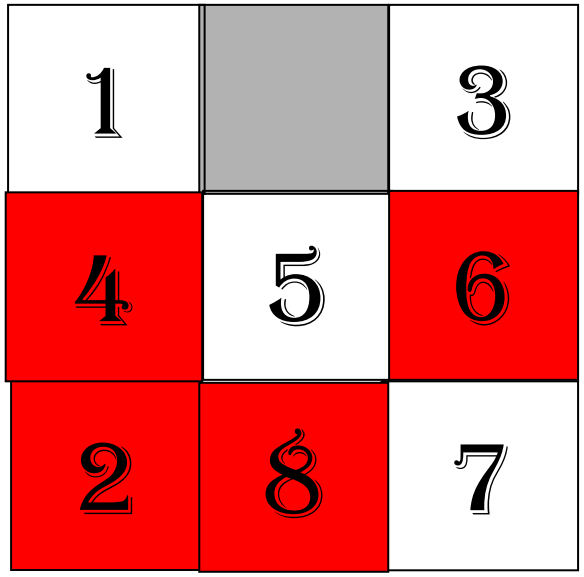
VR = 2


1 2 3
4 5 6
7 8
h = 3
VR = 2
1 2 3
5 6
4 7 8
1 2 3
4 5 6
7 8
1 2 3
4 5 6
7 8
1 2 3
4 6
7 5 8
1 2 3
4 5 6
7 8

Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



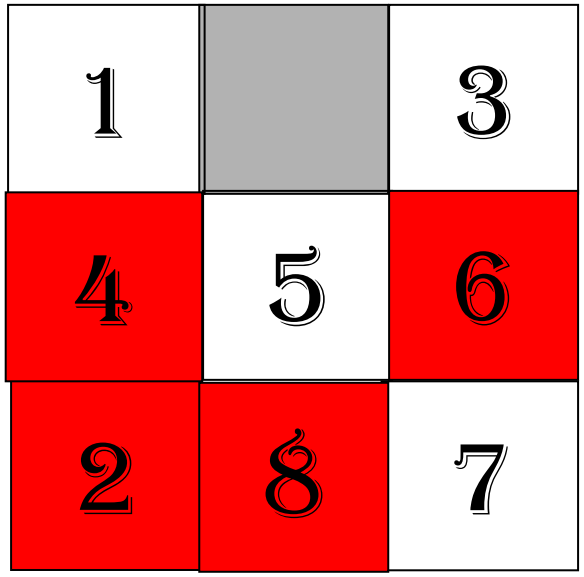
h = ?

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



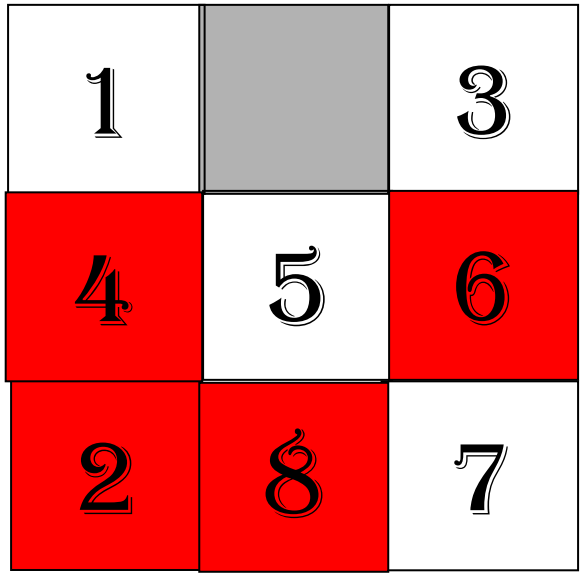
h = 3

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (incluyendo la placa vacía)

Cada movimiento de la placa vacía cuesta 1



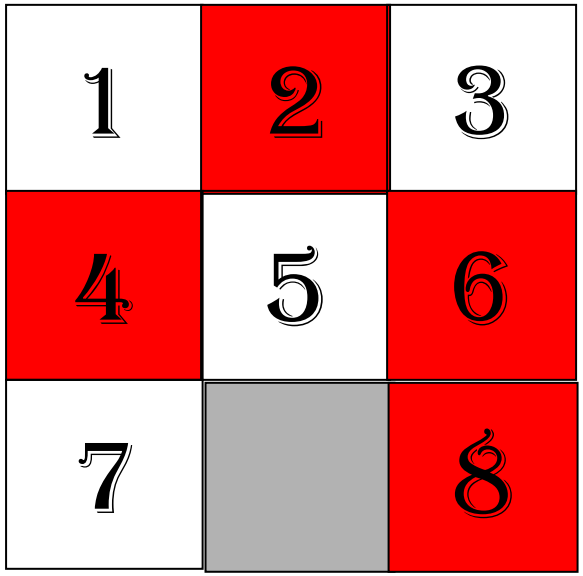
h = 3

VR = no lo sé, pero seguro son más de 3 movimientos

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

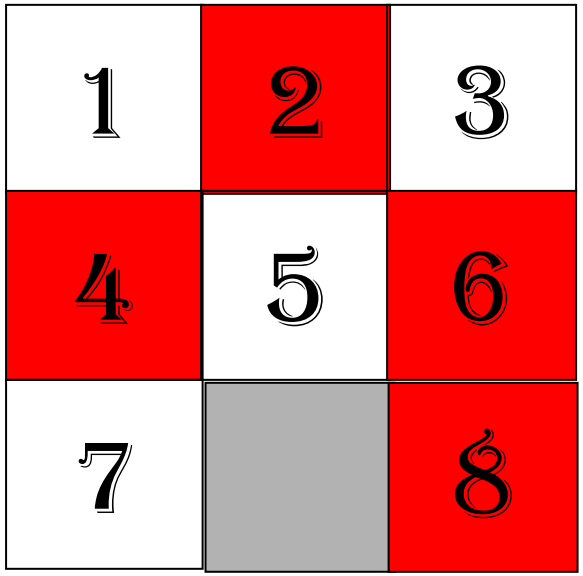
Cada movimiento de la placa vacía cuesta 1



Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



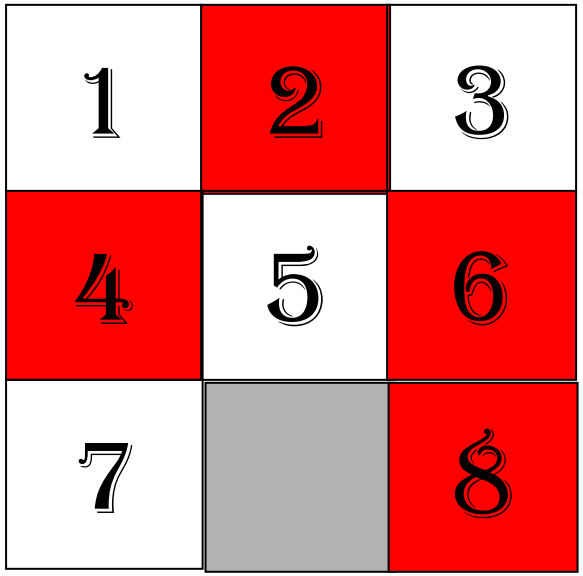
h = ?

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



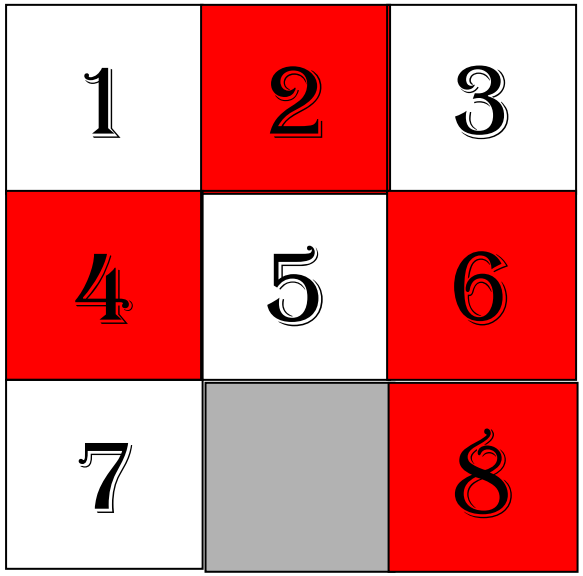
h = 1

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



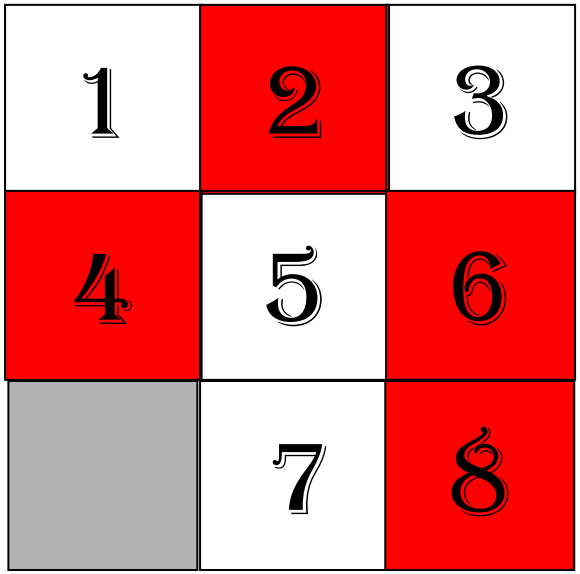
h = 1

VR = 1

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



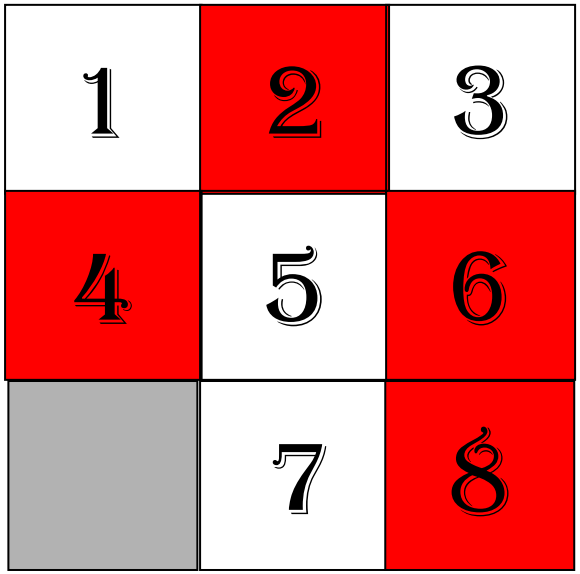
h = ?

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



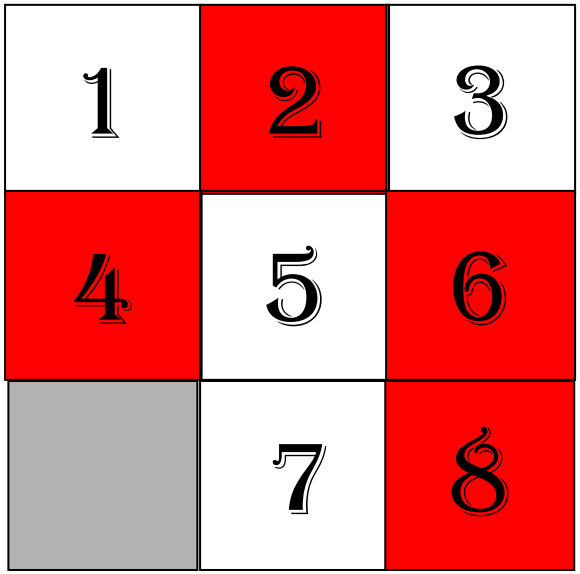
h = 2

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



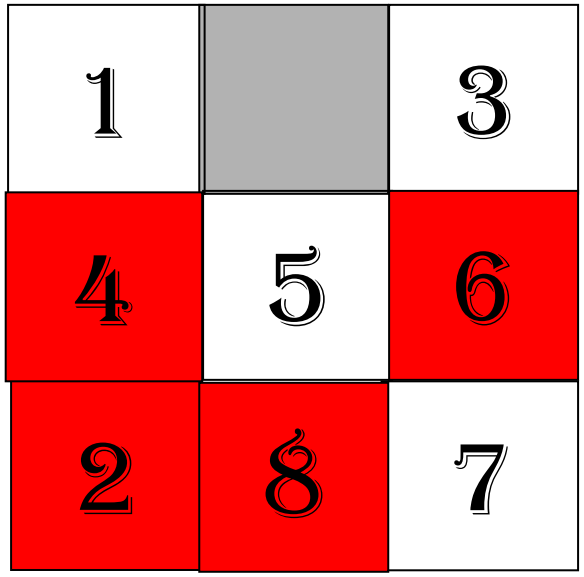
h = 2

VR = 2

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



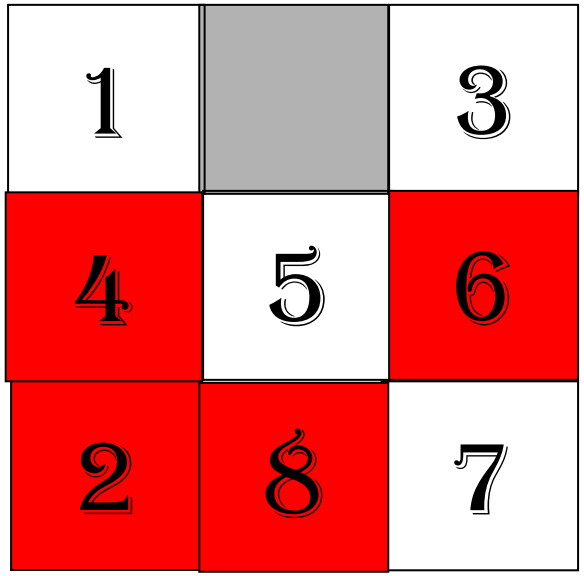
h = 2

VR = ?

Búsqueda informada

h=cantidad de placas en la posición incorrecta (**sin incluir la placa vacía**)

Cada movimiento de la placa vacía cuesta 1



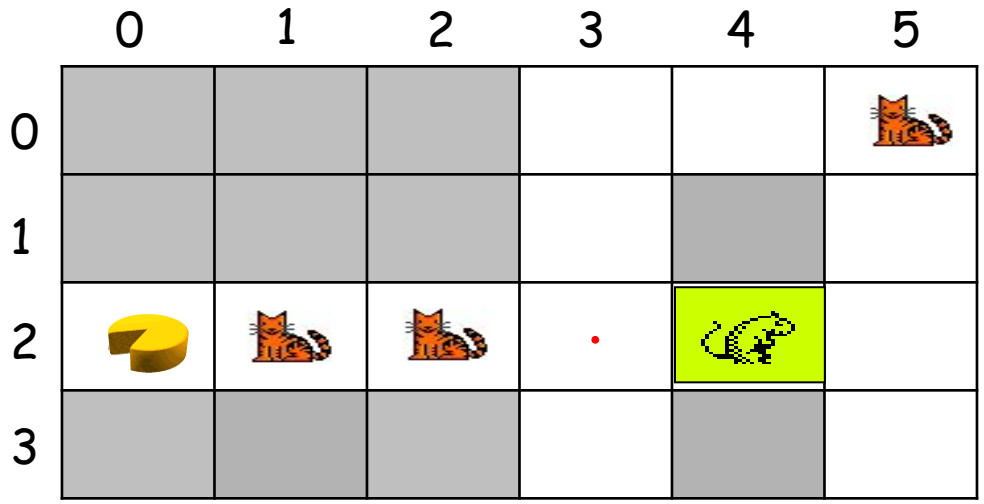
h = 2

VR = no lo sé, pero seguro son más de 2 movimientos

Búsqueda informada

h(n): distancia de Manhattan al queso

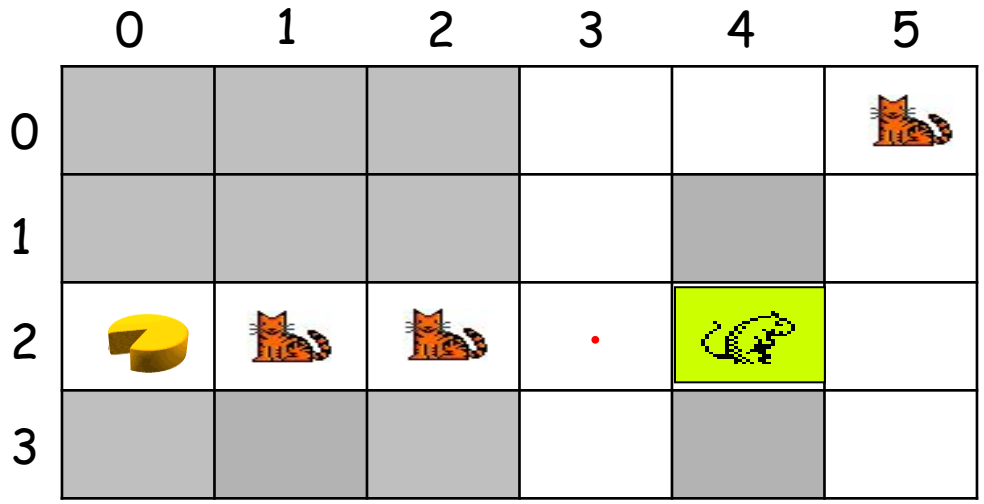
Cada movimiento cuesta 1 y pasar por el gato cuesta 5



Búsqueda informada

h(n): distancia de Manhattan al queso

Cada movimiento cuesta 1 y pasar por el gato cuesta 5



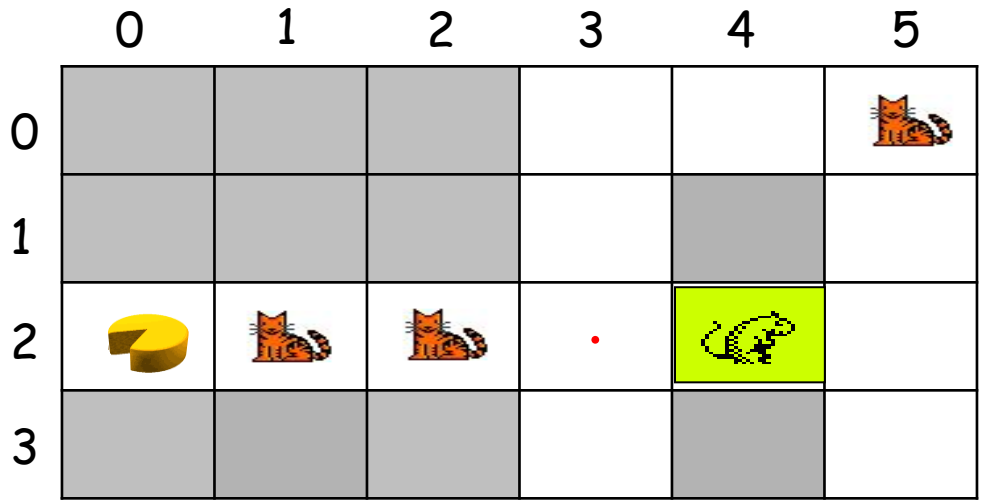
h = ?

VR = ?

Búsqueda informada

h(n): distancia de Manhattan al queso

Cada movimiento cuesta 1 y pasar por el gato cuesta 5



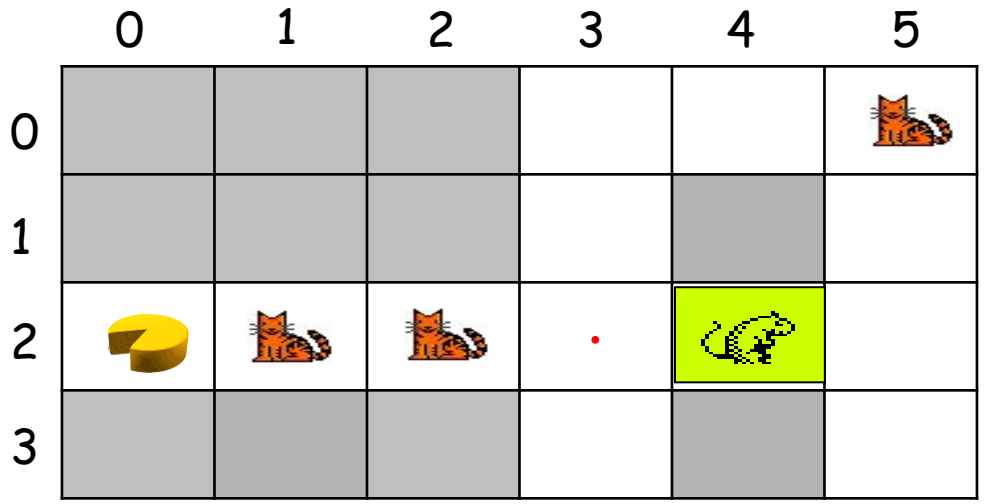
h = 3

VR = ?

Búsqueda informada

h(n): distancia de Manhattan al queso

Cada movimiento cuesta 1 y pasar por el gato cuesta 5



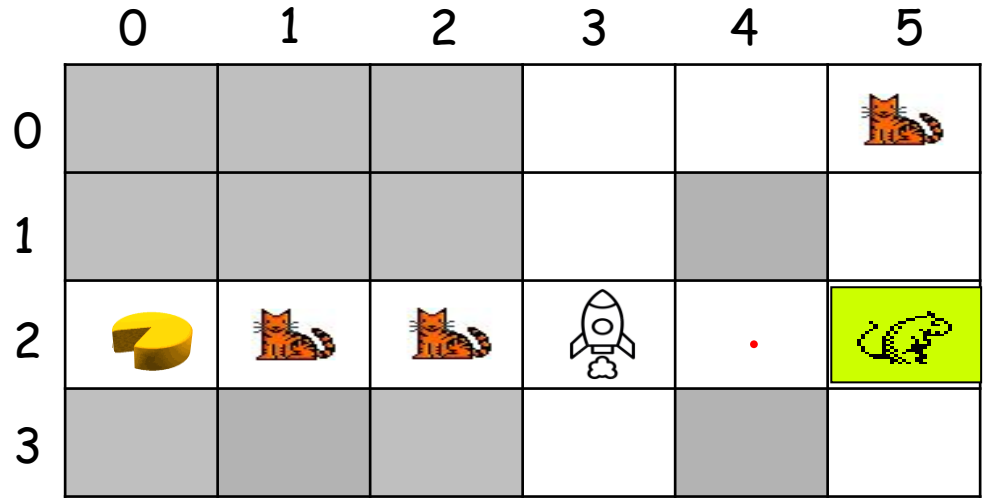
h = 3

VR = 11

Búsqueda informada

h(n): distancia de Manhattan

Cada movimiento cuesta 1. Llegar a una casilla con gato cuesta 5 (está incluido el movimiento). Si tiene la nave, cada movimiento cuesta 1/2 y no lo afectan los gatos



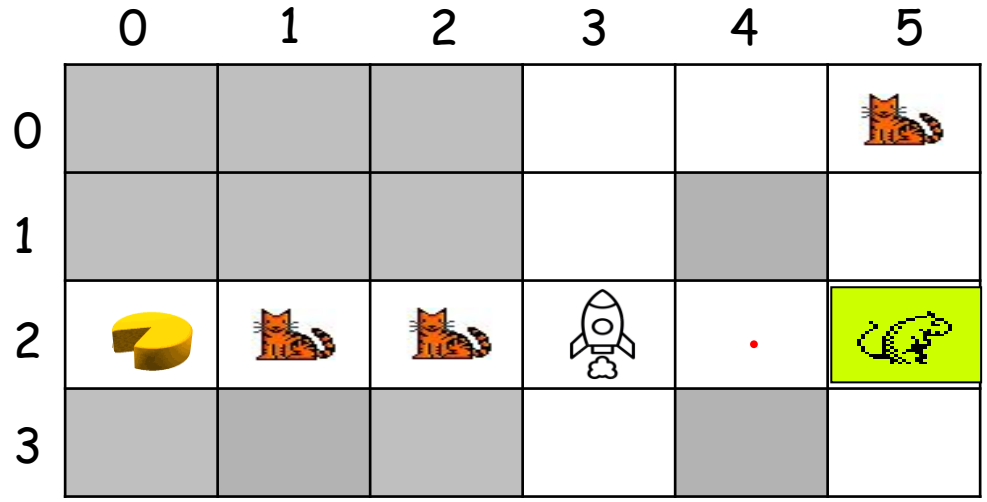
h = ?

VR = ?

Búsqueda informada

h(n): distancia de Manhattan

Cada movimiento cuesta 1. Llegar a una casilla con gato cuesta 5 (está incluido el movimiento). Si tiene la nave, cada movimiento cuesta 1/2 y no lo afectan los gatos



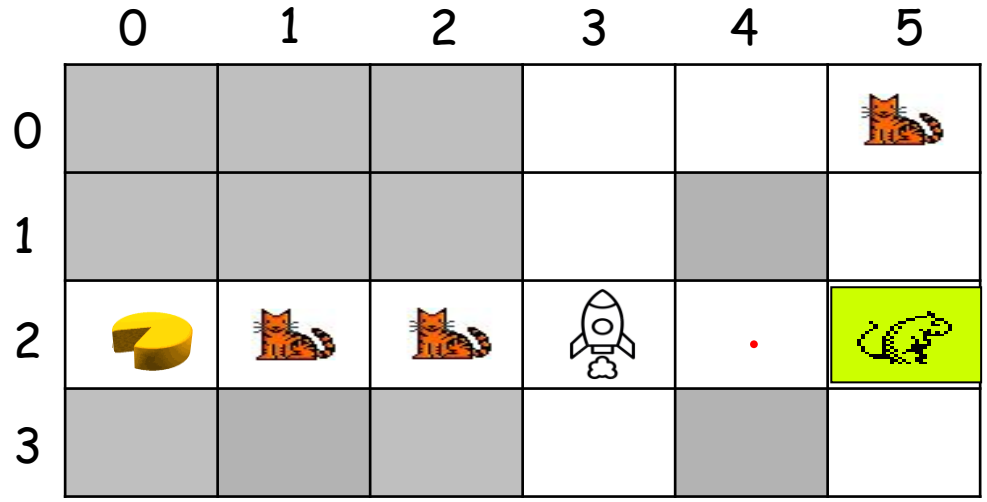
h = 4

VR = ?

Búsqueda informada

h(n): distancia de Manhattan

Cada movimiento cuesta 1. Llegar a una casilla con gato cuesta 5 (está incluido el movimiento). Si tiene la nave, cada movimiento cuesta 1/2 y no lo afectan los gatos



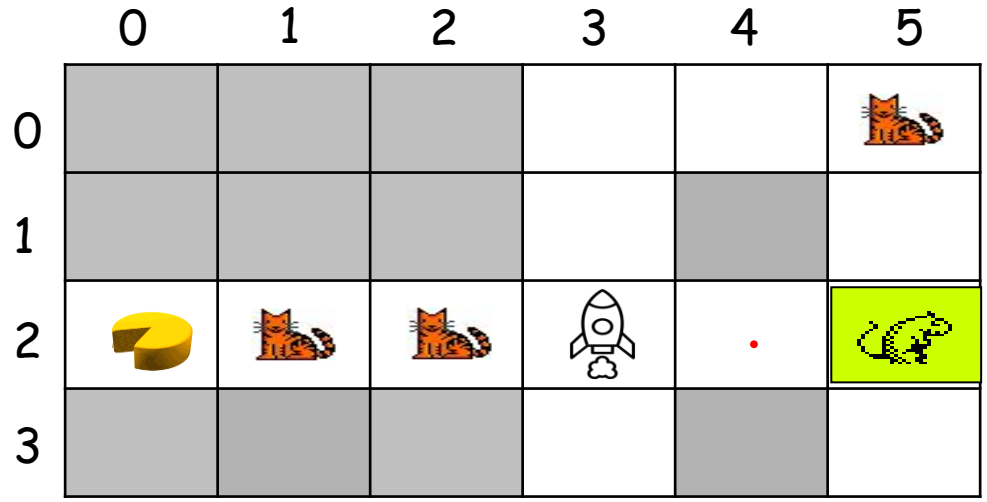
h = 4

VR = 2.5

Búsqueda informada

h(n): (distancia de Manhattan)/2 ← Se divide sobre 2 para que sea admisible

Cada movimiento cuesta 1. Llegar a una casilla con gato cuesta 5 (está incluido el movimiento). Si tiene la nave, cada movimiento cuesta 1/2 y no lo afectan los gatos



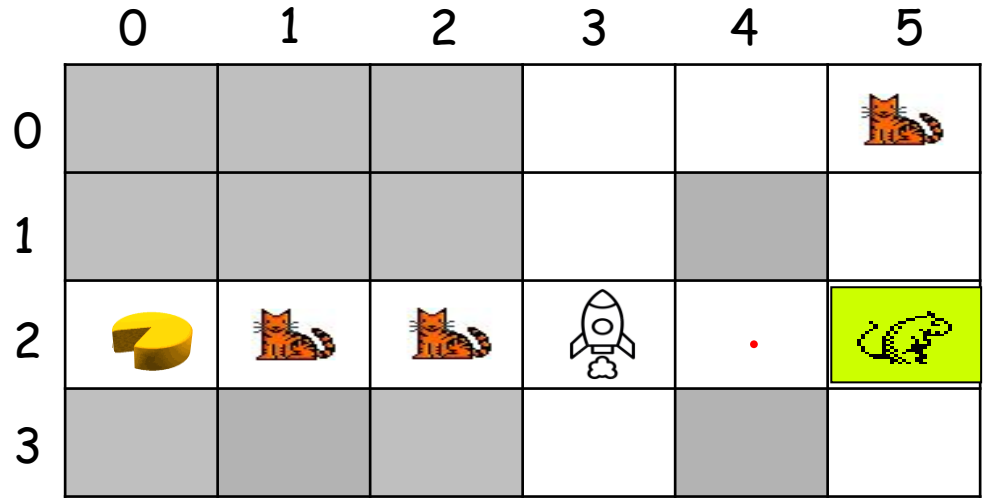
h = ?

VR = 2.5

Búsqueda informada

h(n): (distancia de Manhattan)/2 ← Se divide sobre 2 para que sea admisible

Cada movimiento cuesta 1. Llegar a una casilla con gato cuesta 5 (está incluido el movimiento). Si tiene la nave, cada movimiento cuesta 1/2 y no lo afectan los gatos

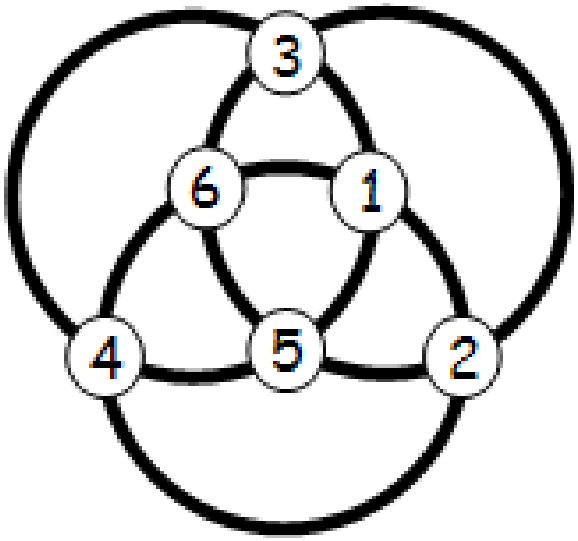


h = 2

VR = 2.5

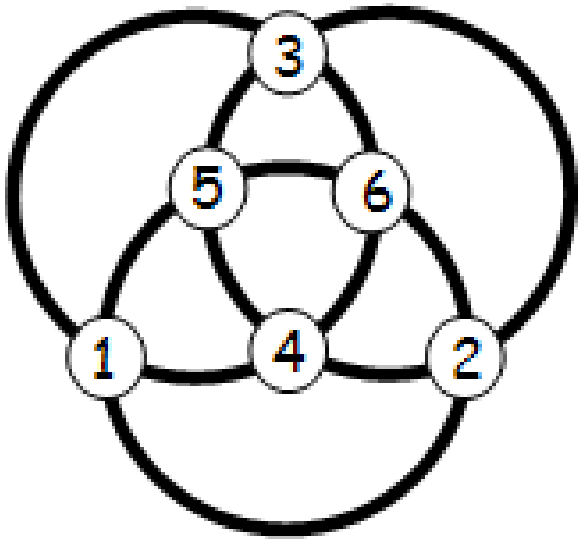
Diseño de heurísticas

- **Los aros mágicos**. Utilizando el operador *intercambio(a,b)* que cambia de posición los números a y b, colocar los cuatro números sobre cada aro de tal forma que la suma sea 14. Cada intercambio cuesta 1

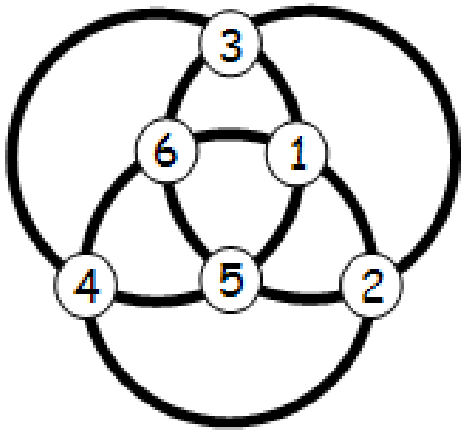


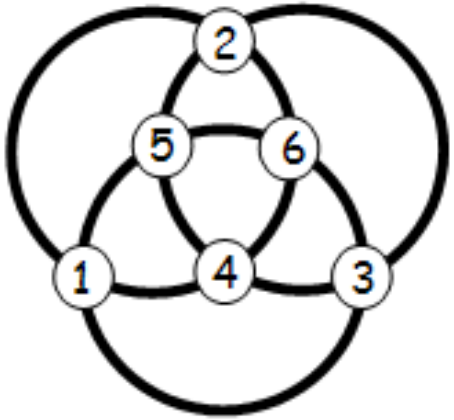
Búsqueda informada

Diseño de heurísticas



meta



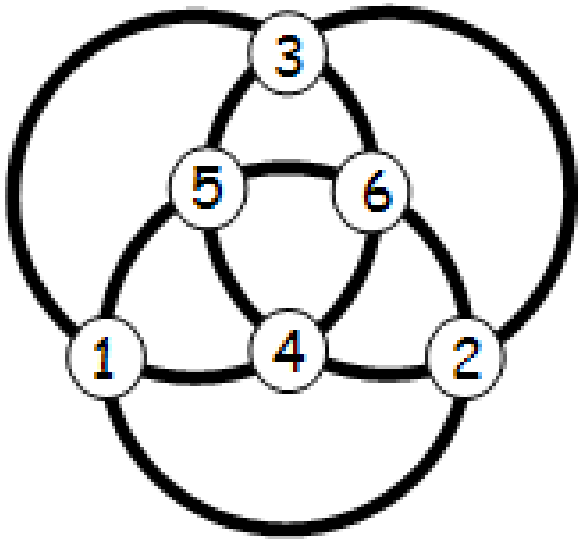


¿Cuál está más distante de la meta?

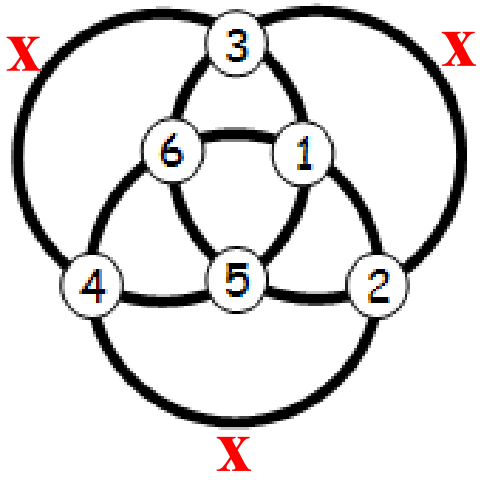
¿Cómo lo sabe?

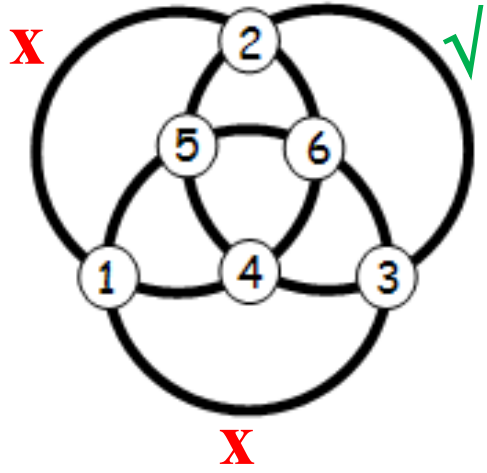
Búsqueda informada

Diseño de heurísticas



meta



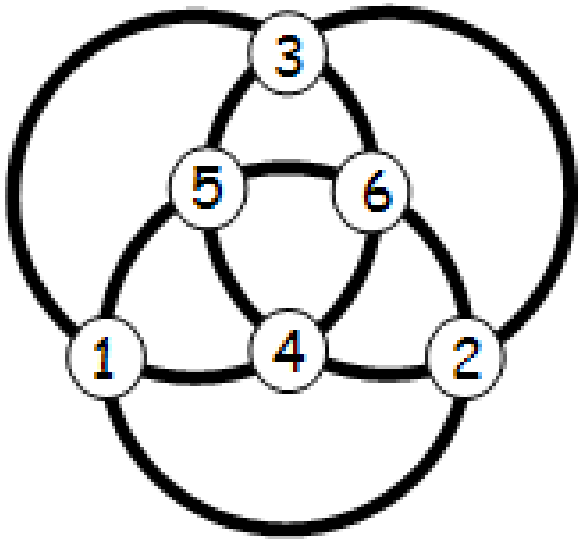


¿Cuál está más distante de la meta?

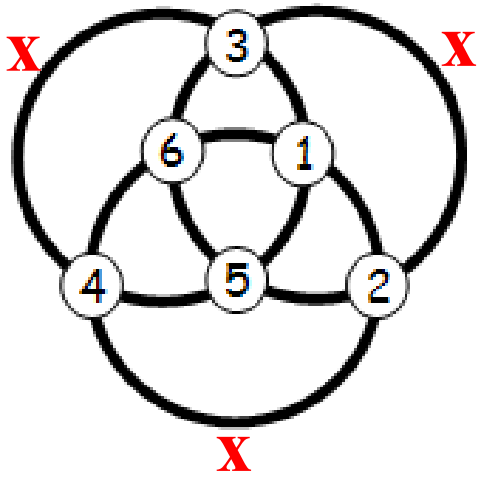
¿Cómo lo sabe?

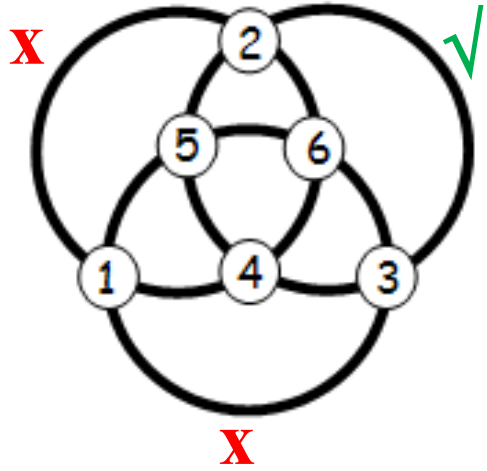
Búsqueda informada

Diseño de heurísticas



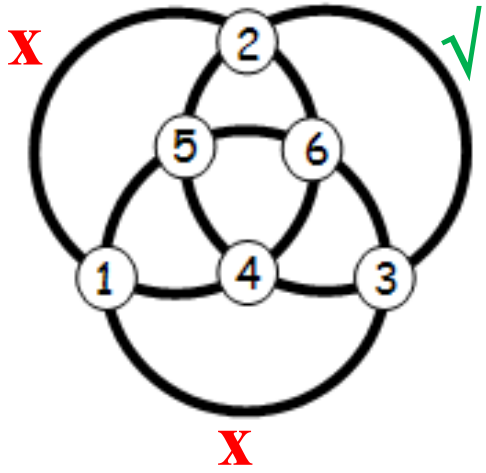
meta





h(n) = cantidad de aros que no suman 14

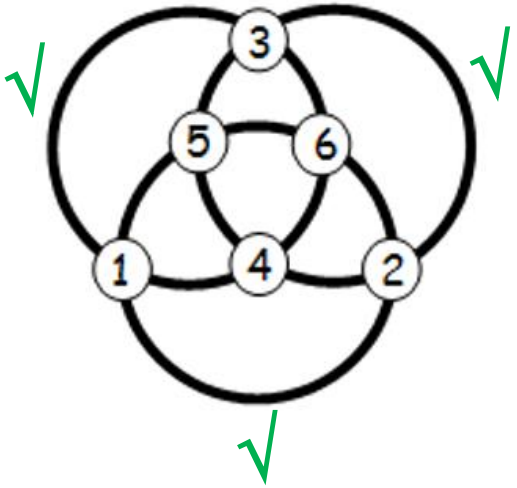
es admisible?



h(n) = 2

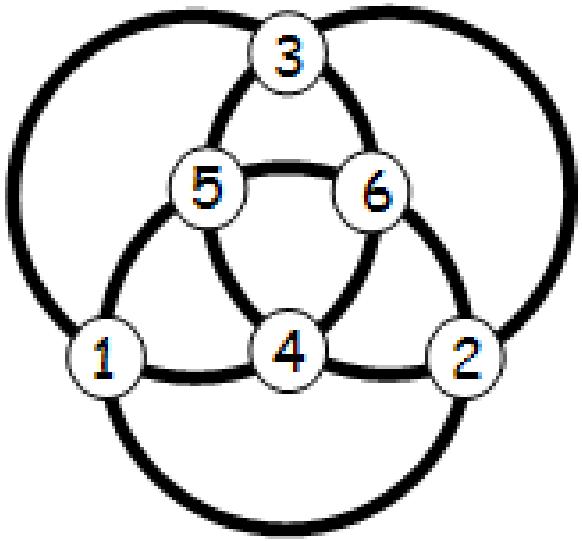
VR = 1

intercambio(2,3)

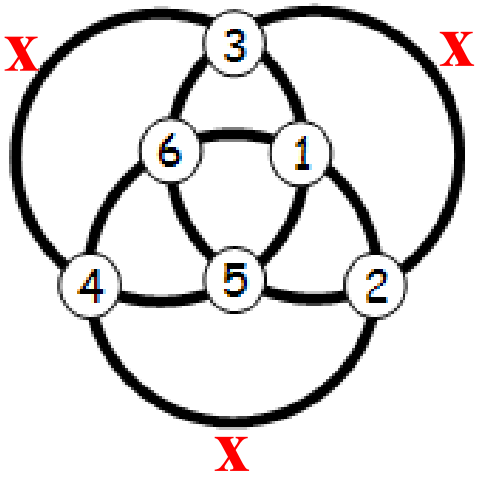


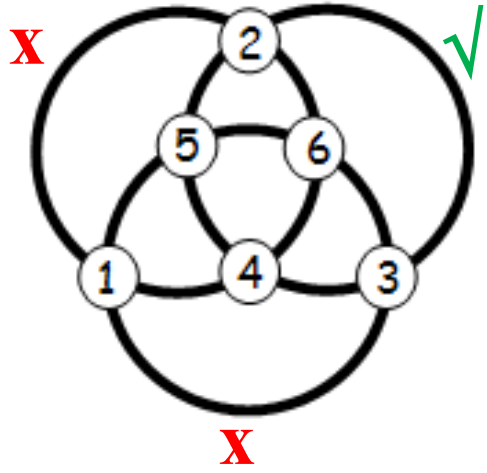
Búsqueda informada

Diseño de heurísticas



meta



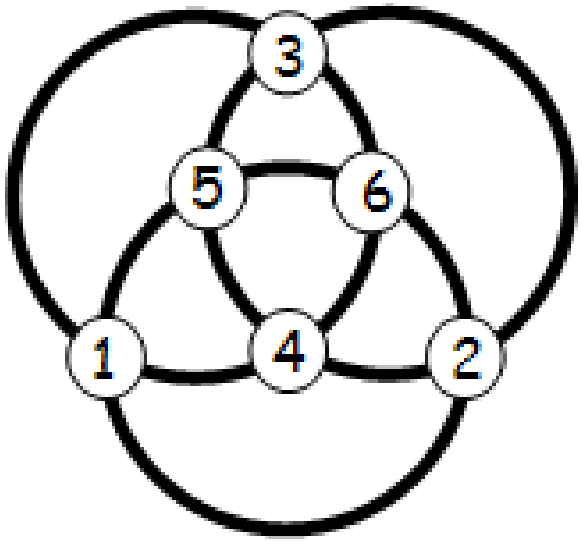


h(n) = cantidad de aros que no suman 14

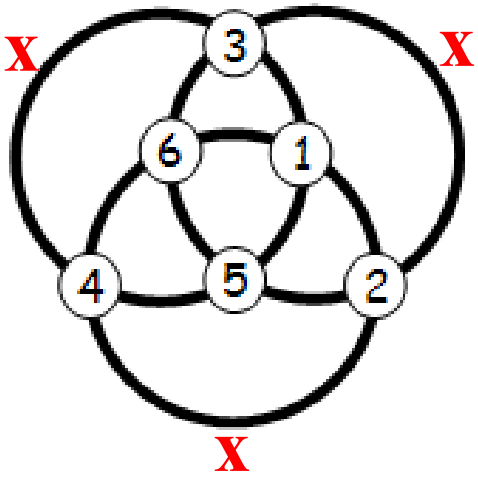
?

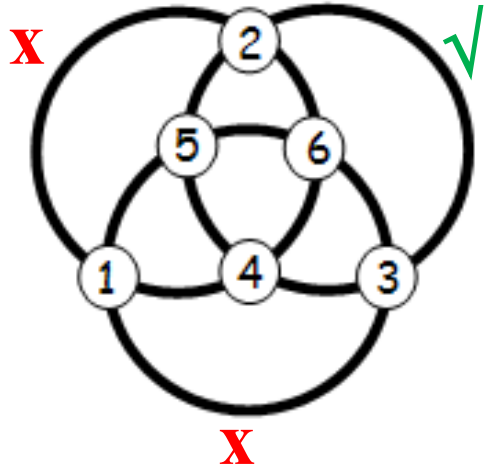
Búsqueda informada

Diseño de heurísticas



meta

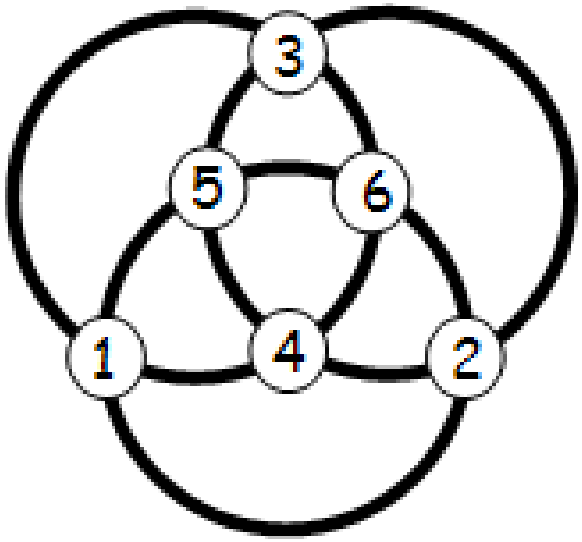




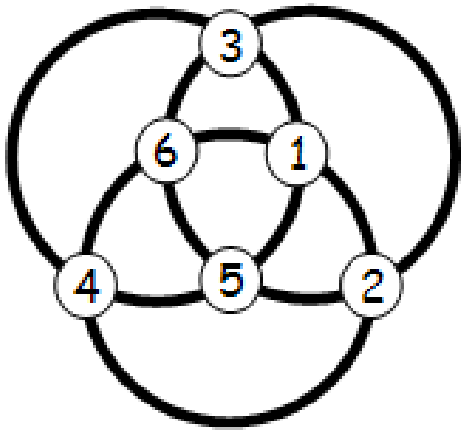
$$h(n) = \frac{\text{cantidad de aros que no suman 14}}{2}$$

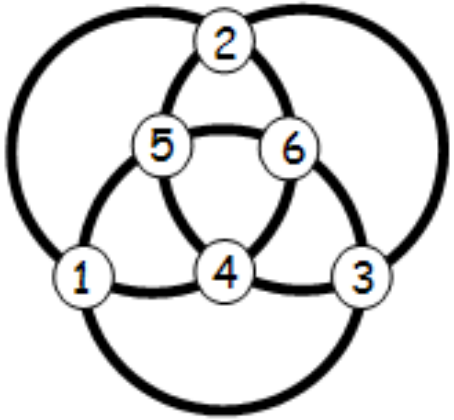
Búsqueda informada

Diseño de heurísticas



meta



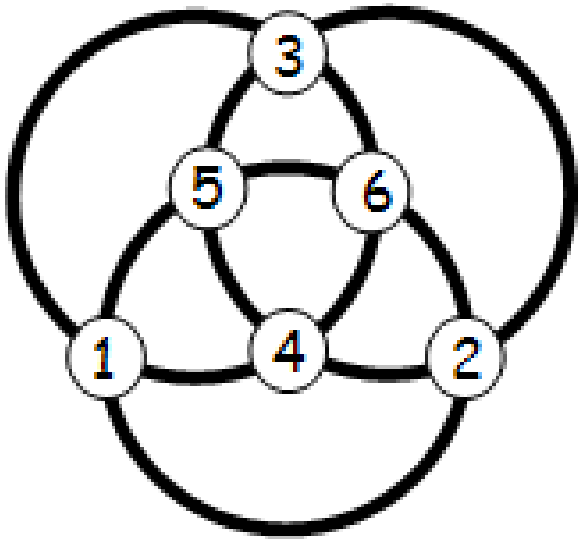


h(n) = $\frac{\text{cantidad de aros que no suman 14}}{2}$

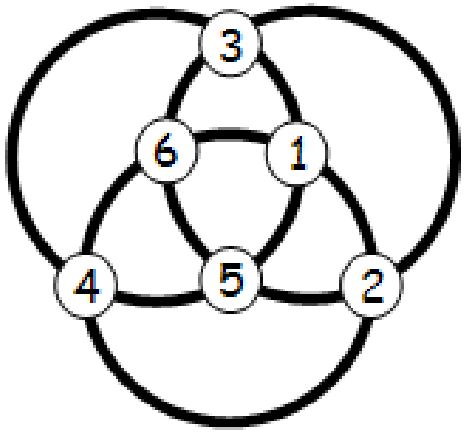
Se divide entre el mejor avance posible en un solo paso

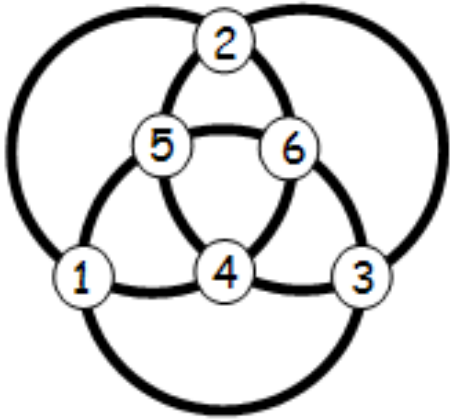
Búsqueda informada

Diseño de heurísticas



meta

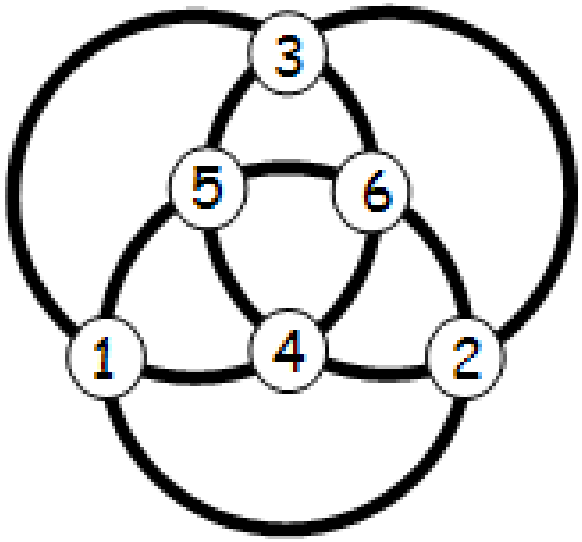




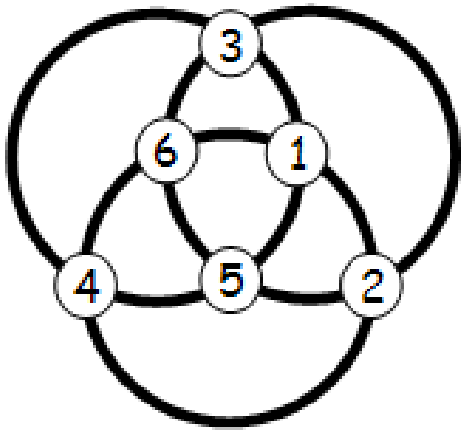
$$h(n) = \frac{\text{cantidad de aros que no suman 14}}{100}$$

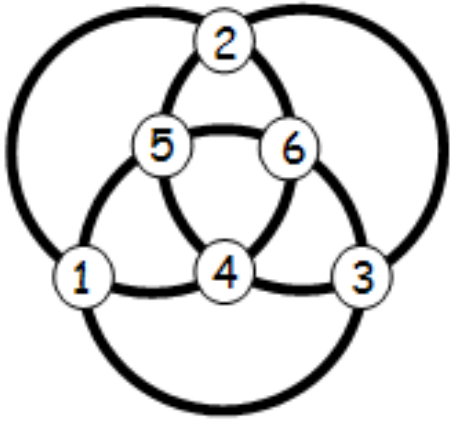
Búsqueda informada

Diseño de heurísticas



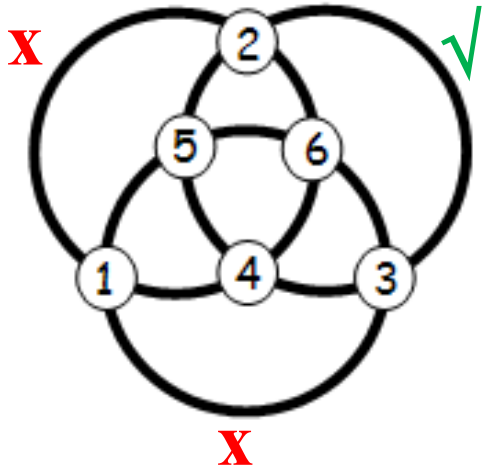
meta





$h_1(n) = \frac{\text{cantidad de aros que no suman 14}}{2}$

$h_2(n) = \frac{\text{cantidad de aros que no suman 14}}{3}$

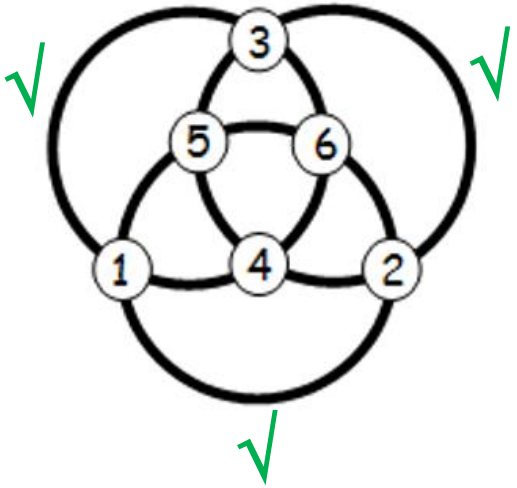


$h_1(n) = (2/2) = 1$

$h_2(n) = (2/3) = 0.66$

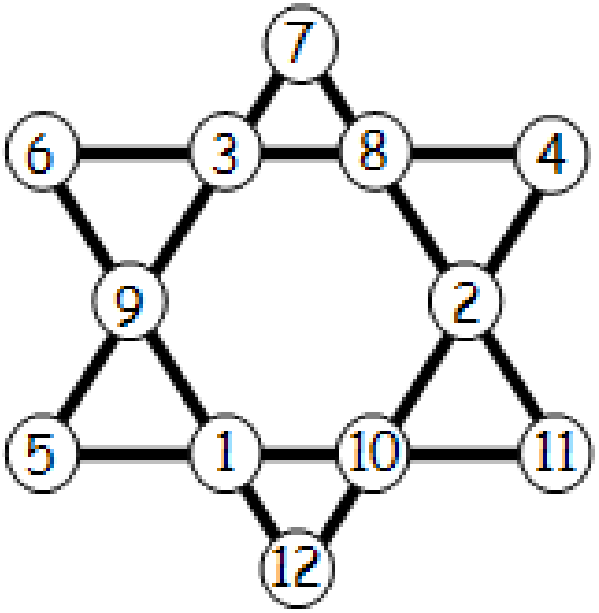
VR = 1

intercambio(2,3)



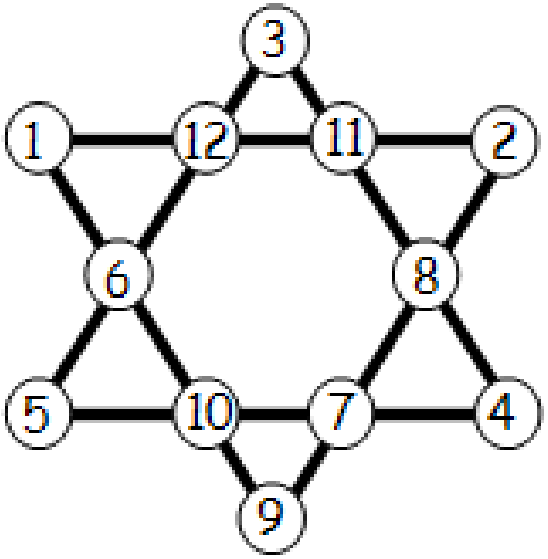
Diseño de heurísticas

- **El número 26**. Utilizando el operador *intercambio(a,b)* que cambia de posición los números a y b, colocar los números de tal forma que sobre cada lado la suma sea 26

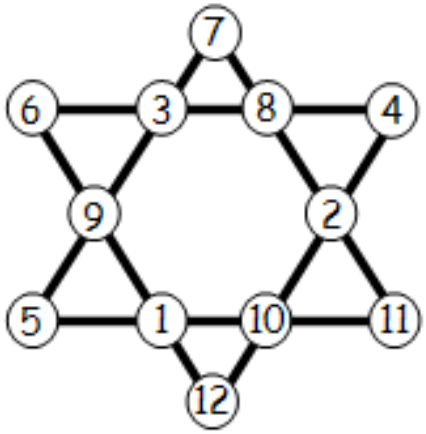


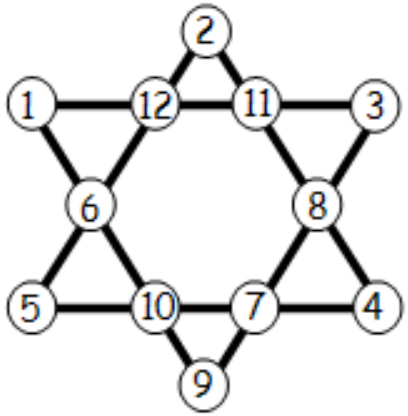
Búsqueda informada

Diseño de heurísticas



meta



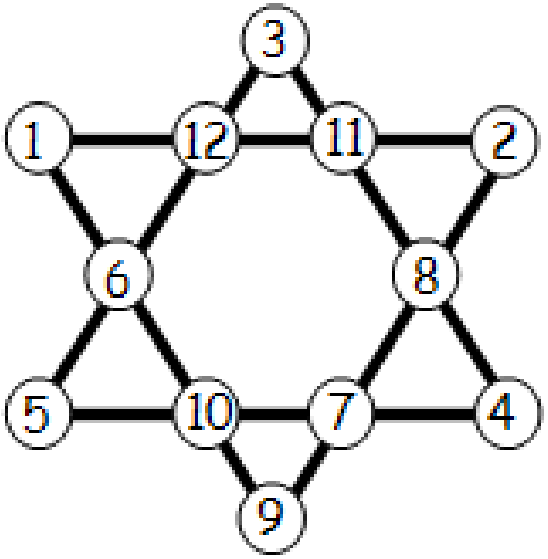


¿Cuál está más distante de la meta?

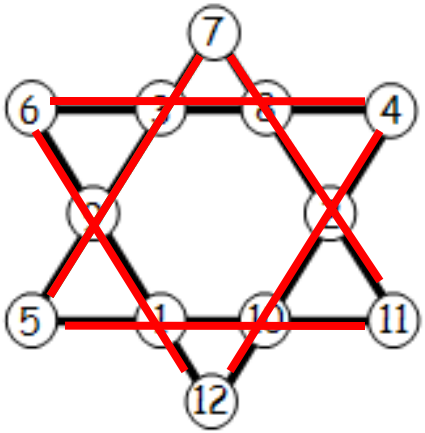
¿Cómo lo sabe?

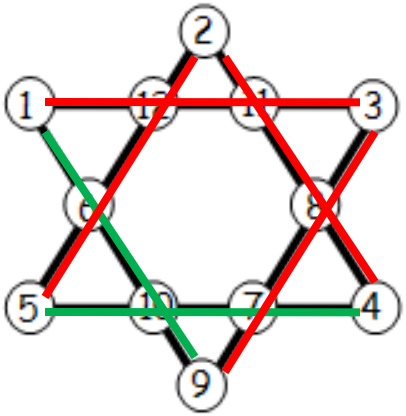
Búsqueda informada

Diseño de heurísticas



meta



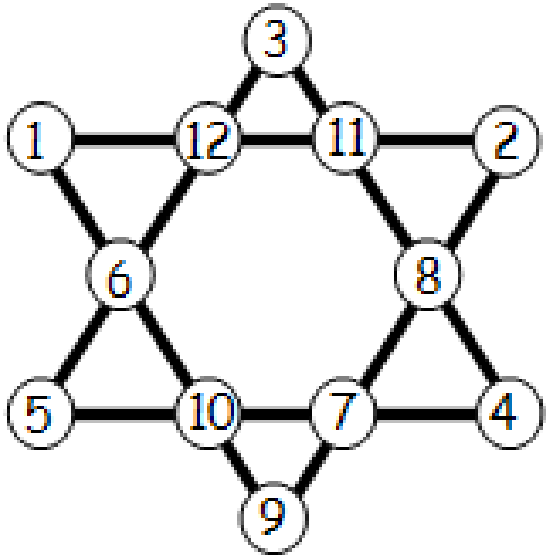


¿Cuál está más distante de la meta?

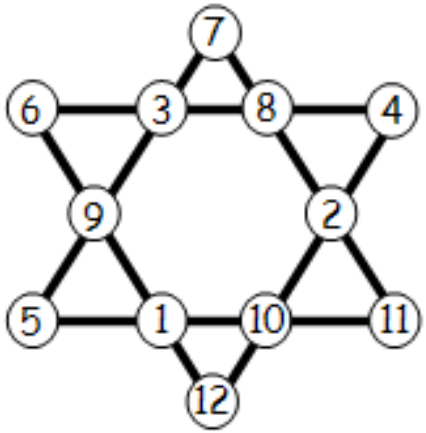
¿Cómo lo sabe?

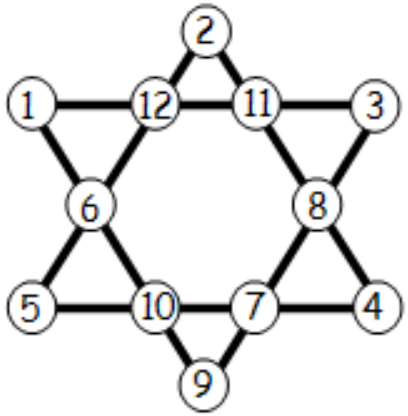
Búsqueda informada

Diseño de heurísticas



meta



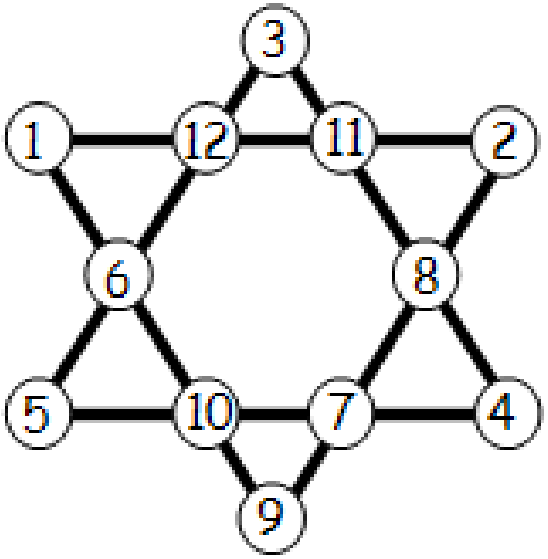


h(n) = cantidad de lados que no suman 26

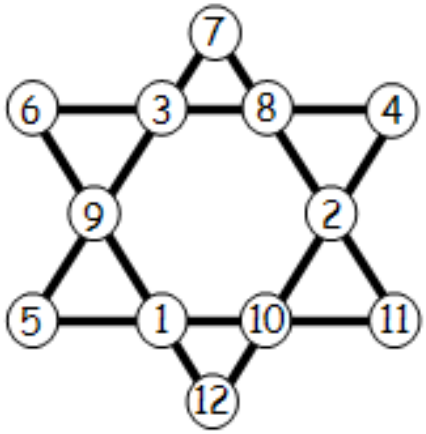
?

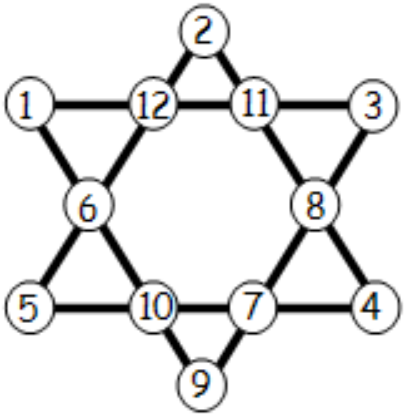
Búsqueda informada

Diseño de heurísticas



meta

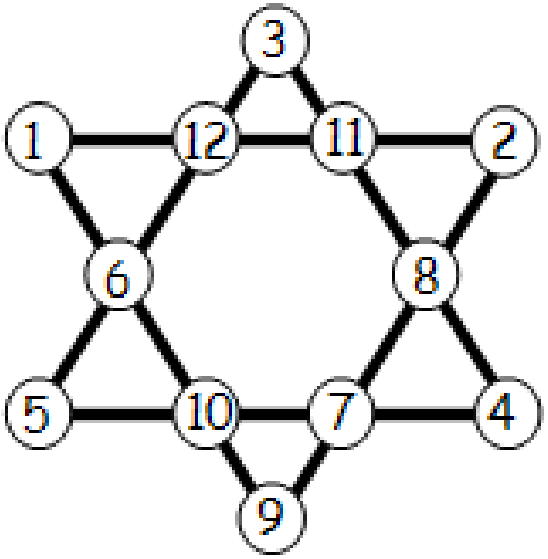




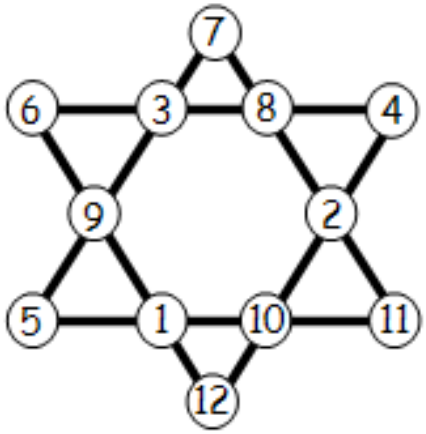
$$h(n) = \frac{\text{cantidad de lados que no suman 26}}{4}$$

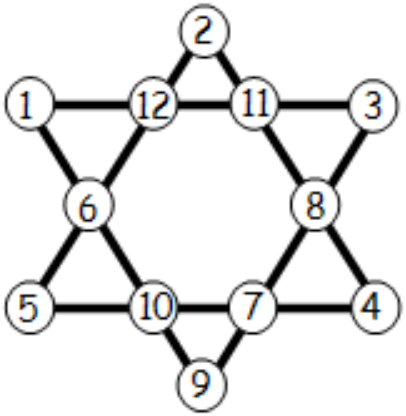
Búsqueda informada

Diseño de heurísticas



meta

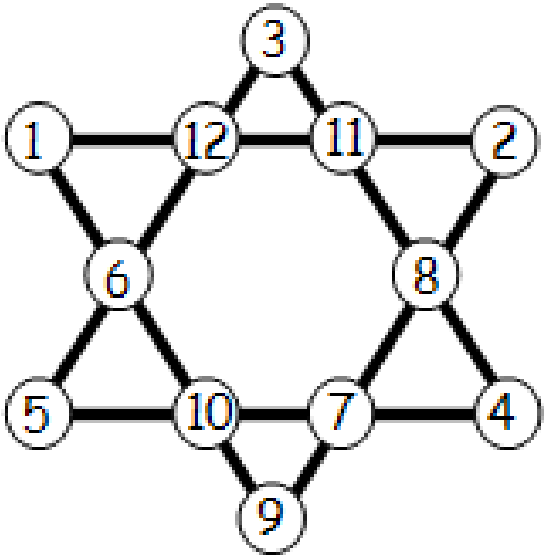




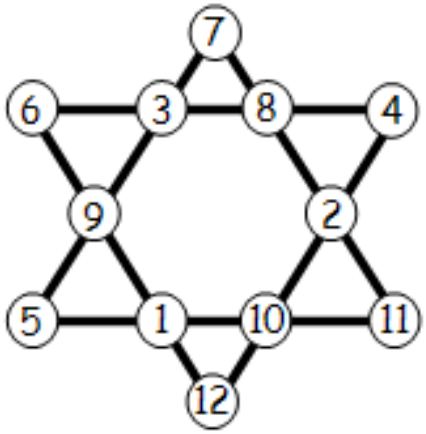
$$h(n) = \frac{\text{cantidad de lados que no suman 26}}{100}$$

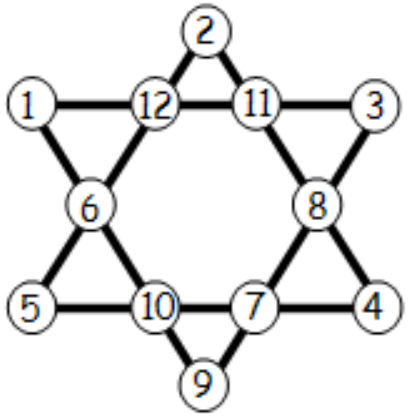
Búsqueda informada

Diseño de heurísticas



meta

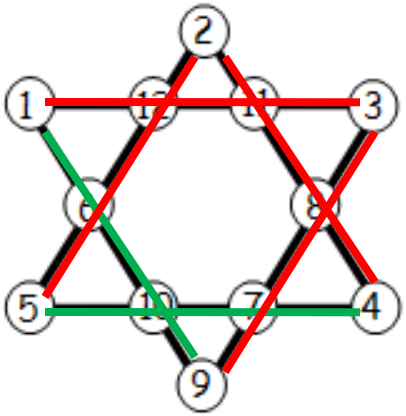




$h_1(n) = \frac{\text{cantidad de lados que no suman 26}}{4}$

$h_2(n) = \frac{\text{cantidad de lados que no suman 26}}{3}$

$h_3(n) = \frac{\text{cantidad de lados que no suman 26}}{5}$



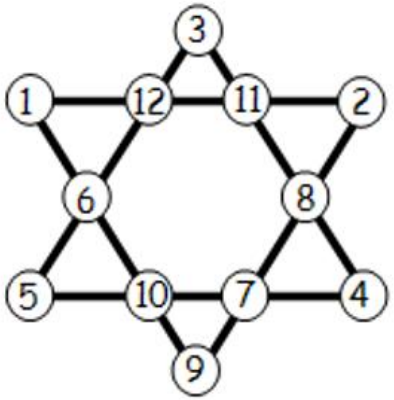
$h_1(n) = (4/4) = 1$

$h_2(n) = (4/3) = 1.33$

$h_3(n) = (4/5) = 0.8$

VR = 1

intercambio(2,3)



Búsqueda informada

Problema. El cuadrado mágico

La suma de cada fila, columna y las diagonales debe ser 15. Los números en las celdas coloreadas no se pueden mover. En cada avance del juego se puede realizar la operación **intercambiar**(x,y) que permite intercambiar de posición los números x y y. Cada avance tiene costo de una unidad.

7	1	8
2	5	3
6	9	4

Búsqueda informada

Problema. El cuadrado mágico

6	1	8
7	5	3
2	9	4

meta

7	1	2
8	5	3
6	9	4

6	1	7
8	5	3
2	9	4

¿Cuál está más distante de la meta?

¿Cómo lo sabe?

Búsqueda informada

Problema. El cuadrado mágico

6	1	8
7	5	3
2	9	4

meta

	x	x	√	x	x
x	7	1	2	
x	8	5	3	
x	6	9	4	

√	x	√	x	x
x	6	1	7	
x	8	5	3	
√	2	9	4	

¿Cuál está más distante de la meta?

¿Cómo lo sabe?

Búsqueda informada

Problema. El cuadrado mágico

6	1	8
7	5	3
2	9	4

meta

	x	x	√	x	x
x	7	1	2	
x	8	5	3	
x	6	9	4	

√	x	√	x	x
x	6	1	7	
x	8	5	3	
√	2	9	4	

$$h(n) = \frac{\text{cantidad de filas, columnas y diagonales que no suman 15}}{?}$$

Búsqueda informada

Problema. El cuadrado mágico

6	1	8
7	5	3
2	9	4

meta

	x	x	√	x	x
x	7	1	2	
x	8	5	3	
x	6	9	4	

√	x	√	x	x
x	6	1	7	
x	8	5	3	
√	2	9	4	

$$h(n) = \frac{\text{cantidad de filas, columnas y diagonales que no suman 15}}{5}$$

Búsqueda informada

Problema. El cuadrado mágico

6	1	8
7	5	3
2	9	4

meta

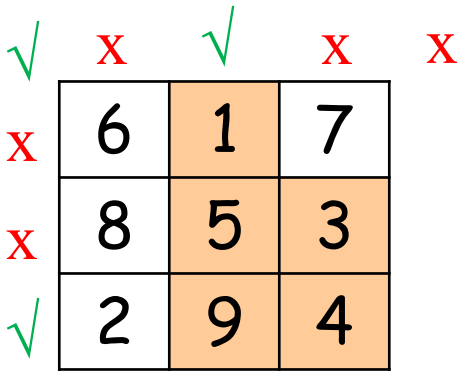
	x	x	√	x	x
x	7	1	2	
x	8	5	3	
x	6	9	4	

√	x	√	x	x
x	6	1	7	
x	8	5	3	
√	2	9	4	

$$h_1(n) = \frac{\text{fil, col, diag que no suman 15}}{5}$$

$$h_2(n) = \frac{\text{fil, col, diag que no suman 15}}{4}$$

$$h_3(n) = \frac{\text{fil, col, diag que no suman 15}}{6}$$



$h_1(n) = (5/5) = 1$

$h_2(n) = (5/4) = 1.25$

$h_3(n) = (5/6) = 0.83$

VR = 1

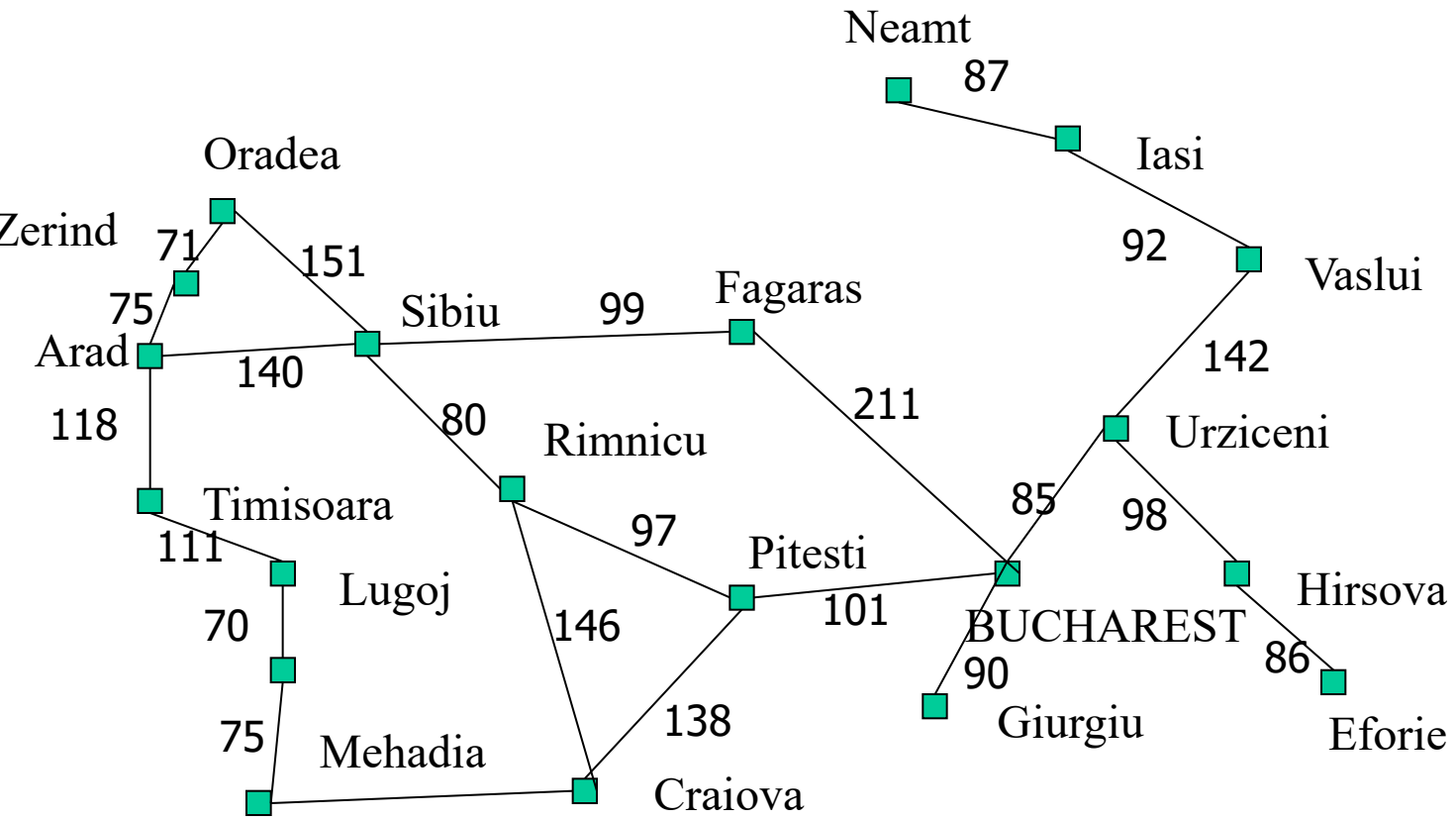
intercambio(7,8)

6	1	8
7	5	3
2	9	4

Búsqueda informada

Heurísticas y desempeño

Búsqueda informada

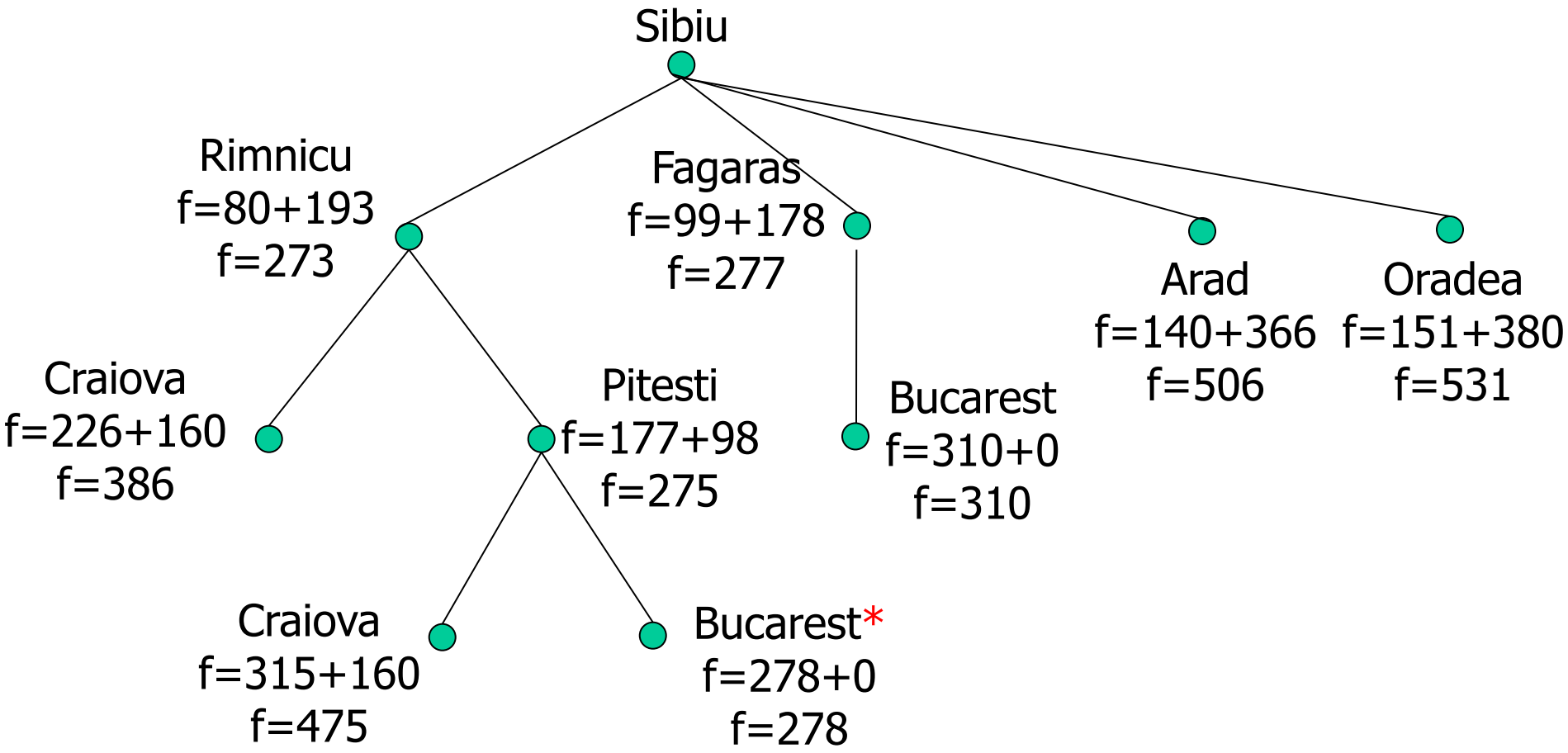


Problema: ir de Sibiu a Bucarest

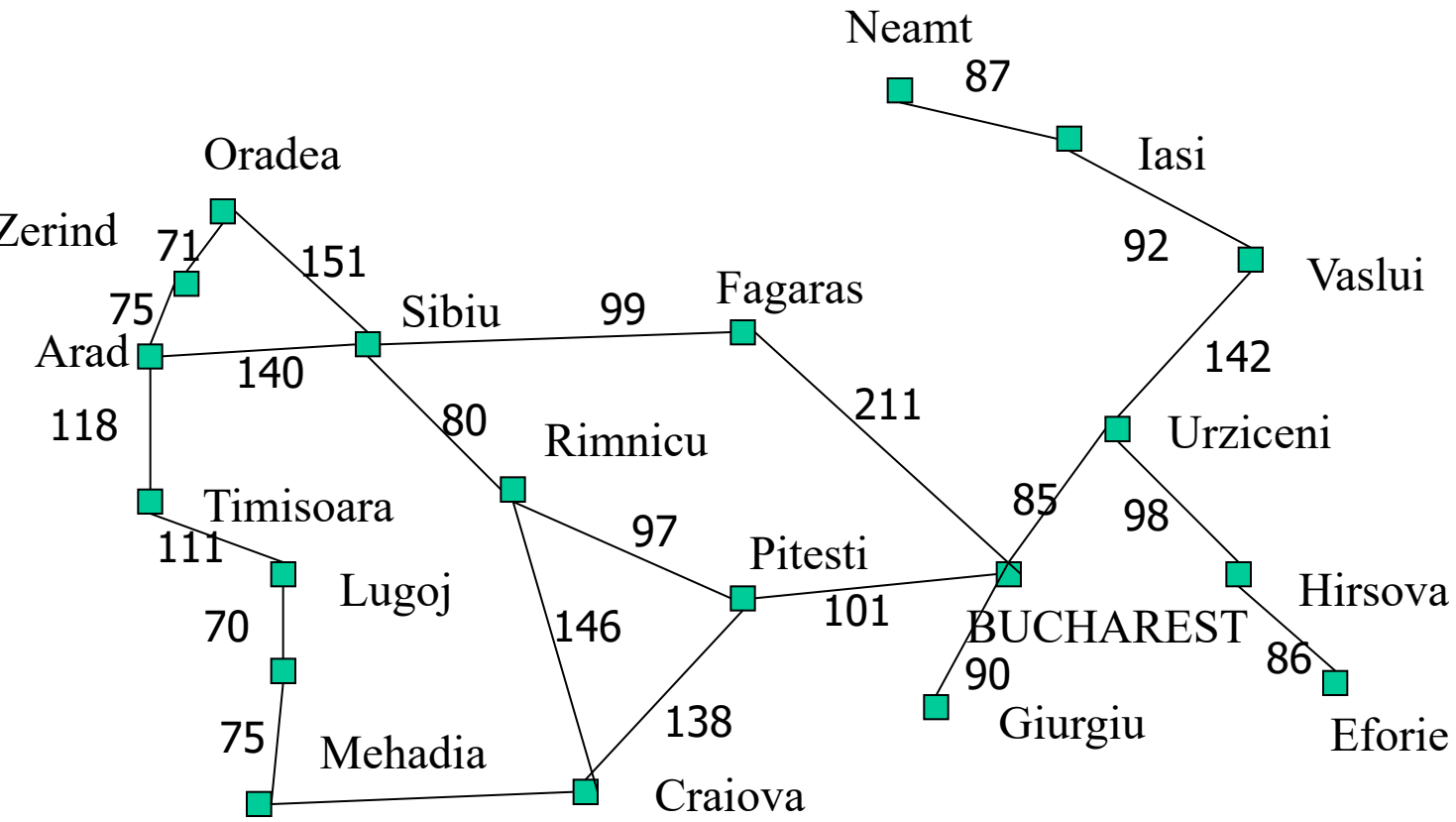
Distancia en línea recta a Bucarest:

Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Búsqueda informada



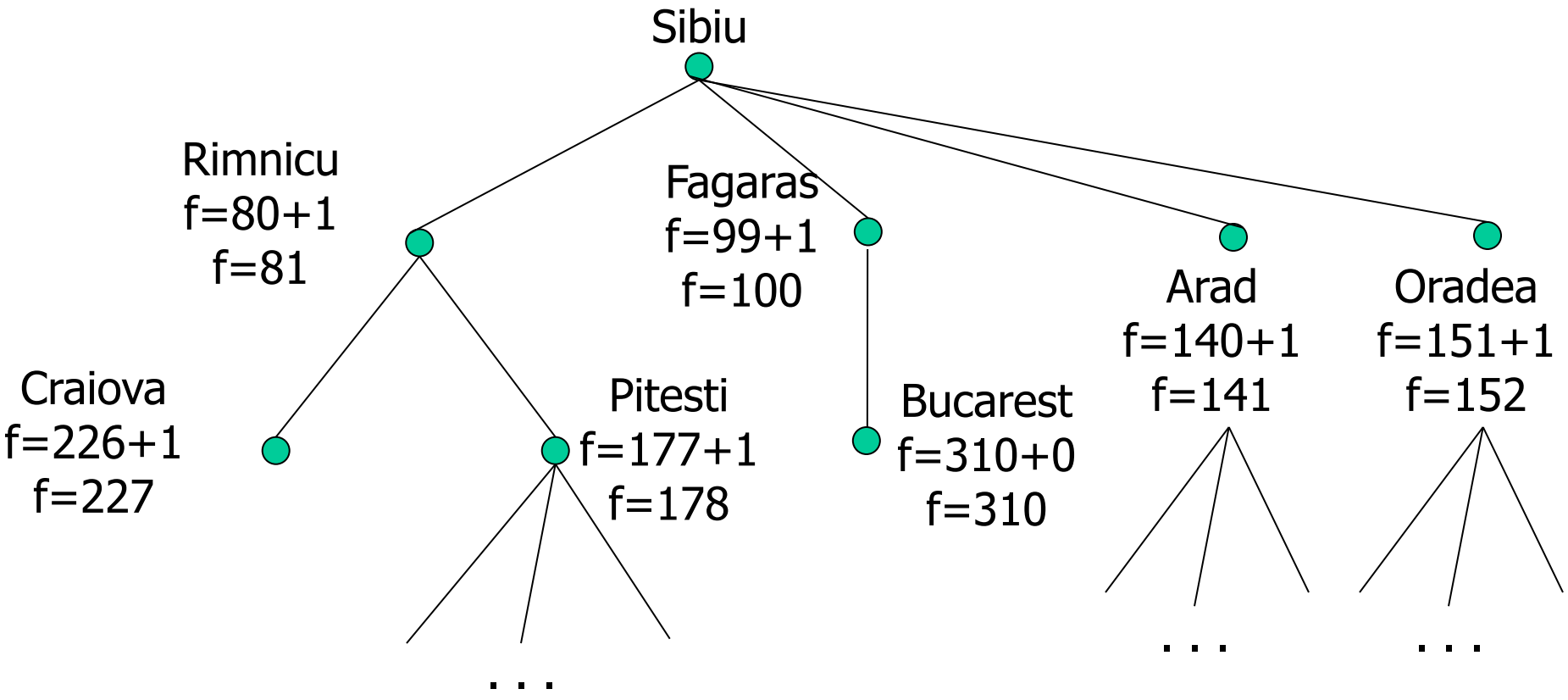
Búsqueda informada



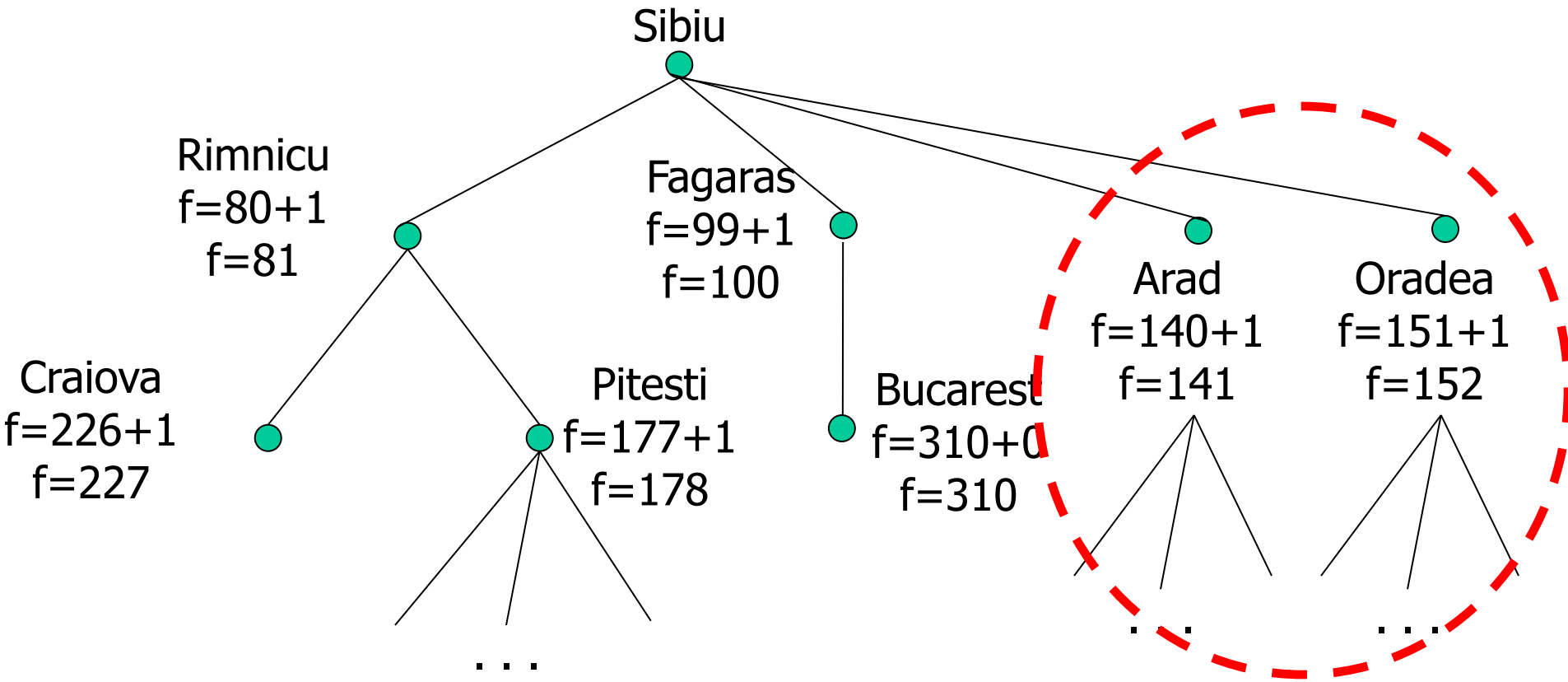
Distancia mínima a Bucarest:

Arad	1
Bucarest	0
Craiova	1
Dobreta	1
Eforie	1
Fagaras	1
Giurgiu	1
Hirsova	1
Iasi	1
Lugoj	1
Mehadia	1
Neamt	1
Oradea	1
Pitesti	1
Rimnicu	1
Sibiu	1
Timisoara	1
Urziceni	1
Vaslui	1
Zerind	1

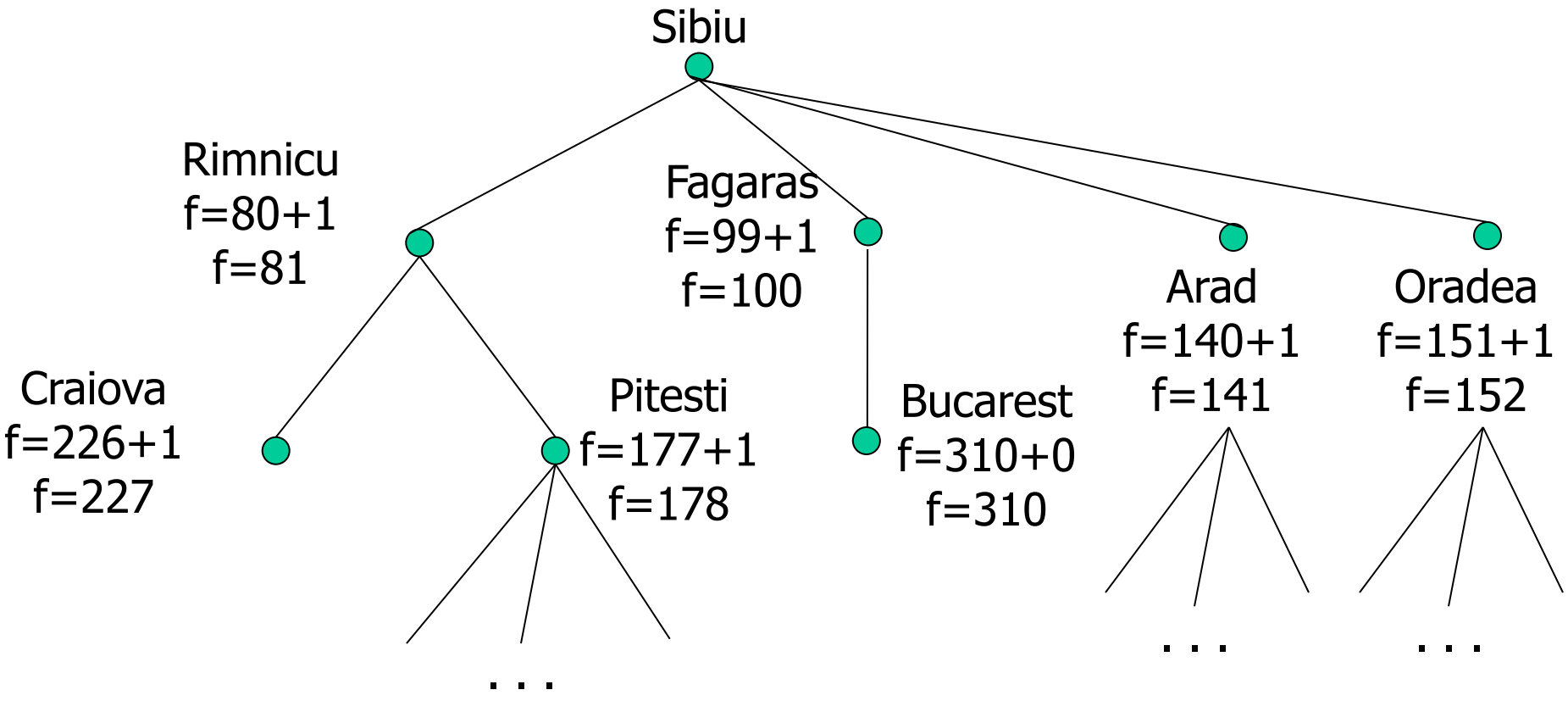
Búsqueda informada



Búsqueda informada



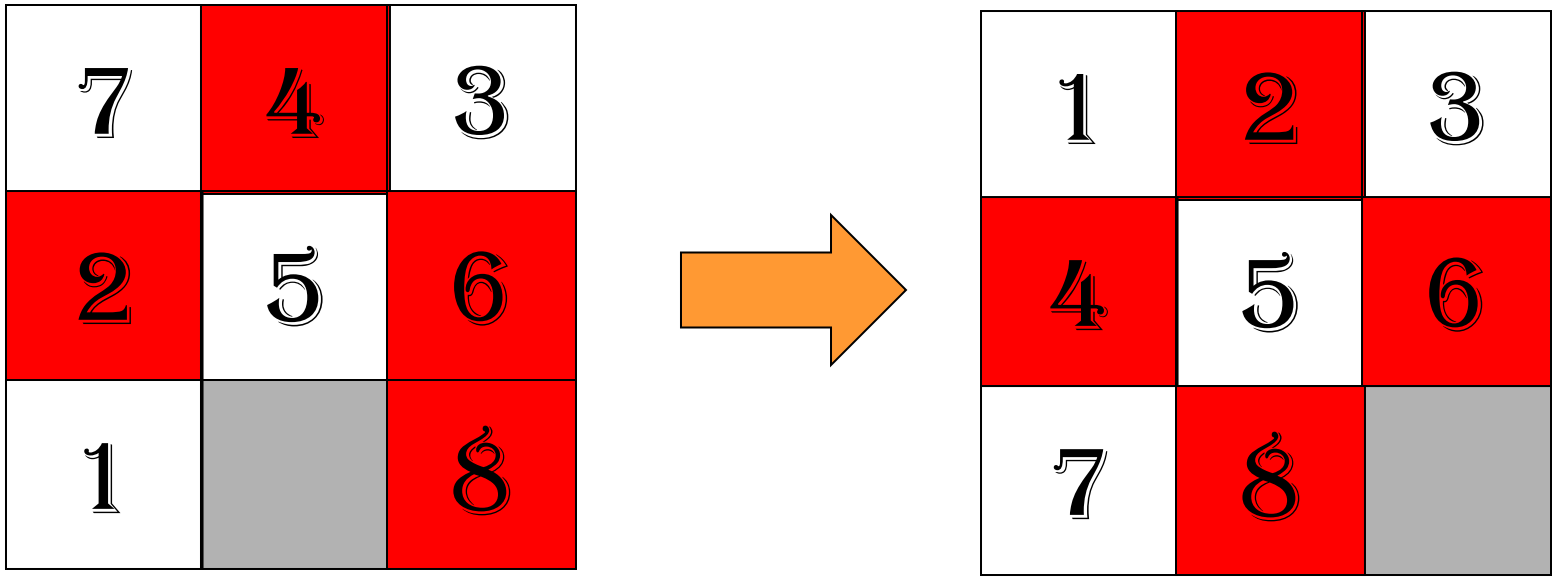
Búsqueda informada



Si los valores de la heurística son muy pequeños, A* se vuelve como Costo uniforme

Búsqueda informada

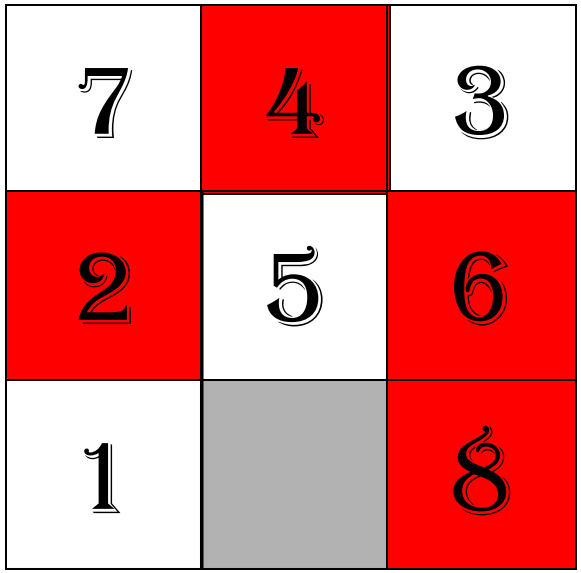
8-puzzle

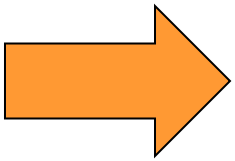


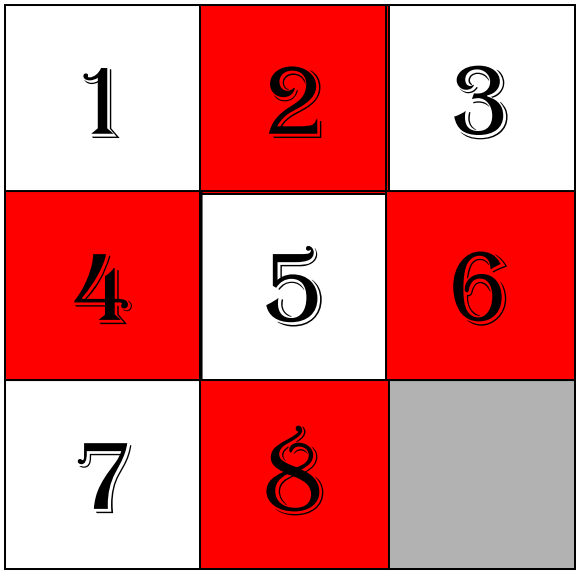
$h_1(n)$: cantidad de piezas ubicadas en el lugar incorrecto (sin incluir la placa vacía)

Búsqueda informada

8-puzzle



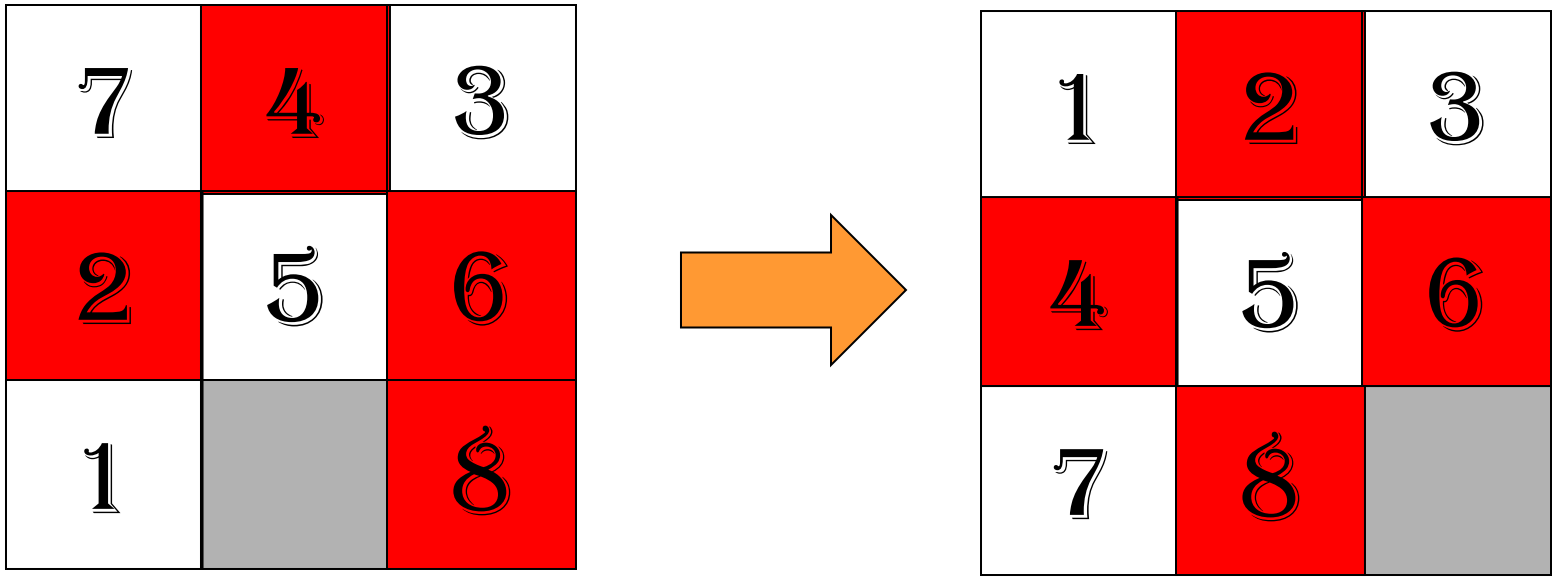




$h_1(n)$: 5

Búsqueda informada

8-puzzle

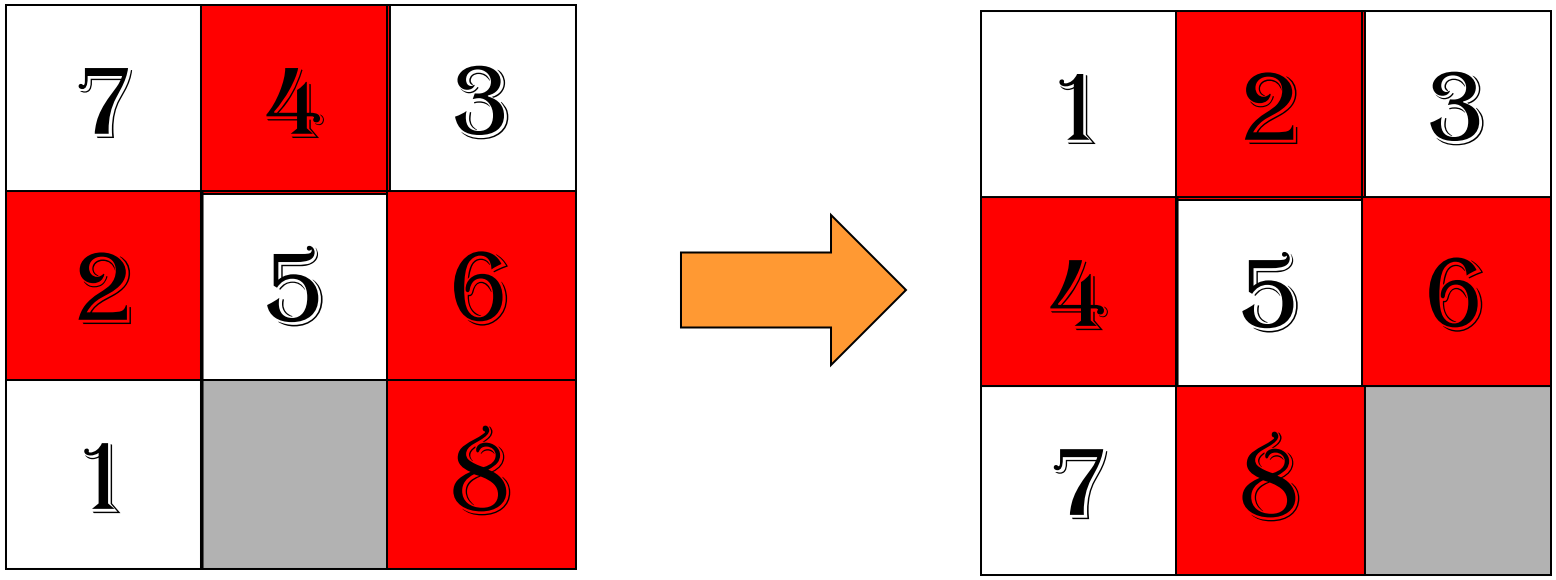


$h_1(n)$: 5

Realmente son muchos más que 5 movimientos
El valor estimado está muy por debajo del valor real

Búsqueda informada

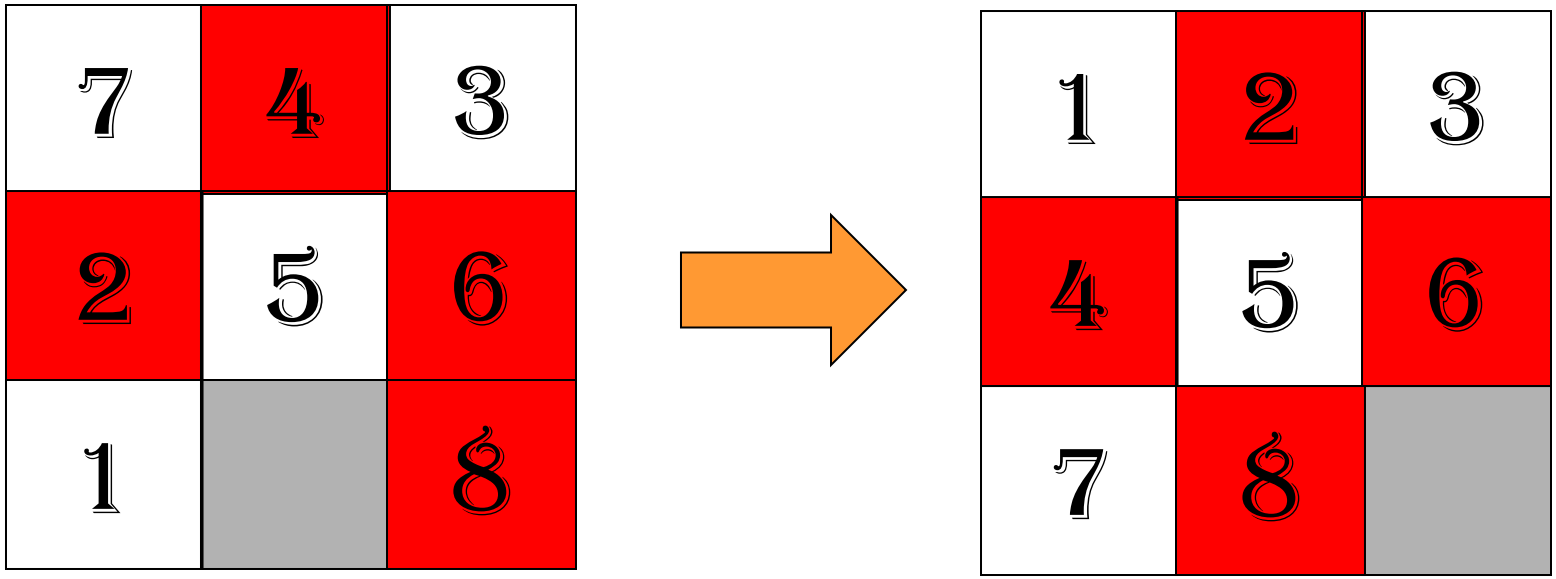
8-puzzle



$h_2(n)$: suma de las distancias de Manhattan de cada pieza a su posición correcta (sin incluir la placa vacía)

Búsqueda informada

8-puzzle



$h_2(n)$: 2+2+0+2+0+0+2+1=9

Búsqueda informada

8-puzzle

$h_1(n)$: cantidad de piezas ubicadas en el lugar incorrecto

$h_2(n)$: suma de la distancias horizontales y verticales que separan a las placas de las posiciones meta (excluyendo la vacía). **Distancia de Manhattan**

Búsqueda informada

8-puzzle

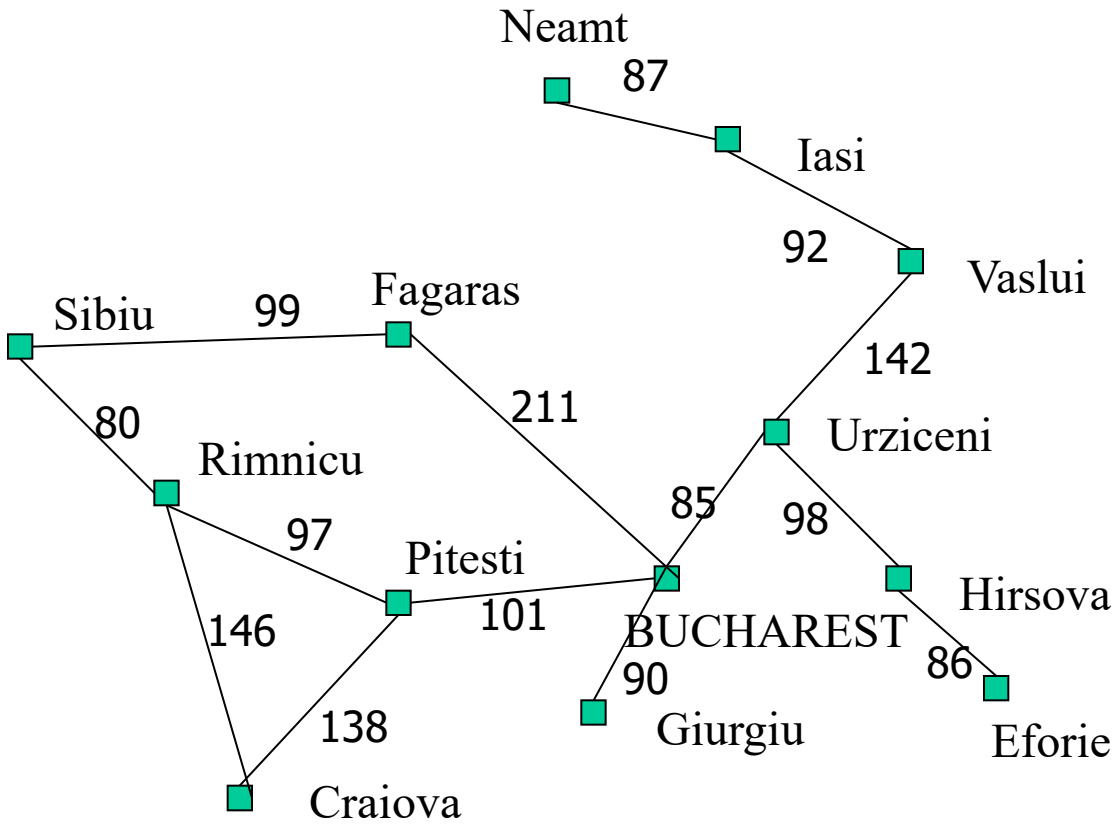
$h_1(n)$: cantidad de piezas ubicadas en el lugar incorrecto
$h_2(n)$: suma de las distancias de **Manhattan**

	Nodos expandidos		Factor de ramificación efectivo	
d	$A^*(h_1)$	$A^*(h_2)$	$A^*(h_1)$	$A^*(h_2)$
2	6	6	1.79	1.79
4	13	12	1.48	1.45
6	20	18	1.34	1.30
8	39	25	1.33	1.24
10	93	39	1.38	1.22
12	227	73	1.42	1.24
14	539	113	1.44	1.23
16	1,301	211	1.45	1.25
18	3,056	363	1.46	1.26
20	7,276	676	1.47	1.27
22	18,094	1,219	1.48	1.28
24	39,135	1,641	1.48	1.26

Heurísticas y desempeño

- Si para todo nodo n, $h_2(n) \geq h_1(n)$, se dice que **h_2 domina a h_1**
- En general, conviene más emplear una función heurística con valores más altos, siempre y cuando no dé lugar a una sobreestimación

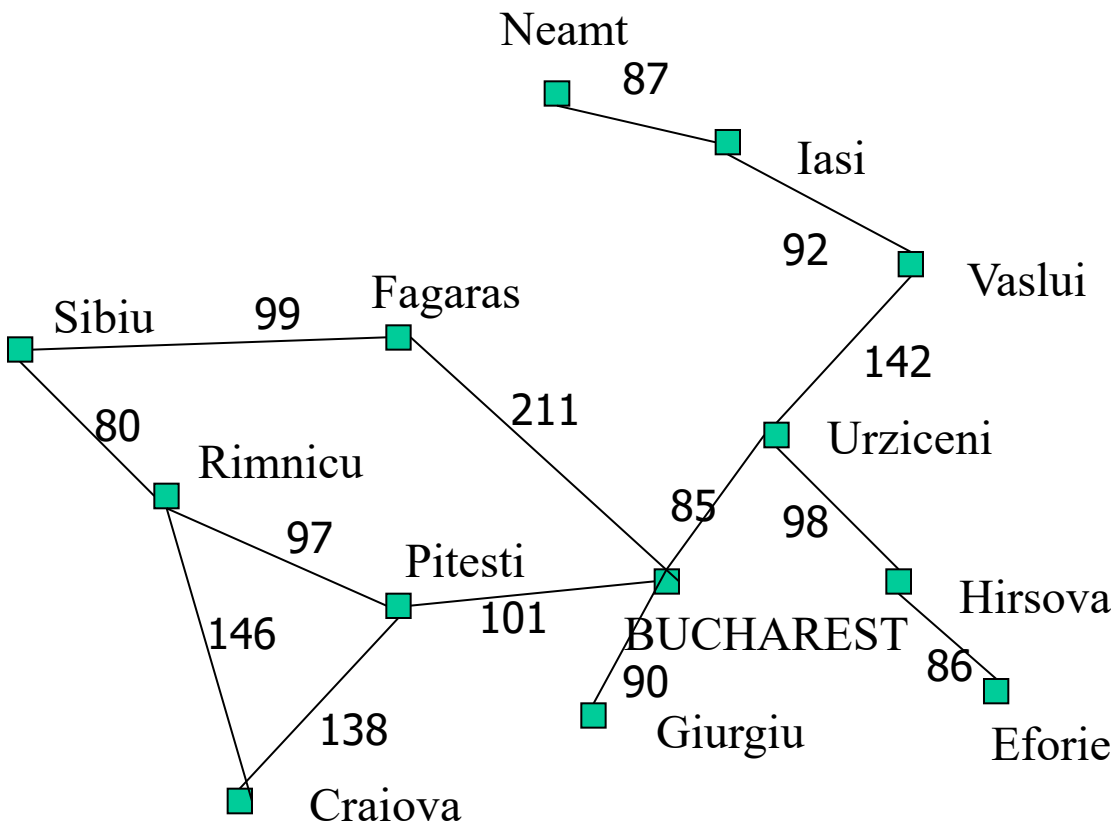
Búsqueda informada



Distancia en línea recta a Bucarest:	
Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Distancia mínima Bucarest:	
Arad	1
Bucarest	0
Craiova	1
Dobreta	1
Eforie	1
Fagaras	1
Giurgiu	1
Hirsova	1
Iasi	1
Lugoj	1
Mehadia	1
Neamt	1
Oradea	1
Pitesti	1
Rimnicu	1
Sibiu	1
Timisoara	1
Urziceni	1
Vaslui	1
Zerind	1

Búsqueda informada



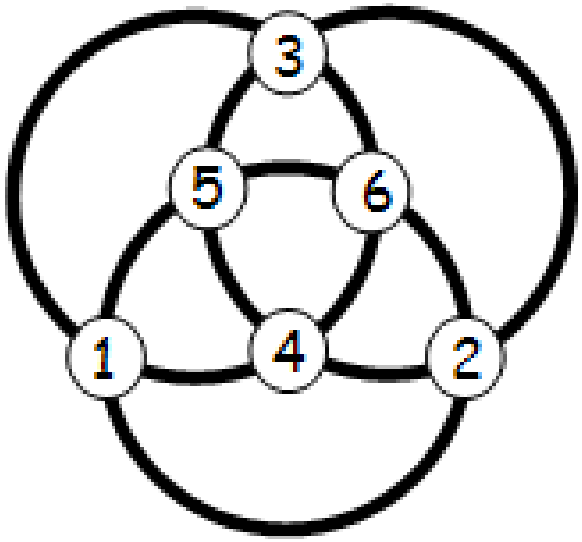
h_1 domina a h_2

Distancia en línea recta a Bucarest:	
Arad	366
Bucarest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
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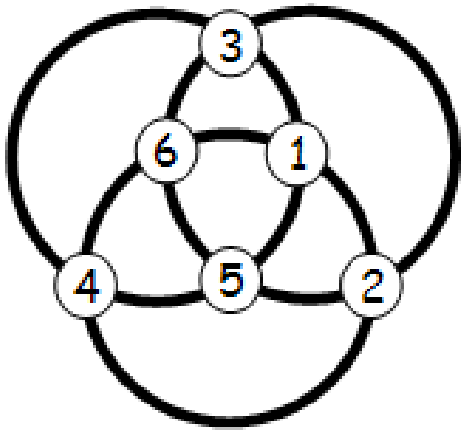
Distancia mínima Bucarest:	
Arad	1
Bucarest	0
Craiova	1
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Iasi	1
Lugoj	1
Mehadia	1
Neamt	1
Oradea	1
Pitesti	1
Rimnicu	1
Sibiu	1
Timisoara	1
Urziceni	1
Vaslui	1
Zerind	1

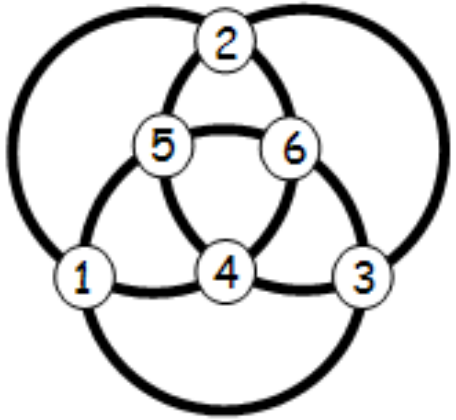
Búsqueda informada

Heurísticas dominantes



meta



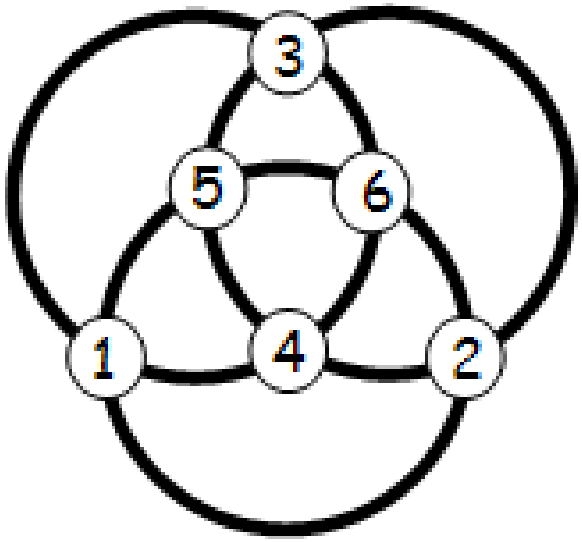


$$h_1(n) = \frac{\text{cantidad de aros que no suman 14}}{2}$$

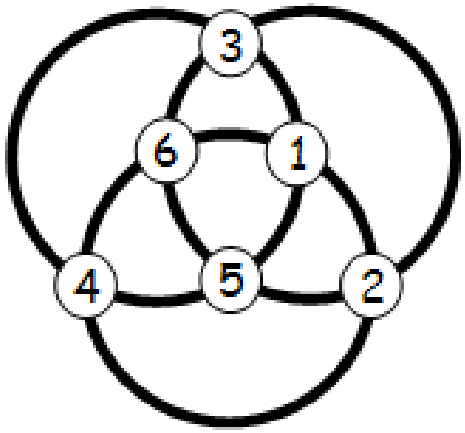
$$h_2(n) = \frac{\text{cantidad de aros que no suman 14}}{100}$$

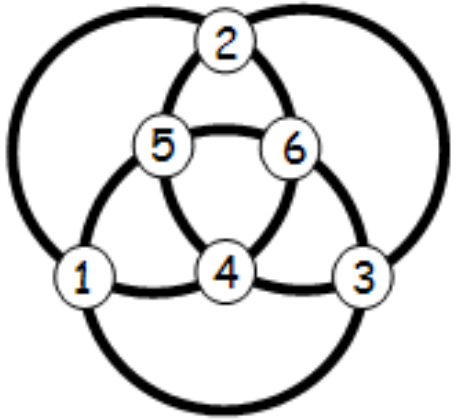
Búsqueda informada

Heurísticas dominantes



meta





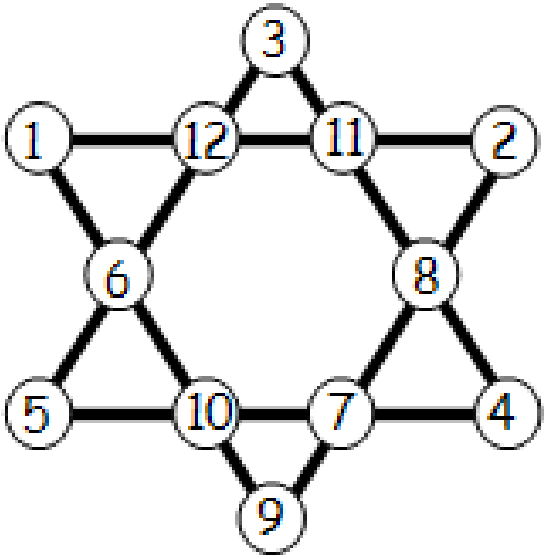
$$h_1(n) = \frac{\text{cantidad de aros que no suman 14}}{2}$$

$$h_2(n) = \frac{\text{cantidad de aros que no suman 14}}{100}$$

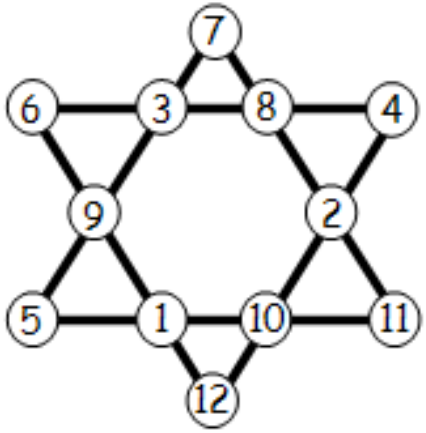
h_1 domina a h_2

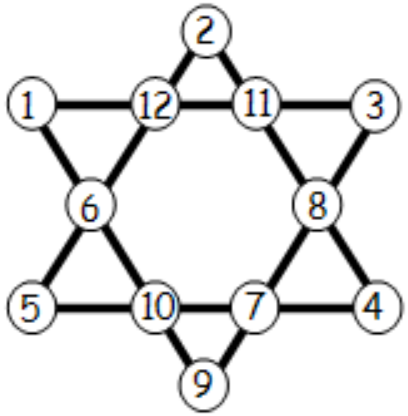
Búsqueda informada

Heurísticas dominantes



meta



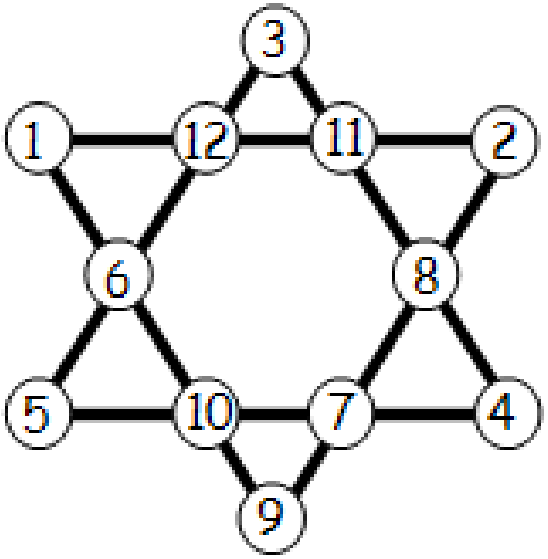


$$h_1(n) = \frac{\text{cantidad de lados que no suman 26}}{4}$$

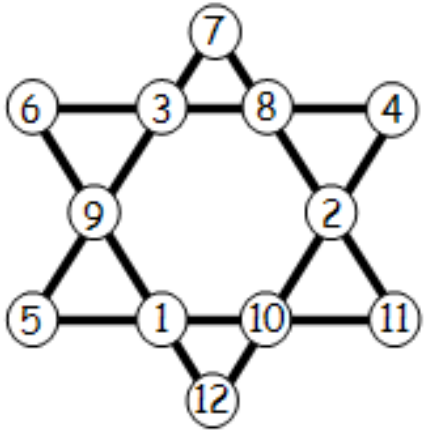
$$h_2(n) = \frac{\text{cantidad de lados que no suman 26}}{5}$$

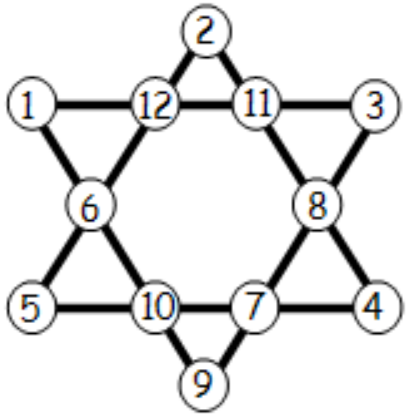
Búsqueda informada

Heurísticas dominantes



meta





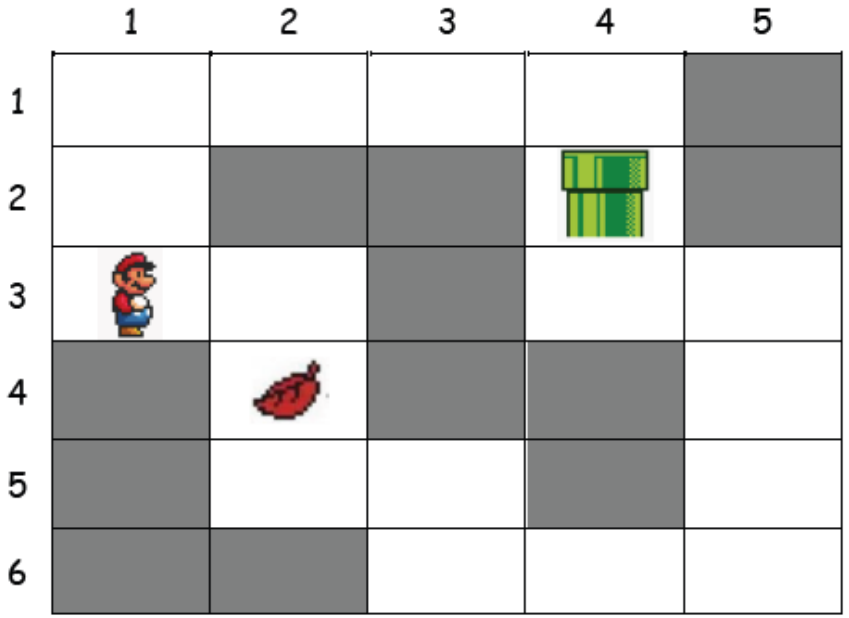
$h_1(n) = \frac{\text{cantidad de lados que no suman 26}}{4}$

$h_2(n) = \frac{\text{cantidad de lados que no suman 26}}{5}$

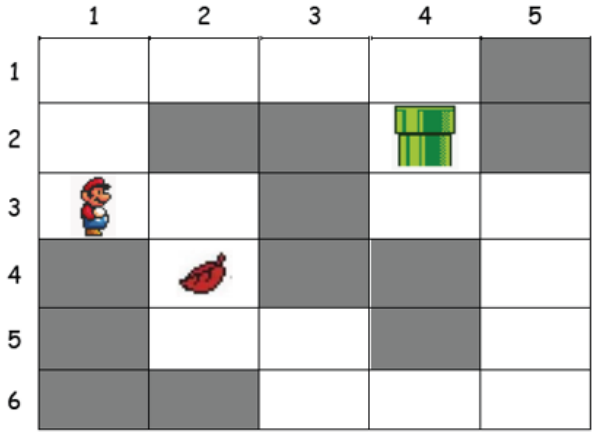
h_1 domina a h_2

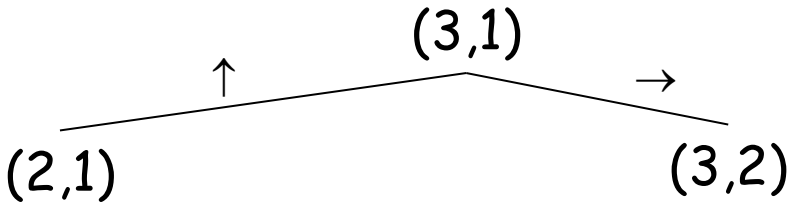
4. Búsqueda informada. Considere el siguiente ambiente representado por una matriz de 6x5 en la cual Mario inicia en la posición (3,1) y debe llegar a la posición (2,4). Los operadores que puede aplicar el agente son arriba, abajo, izquierda, y derecha. El costo de cada movimiento del agente es 1. Sin embargo, si Mario llega a la casilla (4,2) tomará la hoja que le permite volar y de allí en adelante el costo de cada movimiento será 1/3. Para resolver el problema se va a utilizar el algoritmo A* evitando devolverse. Tenga en cuenta que si el agente se encuentra en la posición (3,2) y aplica el operador "abajo", luego sí podrá regresar a la casilla (3,2) porque el estado del agente es diferente, antes no podía volar y ahora sí.

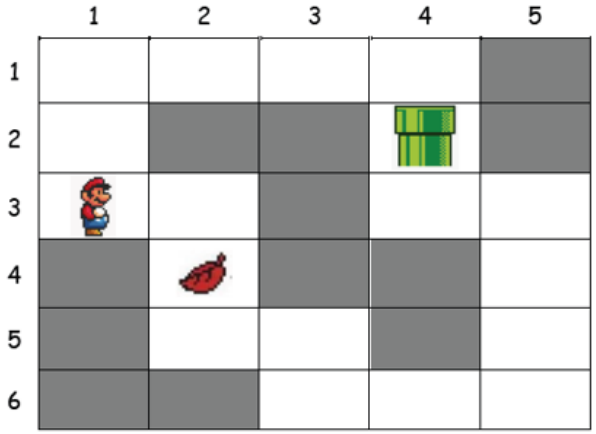
- **[4pts]** Defina una heurística que sea admisible para el ambiente que se muestra en la figura
- **[8pts]** Muestre el <u>árbol</u> y la <u>solución</u> que se encuentra con A*. Debe aplicar los operadores en el orden →, ↓, ↑, ←. Indique los valores de g y h en cada nodo. Evite devolverse

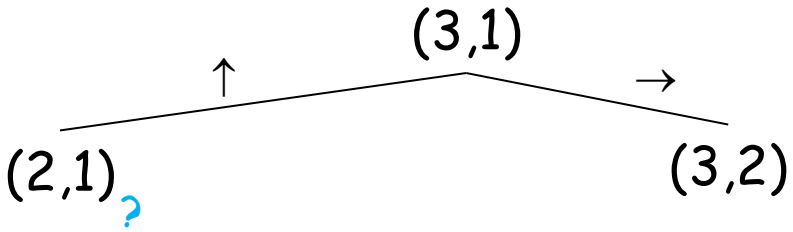


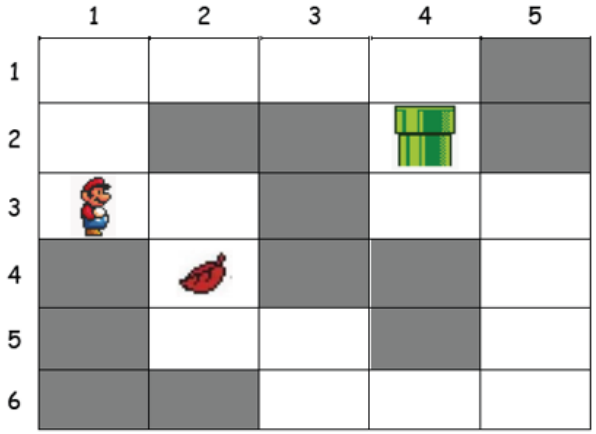
(3,1)

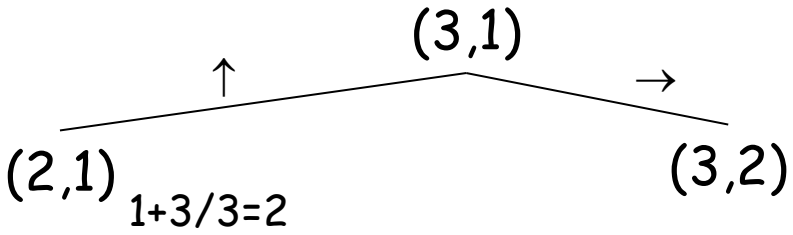


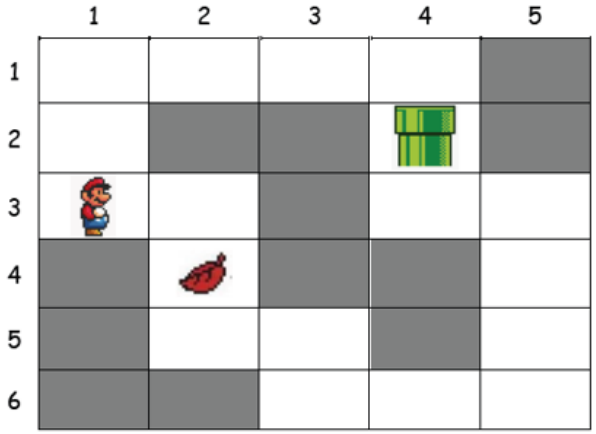

(3,1)
(2,1)
(3,2)

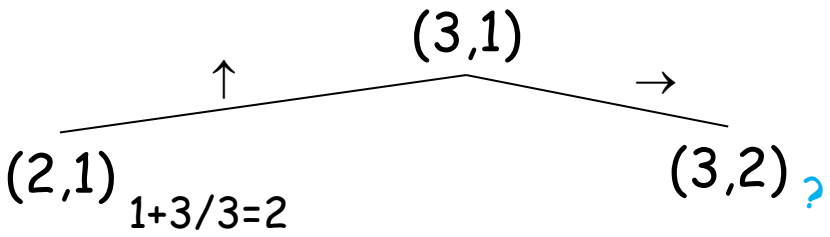

1
2
3
4
5
1
2
3
4
5
6

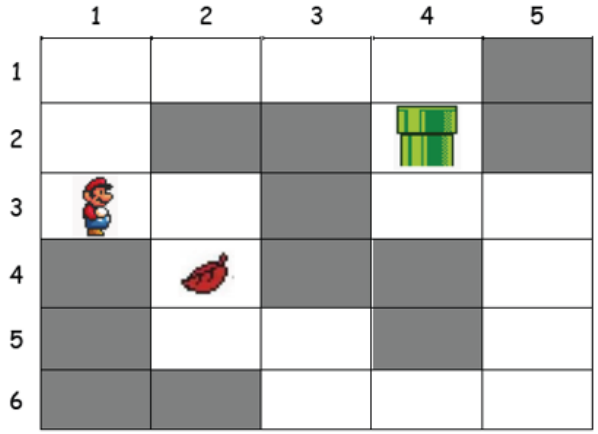

(3,1)
(2,1)
?
(3,2)

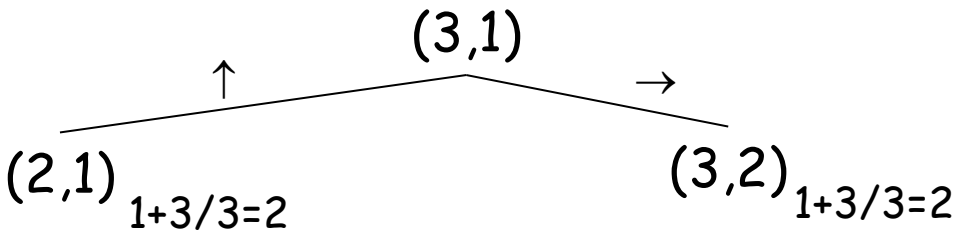

1
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1
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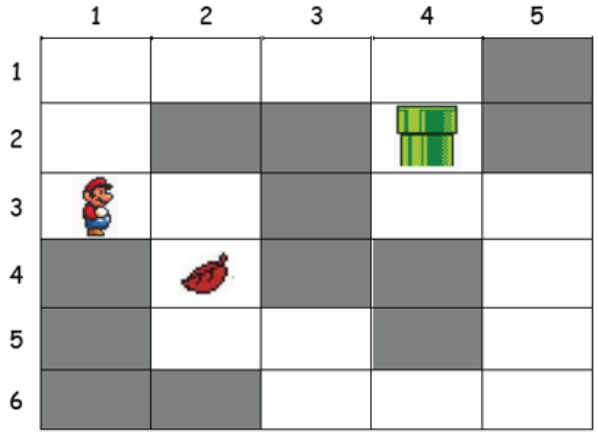

(3,1)
↑
→
(2,1)
(3,2)
1+3/3=2

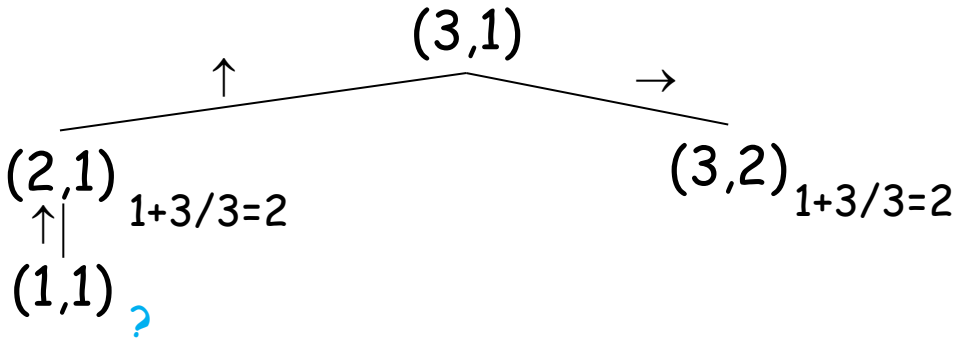

1
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1
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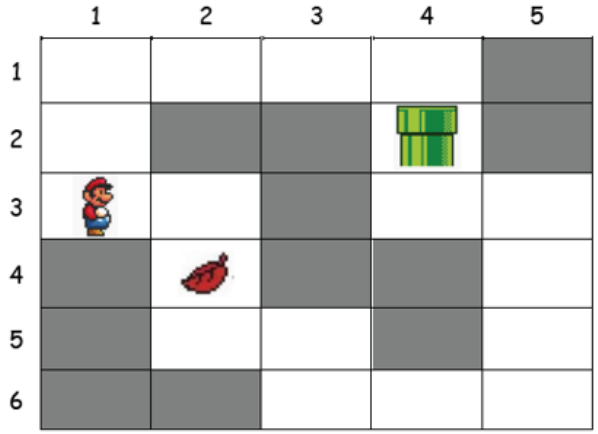

(3,1)
↑
→
(2,1)
(3,2)
1+3/3=2
?

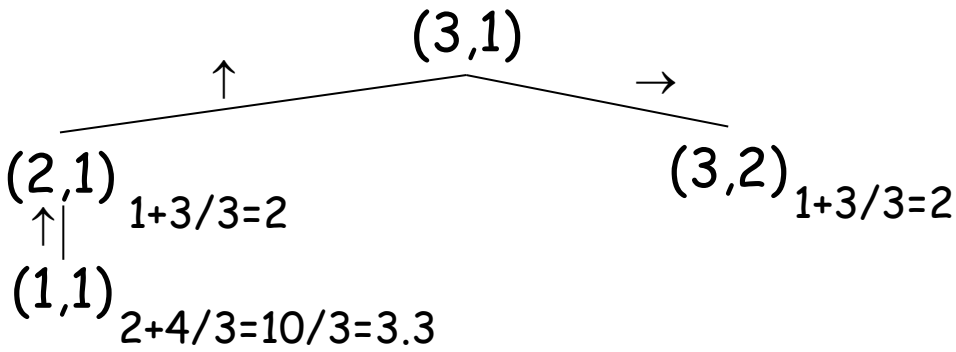

1
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1
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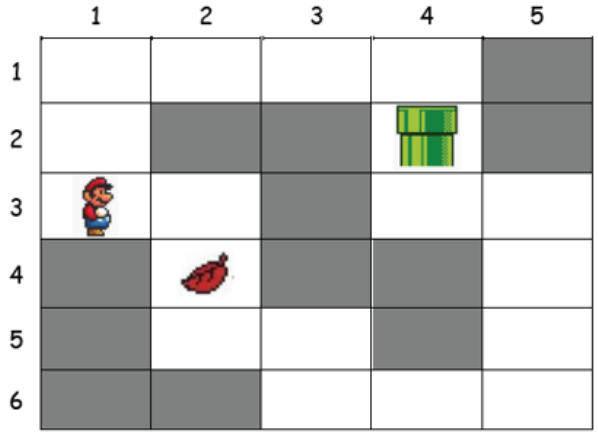

(3,1)
↑
→
(2,1)
1+3/3=2
(3,2)
1+3/3=2

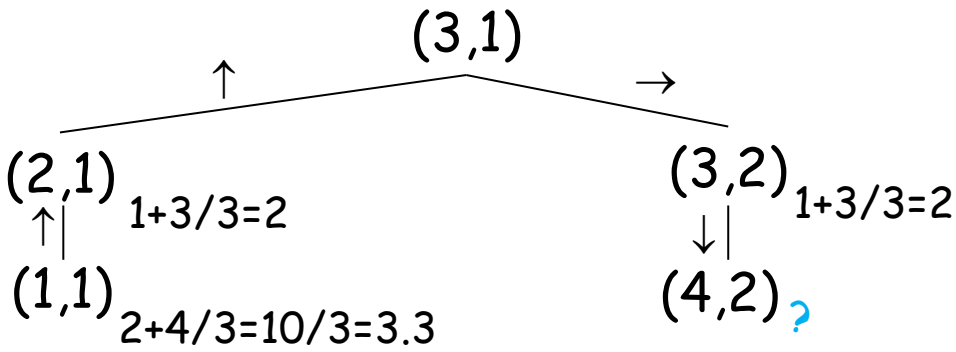

1
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1
2
3
4
5
6

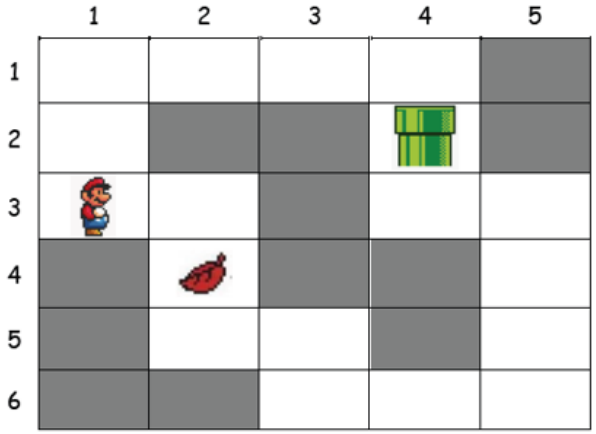

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
?

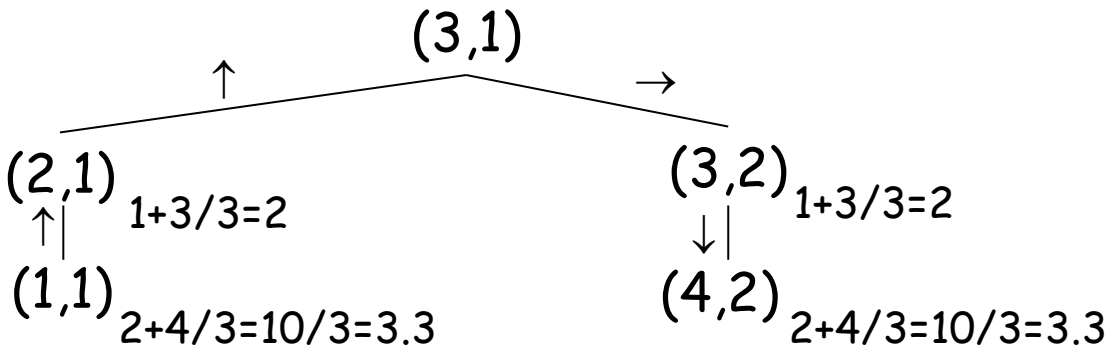

1
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4
5
1
2
3
4
5
6

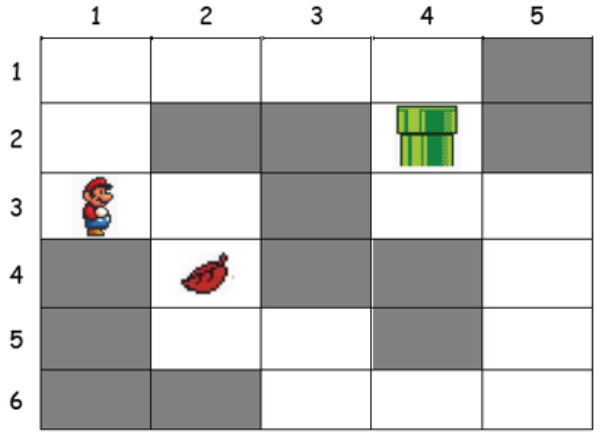

(3,1)
↑
→
(2,1)
1+3/3=2
(3,2)
1+3/3=2
↑
(1,1)
2+4/3=10/3=3.3

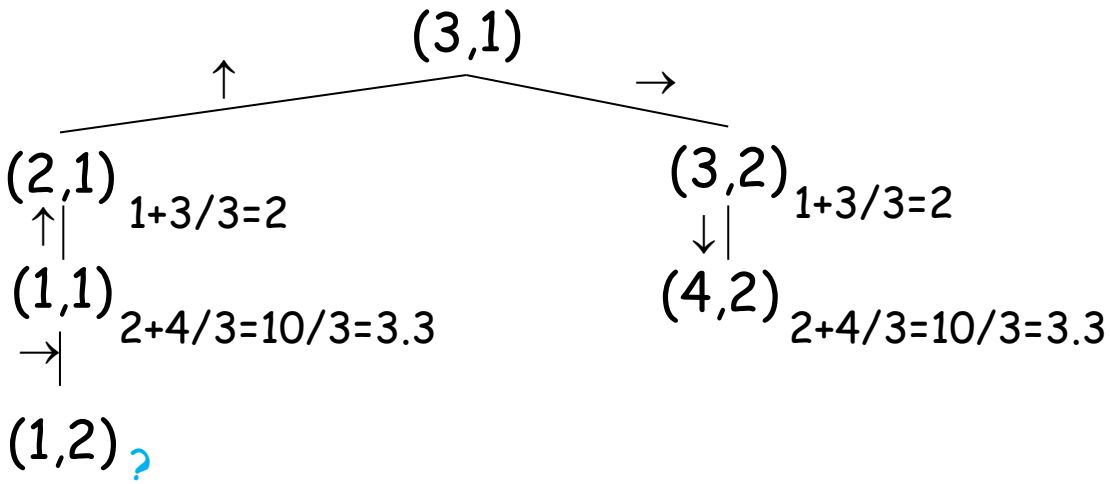

1
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1
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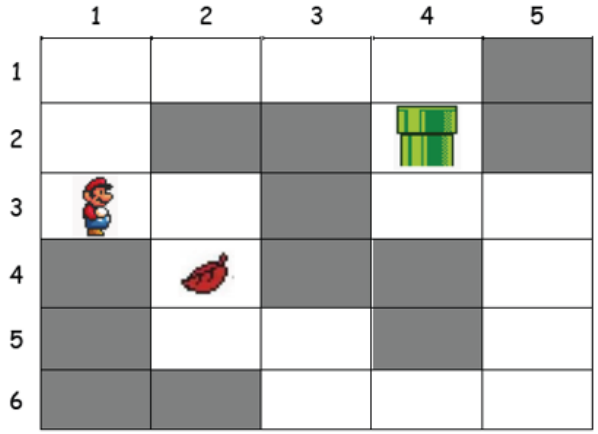

(3,1)
↑
→
(2,1)
1+3/3=2
↑
(1,1)
2+4/3=10/3=3.3
(3,2)
1+3/3=2
↓
(4,2)
?

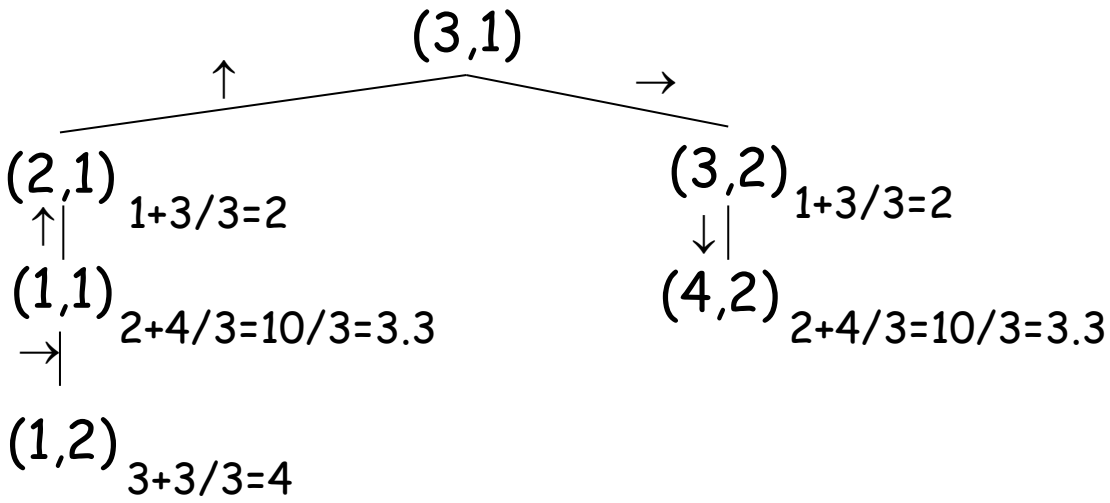

1
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1
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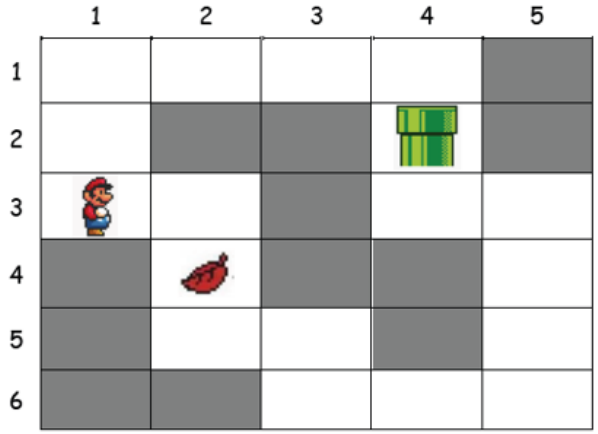

(3,1)
↑
→
(2,1)
1+3/3=2
↑
(1,1)
2+4/3=10/3=3.3
(3,2)
1+3/3=2
↓
(4,2)
2+4/3=10/3=3.3

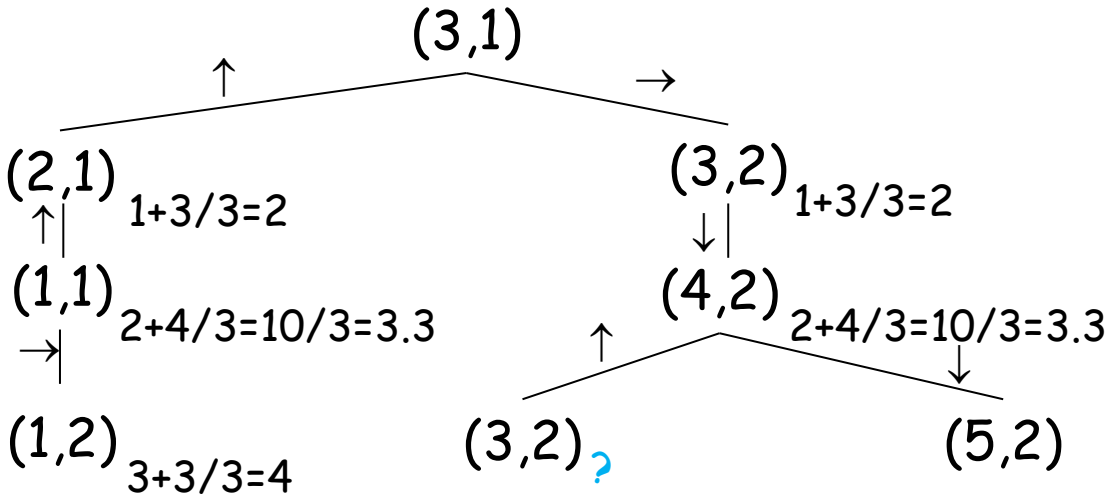

1
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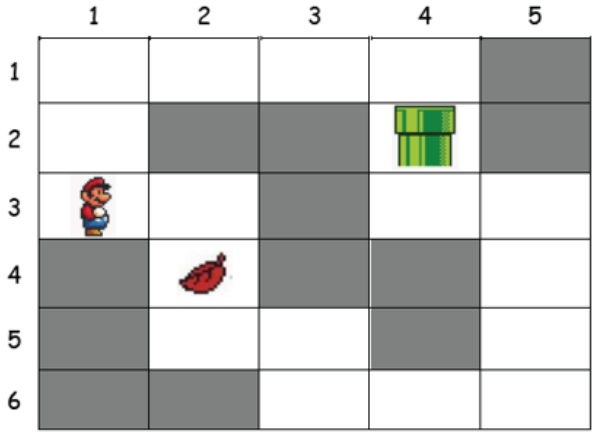

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
?

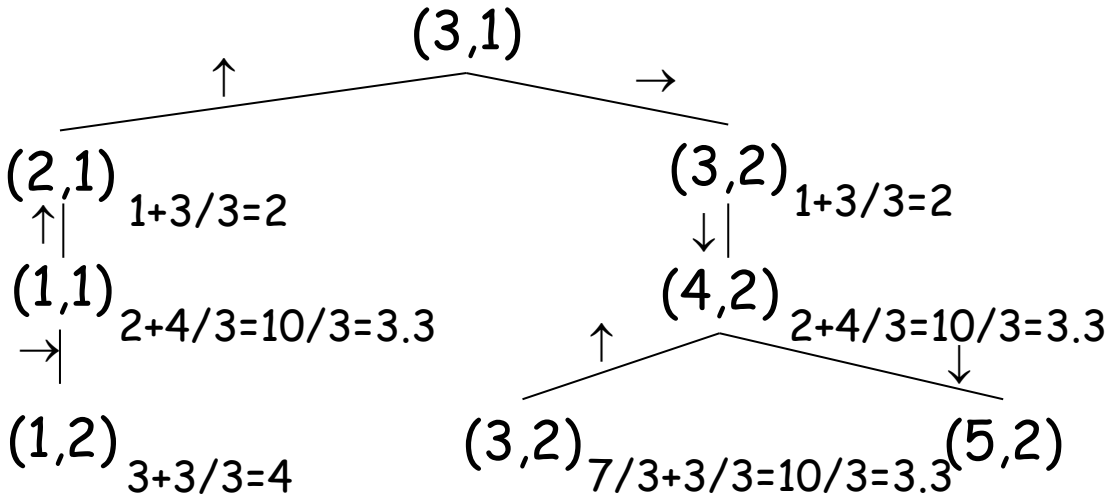

1
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1
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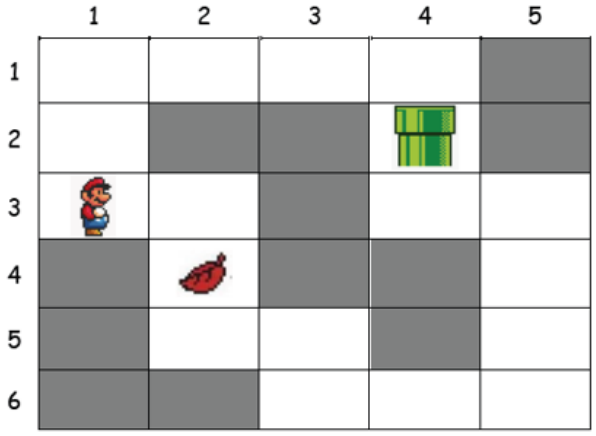

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4

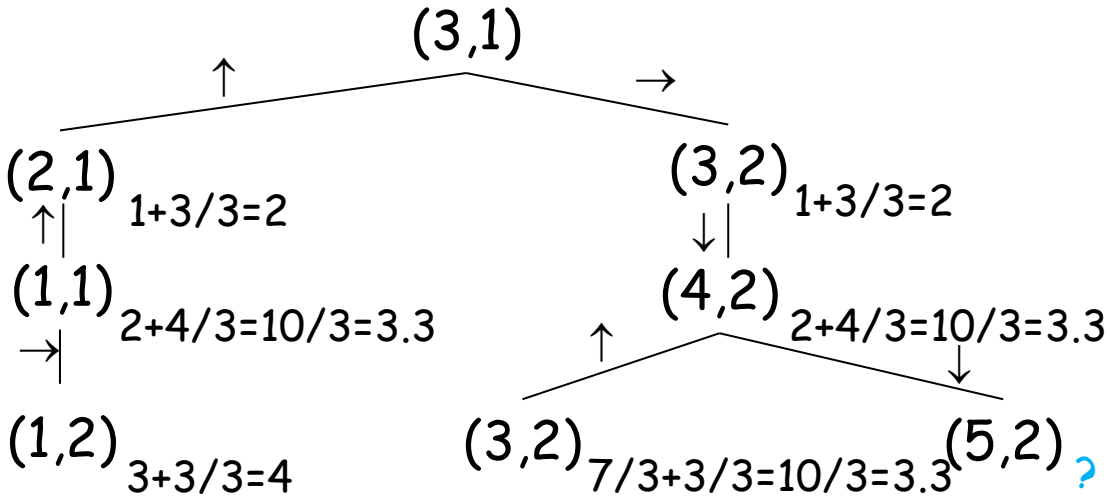

1
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1
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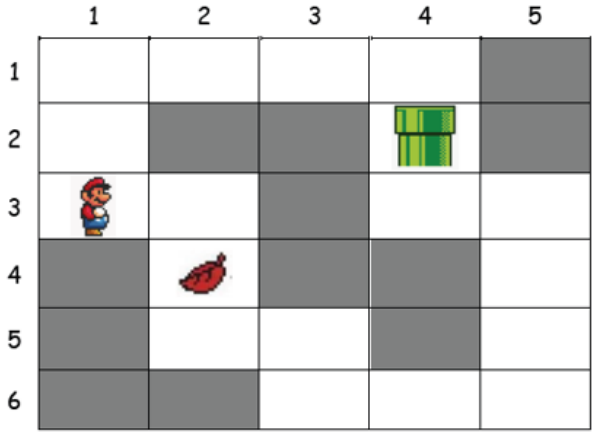

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
?
(5,2)

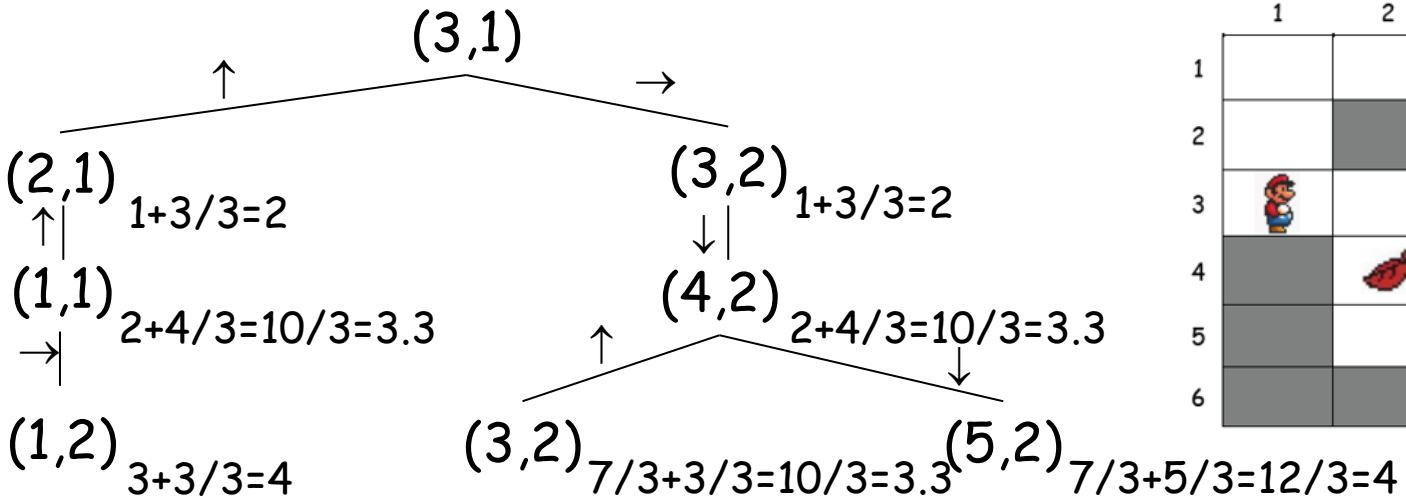

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1
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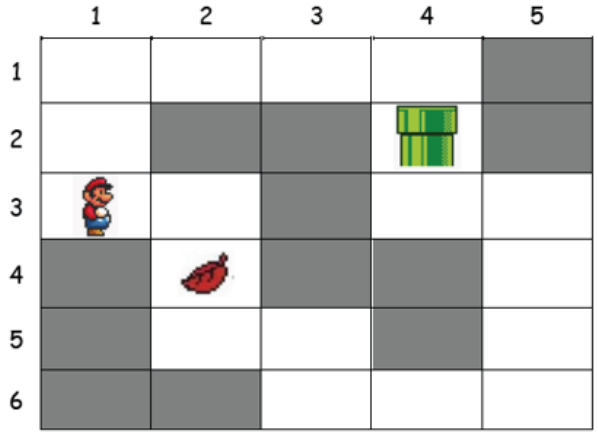

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)

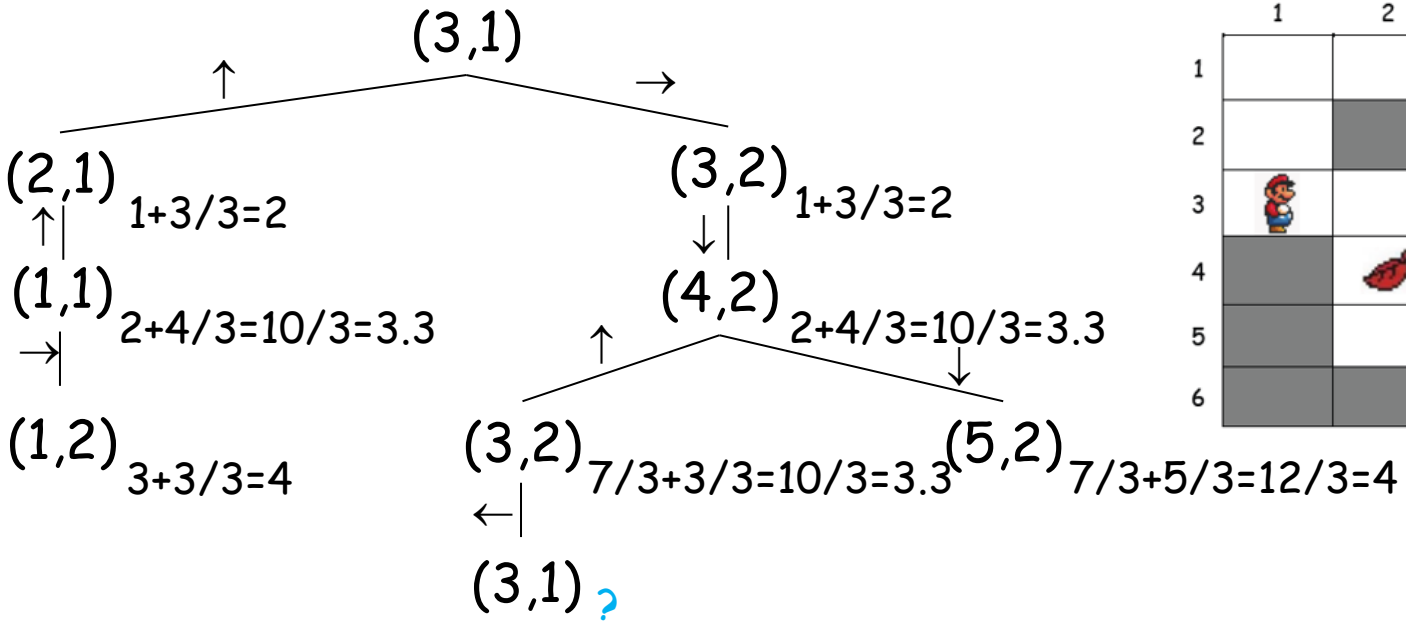

1
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1
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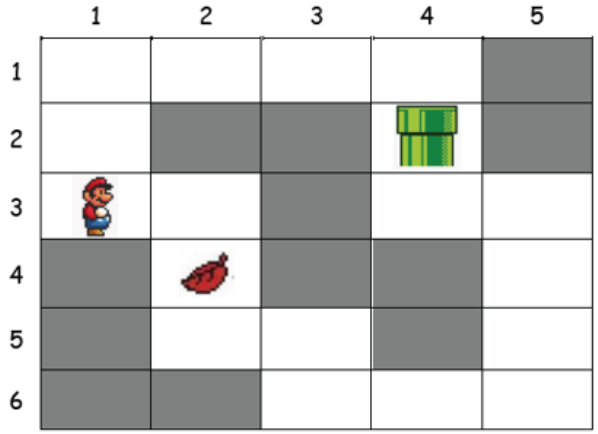

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)
?

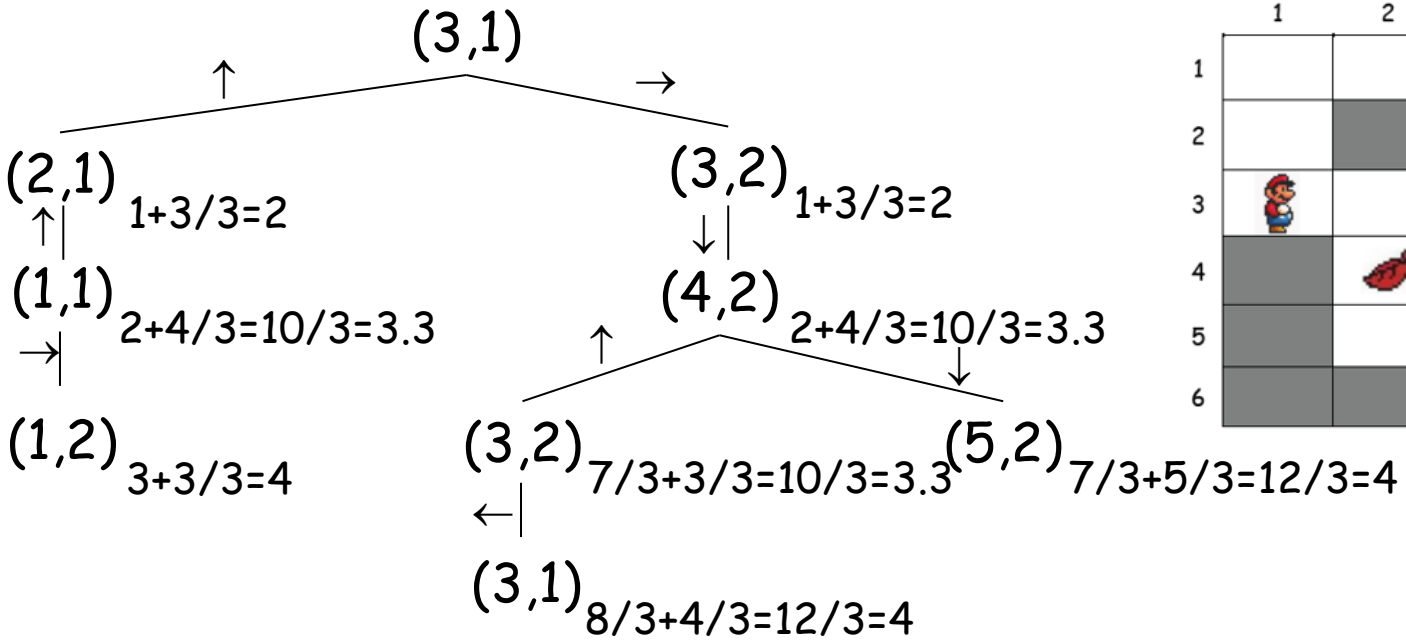

1
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1
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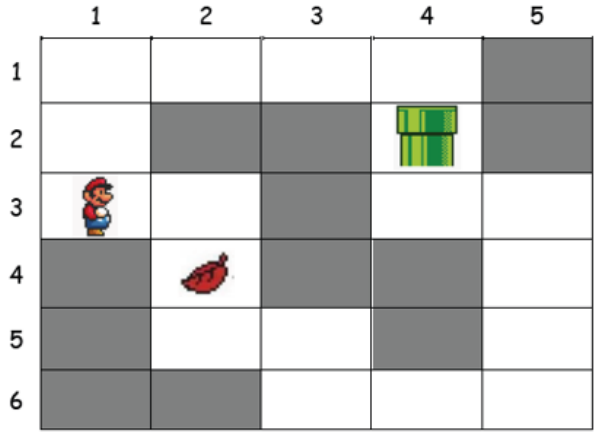

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)
7/3+5/3=12/3=4

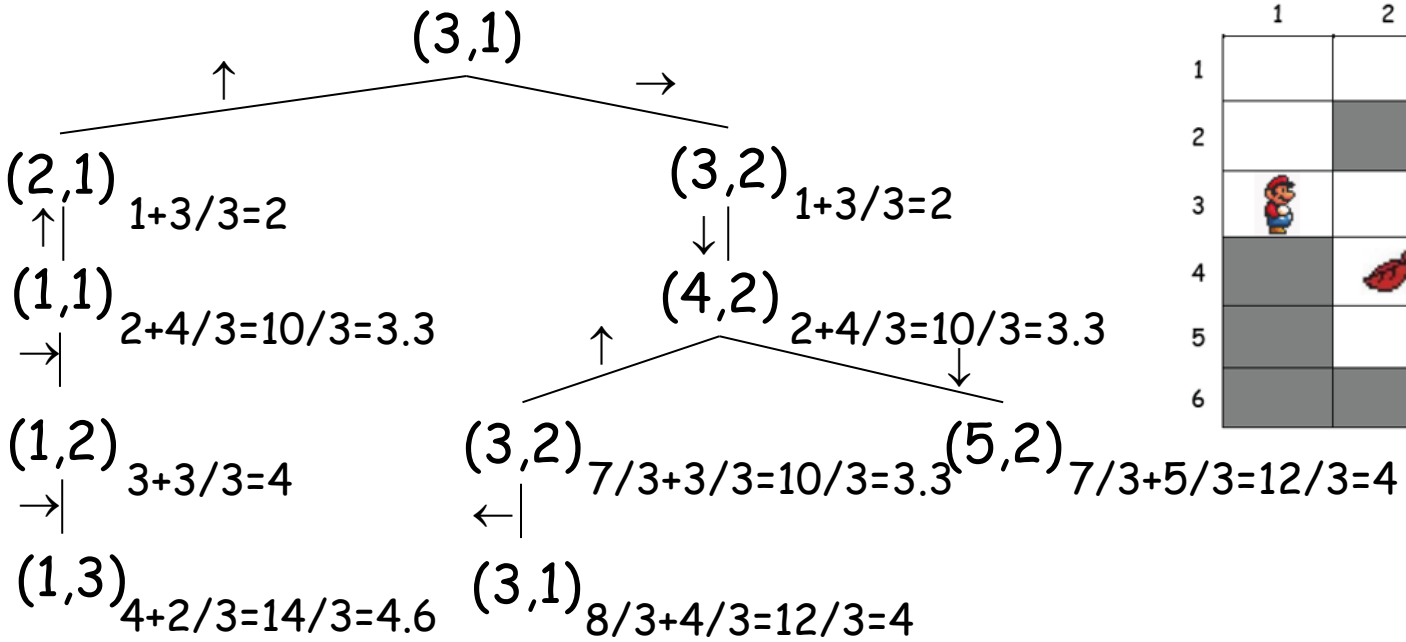

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1
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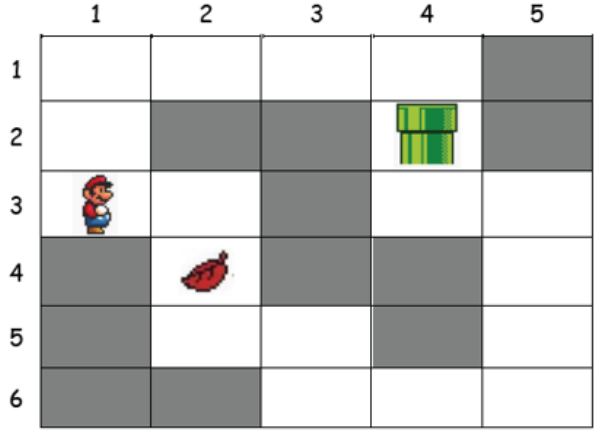

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)
7/3+5/3=12/3=4
(3,1) ?

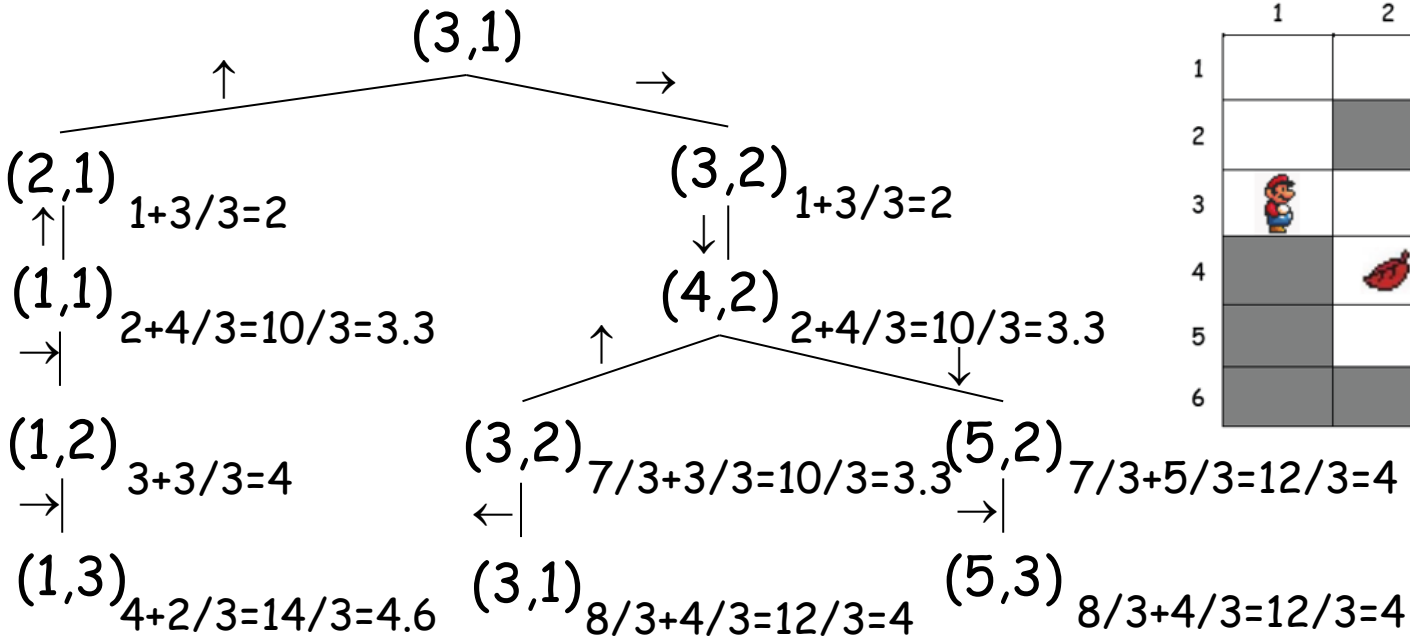

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1
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5
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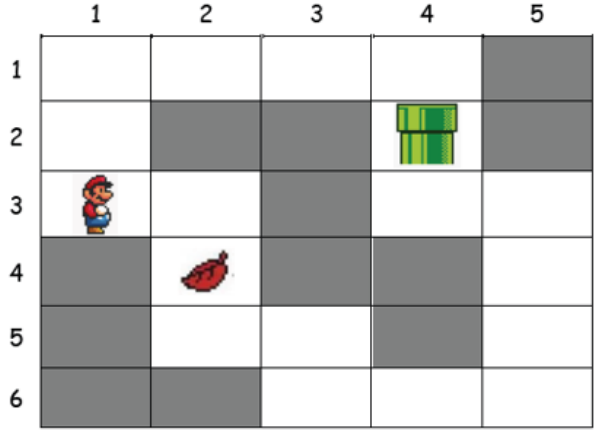

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)
7/3+5/3=12/3=4
(3,1)
8/3+4/3=12/3=4

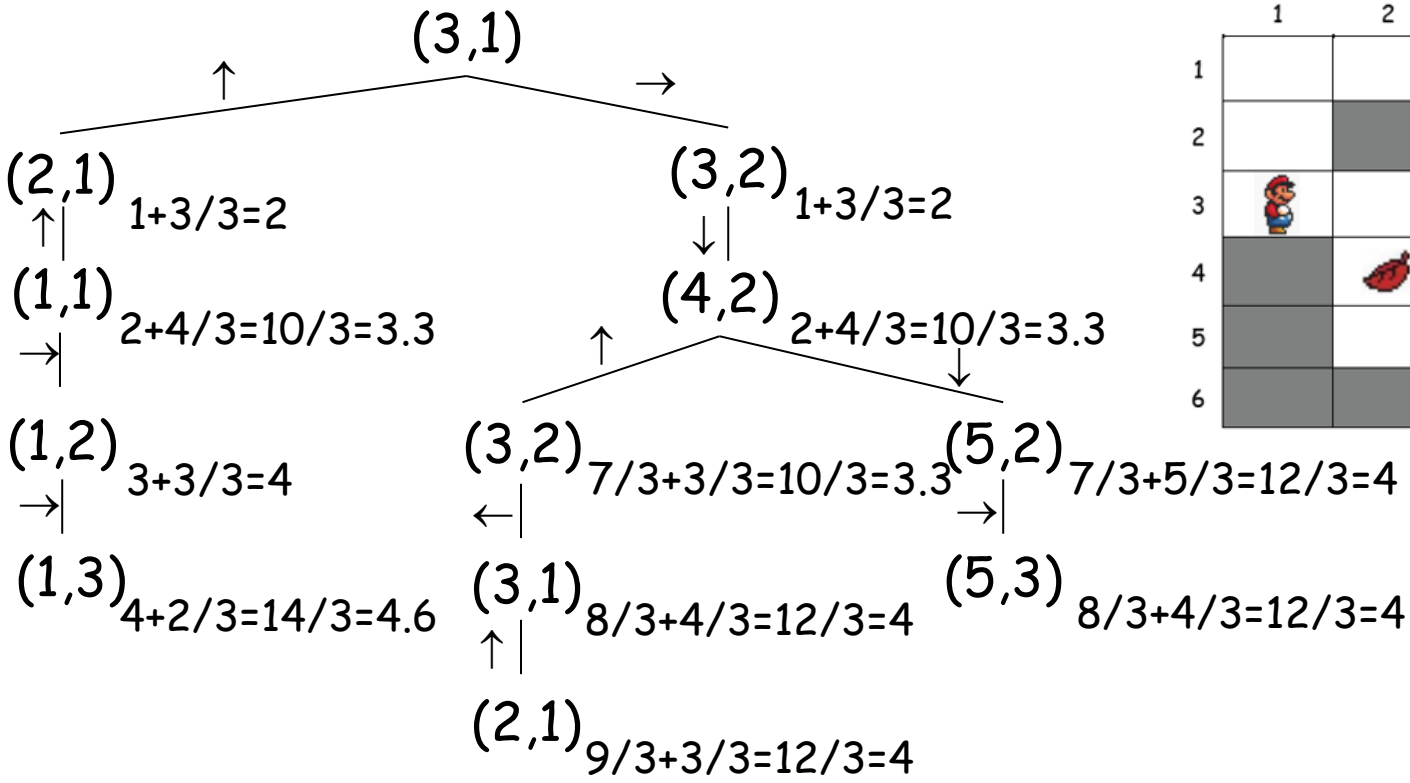

1
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1
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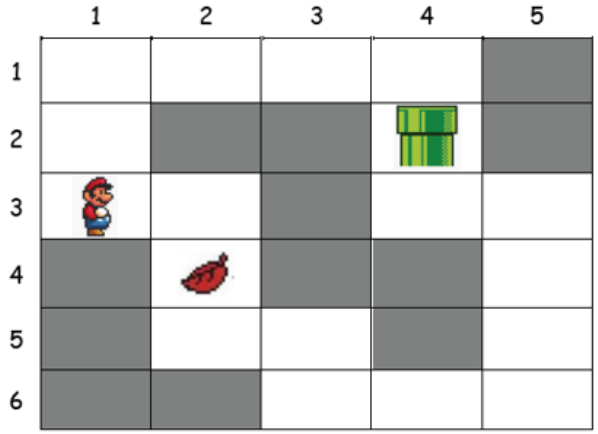

(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)
7/3+5/3=12/3=4
(1,3)
4+2/3=14/3=4.6
(3,1)
8/3+4/3=12/3=4

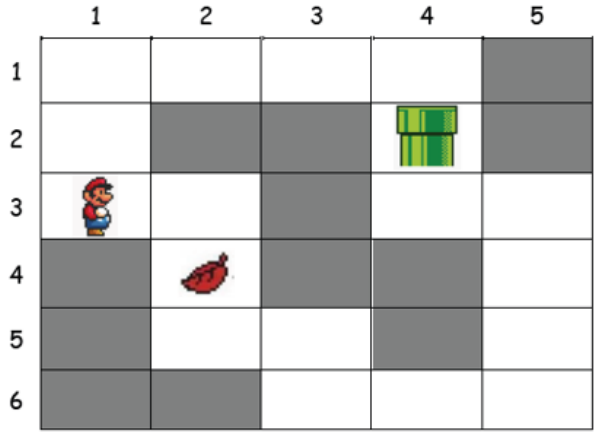

1
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4
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1
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3
4
5
6


(3,1)
(2,1)
1+3/3=2
(3,2)
1+3/3=2
(1,1)
2+4/3=10/3=3.3
(4,2)
2+4/3=10/3=3.3
(1,2)
3+3/3=4
(3,2)
7/3+3/3=10/3=3.3
(5,2)
7/3+5/3=12/3=4
(1,3)
4+2/3=14/3=4.6
(3,1)
8/3+4/3=12/3=4
(5,3)
8/3+4/3=12/3=4


1
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1
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3
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5
6


(3,1)
(2,1) 1+3/3=2
(3,2) 1+3/3=2
(1,1) 2+4/3=10/3=3.3
(4,2) 2+4/3=10/3=3.3
(1,2) 3+3/3=4
(3,2) 7/3+3/3=10/3=3.3
(5,2) 7/3+5/3=12/3=4
(1,3) 4+2/3=14/3=4.6
(3,1) 8/3+4/3=12/3=4
(5,3) 8/3+4/3=12/3=4
(2,1) 9/3+3/3=12/3=4


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6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

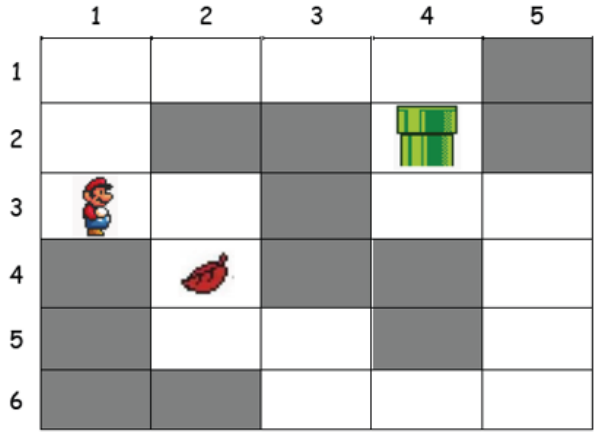
(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

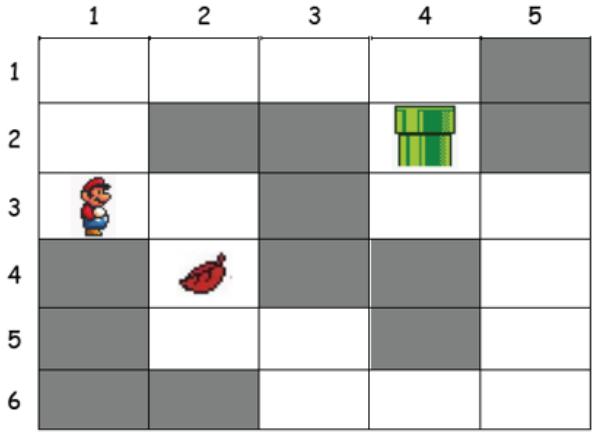
(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

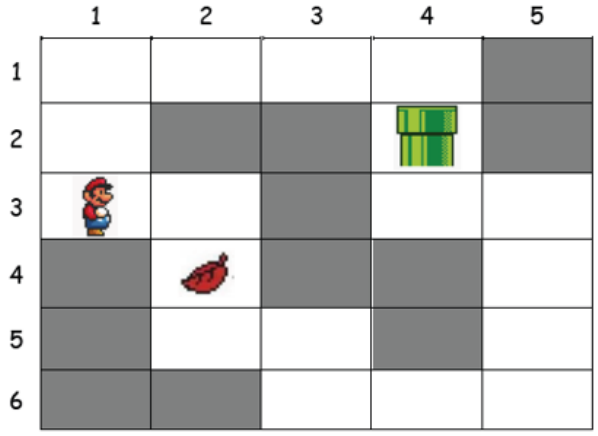
(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

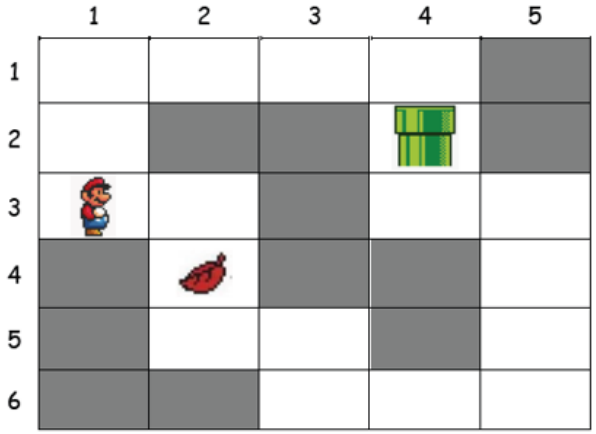
(1,1) 10/3+4/3=14/3=4.6

(1,2) 11/3+3/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

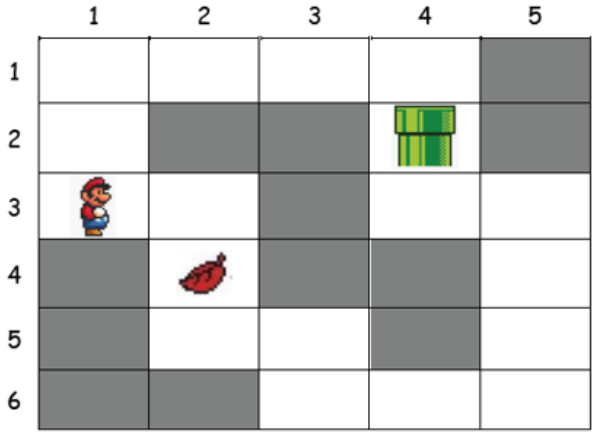
(1,2) 11/3+3/3=14/3=4.6

(1,3) 12/3+2/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6



(3,1)

↑ (2,1) 1+3/3=2

↑ (1,1) 2+4/3=10/3=3.3

→ (1,2) 3+3/3=4

→ (1,3) 4+2/3=14/3=4.6

→ (1,4) 5+1/3=16/3=5.3

→ (3,2) 1+3/3=2

↓ (4,2) 2+4/3=10/3=3.3

↑ (3,2) 7/3+3/3=10/3=3.3

← (3,1) 8/3+4/3=12/3=4

↑ (2,1) 9/3+3/3=12/3=4

→ (1,1) 10/3+4/3=14/3=4.6

→ (1,2) 11/3+3/3=14/3=4.6

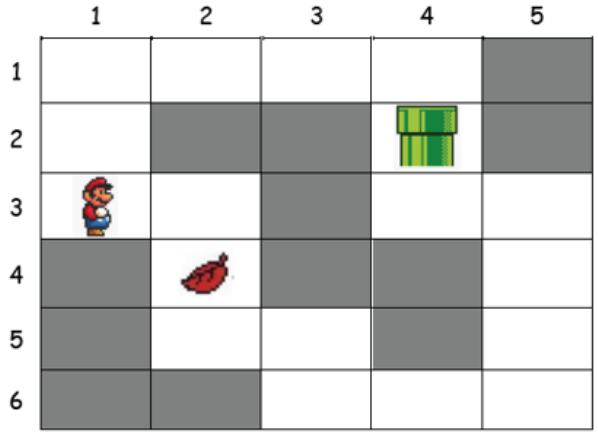
→ (1,3) 12/3+2/3=14/3=4.6

→ (1,4) 13/3+1/3=14/3=4.6

→ (5,2) 7/3+5/3=12/3=4

↓ (5,3) 8/3+4/3=12/3=4

↓ (6,3) 9/3+5/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(1,2) 11/3+3/3=14/3=4.6

(1,3) 12/3+2/3=14/3=4.6

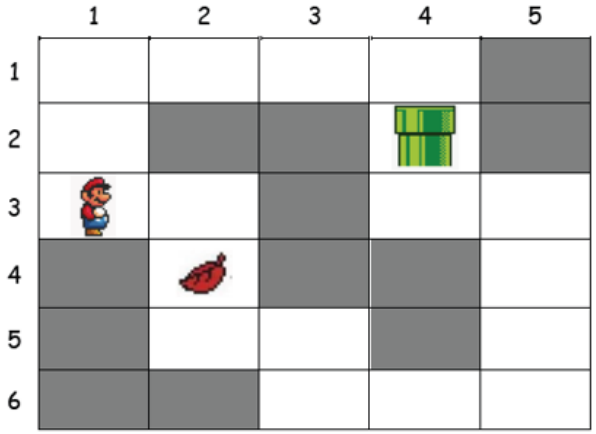
(1,4) 13/3+1/3=14/3=4.6

(2,4) 14/3+0/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(1,2) 11/3+3/3=14/3=4.6

(1,3) 12/3+2/3=14/3=4.6

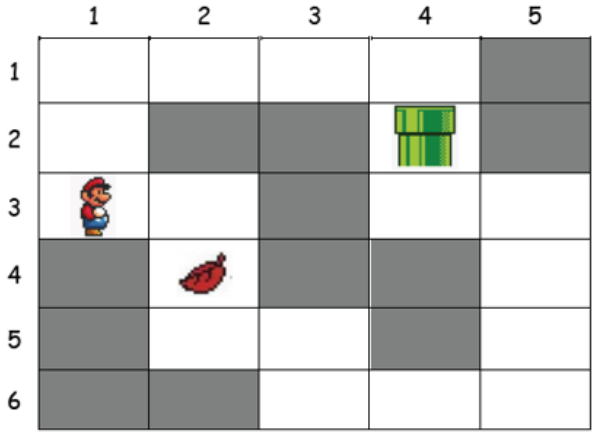
(1,4) 13/3+1/3=14/3=4.6

(2,4)* 14/3+0/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(1,2) 11/3+3/3=14/3=4.6

(1,3) 12/3+2/3=14/3=4.6

(1,4) 13/3+1/3=14/3=4.6

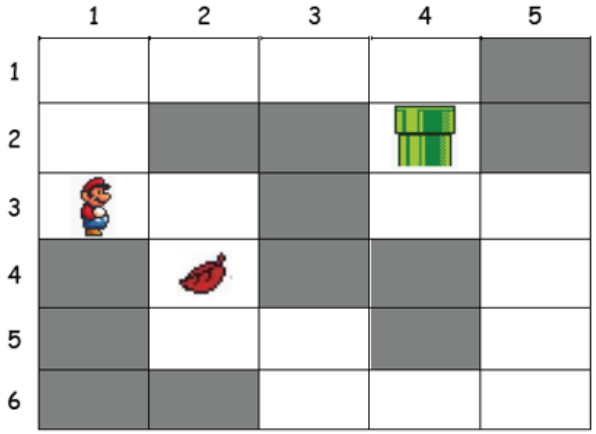
(2,4) 14/3+0/3=14/3=4.6

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6

(6,4) 10/3+4/3=14/3=4.6



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(1,2) 11/3+3/3=14/3=4.6

(1,3) 12/3+2/3=14/3=4.6

(1,4) 13/3+1/3=14/3=4.6

(2,4) 14/3+0/3=14/3=4.6

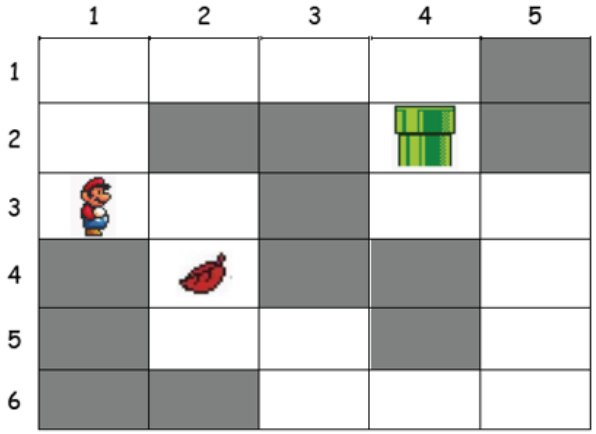
(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

(6,3) 9/3+5/3=14/3=4.6

(6,4) 10/3+4/3=14/3=4.6

(6,5) 11/3+5/3=16/3=5.3



(3,1)

(2,1) 1+3/3=2

(1,1) 2+4/3=10/3=3.3

(1,2) 3+3/3=4

(1,3) 4+2/3=14/3=4.6

(1,4) 5+1/3=16/3=5.3

(3,2) 1+3/3=2

(4,2) 2+4/3=10/3=3.3

(3,2) 7/3+3/3=10/3=3.3

(3,1) 8/3+4/3=12/3=4

(2,1) 9/3+3/3=12/3=4

(1,1) 10/3+4/3=14/3=4.6

(1,2) 11/3+3/3=14/3=4.6

(1,3) 12/3+2/3=14/3=4.6

(1,4) 13/3+1/3=14/3=4.6

(2,4) 14/3+0/3=14/3=4.6*

(5,2) 7/3+5/3=12/3=4

(5,3) 8/3+4/3=12/3=4

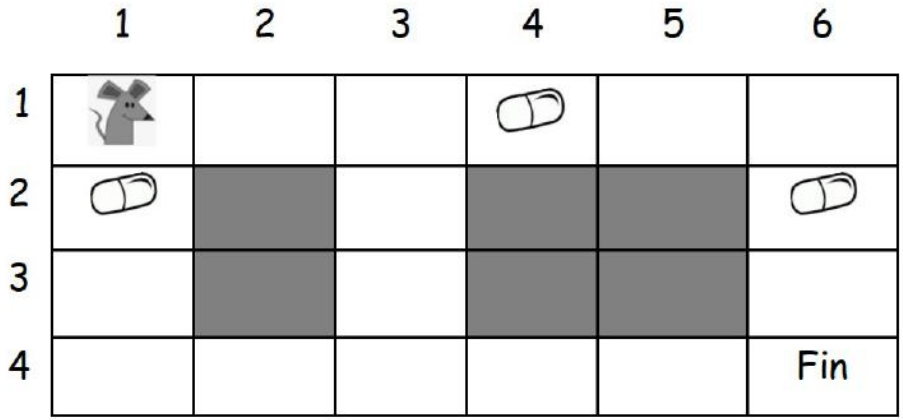
(6,3) 9/3+5/3=14/3=4.6

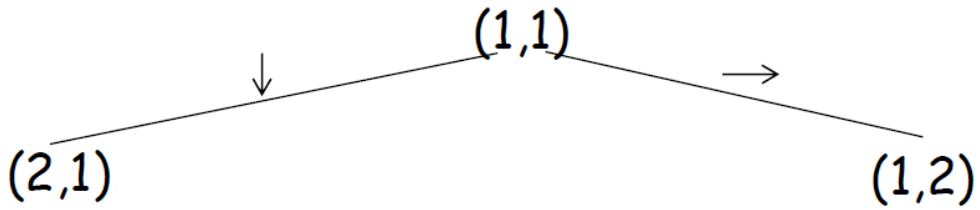
(6,4) 10/3+4/3=14/3=4.6

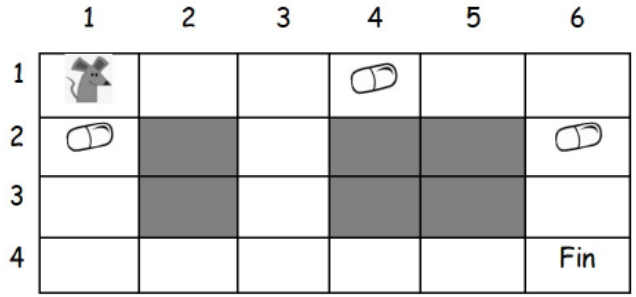
(6,5) 11/3+5/3=16/3=5.3

4. Búsqueda informada. El ratón y las píldoras. Considere el siguiente ambiente representado por una matriz de 4x6 donde se tiene un ratón que inicia en la posición (1,1) y debe llegar a la posición (4,6). Los operadores que puede aplicar el agente son arriba, abajo, izquierda, y derecha. Cada movimiento tiene costo 1. En el ambiente hay tres píldoras. Cuando el ratón llega a una casilla con píldora, sus dos siguientes movimientos son más rápidos y por lo tanto el costo de cada desplazamiento es 1/2. Tenga en cuenta que cada píldora tiene efecto solamente en los dos movimientos que realiza el ratón justo después de tomar una píldora. Después de que pase el efecto de la píldora el costo de cada movimiento será 1 nuevamente. En el recorrido que realice el ratón se pueden usar varias píldoras.

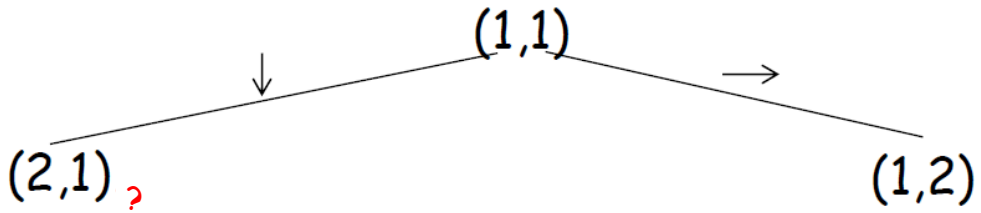
- **[I.L 1.5 (2pts)]** Defina una heurística que sea admisible para el ambiente que se muestra en la figura.
- **[I.L 1.6 (7pts)]** Muestre el <u>árbol</u> y la <u>solución</u> que se encuentra con A*. Debe aplicar los operadores en el orden, ↓, →, ↑, ←. En cada nodo debe indicar los valores de g y h. Evite devolverse en todos los casos. Incluso cuando el ratón tome una píldora, evite que se devuelva.

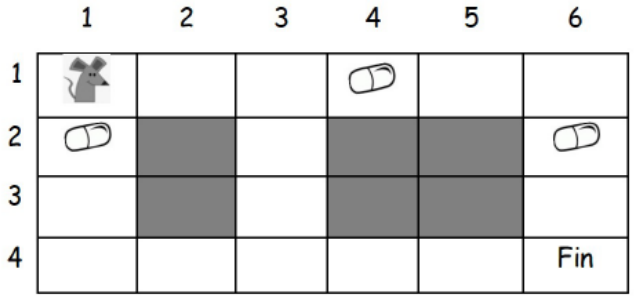



(1,1)
(2,1)
(1,2)

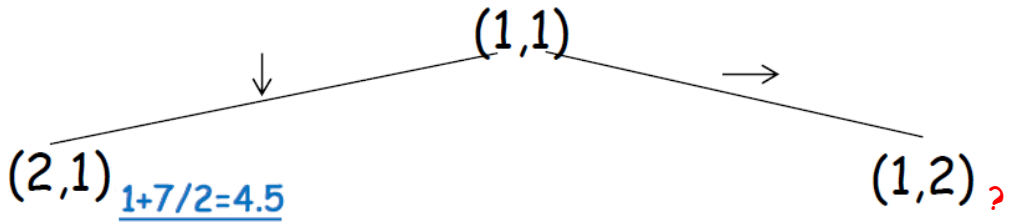

1
2
3
4
5
6
1
2
3
4
Fin

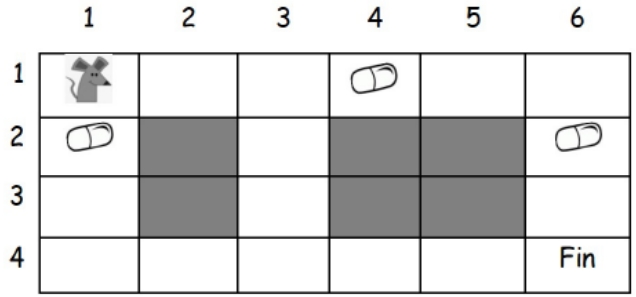
↓, →, ↑, ←.


(1,1)
(2,1)
?
(1,2)

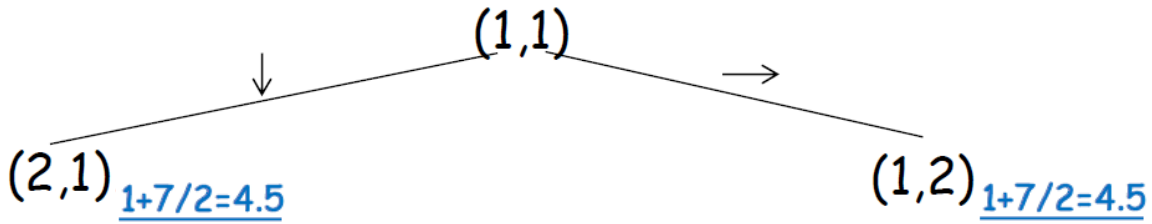

1
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3
4
5
6
1
2
3
4
Fin

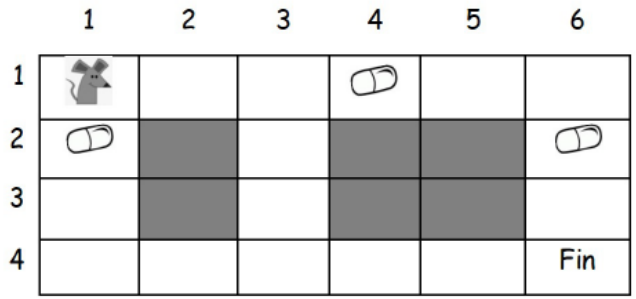
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
?

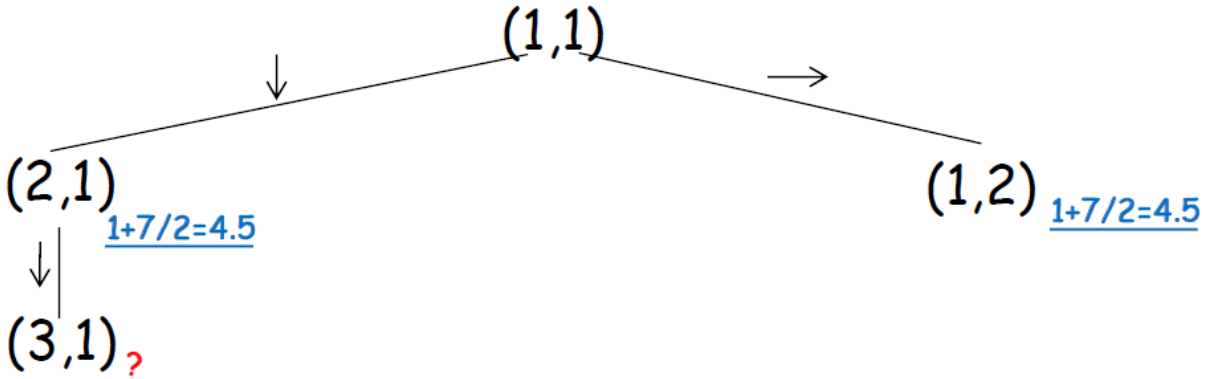

1
2
3
4
5
6
1
2
3
4
Fin

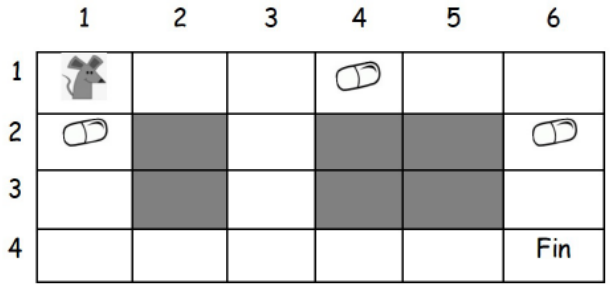
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5

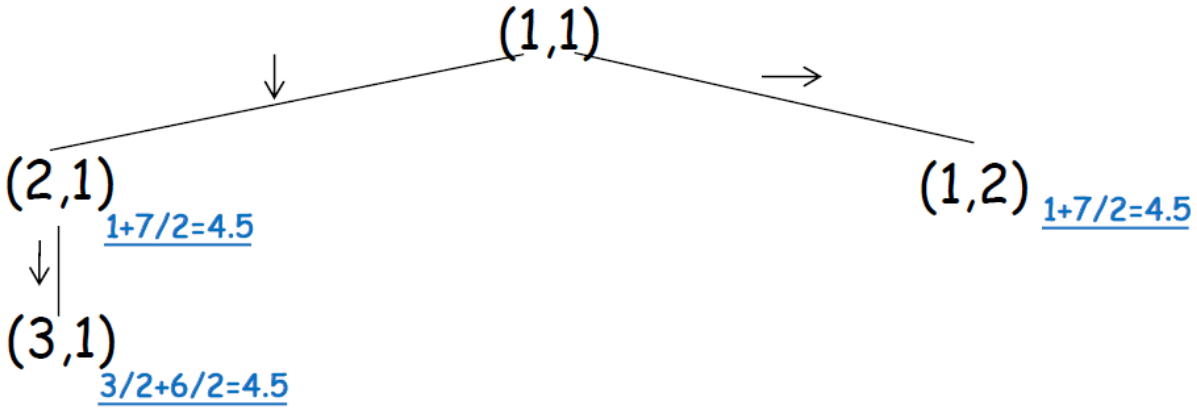

1
2
3
4
5
6
1
2
3
4
Fin

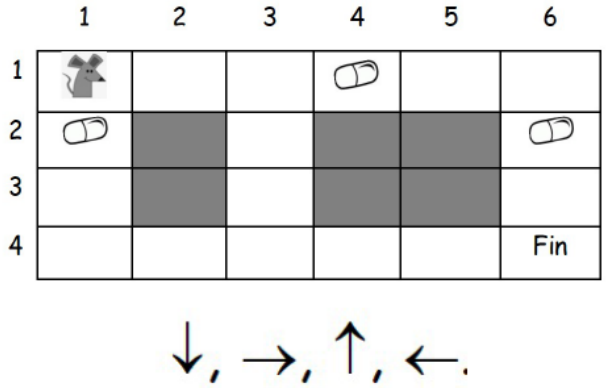
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
?

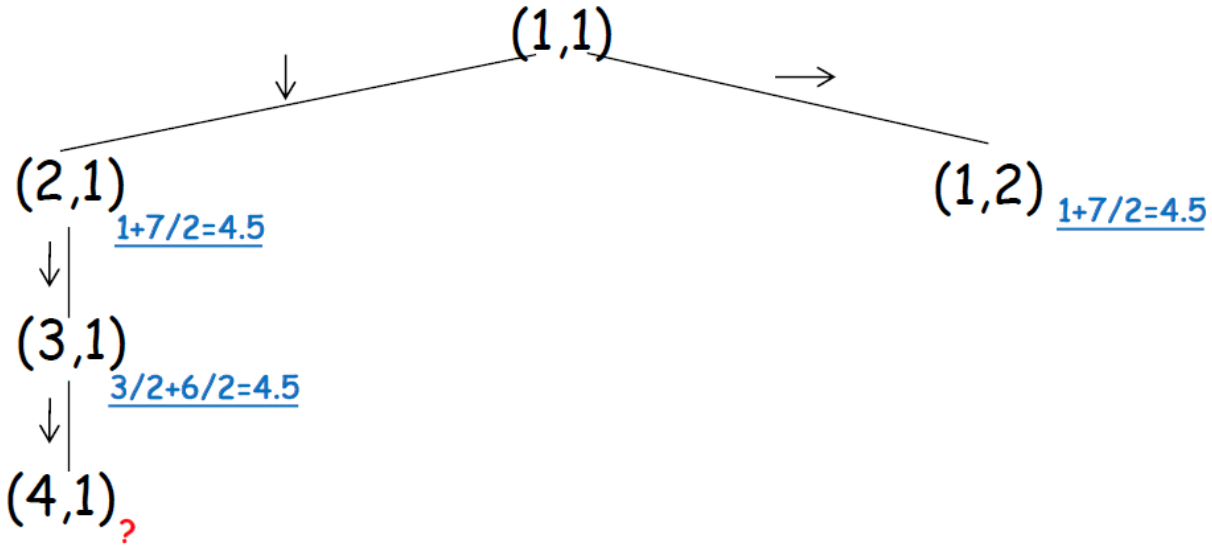

1
2
3
4
5
6
1
2
3
4
Fin

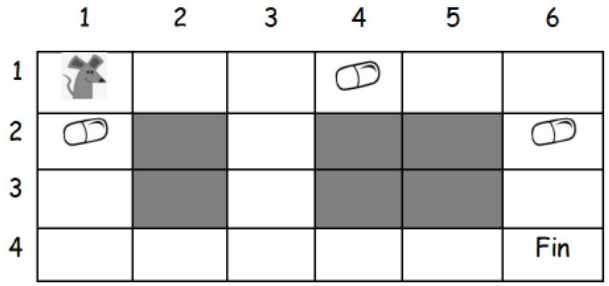
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5

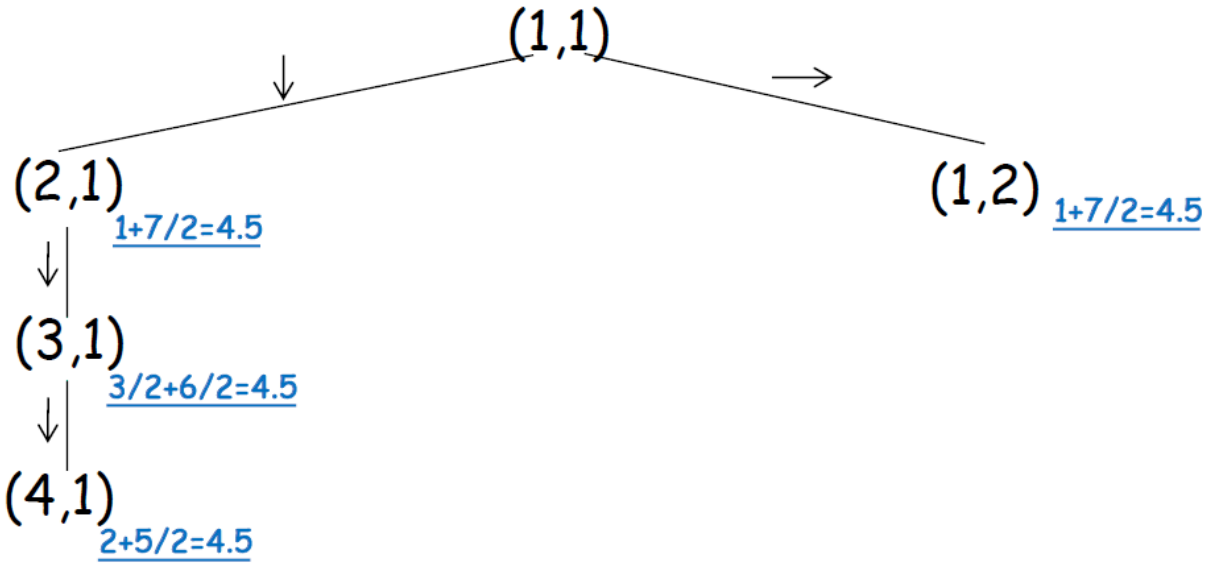

1
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3
4
5
6
Fin

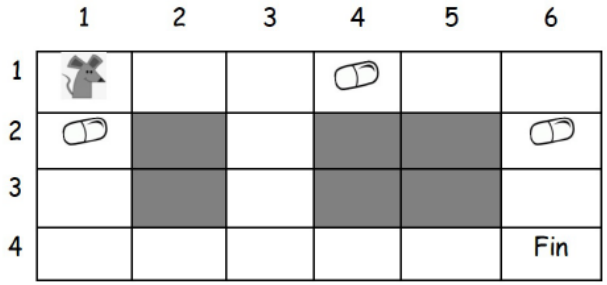
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(4,1)
?

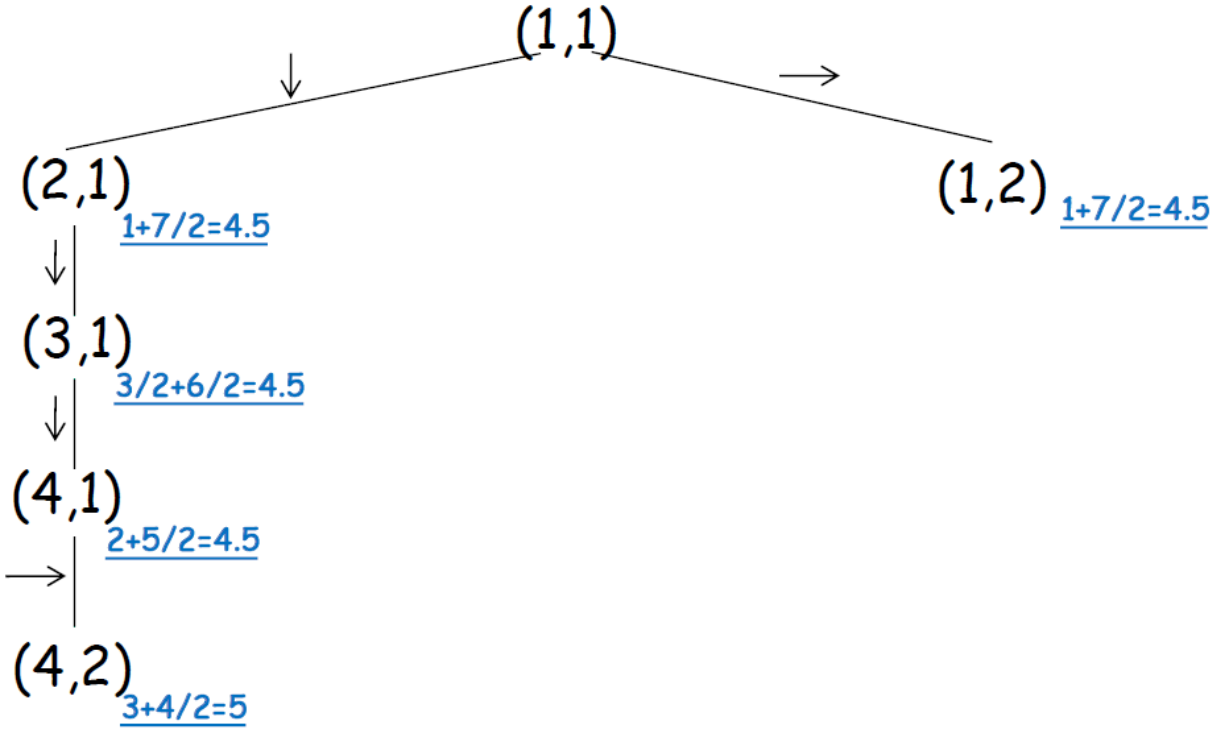

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4
5
6
Fin

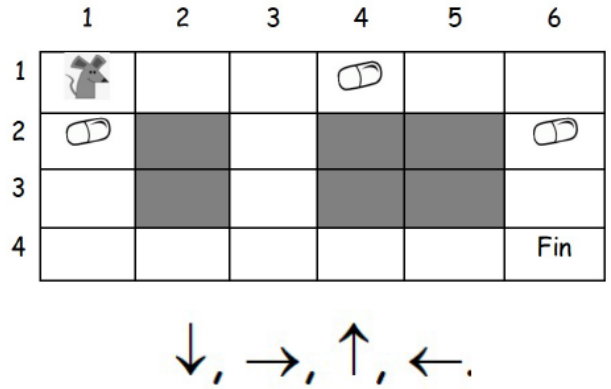
↓, →, ↑, ←.

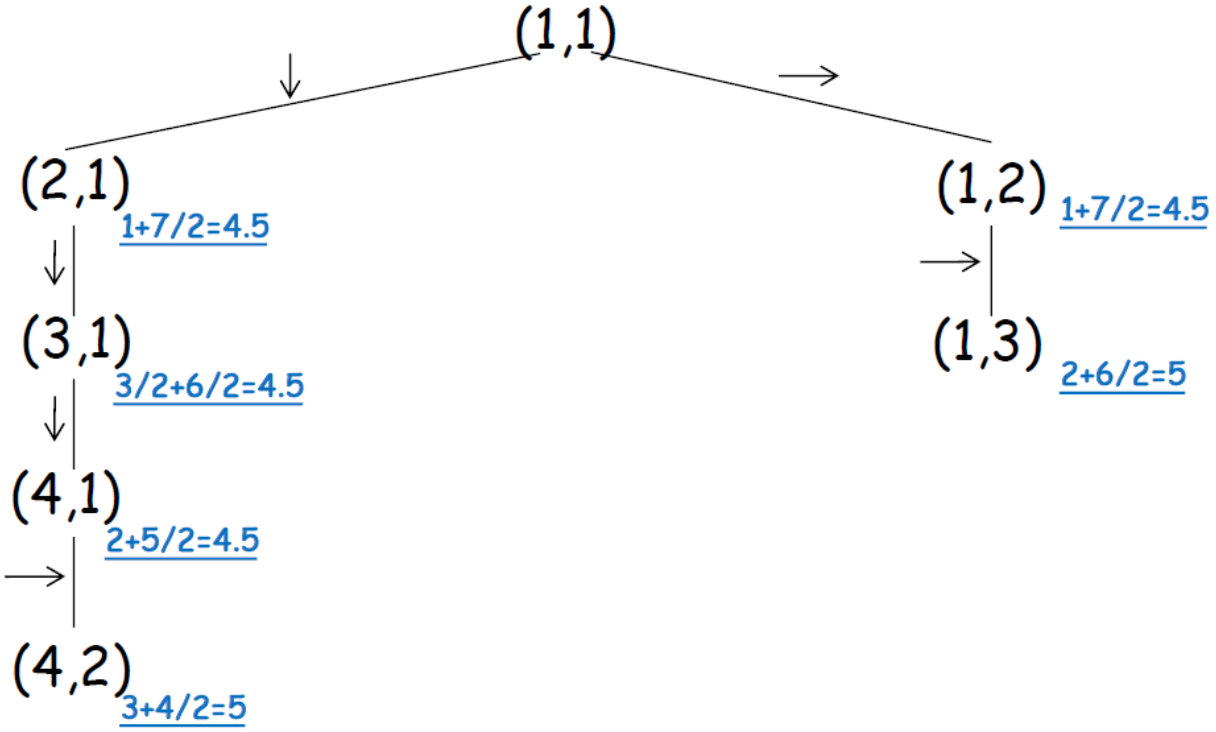

(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(4,1)
2+5/2=4.5

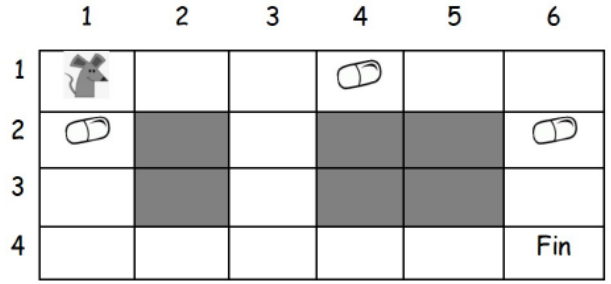

1
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Fin

↓, →, ↑, ←.

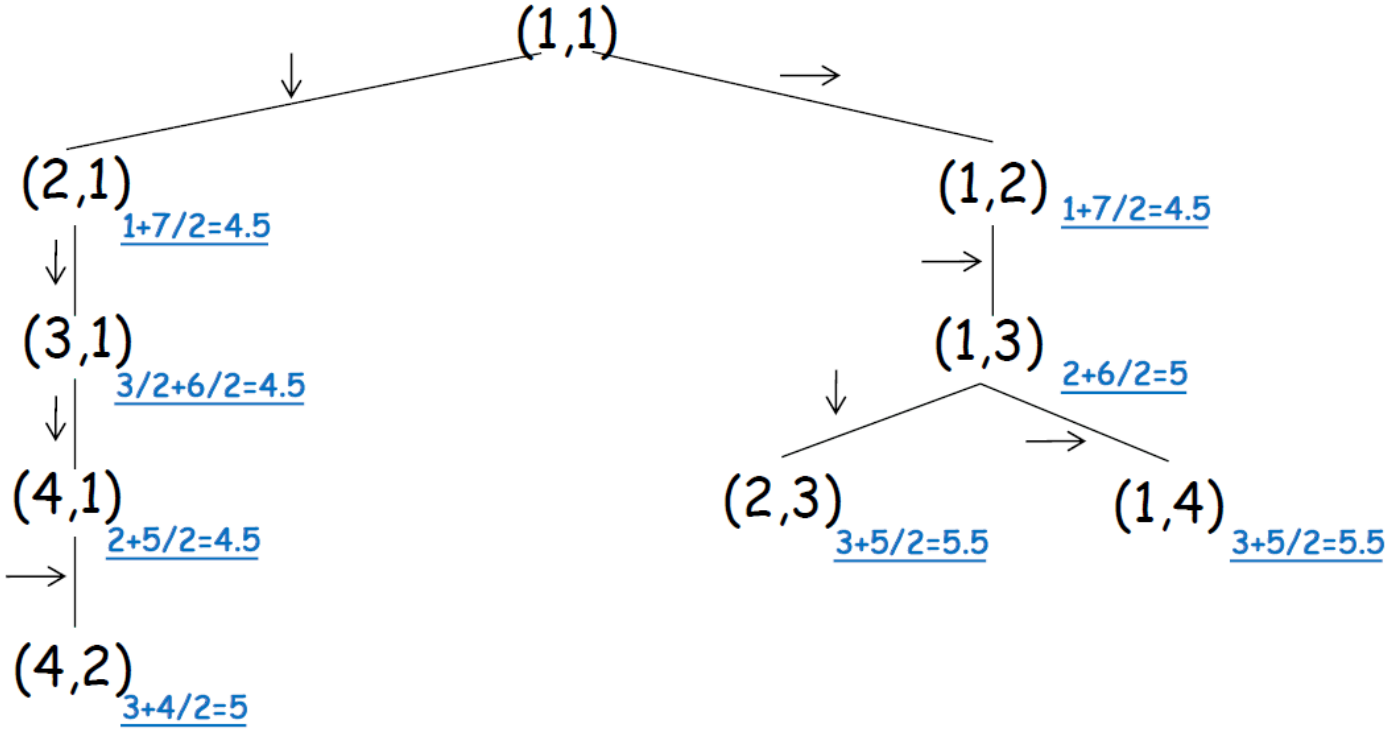

(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(4,1)
2+5/2=4.5
(4,2)
3+4/2=5

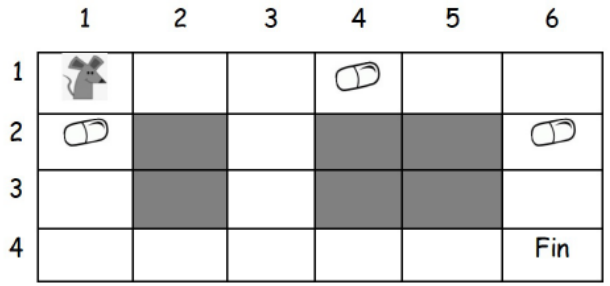

1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(4,2)
3+4/2=5

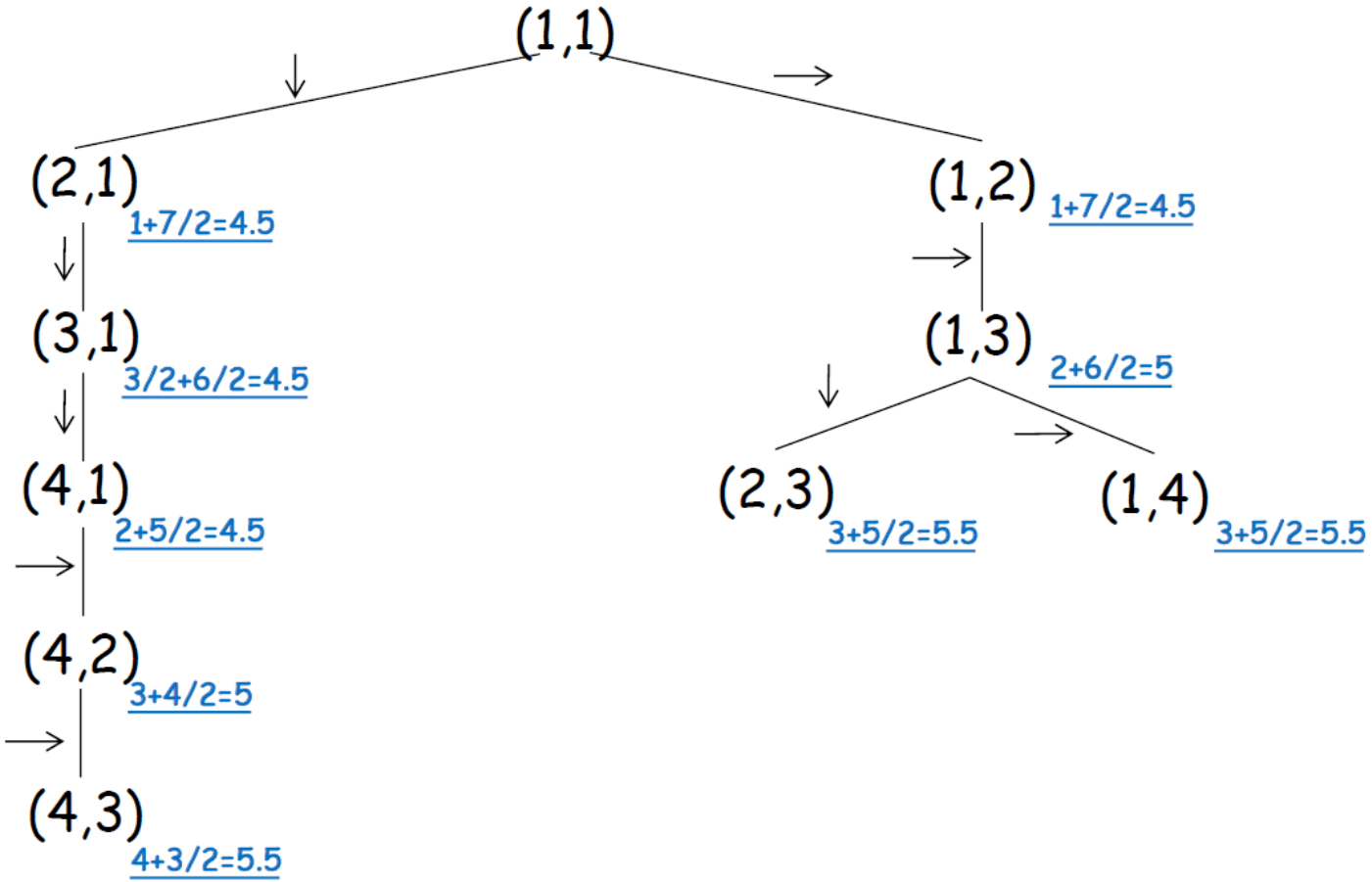

1
2
3
4
5
6
Fin

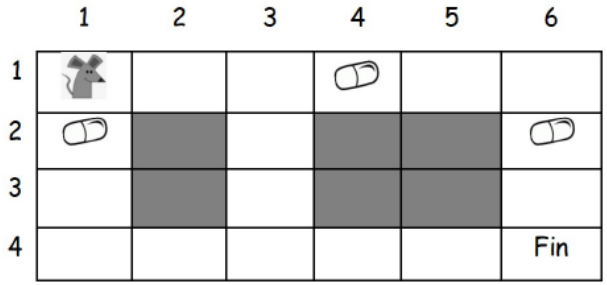
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5

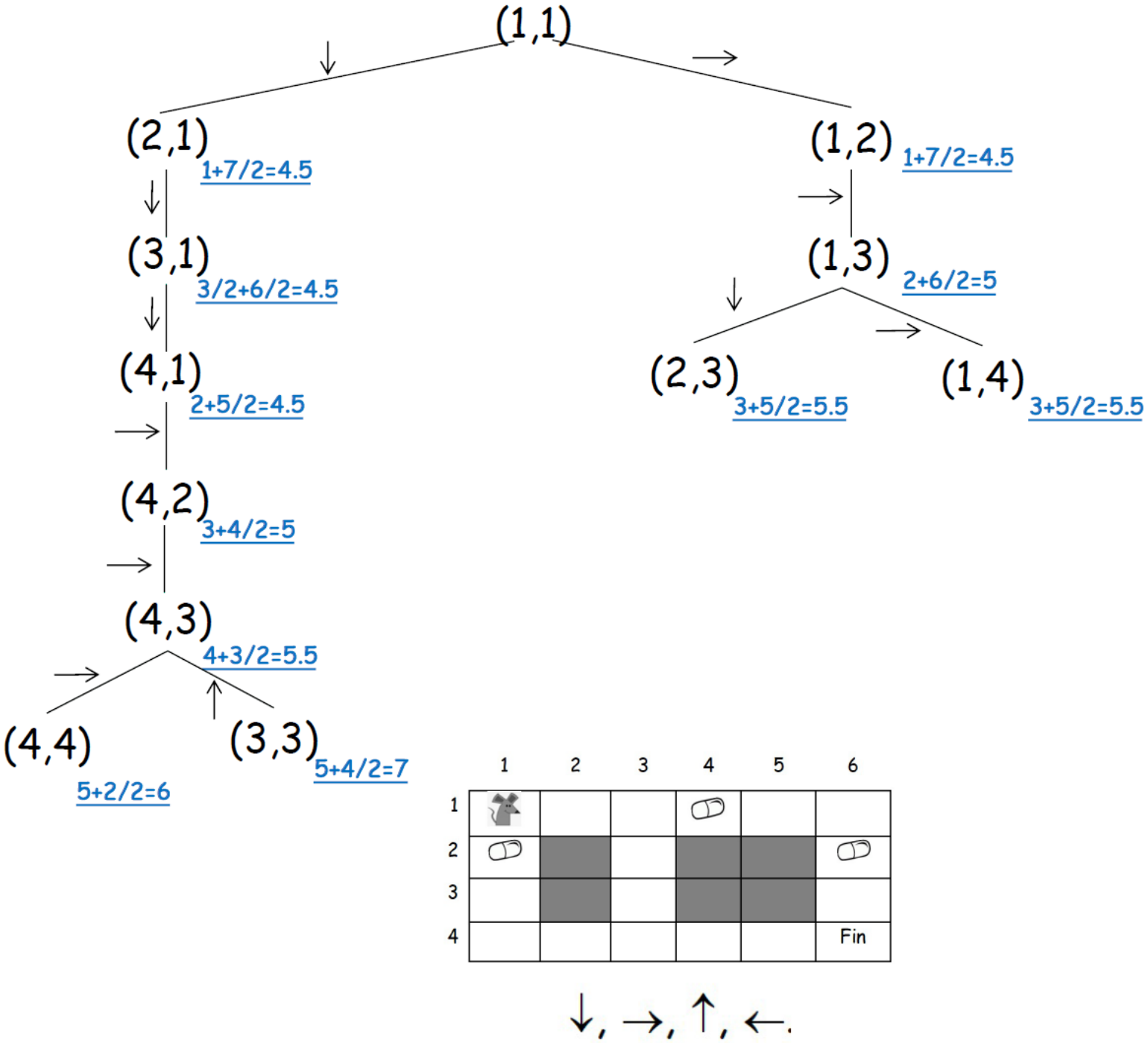

1
2
3
4
5
6
Fin

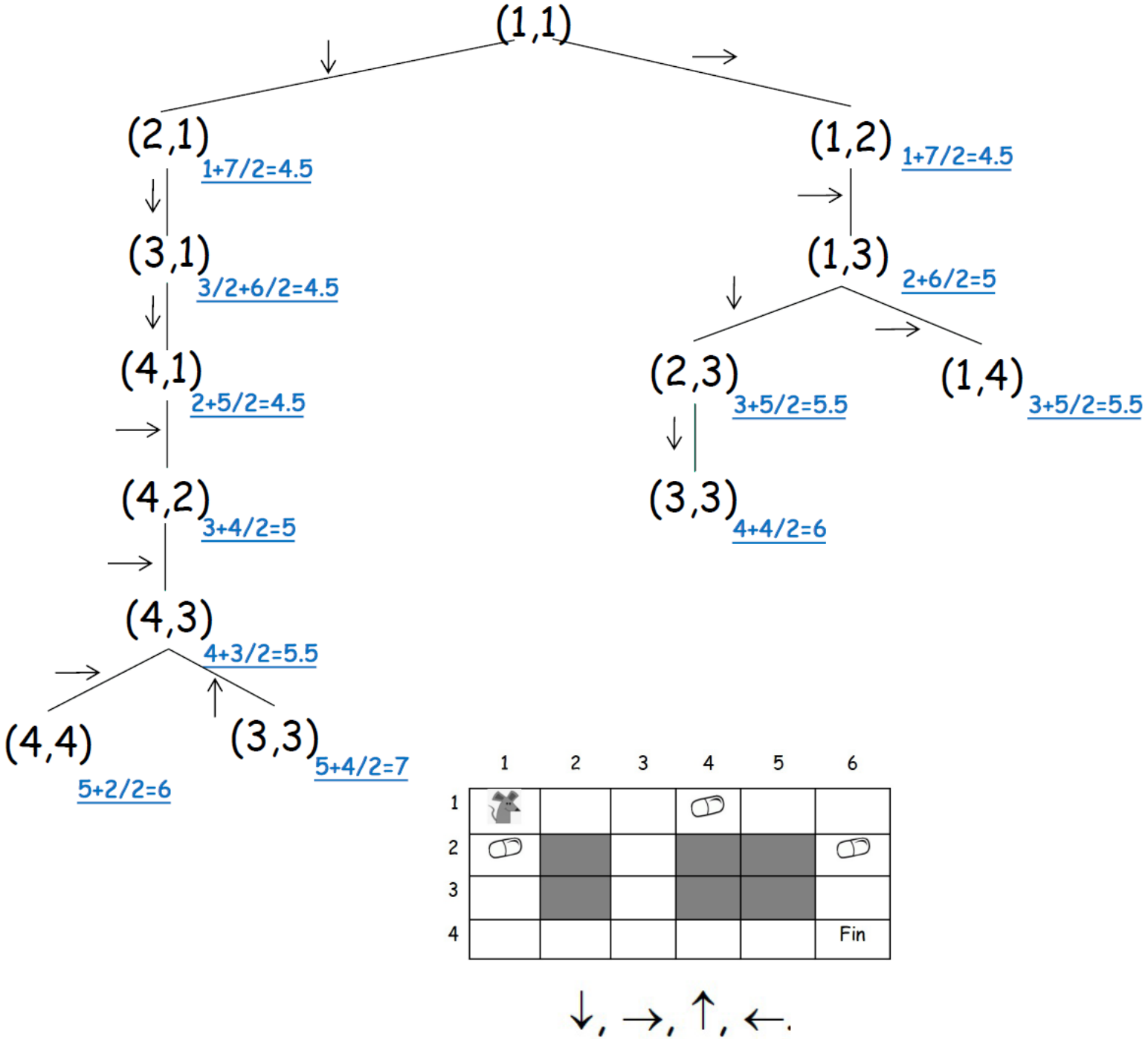
↓, →, ↑, ←.

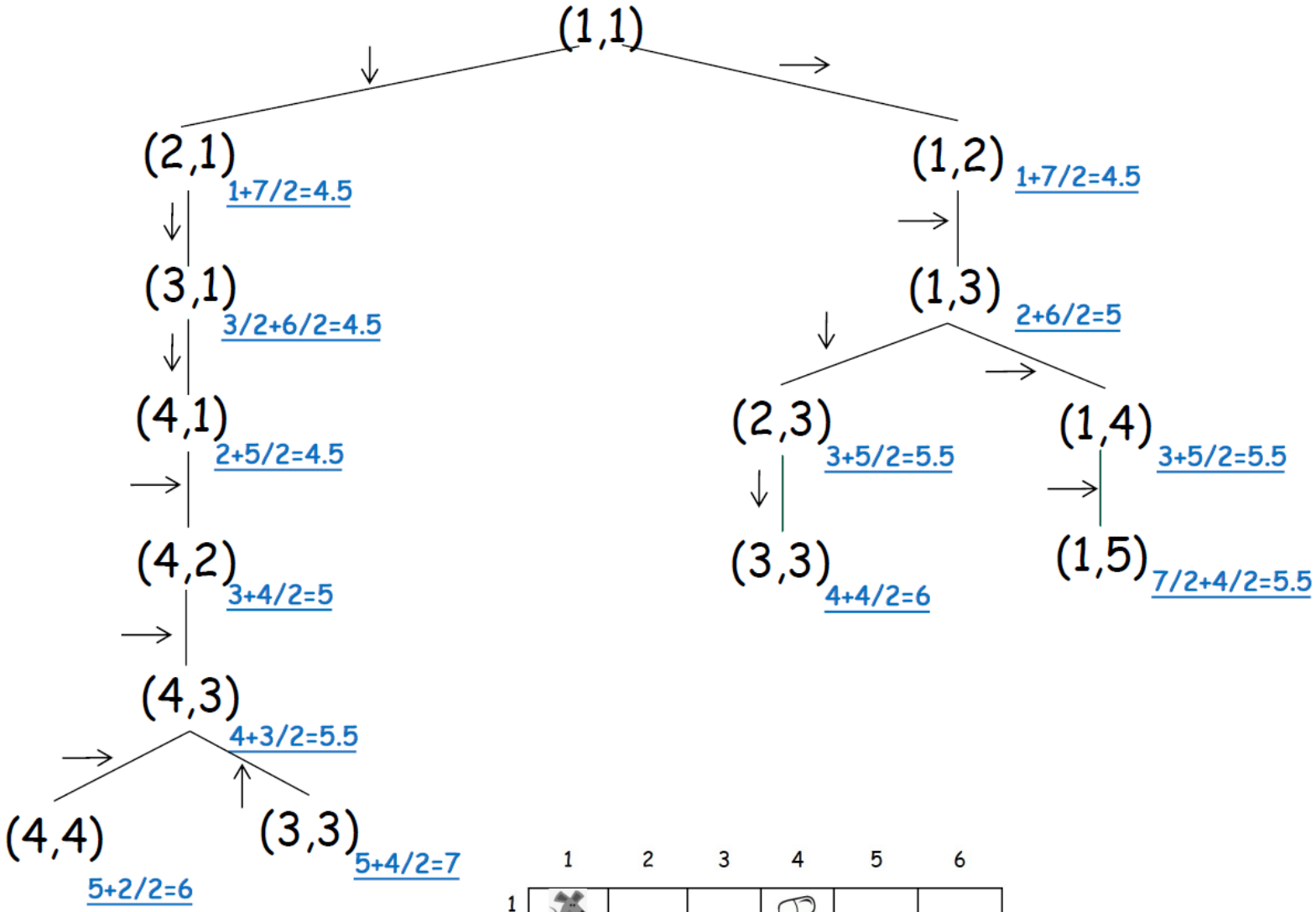

(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(4,3)
4+3/2=5.5

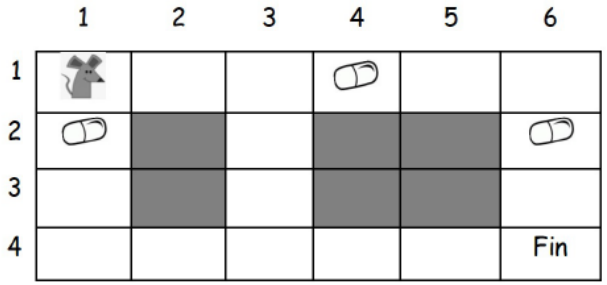

1
2
3
4
5
6
Fin

↓, →, ↑, ←.

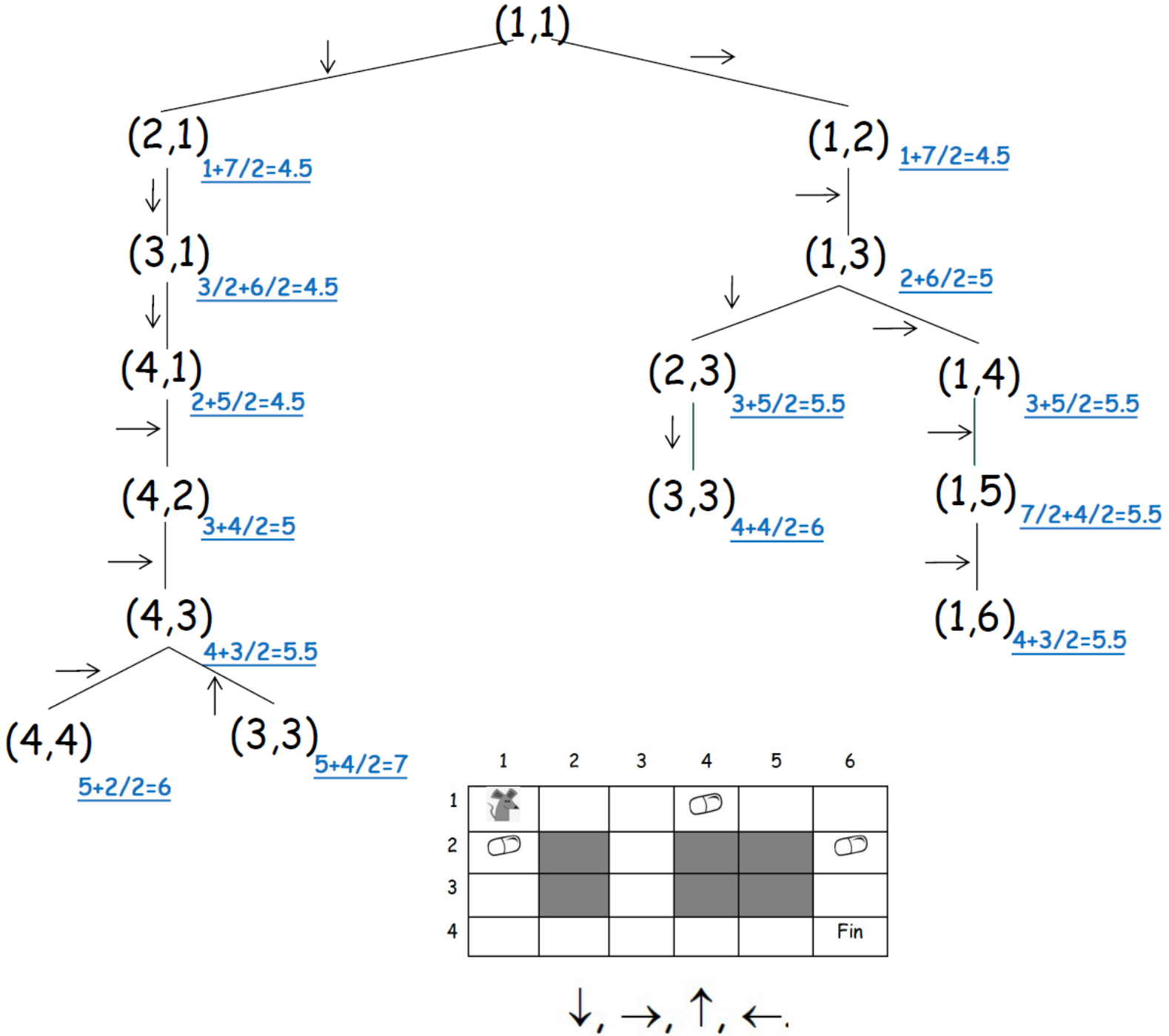

(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(4,3)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.

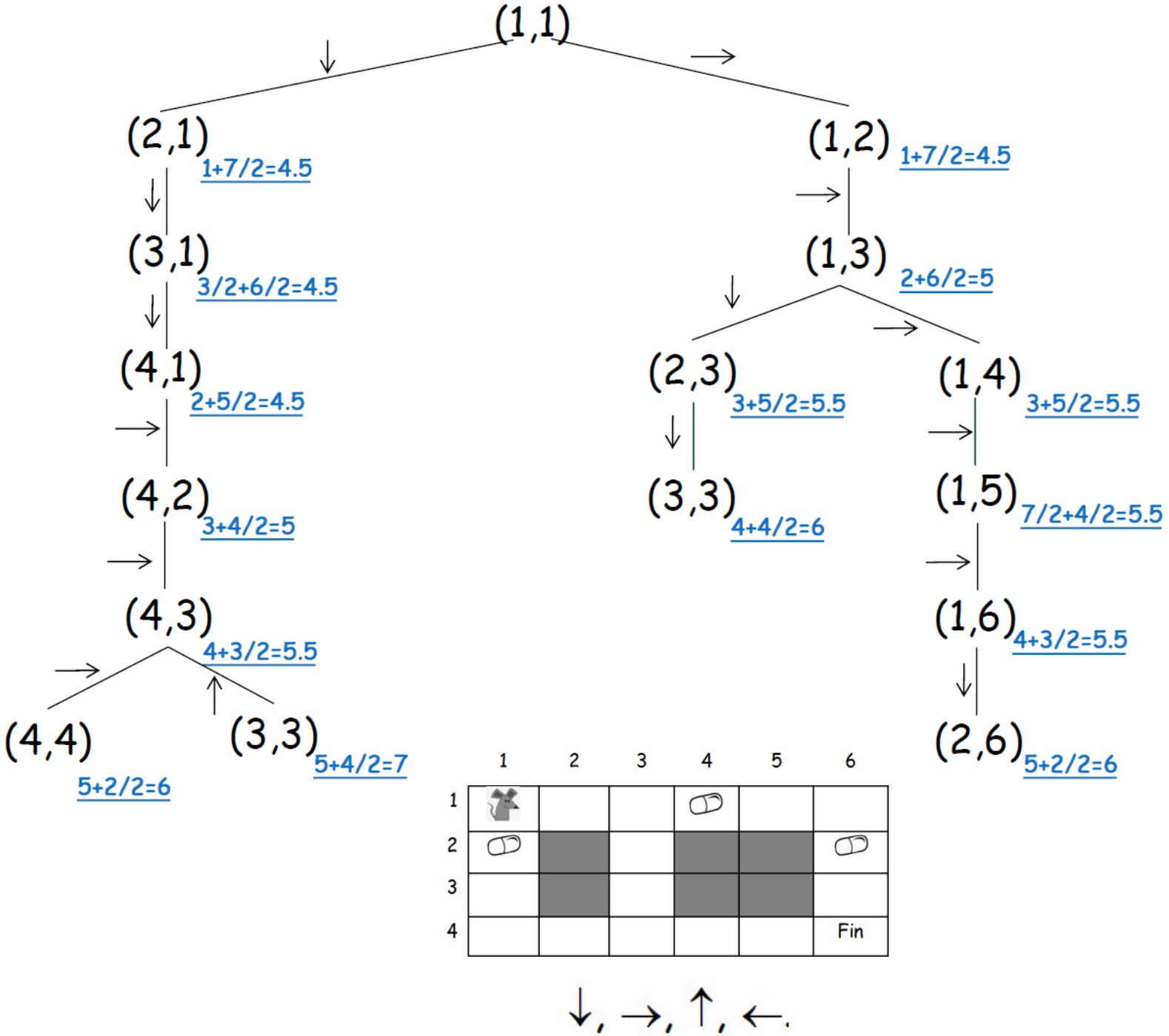

(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(4,3)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.

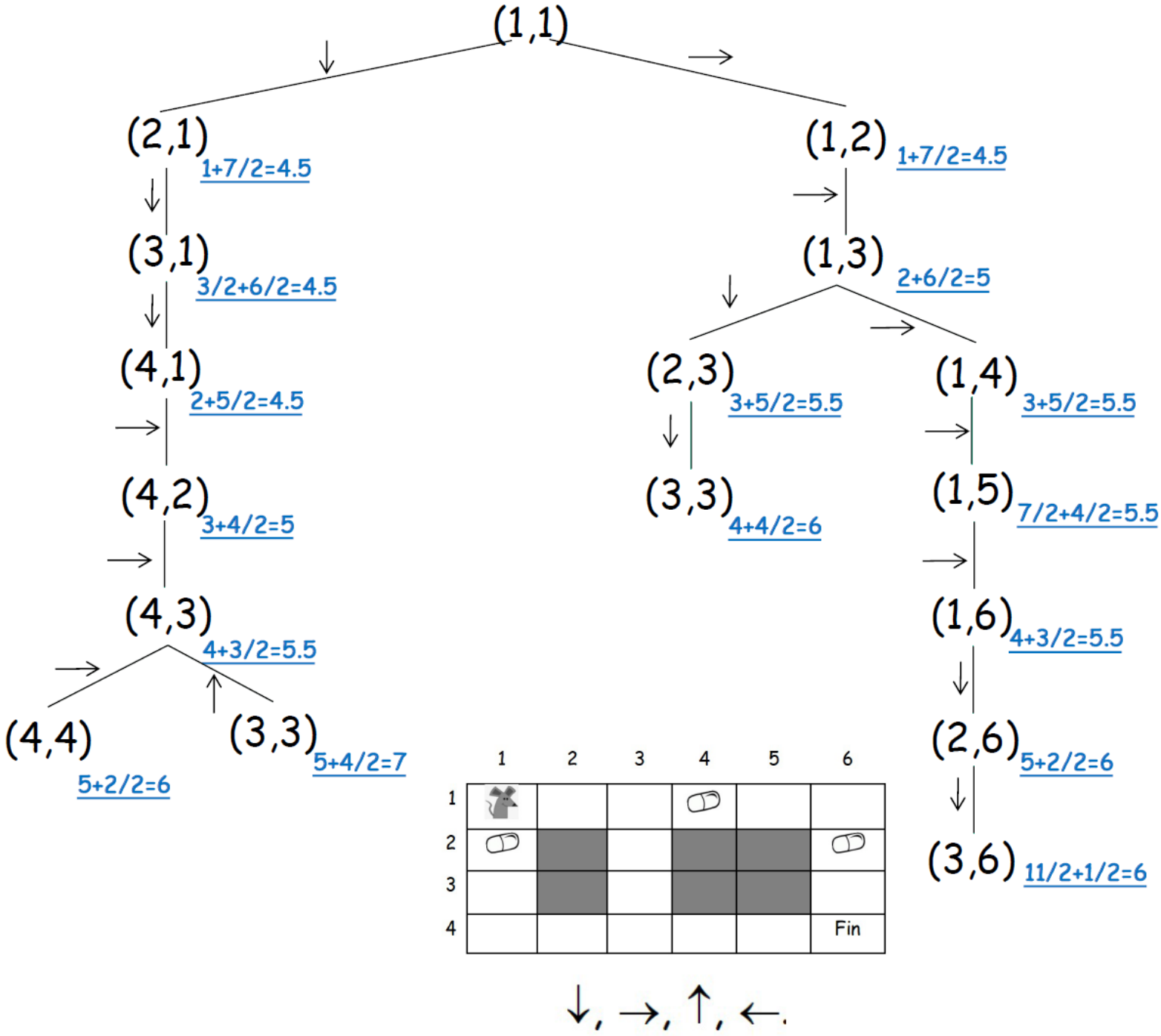

(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(1,5)
7/2+4/2=5.5
(4,3)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7

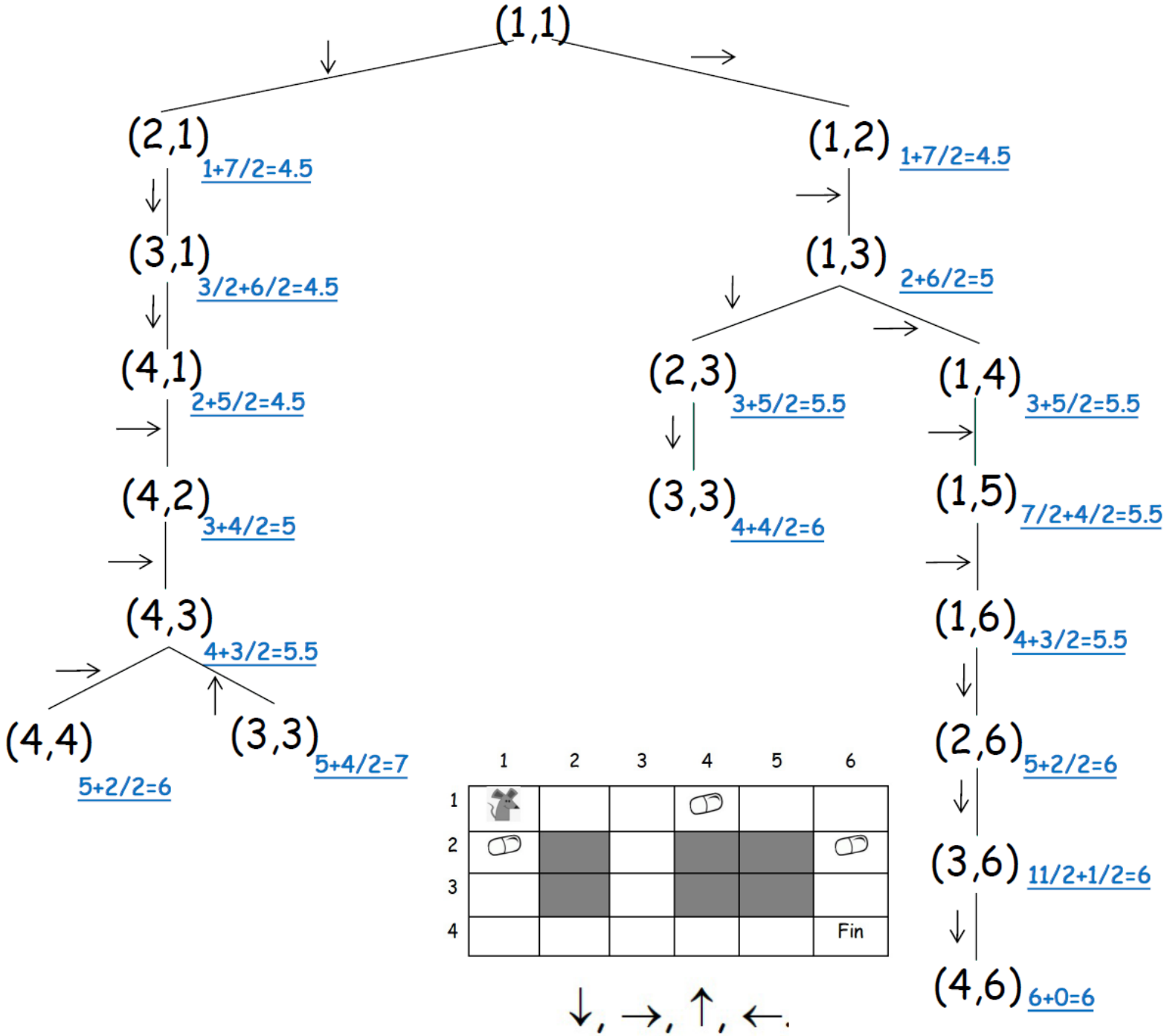

1
2
3
4
5
6
1
2
3
4
Fin

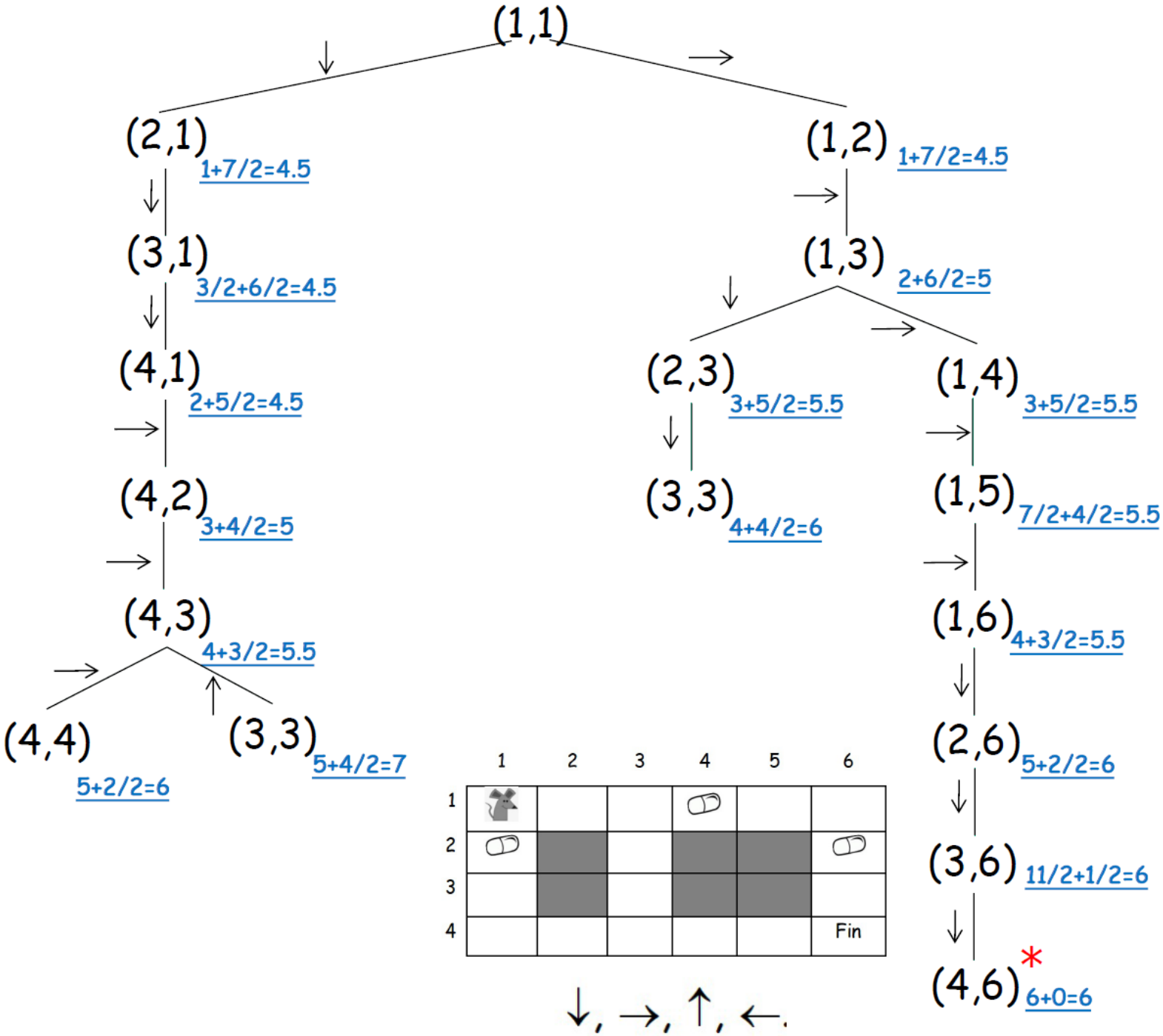
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(1,5)
7/2+4/2=5.5
(4,3)
4+3/2=5.5
(1,6)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(1,5)
7/2+4/2=5.5
(4,3)
4+3/2=5.5
(1,6)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
(2,6)
5+2/2=6
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(1,5)
7/2+4/2=5.5
(4,3)
4+3/2=5.5
(1,6)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
(2,6)
5+2/2=6
(3,6)
11/2+1/2=6
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(1,5)
7/2+4/2=5.5
(4,3)
4+3/2=5.5
(1,6)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
(2,6)
5+2/2=6
(3,6)
11/2+1/2=6
(4,6)
6+0=6
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.


(1,1)
(2,1)
1+7/2=4.5
(1,2)
1+7/2=4.5
(3,1)
3/2+6/2=4.5
(1,3)
2+6/2=5
(4,1)
2+5/2=4.5
(2,3)
3+5/2=5.5
(1,4)
3+5/2=5.5
(4,2)
3+4/2=5
(3,3)
4+4/2=6
(1,5)
7/2+4/2=5.5
(4,3)
4+3/2=5.5
(1,6)
4+3/2=5.5
(4,4)
5+2/2=6
(3,3)
5+4/2=7
(2,6)
5+2/2=6
(3,6)
11/2+1/2=6
(4,6)
6+0=6
1 2 3 4 5 6
1
2
3
4
Fin
↓, →, ↑, ←.